

FIFTH
EDITION

A Coursebook on
**Aphasia and Other
Neurogenic
Language Disorders**

M. N. Hegde

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Preface to the Fifth Edition



The first edition of this book on aphasia and other neurogenic communication disorders was one of the first to be developed as a coursebook — a new format for teaching and learning. Instructors and students alike have liked this format because it makes both teaching and learning a bit easier and more efficient than the traditional lecture method and classroom note-taking.

The coursebook format was originally designed as an effective instructional package that reduced the amount of note-taking needed. The coursebook also reduced the variability in the accuracy and completeness of notes students take. The use of this type of book promotes class discussion as the students are not as busy taking notes in the class as they otherwise would be.

My students who have used this book for the first time in a course I taught on aphasia and related communication disorders gave me much positive feedback. Students have found the coursebook a valuable means of integrating textbook information with class notes they take. They have a single source of information that is easier to study than a text on the (literally) one hand and the class notebook on the other. They have expressed a preference for this type of book for all of their courses. I would like to thank them for their comments and suggestions. I also welcome feedback from instructors who use this coursebook.

This fifth edition is not only a coursebook but also a stand-alone textbook on courses in aphasia and other neurologically based communication disorders, whether offered at the undergraduate or graduate level. This new textbook retains the coursebook format so the instructors can make lecture notes and students can write down instructor's notes on the right half of each page of the text.

The text offers a comprehensive description and critical review of basic and applied information on aphasia, right hemisphere disorder, traumatic brain injury, and dementia — the four major language and communication disorders associated with neurological pathologies. The relationship between the brain and language, major features of aphasia and other disorders, their assessment, and treatment have been described in streamlined and clinician-friendly language. Critical review of theories, assessment, and treatment research helps speech-language pathologists distinguish what is valid from the questionable in the professional and scientific literature. All assessment and treatment chapters give an outline of comprehensive and practical procedures, integrating current practices the clinicians might readily use.

New to the Fifth Edition

The new edition is a thorough revision of both the structure and content of the book. Several new topics and areas of research in understanding and treating the disorders are reviewed and critically evaluated:

- Part I of the book has been restructured under the heading “Brain and Language” to describe the neuroanatomical bases of language and language impairments associated with neuropathological variables. Research on brain and language continues to use newer technologies producing a dynamic field of theories and applications. The chapter on neurodiagnostics has been revised and expanded to include a variety of surgical, radiological, and imaging procedures that help understand the relation between the brain structure and function and their involvement in language production and comprehension.
- Part II has been reorganized into three chapters on aphasia. [Chapter 3](#) offers a comprehensive review of aphasia prevalence, definition, and classification. Newer perspectives on intraoperative cortical brain mapping and alternative classification of aphasia based on recent research on the dual-stream hypotheses related to brain and language have been reviewed with critical evaluation to help the clinicians.

Chapters 4 and 5 on assessment and treatment of aphasia offer a more comprehensive review of established and newer procedures. The chapter on treatment is expanded to include telerehab, drug treatment, brain stimulation, and technologically based interventions. All major language intervention techniques are reviewed with outlined recommendations for clinicians.

- Part III on the right hemisphere disorder consists of two chapters that have been revised to reflect current terminology, research, and clinical practice issues. Sections on neglect, deficit awareness, social communication, and abstract language intervention have been updated.
- Part IV offers the most recent research on traumatic brain injury rehabilitation. Research on teaching compensatory strategies, group therapy, and community reentry has been updated with clinical recommendations.
- Part V consists of two chapters on dementia with new information on changing incidence and prevalence patterns of dementia, infectious and rapidly progressive dementias, frontotemporal dementias, primary progressive aphasia, and HIV-associated neurocognitive disorders. An expanded clinical management and caregiver programs are included.

I am gratified that many instructors and students have found this innovative tool of teaching and learning useful and I am thankful for the many positive comments I have received from instructors across the country.

I am very pleased that this new edition is being published by Plural Publishing, the leading publishing house in communication disorders and related medical specialties. I would like to thank the editorial department headed by an able editor, Valerie Johns. Her kind and continuous help is greatly appreciated. I would also like to thank Lori Asbury, Production Manager and Jessica Bristow, Production Editor for their excellent help throughout the development and production process. Angie Singh, the president of Plural, has been a friend and supporter for decades. I thank her and all her efficient staff at Plural Publishing.

About the Author

M. N. (Giri) Hegde, PhD, Professor Emeritus of Communication Sciences and Disorders at California State University–Fresno, holds a master's degree in experimental psychology from the University of Mysore, India; a post-master's diploma in medical (clinical) psychology from Bangalore University, India; and a doctoral degree in speech-language pathology from Southern Illinois University Carbondale.

Dr. Hegde is a specialist in fluency disorders, language disorders, research methods, and treatment procedures in communicative disorders. He has made numerous presentations to national and international audiences on various basic and applied topics in communicative disorders and experimental and applied behavior analysis. With his deep and wide scholarship, Dr. Hegde has authored several highly regarded and widely used scientific and professional books, including *Clinical Research in Communicative Disorders* (with P. Salvatore), *Introduction to Communicative Disorders, A Coursebook on Aphasia and Other Neurogenic Language Disorders*, *Hegde's PocketGuide to Communication Disorders*, *Hegde's PocketGuide to Treatment in Speech-Language Pathology*, *Hegde's PocketGuide to Assessment in Speech-Language Pathology*, *Assessment of Communication Disorders in Children* (with F. Pomaville), *Assessment of Communication Disorders in Adults* (with D. Freed), *Clinical Methods and Practicum in Speech-Language Pathology* (with K. Kuyunjian), and several others. He has served on the editorial boards of scientific and professional journals and continues to serve as an editorial consultant to the *Journal of Fluency Disorders*.

Dr. Hegde is a recipient of various honors, including the Outstanding Professor Award from California State University–Fresno, CSU-Fresno Provost's Recognition for Outstanding Scholarship and Publication, Distinguished Alumnus Award from the Southern Illinois University

Department of Communication Sciences and Disorders, and Outstanding Professional Achievement Award from District 5 of the California Speech-Language-Hearing Association. Dr. Hegde is a Fellow of the American Speech-Language-Hearing Association.



PART I

BRAIN AND LANGUAGE



1

OVERVIEW OF NEUROANATOMY AND NEUROPHYSIOLOGY

Chapter Outline

- The Human Nervous System
- Neurons and Nerves
- The Peripheral Nervous System (PNS)
- The Central Nervous System (CNS)
- The Protective Layers of the Brain
- Cerebral Blood Supply
- References

Learning Objectives

After reading this chapter, the reader will:

- Define the terms of anatomic orientation
- Describe the central and the peripheral nervous system
- Describe and distinguish the various types of neural cells
- Distinguish cranial and spinal nerves both structurally and functionally
- Describe the surface structure of the cerebral cortex

- Describe the cerebral locations of language, motor, and auditory areas
- Describe the relevant subcortical structures of the brain
- Distinguish the different protective layers of the brain
- Describe the arteries that supply the particular regions of the brain

PRINTED NOTES

Communication disorders described in this book follow recent brain trauma or diseases that develop over time. These disorders include aphasia, dementia, traumatic brain injury, and right hemisphere disorder. Therefore, to appreciate the physical basis of these communication disorders, a basic understanding of the central nervous system is essential. Also essential is a familiarity with the medical neurodiagnostic methods. Speech-language pathologists (SLPs) need this knowledge to effectively assess and treat communication disorders associated with brain diseases or trauma. The knowledge will also promote more effective interprofessional collaboration with other professionals, including the primary care physicians and neurologists, who are also involved in the care of persons with communication disorders associated with brain diseases or trauma.

This chapter offers only a simplified overview of neuroanatomy and neurophysiology. It is expected that students will have had detailed coursework in anatomy, physiology, and neurology of speech and language. Students are referred to other sources for details (Andreatta, 2020; Bear et al., 2020; Bhatnagar, 2017; Blake & Hoepner, 2023; Haines & Mihailoff, 2017; Seikel et al., 2020; Vanderah & Gould, 2020).

The Human Nervous System

Anatomically, the nervous system is divided into the central nervous system (CNS) and the peripheral nervous system (PNS). The CNS includes the cerebrum, diencephalon, midbrain, cerebellum, pons, medulla, and the spinal cord.

The PNS is a network of cranial and spinal nerves that connects the brain with the peripheral organs. An overview of these structures follows. [Figure 1–1](#)

shows the main division of the nervous system.

Neurons (nerve cells) and nerves (neural fibers) are the basic building blocks of the nervous system. We begin with these two structures.

Neurons and Nerves

Neurons send and receive chemically mediated electrical signals, often referred to as *neural messages* or *information*. Neurons receive such signals from other neurons through their dendrites and transmit the same through their axons to other neurons.

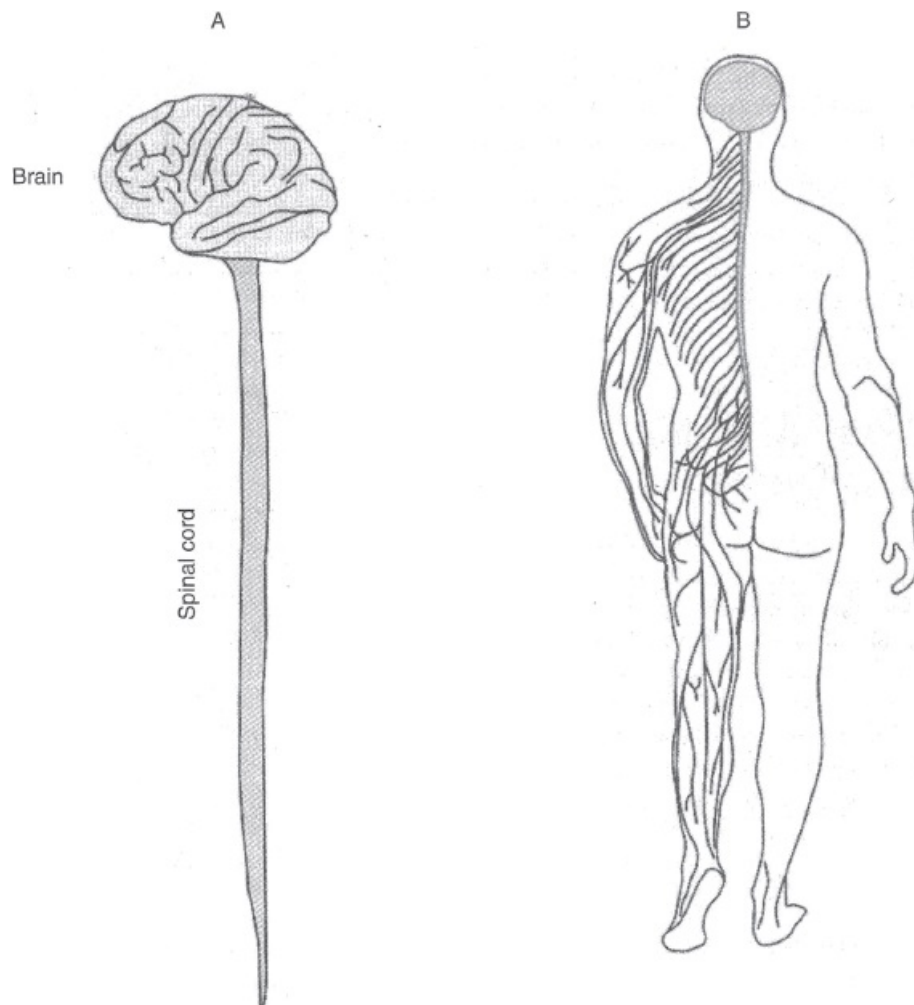


Figure 1-1. The two main divisions of the nervous system: A. The central nervous system. B. The peripheral nervous system, which includes the cranial and spinal nerves.

The size and shape of neurons vary a great deal as there are a variety of neurons. [Figure 1-2](#) shows a typical neuron and some common variations. Neurons have three parts: soma (cell body), dendrites, and axon. Collectively, dendrites and axons are called neurites.

The **cell body** (*soma*; *somata*, plural) contains the *nucleus* and *cytoplasm*. The cytoplasm is enclosed with neuronal membrane, which regulates the flow of chemicals (and chemically transmitted signals) to and from the cell.

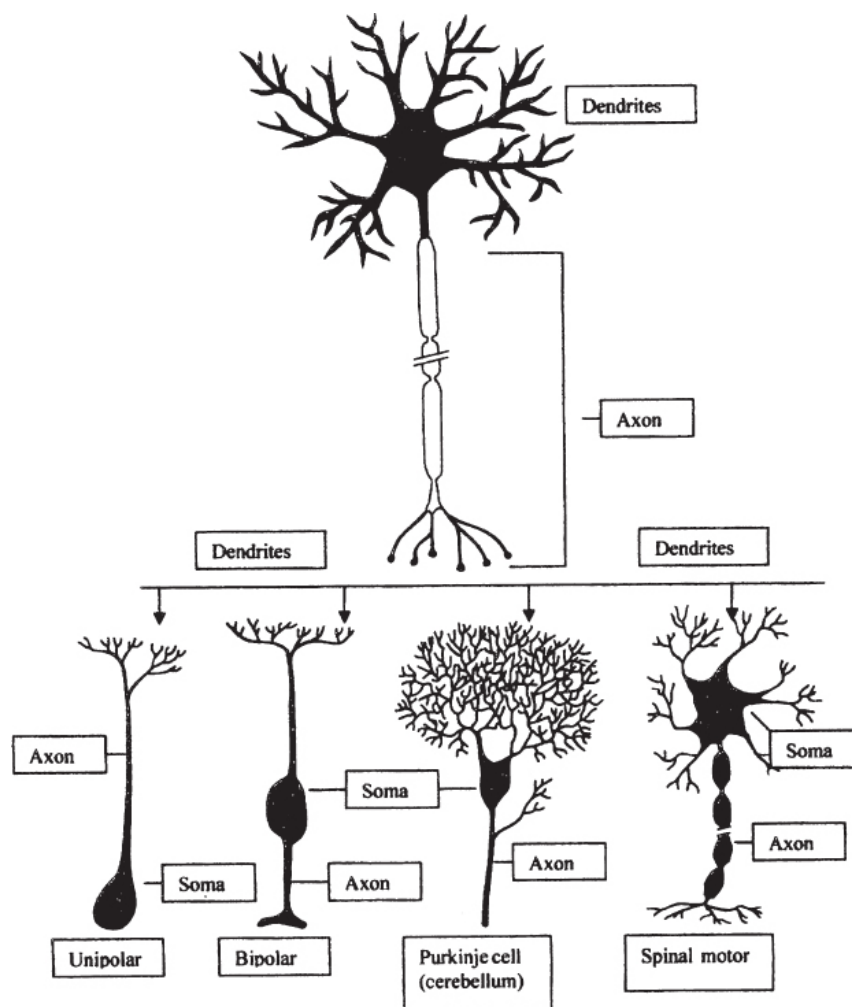


Figure 1-2. Varieties of neurons.

Cytoplasm includes everything the soma contains except for the nucleus. The term *protoplasm* includes both the nucleus and cytoplasm. The chromosomes, which contain the genetic material called deoxyribonucleic acid (DNA), are contained within the nucleus.

Dendrites and axons are two kinds of nerve fibers. **Dendrites** (*tree* in Greek) are short, unmyelinated fibers that extend from the cell body. Each neuron may contain several dendrites. The cell is *unipolar* when only one dendrite extends from the cell body, it is *bipolar* when two dendrites extend from it, and it is *multipolar* when several dendrites extend from it.

Dendrites receive information from axons of other cells. Hence, they are classified as *afferent* (receptive). Dendrites transmit information thus received to the cell body.

While cells may have multiple dendrites, they only have a single axon (*axle* in Greek). An **axon** is the nerve fiber that is longer than dendrites and originates from soma at a cone-shaped region called the *axon hillock*. Looking like a string of sausages, the tubular axon is thinner at its origin and becomes thicker as it becomes longer. Axons are not structurally intact; there are structural gaps in them (Vanderah & Gould, 2020).

Axons may be myelinated or unmyelinated, although the thicker, longer axons are *myelinated* (wrapped in a thin layer of white, protective, and fatty material called **myelin**). The myelin sheath helps increase the speed of conduction of neural impulses across the axon. Some demyelinating diseases degrade or destroy this protective sheath, resulting in slow or disrupted conduction of neural messages, causing motor, visual, sensory, or cognitive dysfunctions. A prominent demyelinating disease is *multiple sclerosis*.

An axon finally branches into many small filaments called **telodendria**. The terminal points of an axon are capped with small structures called **terminal buttons**. The terminal buttons are the functional contact points between neurons. Axons are functionally *efferent* in that they transmit signals away from the cell body to other neurons or muscles.

To communicate with each other, an axon of one neuron makes contact with another neuron through multiple methods. An axon may make a contact (*synapse*) with (a) a dendrite of another neuron (*axodendritic synapse*), (b) the

soma of another neuron (*axosomatic synapse*), (c) another axon of a neuron (*axoaxonic synapse*), or (d) an *effector cell* such as a skeletal muscle. The point at which two neurons come in contact with each other is called a **synapse**, a neural junction. The several smaller branches of an axon at its end form synaptic terminals that come in contact with a dendrite, the soma, or an axon of another neuron or motor muscle. The contact, however, is not physically continuous as the neural junction has a small space or a gap, called the **synaptic cleft**. The synaptic cleft is a structural gap, not a chemical gap. The cleft is filled with a specialized form of protein.

Once injured or destroyed by a disease, axons of the central nervous system do not regenerate to recover their function. Dead neural cells leave a clump of debris. Consequently, in cases of severe brain injury or tissue loss due to disease, the cognitive functions may not be recovered fully.

The **glial cells** (also known as glia, meaning glue) are nonneural cells of the nervous system. Also called neuroglia, these nonneural cells provide a structural framework for the neural cells. Glia outnumber neural cells by at least 5:1 and occupy half the volume of the brain. Glial cells do not receive or send messages. Instead, they help maintain neural metabolic activity and regulate concentrations of neurotransmitters.

Normal transmission of signals within the neural network is essential for both typical communication and cognitive functions. Various neuropathologies disrupt the efficient and speedy neural transmission of messages, giving rise to several communication problems and general behavioral deficits.

Because of the synaptic cleft and other structural gaps in the human nervous system, neural transmission of information is unlike a pure physical transmission over uninterrupted lines of fibers. Neural transmission across the synaptic gaps is an electrochemical process of information exchange.

Neurotransmitters

Chemical compounds known as **neurotransmitters** contained within the axon terminal buttons establish contact between two cells by diffusing themselves across the synaptic space. This diffused neurotransmitter becomes bound to receptors in the postsynaptic membrane (neuron receiving the impulse). The diffused neurotransmitter may either excite or inhibit the next neuron in a chain.

Among the several varieties of neurotransmitter chemical compounds, amino acids and amines are critically important. **Amino acids** include gamma-aminobutyric acid (GABA), glutamate, and glycine. **Amines** include acetylcholine (Ach), dopamine, epinephrine, histamine, norepinephrine, and serotonin. Each neuron specializes in releasing a certain kind of neurotransmitter. For instance, most CNS synapses release amino acids (glutamate, GABA, and glycine) to accomplish fast transmission of messages. At neuromuscular junctions (and throughout the peripheral nervous system), Ach mediates fast message transmission; its action in the CNS is somewhat slower. Ach primarily mediates voluntary movement. All varieties of neurotransmitters are involved in slower forms of message transmission.

Various neurological diseases and trauma affect neurotransmission, giving rise to many neurological and psychiatric disorders. They are involved in both motor and cognitive deficits associated with neurological disorders. Dopamine deficiency due to the death of dopamine producing cells in the brainstem are related to the tremor and poor control of movements in Parkinson's disease. Alzheimer's disease may be associated with deficiencies in Ach in the hippocampus and frontal cortex. On the other hand, excessive production of dopamine in the forebrain may be involved in schizophrenia.

GABA, an amino acid and a derivative of glutamate, is the main neurotransmitter in the brain. While glutamate is excitatory, GABA is inhibitory. Death or degradation of neurons that produce GABA in basal ganglia (especially in the caudate and putamen) causes the involuntary movements of Huntington's chorea.

Afferent and Efferent Nerves

Bundles of axons, dendrites, or both, specializing in certain functions, are called **nerves**. Two major kinds of nerves of interest to us are motor nerves and sensory nerves.

Motor nerves, or **motor neurons**, cause muscle contractions (movement) or glandular secretions. Because the motor nerves transmit impulses away from the central nervous system, they also are called **efferent nerves**. Efferent nerves terminate at an effector, which leads to glandular secretion or muscle movement. Motor or efferent neurons have their origins in the brain and terminate in a gland or muscle. In essence, the brain issues commands to the glands and muscles through the efferent neurons.

Sensory nerves carry sensory impulses from the peripheral sense organs toward the brain. Because they carry information toward the center, sensory nerves also are known as **afferent nerves**. Afferent nerves help the brain gather information about the body's internal (visceral) environment as well as its external, physical environment. Sensory organs (e.g., eyes and ears) send the information they gather to the brain through afferent nerves. Information thus brought to the brain may generate efferent impulses that lead to some action dictated by the received sensory information.

The Peripheral Nervous System (PNS)

The **peripheral nervous system** includes all of the nervous system except for the brain and the spinal cord. The system is *peripheral* because it connects the brain (and the spinal cord) with the structures that lie away from the brain. Cranial nerves, spinal nerves, and parts of the autonomic (visceral) system constitute the peripheral nervous system.

Cranial Nerves

Cranial nerves emerge from the brainstem; they are attached to the base of the brain (see a later section on brainstem for a graphic representation of the brainstem and the cranial nerves that emerge from it). Cranial nerves innervate

(supply and stimulate) larynx, tongue, pharynx, and muscles of the face, neck, and head, and are therefore important for phonation, voice, and speech. There are 12 pairs of cranial nerves:

- I. Olfactory: Sensory; sense of smell
- II. Optic: Sensory; vision
- III. Oculomotor: Motor; eye movements
- IV. Trochlear: Motor; eye movement
- V. Trigeminal: Mixed; face, sensory; jaw, motor
- VI. Abducens: Motor; eye movement
- VII. Facial: Mixed; tongue and nasopharynx, sensory; face, motor
- VIII. Vestibular acoustic: Sensory; hearing and balance
- IX. Glossopharyngeal: Mixed; tongue and pharynx, sensory; pharynx, motor
- X. Vagus: Mixed; larynx, pharynx, cardiac, and gastrointestinal systems, both sensory and motor
- XI. Accessory: Motor; shoulder, arm, throat movements
- XII. Hypoglossal: Motor; mostly tongue movements

Some cranial nerves carry sensory or motor messages while mixed nerves carry both. Cranial nerves are not critical in language comprehension and formulation. Those cranial nerves that innervate face, tongue, larynx, pharynx, and so forth are important for speech production. Cranial nerve damage is associated with dysarthria, a motor speech disorder. Some persons who have aphasia and those with traumatic brain injury may have associated speech production problems.

Spinal Nerves

Spinal nerves arise from the spinal cord. Of the 31 pairs, 8 pairs are cervical (C1 through C8), 12 pairs are thoracic (T1 through T12), 5 pairs are lumbar (L1 through L5), 5 pairs are sacral (S1 through S5), and a single pair is coccygeal. The nerves are named after the specific portions of the spinal cord to

which they are attached (cervical, thoracic, lumbar, sacral, and coccygeal). Being mixed in function, spinal nerves carry sensory information to the CNS and motor information to the skeletal muscles. These spinal nerves are shown in [Figure 1–4](#), in a later section on *Spinal Cord*.

Spinal nerves are not significant for language comprehension and formulation. However, cervical and thoracic spinal nerves are important in speech production because they control respiration, essential for normal speech production. Damage to these nerves, especially the phrenic (C1–C5) nerves, weakens the diaphragm and tends to cause dysarthria, a motor speech disorder (Freed, 2020).

The Central Nervous System (CNS)

The **central nervous system** includes the brain and the spinal cord, both encased in bone. The brain is encased within the cranial structure (skull) and the spinal cord is encased within the bony vertebral column (spinal column).

Structures within the CNS that are especially relevant to a study of neurogenic communication disorders include the (1) the spinal cord, (2) the brainstem, (3), the cerebellum, and (4) the cerebrum. [Figure 1–3](#) shows a gross representation of the major parts of the CNS.

Spinal Cord

The spinal cord is a part of the CNS, although spinal nerves are a part of the PNS. The **spinal cord**, a bundle of nerve fibers within the vertebral column, is a cylindrical and caudal (lower) continuation of the medulla oblongata, which is the lowest structure of the brainstem. [Figure 1–4](#) shows a lateral view of the central nervous system with the spinal cord.

As noted earlier, the spinal nerves carry motor impulses to various organs, including muscles and glands, and carry such sensory impulses as pain, touch,

temperature, and pressure from peripheral organs to the brain. The spinal cord is also responsible for all reflexive activity.

The Brainstem

The upper end of the spinal cord is continuous with the lower end of the brainstem. The **brainstem**, illustrated in [Figure 1-5](#), includes **medulla**, **pons**, and **midbrain**. The medulla is the lowest (caudal) part of the brainstem and the midbrain is the highest; the pons is in the middle.

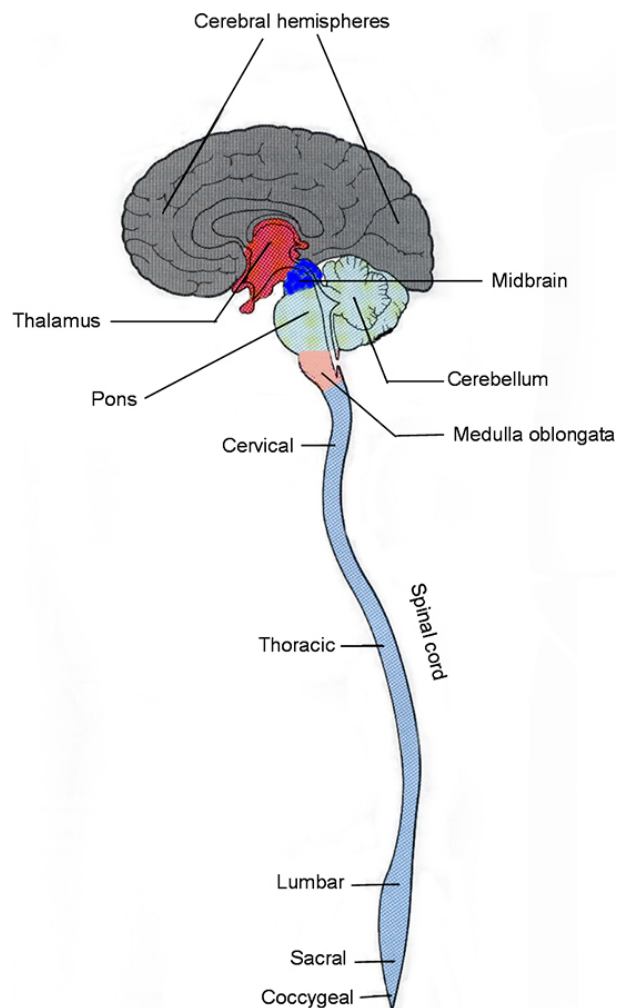


Figure 1-3. The central nervous system.

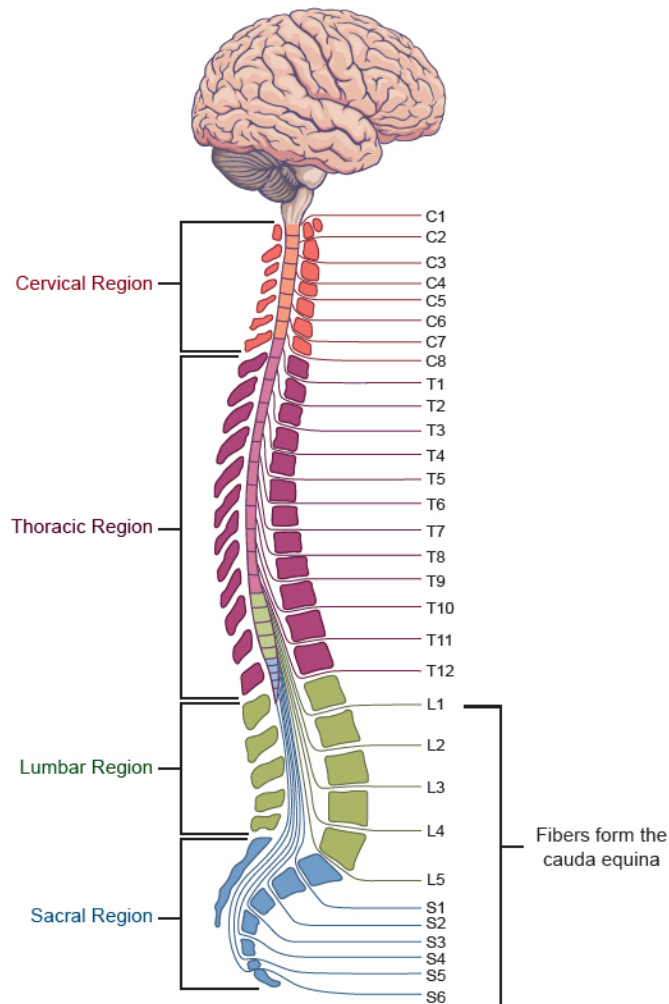


Figure 1-4. The spinal cord. From *Neuroscience Fundamentals for Communication Sciences and Disorders* by R. D. Andreatta, 2020 (p. 100, Figure 4-14). Plural Publishing. Reprinted with permission.

The **medulla** (also called medulla oblongata and myelencephalon) is the upward extension of the spinal cord as it passes through the foramen magnum at the base of the skull. The spinal cord ends at the beginning of the medulla. The medulla contains all the fibers that originate in the brain and cerebellum and move down to form the spinal cord. The medulla includes several centers that control such autonomic functions as digestion, breathing (affecting speech), blood pressure, and heart rate.

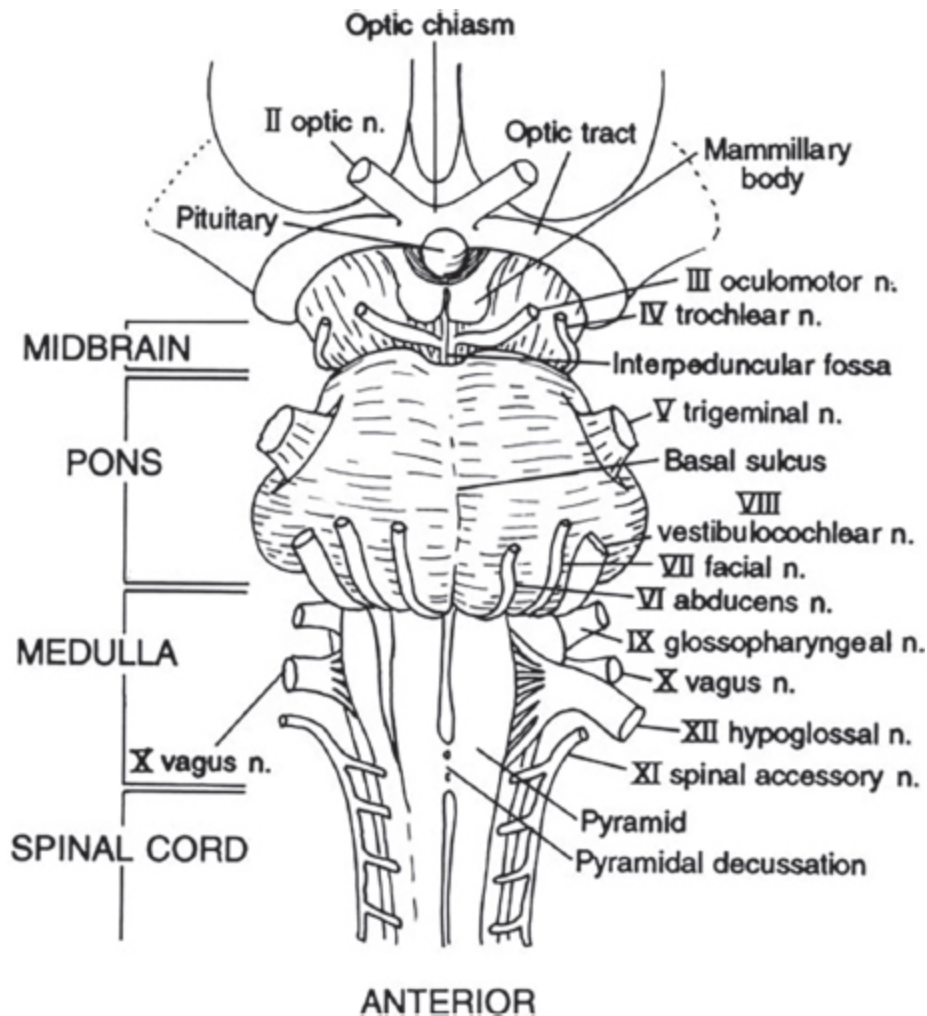


Figure 1-5. The brainstem. Used with permission from J. Anthony Seikel.

The **pons**, a part of metencephalon (which includes the cerebellum, not a part of the brainstem), is a bridge to the hemispheres of the cerebellum (not cerebrum). The pons is concerned with hearing and balance; some cranial nerves (trigeminal, facial) originate here. The pons transmits information relative to movement from the cerebral hemispheres to the cerebellum.

The **midbrain**, also called mesencephalon, lies above the pons. Cranial nerves III and IV originate in the midbrain. The midbrain controls many sensory and motor functions including eye movements, postural reflexes, and coordination of visual and auditory reflexes. Structures at the midbrain-diencephalon junction provide transition from midbrain to thalamus.

Cerebellum

The **cerebellum**, illustrated in [Figure 1–6](#), is the major portion of the hindbrain, covering most of the posterior (back) and lower side of the brainstem. Located at the base of the brain (dorsal to the pons and medulla), the cerebellum may be aptly named the “little brain” because it contains as many neurons as the rest of the CNS. Like the cerebrum, cerebellum has two hemispheres called cerebellar hemispheres (not to be confused with cerebral hemispheres).

The cerebellum has fissures like the cortex and its surface is full of small ridges called *folia*; it has gray matter on the surface and white matter on the inside. The core white matter contains important nerve fibers that connect to the cerebellar cortex.

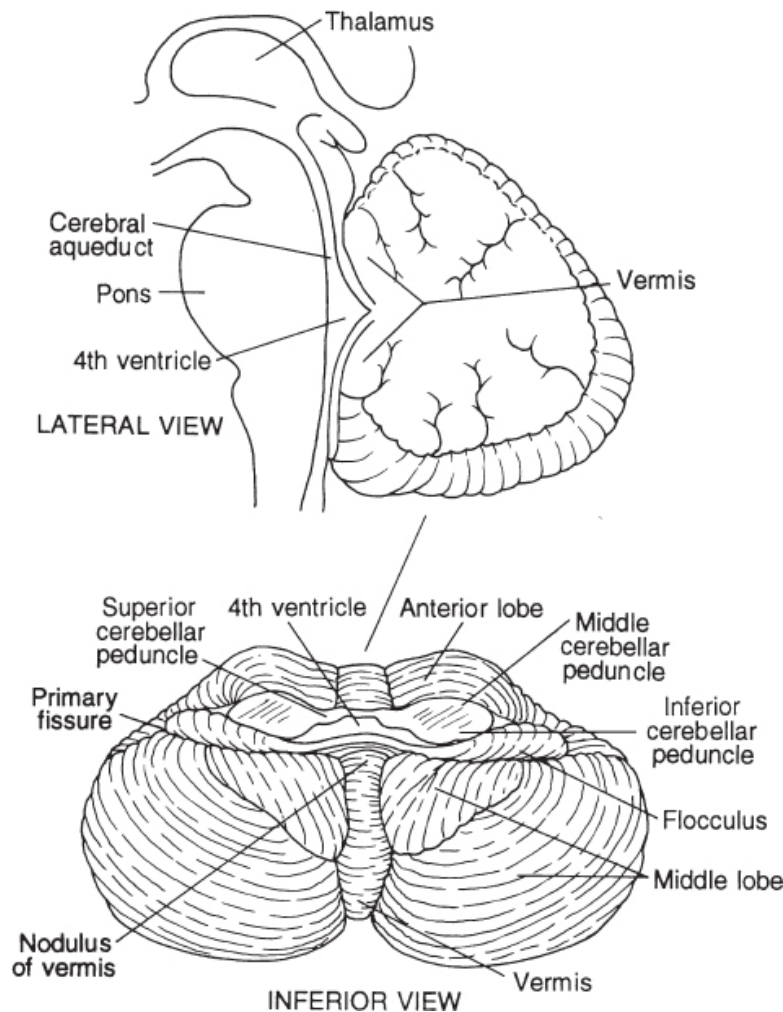


Figure 1-6. The cerebellum. Used with permission from J. Anthony Seikel.

The cerebellum is a part of the motor system. It coordinates and modulates the force and range of body movement. Although it does not initiate movement, the cerebellum coordinates actions to produce smooth and rhythmic movements, including those involved in speech. Damage to the cerebellum is associated with ataxic dysarthria, a form of motor speech disorder, with dominant articulatory and prosodic problems. Cerebellar damage also results in uncoordinated movements, impaired postural control, and abnormal voluntary movements.

Diencephalon

The **diencephalon** is a structure in between the brainstem and the cerebral hemispheres, located above the midbrain and below the cerebral cortex. Within it is the third ventricle, a narrow and tall space filled with the cerebrospinal fluid. The **thalamus**, **epithalamus**, **subthalamus**, and **hypothalamus** are its four major structures. Of these, the thalamus, shown in [Figure 1–7](#), is especially relevant for our purposes.

The thalamus integrates sensory experiences and relays them to cortical areas; it also receives information about motor impulses from the basal ganglia and cerebellum and relays these to motor areas of the cortex. The thalamus plays a major role in maintaining consciousness and alertness.

Basal Ganglia

Found deep within the brain, the **basal ganglia** are structures located near the thalamus. These subcortical structures are important in understanding certain motor speech disorders. Dementia associated with Parkinson's disease, for example, is related (among other neuropathological conditions) to neurochemical disorders of basal ganglia.

Basal ganglia is a collection of structures that include the **caudate nucleus**, the **putamen**, and the **globus pallidus**, all interconnected. In addition, substantia nigra and subthalamic nucleus — essentially brainstem structures — have a functional relationship with basal ganglia. See [Figure 1–7](#) for the structures of basal ganglia and their spatial relation to the thalamus.

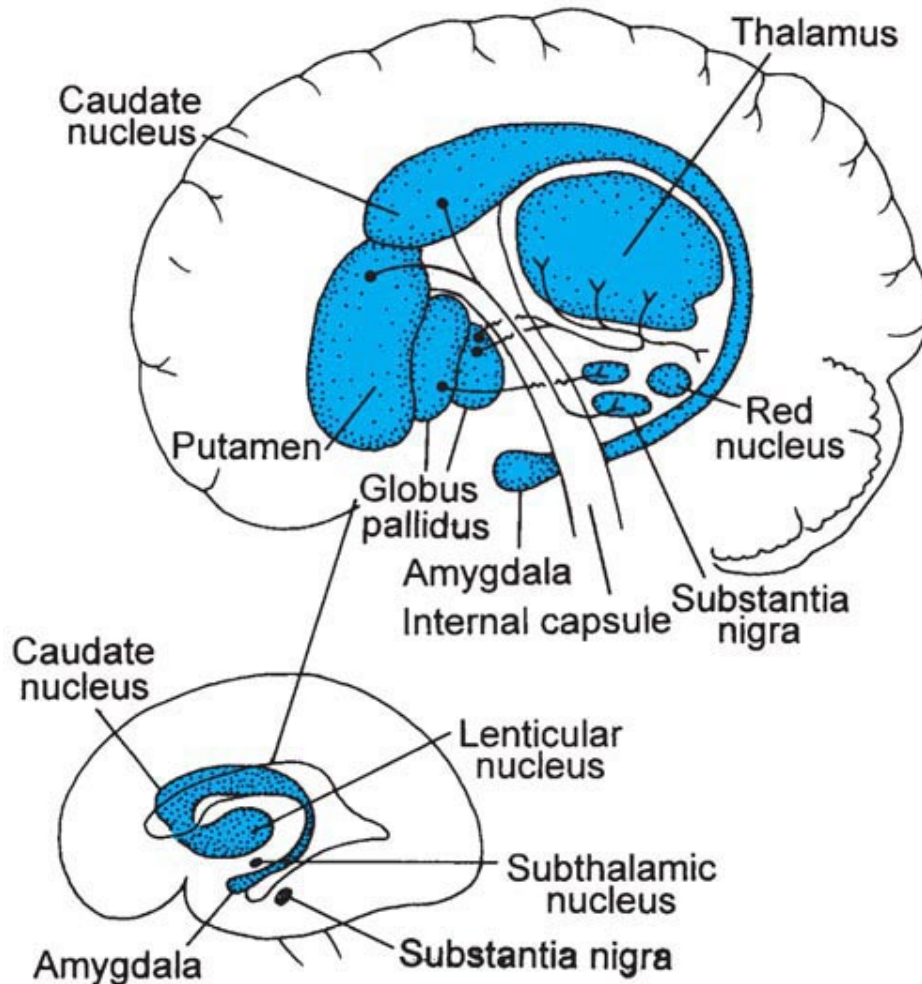


Figure 1-7. The thalamus in relation to basal ganglia. Used with permission from J. Anthony Seikel.

Basal ganglia receive input mostly from the frontal lobe and send information back to the higher centers in the brain through the thalamus. Basal ganglia play an important role in modulating movement because they produce critical neurotransmitters including dopamine, GABA, and acetylcholine. Being inhibitory in their effects, these neurotransmitters help regulate and control movements.

Damaged or diseased basal ganglia deplete neurotransmitters. Both Parkinson's disease and Huntington's disease are associated with damaged or diseased basal ganglia. Eventual dementia is likely. Huntington's disease is associated with hyperkinetic dysarthria. Impaired functioning of basal ganglia may cause **dyskinesias** (such involuntary movements as tremors), **hypokinesia**

(restricted range of movement), and **bradykinesia** (slowness of movement). Unusual postures, impaired muscle tone, and hypokinetic dysarthria also may result.

White Matter Connecting Fibers in the CNS

An extensive network of fibers found deep within the white matters of the brain connects cerebral structures to each other. Also, any information coming into or going out of the brain has to pass through the subcortical white matter. Various bundles of connecting fibers keep the information flowing throughout the brain so that information about sensory stimuli received, and actions planned and performed, are coordinated within various brain structures (Chang et al., 2015).

The connecting fibers in the white matter core of the brain are classified as *projection fibers*, *commissural fibers*, and *association fibers*.

Projection fibers are a band of fibers that transmit sensory information to the brain and motor information to the muscles and glands. These fibers form a connection between the cortical structures at the top and the brainstem and spinal cord at the lower level. The projection fibers are compact and concentrated as they pass through the internal capsule near the brainstem. As they move toward the upper regions of the brain, projection fibers fan out in a structure called the corona radiata, through which information is transmitted to other regions of the brain. [Figure 1–8](#) illustrates the corona radiata.

Efferent projection fibers pass through the thalamus and basal ganglia. These fibers transmit motor commands to muscles and glands. Afferent projection fibers transmit sensory information from the peripheral sense organs to the brain.

Association fibers connect the different areas within a hemisphere of the brain. They may be short or long. Shorter association fibers connect adjacent structures, and the longer fibers connect more distant areas within the same hemisphere.

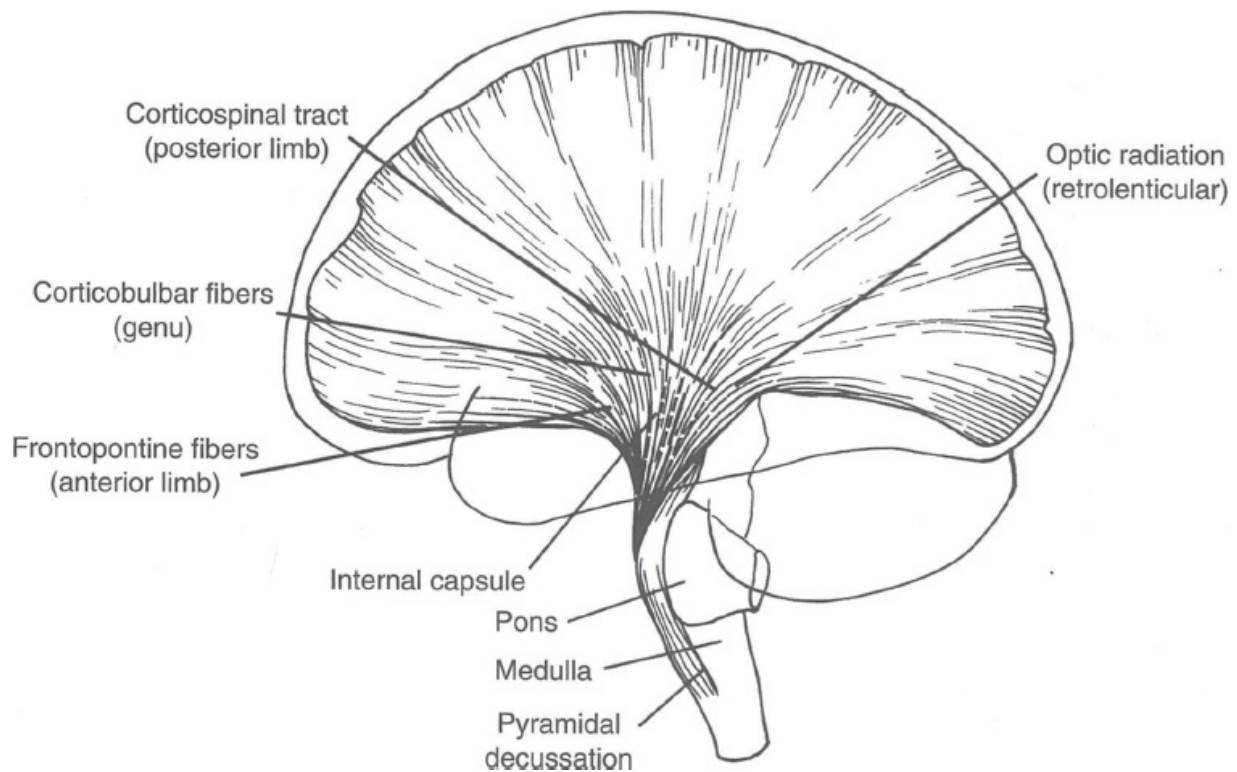


Figure 1-8. Corona radiata. Used with permission from J. Anthony Seikel.

Longer association fibers that connect more distant parts of a hemisphere are called **fasciculi** (plural; *fasciculus*, singular). Most of these association fibers are important for speech and language functions. More than localized damage to such areas as Broca's and Wernicke's, damage to these long connecting fibers in the white matter is now considered essential for the varied and serious speech language symptoms in brain trauma or disease (Chang et al., 2015; Fridriksson et al., 2018; Hickok, 2022). Some of the most important fasciculi include the following:

- The **superior longitudinal fasciculus** connects the frontal, parietal, occipital, and temporal lobes (Andreatta, 2020; Bhatnagar, 2017). A curved portion of the superior longitudinal that connects Broca's motor speech area with Wernicke's speech comprehension area is often described as the **arcuate fasciculus**. These fibers are important for language acquisition, speech sound and syntactic production, and language comprehension.

- The **inferior longitudinal fasciculus**, lying under the sylvian fissure, connects temporal and occipital lobes. This bundle is important for word meaning and comprehension.
- The **uncinate fasciculus** connects frontal and temporal lobes. In addition to aiding word meaning and comprehension, these fibers are also important for syntactic skills.

The four major fasciculi that connect the different parts of the brain are shown in [Figure 1-9](#). Some of the other fasciculi not shown in [Figure 1-9](#) include the *extreme capsule* (frontal- temporal connections), the *middle longitudinal* (temporal-parietal connections), and the *inferior frontal occipital* (frontal-occipital connection).

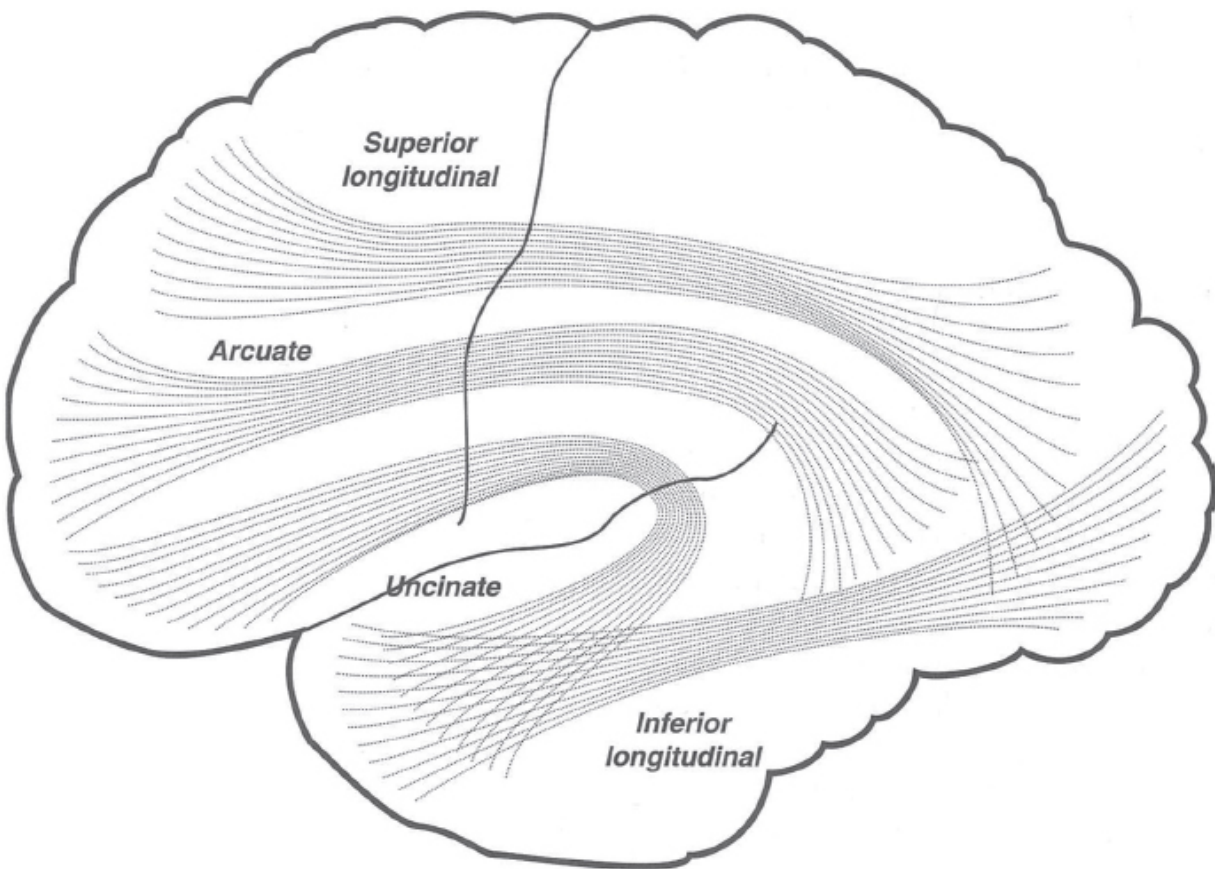


Figure 1-9. Four major fasciculi that connect the different parts of the same hemisphere. From *The Speech Sciences* by R. D. Kent (1998). Used with permission from Raymond D. Kent.

The **commissural fibers** connect the corresponding areas of the two hemispheres that are divided by the median longitudinal fissure. Therefore, they are interhemispheric connectors, whereas association fibers are intrahemispheric. The **corpus callosum** is the most important of the commissural fibers. Found at the base of the hemispheres and forming the roof of most of the lateral ventricles, the corpus callosum is a broad, thick, and the largest band of 300 to 400 million fibers that connect the two hemispheres. Damage to the corpus callosum, or surgical transection as a treatment for intractable epilepsy, disconnects the two hemispheres. The resulting problems in movement, reading, and naming are known as disconnection syndromes. The corpus callosum is shown in [Figure 1-10](#).

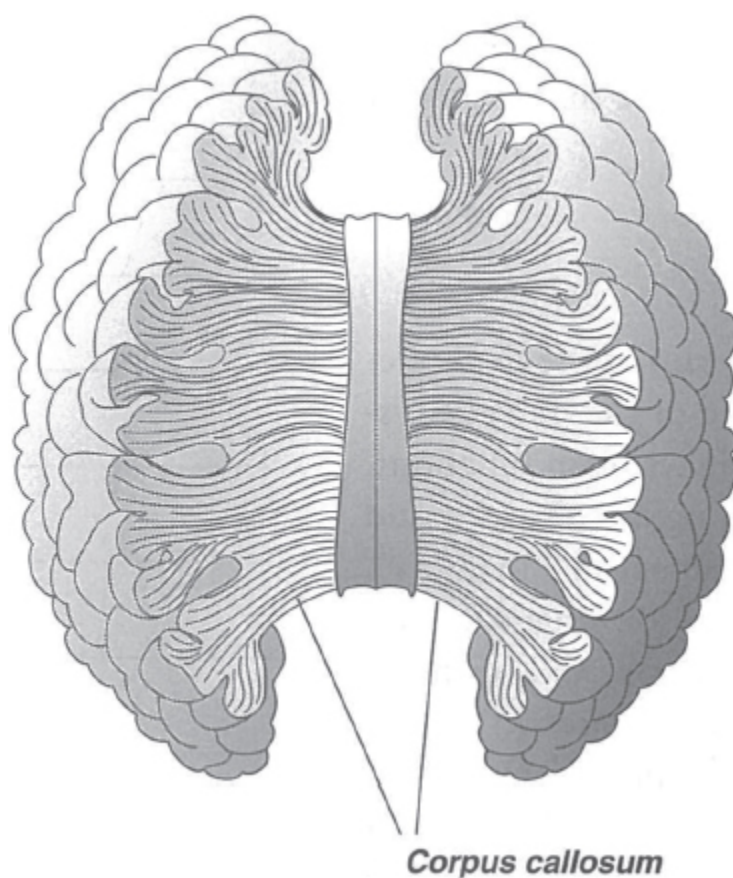


Figure 1-10. Corpus callosum that connects the two hemispheres. From *The Speech Sciences* by R. D. Kent (1998). Used with permission from Raymond D. Kent.

Cerebral Cortex

The **cerebral cortex** or **cerebrum** is the integrative and executive structure of the nervous system. It is responsible for all higher brain functions including everyday thinking; logical, abstract, and mathematical reasoning; remembering (memory); speech and language comprehension and production; creative activities, including all artistic and scientific achievements; and judgment, emotional experience, and so forth. It is a thin (2 to 4 mm) outer layer of the brain. Some basic facts about the human brain include the following:

- The term *cortex* means the bark of a tree because the cortex does resemble bark. The brain contains about 10 to 14 billion neurons. Two thirds of this massive number is buried in the various fissures.
- An average human brain weighs about 3 pounds.
- Most regions of the cortex are structured in six layers; the outermost surface is gray.
- A thick band of long commissural fibers connects the two hemispheres
- Shorter association fibers connect the regions within a hemisphere.
- The surface is a folded mass of rolling “hills” and “valleys.”

The hills, folds, convolutions, or elevated masses are called **gyri** (*gyrus*, singular). The various gyri help distinguish different landmarks on the cortex. The grooves, or valleys, are called **sulci** (*sulcus*, singular) or **fissures**. [Figure 1–11](#) shows the major gyri and sulci on the right hemisphere.

The **longitudinal cerebral fissure** separates the left and the right hemisphere of the cerebrum. This fissure is a groove in the middle of the brain, running from front to back. One can see this fissure in a superior view of the cortex (viewed from above).

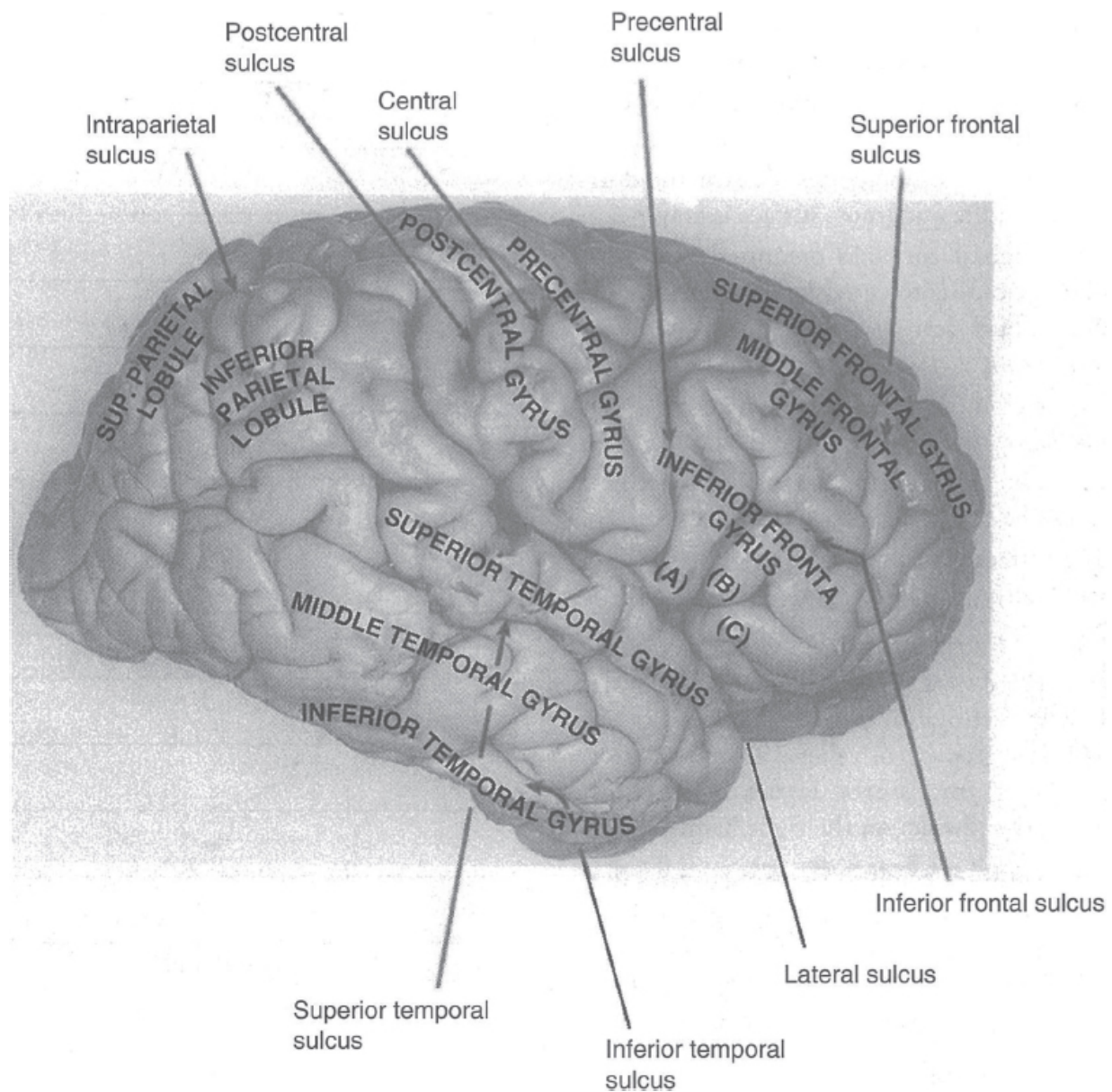


Figure 1-11. Major gyri and sulci of the brain.

At the top of the brain, a major fissure that runs laterally (from one side to the other), downward, and forward is called the **fissure of Rolando** or the **central sulcus**. The central sulcus arbitrarily and roughly divides the anterior half of the brain from the posterior half of the brain. In a top view of the brain, one can see four portions of the brain divided by the longitudinal fissure and the central sulcus, as shown in [Figure 1-12](#).

At the lower (inferior) frontal lobe, a deep fissure begins and moves laterally and upward. This is **the lateral sulcus** (also known as the *sylvian fissure*), shown in [Figure 1-11](#). The regions surrounding the sylvian fissure are especially involved in speech, language, and hearing.

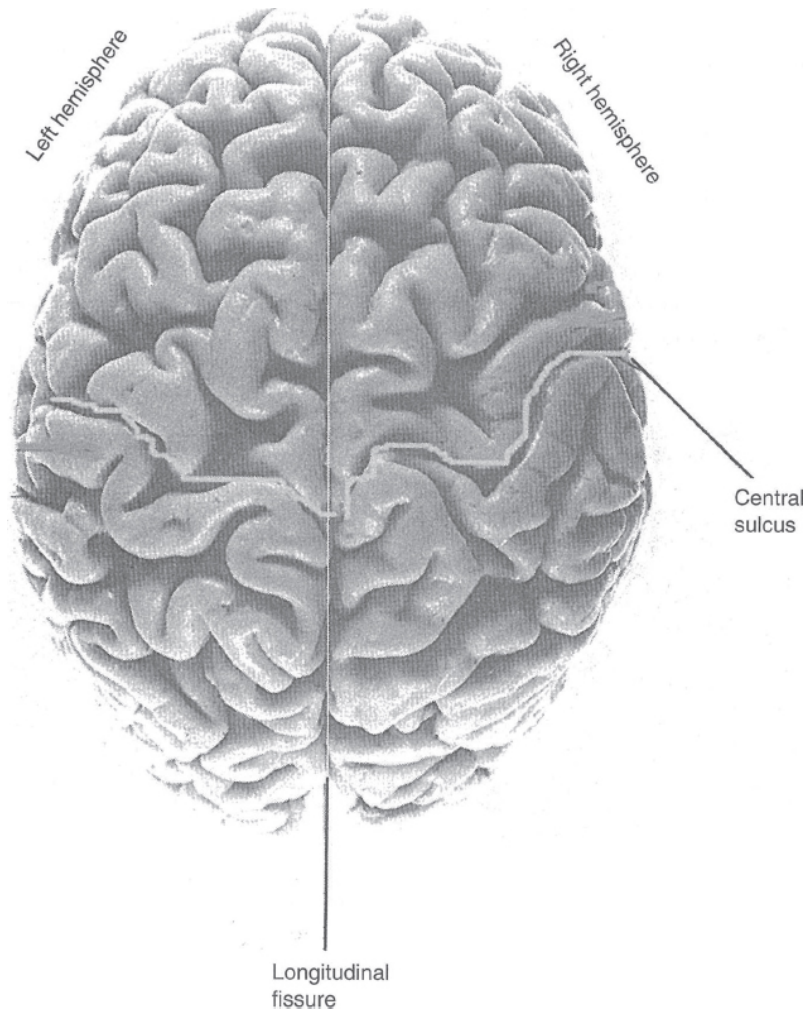


Figure 1-12. The longitudinal fissure and the central sulcus.

Cortical Lobes

Each cerebral hemisphere is divided into four primary lobes: the frontal, parietal, occipital, and temporal. (A secondary lobe is described as the insular

lobe, a small oval region buried in the depths of the lateral fissure.) Each lobe mediates a predominant function.

The cortical lobes are named after the bones of the skull (cranium). The cranium has two parts: the **neurocranium** (or just *cranium*) that houses and protects the brain and the **viscerocranium** that includes the bones of the face. During the embryonic period, the separately developing cranial bones fuse. A bare skull will reveal the sutures of the fused bones. Each cerebral lobe is located behind the bone it is named after: the frontal (unpaired), parietal (paired), occipital (unpaired), and temporal (paired). [Figure 1–13](#) shows the four cranium bones (along with the bones of the face). Notice the suture lines of the cranium bones. [Figure 1–14](#) shows the four lobes of the brain.

The Frontal Lobe

The frontal lobe, situated in front of the central fissure and above the lateral fissure, is important for speech and language. This lobe is all of the cortex in front of the central fissure; being the largest of the lobes, it makes up about one third of the surface area of the cortex. The frontal lobe is identified in [Figure 1–14](#).

In front of the central fissure in the frontal lobe, there is a large bulge called the precentral gyrus (see [Figure 1–11](#)), which constitutes the major portion of the **primary motor cortex** (motor strip) controlling voluntary movements of skeletal muscles on the opposite (contralateral) side of the body. The precentral gyrus controls the movements of the arm, hand, fingers, face, lips, leg, and foot. The mouth, lips, hands, and fingers are controlled by relatively large areas, suggesting the importance of fine motor movements (including speech) at the human level. The motor strip controls movements through a neural pathway called the **pyramidal system**. The motor impulses are modified by another system, called the **extrapyramidal system**, with a complex set of indirect relay stations.

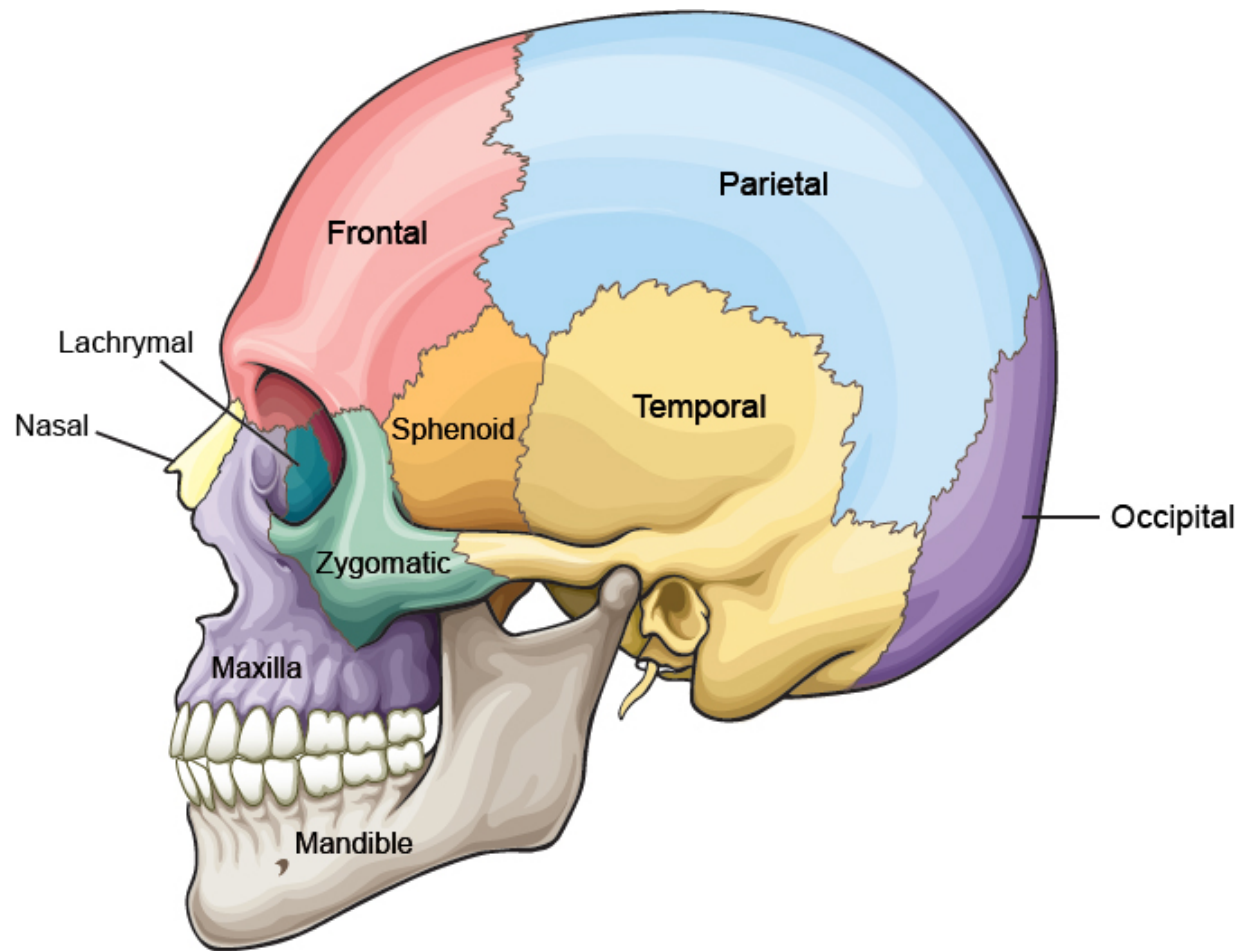


Figure 1-13. The bones of the cranium and face. From *Neuroscience Fundamentals for Communication Sciences and Disorders* by R. D. Andreatta (p. 99, Figure 4-13A). Copyright 2020 by Plural Publishing. Reprinted with permission.

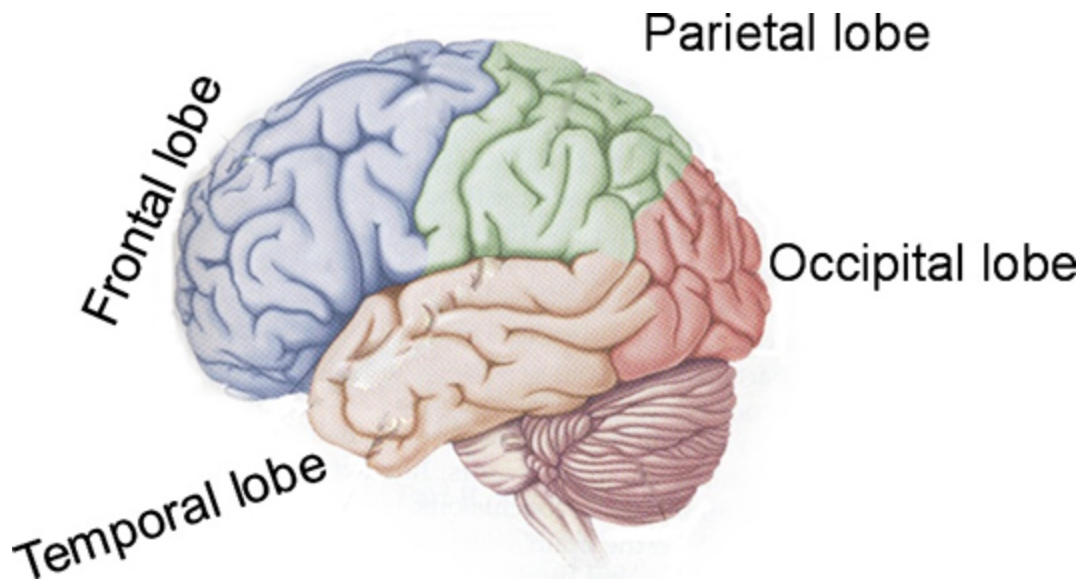


Figure 1-14. The four lobes of the brain.

Just anterior to the primary motor cortex lie the premotor area and the supplementary motor area (Brodmann's area 6), specializing in complex and fine motor skills. The rest of the large anterior section of the frontal lobe is called the prefrontal cortex (Brodmann's areas 10–12), thought to be involved in such intellectual tasks as thinking, reasoning, decision-making, and planning.

Whereas the precentral gyrus is vertical, the other important gyri in the frontal lobe are horizontal. Three such gyri are the superior frontal gyrus, middle frontal gyrus, and the inferior frontal gyrus. Of these three, the inferior frontal gyrus (see [Figure 1-11](#)) in the dominant hemisphere (the left in most individuals) is of importance for speech. In the left, lower, and posterior portion of the inferior frontal gyrus is **Broca's area**, also known as the motor speech area, which controls the motor movements of speech production. Broca's area corresponds to Brodmann's area 44 (and may extend to parts of area 45). In [Chapter 4](#), we learn more about Broca's aphasia, which is thought to be caused by a lesion or lesions in Broca's and surrounding areas, including the underlying white matter fibers.

The Temporal Lobe

The temporal lobe (see [Figure 1–14](#)) is the lowest one third of the brain. The lobe lies behind the temples (temporal bone), starts at the lateral fissure, and ends at the imaginary boundary of the anterior portion of the occipital lobe. The temporal lobe, looking like an extended thumb, is under the frontal and parietal lobes and in front of the occipital lobe. There is no natural demarcation between the temporal lobe and the occipital lobe. The lateral fissure gives a partial demarcation between the temporal and the parietal lobes.

The three important gyri of functional significance found on the lateral surface of the temporal lobe are the (1) superior (upper) temporal gyrus, (2) the middle temporal gyrus, and (3) the inferior (lower) temporal gyrus (all shown in [Figure 1–11](#)). The superior temporal gyrus lies parallel to the lateral fissure; posteriorly, it turns upward in the region of the angular gyrus of the parietal lobe. The middle temporal gyrus is located between the other two temporal gyri. The superior temporal sulcus separates the superior and the middle temporal gyri. The inferior temporal gyrus lies at the undersurface of the middle temporal gyrus.

The **primary auditory cortex** (area), concerned with hearing, is at the border of the superior temporal gyrus and the lateral fissure. The primary auditory area, also known as Heschl's gyrus, corresponds to Brodmann's areas 41 and 42 and is buried within the lateral sulcus. ([Figure 1–18](#) in a later section shows the location of the primary auditory area.) This area is found in both hemispheres, but in a majority of people, it is typically larger in the left hemisphere. This suggests the left dominance of receptive language functions. An area adjacent to the primary auditory area is called the **secondary auditory area** (cortex) and is the same as Brod-mann's area 42.

Located posterior to the primary auditory area, **Wernicke's area** is the posterior two thirds of the superior temporal gyrus in the left (or dominant) hemisphere. ([Figure 1–18](#) in a later section shows the location of Wernicke's area.) Wernicke's area in the left hemisphere may be up to seven times larger than the corresponding area in the right hemisphere. It is close to the intersection of the temporal, parietal, and occipital lobes. As noted earlier, the superior longitudinal fasciculus or its part, the arcuate fasciculus, connects Broca's motor speech area with Wernicke's speech comprehension area. A lesion in the posterior portion of the superior temporal gyrus causes Wernicke's

aphasia with significant language comprehension problems but fluent if meaningless speech. We learn more about Wernicke's aphasia in [Chapter 3](#).

The temporal lobe is the site of auditory reception and interpretation and auditory-visual association. The lobe also is the site of receptive language functions (comprehension of spoken and written material), processing of semantic and syntactic information, and comprehension of nonverbal sounds and music.

The Occipital Lobe

The smallest of the lobes, the occipital is behind the parietal lobe, forming a caudal end of the hemisphere. Its boundaries are mostly imaginary (no sulci separate it from the other lobes) (see [Figure 1–14](#)). The major structures of the occipital lobe are the primary visual cortex and the secondary visual cortex.

The visual cortex in each hemisphere receives visual information from the contralateral visual fields of each eye. The right and the left visual cortices respond to stimulation in the left and the right visual fields, respectively.

The Parietal Lobe

The parietal lobe lies just behind the central fissure and just above the lateral fissure. It is at the back of the frontal lobe, above the temporal lobe, and in front of the occipital lobe. It has an imaginary posterior boundary that separates it from the occipital lobe. Just behind the central sulcus is the postcentral gyrus, better known as the **sensory cortex** (sensory strip). This narrow vertical strip of the parietal lobe is the **primary sensory area**. This area controls and integrates somesthetic sensation (perception of touch, pressure, position in space, and body awareness) (see [Figure 1–14](#)).

The parietal lobe, especially in the nondominant (often the right) hemisphere, also controls understanding of spatial relations and selective attention. Damage to the right parietal lobe may cause **left neglect**, a

condition in which the patient is unaware of objects and persons on the left side. Left neglect and related problems associated with right hemisphere disorder are described in Chapters 6 and 7.

The supramarginal gyrus and the angular gyrus in the parietal lobe are important for language functions. The **supramarginal gyrus** lies above the lateral fissure in the inferior portion of the parietal lobe and its posterior portion curves around the latter fissure. Damage to this gyrus may cause agraphia (writing problems) or a type of aphasia called *conduction aphasia* (described in [Chapter 3](#)).

The **angular gyrus** lies posterior to the supramarginal gyrus. Damage to the angular gyrus can cause naming, reading, and writing difficulties and, in some cases, a type of aphasia known as *transcortical sensory aphasia* (described in [Chapter 3](#)).

The Protective Layers of the Brain

The brain is well protected by a layer of skin, bones of the skull, and three layers of tissue called the meninges (from the Greek word *meninx* for membrane). Cerebrospinal fluid adds a cushioning structure to this protective system. The spinal cord is protected by the bony vertebral column. Because of these protective structures, the brain retains its shape and stays stable when the body and the head moves around. The shape of a brain removed from the skull would soon be distorted because of gravitational forces. The three meninges of the brain are shown in [Figure 1–15](#).

The outermost membrane, called the **dura mater** (“tough mother”), is a tough and thick, leather-like membrane with one side adhering to the skull and the other side to the arachnoid, the second meninx (Greek; singular of *meninges*). The dense connective tissue of the dura mater covers the brain and the spinal cord and protects them from external shock. The dura has two layers, the outer periosteal layer adhering to the skull and an inner meningeal layer attached to the arachnoid.

The middle **arachnoid** (“spider web”) is a thin, semitransparent, nonvascular, delicate, and web-like membrane with the dura mater above and the pia mater below. The arachnoid adheres to the dura without space in between. It is separated from the pia, however, with a subarachnoid space. The cerebrospinal fluid fills the subarachnoid space.

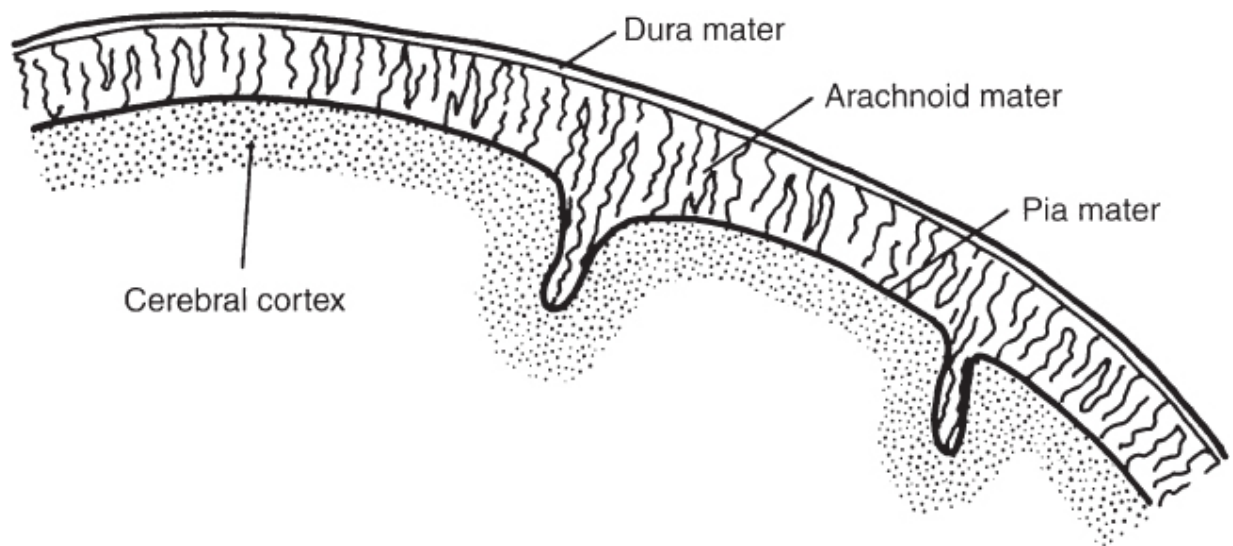


Figure 1-15. The meninges of the brain. Used with permission from J. Anthony Seikel.

The innermost **pia mater** (“tender mother”) is also a thin, delicate, and transparent membrane, which adheres to the brain surface, closely following the surface sulci and gyri. The pia also covers the blood vessels. Many blood vessels enter the brain by penetrating the pia mater.

The meninges may be damaged in cases of traumatic brain injury, tumors that grow within them (called meningiomas), and infections and inflammations (e.g., viral or bacterial meningitis). Brain injuries may cause subdural hemorrhages. Traumatic brain injury and associated clinical and communication problems are described in [Chapter 8](#).

Cerebral Blood Supply

Many neurological and communication problems arise from disrupted blood supply to the brain. In adults, disrupted blood flow within the cortical regions is a major cause of aphasia. Aphasic symptoms take different forms depending on the brain region in which the blood supply is disrupted. Vascular pathologies may either restrict blood flow to a region or result in hemorrhage within the brain.

The brain is a big eater.

- Although it is only 2% of a typical human adult's body weight, the brain receives 17% of the body's blood.
- It consumes 25% of the body's oxygen.

The brain is a vulnerable organ.

- It depends on the supply of blood for nourishment and normal functioning.
- A person loses consciousness within 10 seconds of blood interruption.
- Electrical activity of the brain ceases after 20 seconds of blood interruption.
- Four to 6 minutes of interruption in blood supply may permanently damage the brain.

The Vascular Network

A vascular network of arteries and veins distributes blood throughout the body. While a network of arteries carry oxygenated blood to the brain from the heart, a similar network of veins carries deoxygenated blood from the brain back to the heart for reoxygenation through the lungs. The main structures of the cerebral blood supply are shown in [Figure 1–16](#).

Arteries may be large or small. **Arterioles** are smaller branches of larger arteries. Arterioles branch out into minute hair-like network called *capillaries* from which oxygen and other nutrients in the blood filter into tissue. Metabolic waste from the tissue filters back into the capillaries, which connect

through capillaries to **venues**, which are minute branches of veins. Veins thus receive blood low in oxygen and carry it to the heart.

The Aorta

The main artery of the heart, the **aorta**, carries the blood from the left ventricle to all parts of the body except the lungs. Just above the heart, the aortic arch divides into four branches: the two common carotid arteries (one on each side of the neck) and the two subclavian arteries (one on each side of the neck).

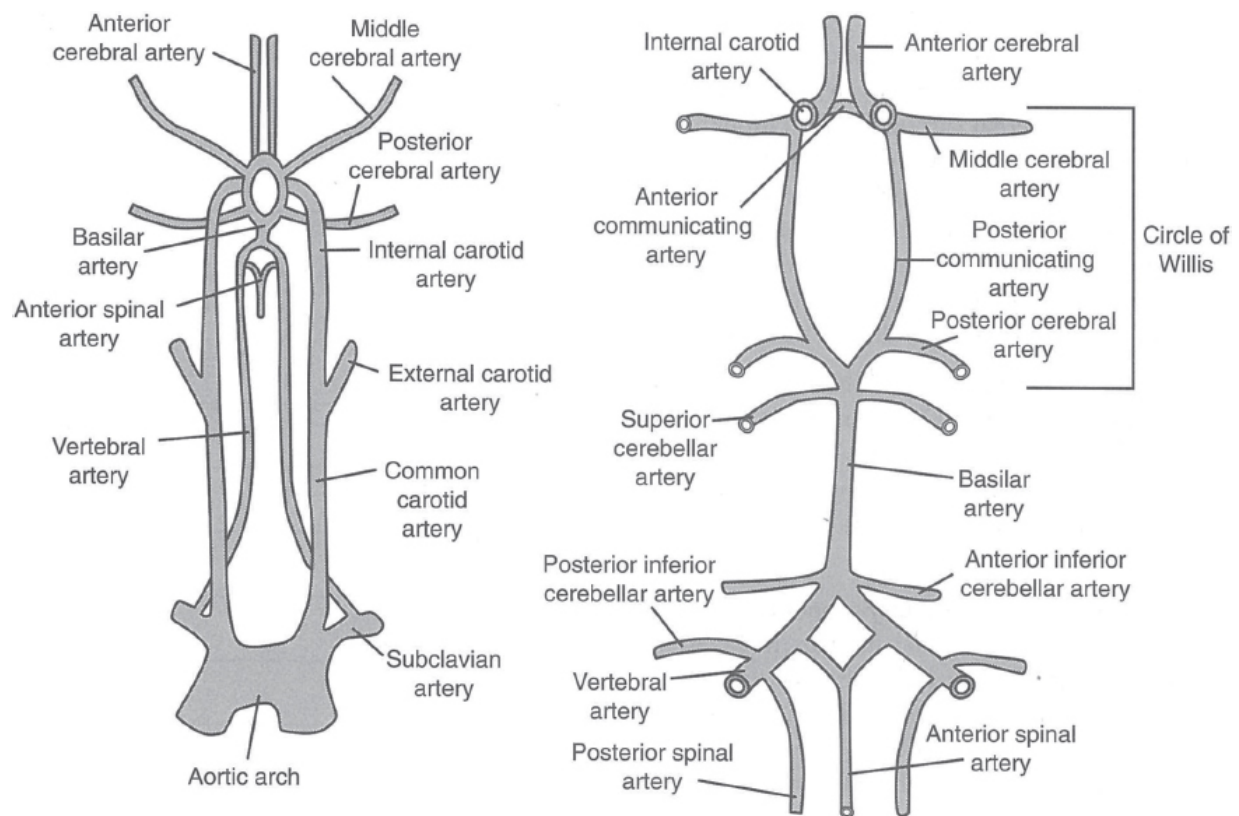


Figure 1-16. The main structures of the cerebral blood supply.

From the two subclavian arteries, two vertebral basilar arteries branch out and ascend to the base of the brain. Several other branches of the subclavian artery supply the thoracic wall, upper arm, and shoulder. The two carotid arteries and the two vertebral basilar arteries supply blood to the brain.

The Carotid Arteries

As they enter the neck, the left and the right common carotid arteries branch into an internal carotid artery and an external carotid artery. Through the base of the skull, the arteries enter the brain through the dura mater and the subarachnoid space. The **external carotid** moves toward the face and branches into smaller arteries. The external carotid supplies blood to muscles of the face and neck, nasal and oral cavities, sides of the head, skull, and dura mater.

The major blood supplier to the brain is the **internal carotid artery**, which branches into several smaller blood vessels supplying various portions of the brain. Its two main branches of special interest are the *anterior cerebral artery* and the *middle cerebral artery*.

The **anterior cerebral artery** supplies mostly the middle portion of the frontal and parietal lobes. In addition, it supplies blood to such structures as the basal ganglia and corpus callosum. Branches of the anterior cerebral artery anastomose (join) with the posterior cerebral artery in the posterior medial areas of the brain. [Figure 1–17](#) shows the distribution of the anterior cerebral artery (as well as the posterior cerebral artery, described later).

Damage to the anterior cerebral artery can cause disruption of the blood to the midsagittal portions of the motor cortex. The motor symptoms of this disruption include paralysis of the legs and feet. Associated deficits, described as *prefrontal lobe symptoms*, include impaired reasoning, judgment, and concentration.

The **middle cerebral artery**, the biggest branch of the internal carotid, supplies the entire lateral surface of the cortex, including the major regions of the frontal lobe. With its several branches (including the lateral, frontal, parietal, and temporal), the middle cerebral supplies blood to major areas concerned with sensory and motor skills, as well as speech, language, and hearing. The areas it supplies include the somatosensory cortex, motor cortex in the precentral gyrus, Broca's area, primary auditory cortex, Wernicke's area, the angular gyrus, and the supramarginal gyrus. Furthermore, its smaller branches serve the putamen, caudate nucleus, globus pallidus, and portions of the thalamus. [Figure 1–18](#) shows the distribution of the middle cerebral artery.

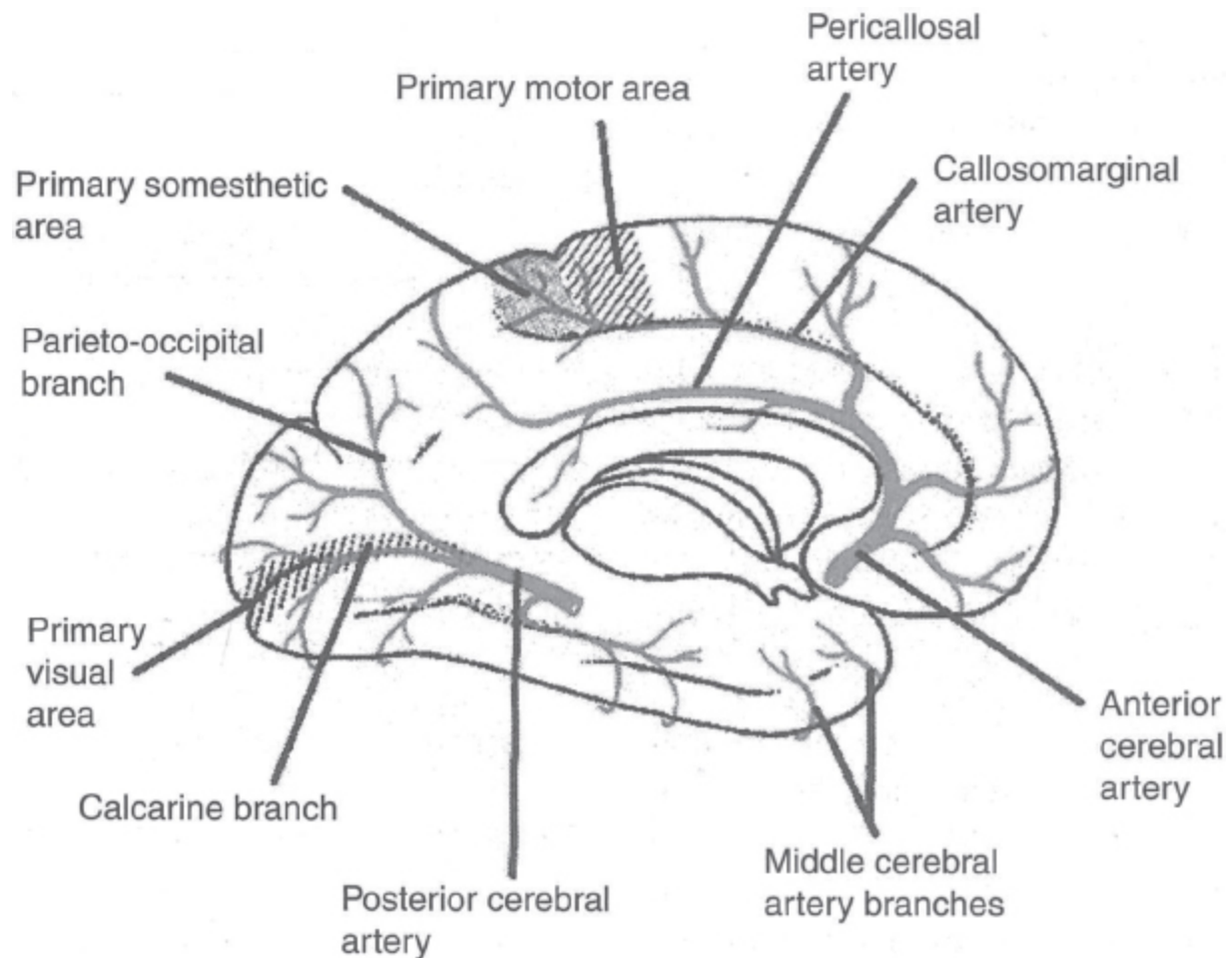


Figure 1-17. The distribution of the anterior and posterior cerebral arteries.

Damage to the middle cerebral artery is a frequent cause of stroke and aphasia. Contralateral hemiplegia — plus impaired sense of touch, position, pain, and temperature — may also result. Reading and writing deficits may be part of the symptom complex.

The Vertebral Arteries

The left and the right **vertebral arteries** are branches of the two subclavian arteries that emerge from the aortic arch (see [Figure 1-16](#)). As noted previously, the other branches of the subclavian supply mostly the upper extremity. The vertebral arteries enter the skull and branch to supply several

organs, including the spinal cord. As they move up to the lower level of the pons, the two vertebral arteries join together to form a single **basilar artery**.

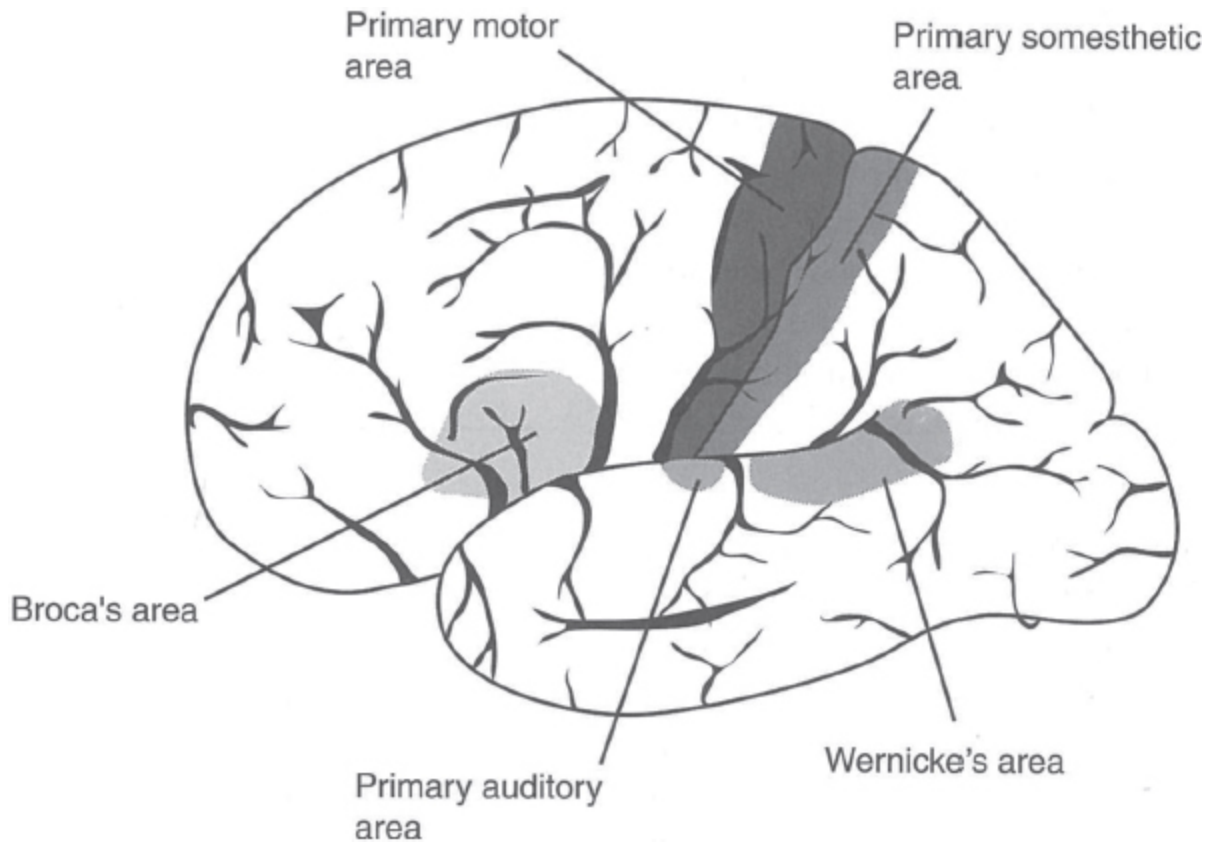


Figure 1-18. Distribution of the middle cerebral artery, supplying blood to Broca's and Wernicke's areas as well as the primary auditory area.

Moving toward the upper portion of the pons, the basilar artery divides again into two **posterior cerebral arteries** that supply blood to the lower and lateral portions of the temporal lobes and the middle and lateral portions of the occipital lobes (please refer to [Figure 1-16](#) for the distribution of the posterior cerebral artery). Several other branches of the basilar artery supply such other structures as the pons, cerebellum, and inner ear.

The Circle of Willis

At the base of the brain, the two carotid and the two vertebral arteries ascending from the heart anastomose (join together) to form the **circle of Willis** (circulus arteriosus). This circle is completed by the two communicating arteries (*anterior communicating* and the *posterior communicating*). The anterior cerebral, the posterior cerebral, and the middle cerebral arteries branch out from the circle.

The circle of Willis provides a common (redundant) blood supply to various cerebral branches. If an artery is blocked below the circle, the damage to the brain is minimal because an alternate channel of blood flow may be maintained. But if the blockage occurs after or above the circle, the brain will not receive the blood due to lack of a common source.

Blood Supply and the Watershed Area of the Brain

The **watershed areas of the brain** are those areas that receive blood from the small end-branches (terminal branches) of the primary arteries that supply blood to the brain: the anterior, middle, and the posterior cerebral. These fine end-branches may anastomose (join together) to supply blood to the region, somewhat inefficiently (Bhatnagar, 2017). [Figure 1–19](#) illustrates the watershed areas of the brain.

If the blood supply to the watershed region is interrupted due to vascular diseases and other potential causes, specific kinds of aphasia, especially transcortical motor aphasia and transcortical sensory aphasia, may result (Benson & Ardila, 1996). Both types are described in [Chapter 3](#).

The Veins and Venous Sinus System

As noted earlier, a system of veins drains deoxygenated blood from organs and carries it back to the heart and eventually to the lungs where the blood is reoxygenated. In essence, the **veins** form a drainage system.



Figure 1-19. The watershed area of the brain.

The venous system includes larger veins and many smaller, deep veins that penetrate the brain structure to collect oxygen-poor blood. Through their venules that are in contact with the capillaries, small veins throughout the brain collect the circulated oxygen-poor blood. The blood thus collected is drained into various sinuses.

The venous system is complex, but just one view of it is provided in [Figure 1-20](#).

Sinuses are channels through which blood or other fluids flow. A network of sinuses in the skull and within the cerebral structures helps collect the oxygen-poor blood from smaller veins. The sinuses in and around the brain are called **dural sinuses** because they are found within the dura. Sinuses channel the blood to the large jugular vein, which returns the blood to the heart.

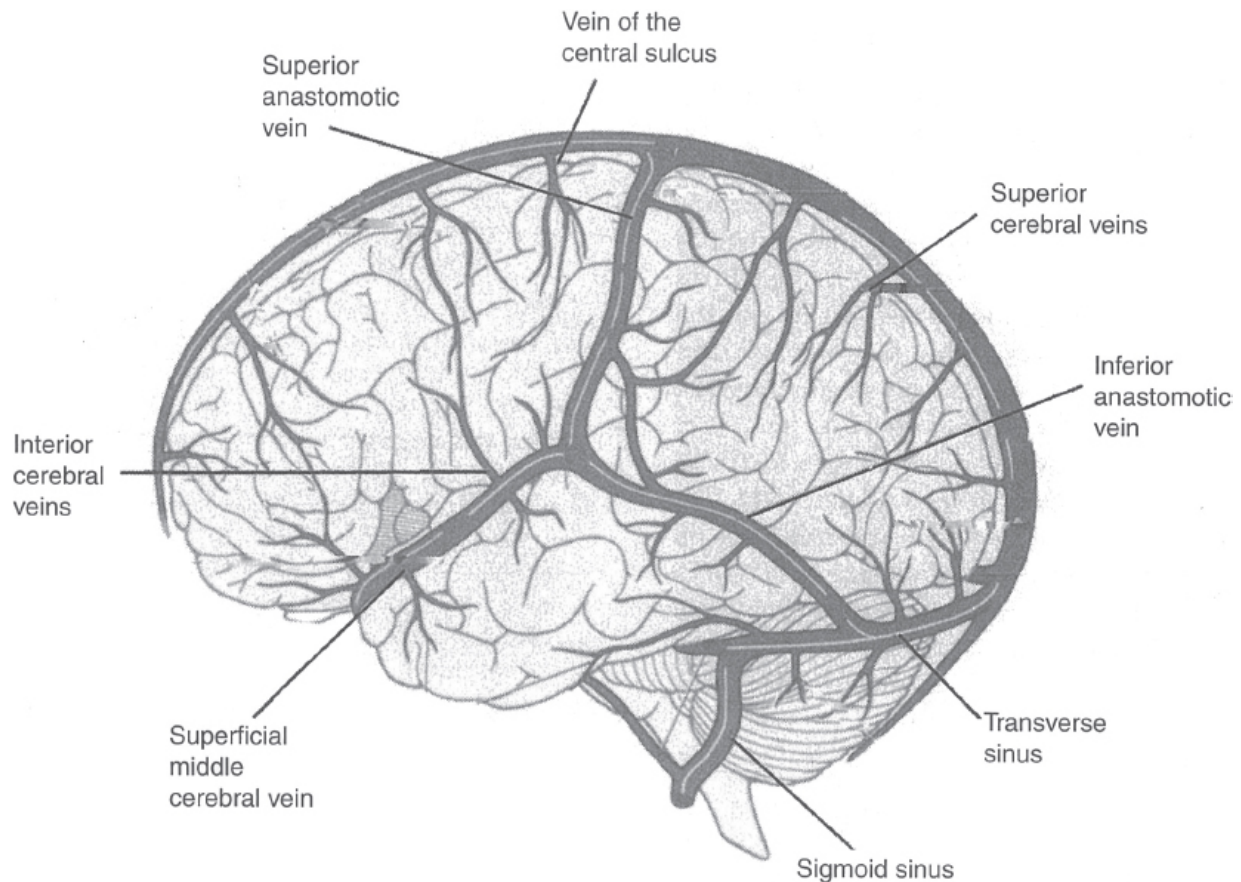


Figure 1-20. A lateral view of cerebral veins and sinuses.

Compared to arterial pathologies, venous pathologies cause communication disorders less frequently. This may be because the venous system is more interconnected and is less likely to develop occlusions and hemorrhages than the arterial system.

The Blood-Brain Barrier

The **blood-brain barrier** refers to a mechanism that prevents the cerebral penetration of harmful chemical substances and infectious microorganisms that may be present in the blood. The walls of the capillaries that help diffuse the blood nutrients to the cerebral cells are formed by endothelial cells that rest on a continuous basement membrane called **basal lamina**. Unlike the cell walls of capillaries that supply blood to the muscles and other organs, the

endothelial cells of the cerebral capillary walls have continuous lining and tightly formed junctions. Such tight junctions do not contain intercellular pores through which substances can pass.

The blood-brain barrier, while preventing harmful serum substances from reaching the neural cells, also prevents helpful medicines from needed cerebral infusion. This barrier is weaker in babies than in adults, thus making the brains of babies more vulnerable to infections. Also, certain diseases break down the blood-brain barrier and thus allow other substances — including harmful infections, helpful medicine, and diagnostic chemicals — to penetrate the cerebral cells. Brain tumors, for example, can break down the blood-brain barrier because their cells develop intercellular pores. This disease-created vulnerability also provides an opportunity to make an accurate diagnosis of the tumor. For instance, injected radioactive amino acid will penetrate the tumor cells but not the normal brain cells. Scanning machines can then identify the tumor site.

CLASS NOTES

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2

NEURODIAGNOSTIC METHODS AND NEUROPATHOLOGY

Chapter Outline

- Neurodiagnostic Methods
- Pathologies of Neurologically Based Communication Disorders
- References

Learning Objectives

After reading the chapter, the reader will:

- Define *neurodiagnostics*
- Define and distinguish between various brain imaging techniques
- Give an overview of neuropathologies
- Describe the neuropathology of communication disorders

PRINTED NOTES

A person is a medical patient soon after a stroke, a head trauma, or being admitted to a hospital for dementia, brain tumor, or brain infection. At this time, various neurodiagnostic procedures are carried out to understand the nature and the extent and location of brain damage, cerebral degeneration, or an infection. How soon a speech-language pathologist will assess the communication skills will depend on the person's physical status. Whereas a quick bedside assessment may be made at the time of admission to the hospital, a communication diagnostic assessment may have to wait until the person's physical condition is stabilized with medical treatment. The bedside speech and language screening or assessment also contributes to the interdisciplinary diagnostic effort because impaired communication skills help distinguish the different kinds of diseases and trauma that affect the health and functioning of the brain.

Assessment and diagnosis of aphasia, right hemisphere disorder, traumatic brain injury, and dementia are described in their respective chapters. This chapter describes the medically administered neurodiagnostic methods aimed at determining the type, location, and extent of cerebral disease, abnormalities, or injury associated with communication disorders. It may be noted that the neurodiagnostic methods are also used to study the relationship between brain and language.

Neurodiagnostic Methods

The methods of diagnosing neural pathology and its various effects are called **neurodiagnostic methods**. Some methods are post hoc, some are invasive, others are noninvasive, and still others are inferential. A general understanding of these methods is essential to speech-language pathologists who work with individuals whose communication disorders have a neurological basis.

Postmortem: The Post Hoc Method

Postmortem is the oldest method of relating particular areas of the brain to specific language functions. Broca and Wernicke used this method to localize certain language functions in specific brain structures. Currently, a definitive neurological diagnosis of dementia requires a postmortem of the brain.

In this method, a person's language problems associated with obvious neurological diseases are carefully noted. Upon the patient's death, a postmortem is performed to see pathological changes in the brain tissue. Observed changes then are related to the previously noted language problems.

Intraoperative Cortical Mapping

Brain surgical methods are used to remove pathological cerebral tissue that causes various neurological symptoms. Before the offending (damaged) brain tissue is removed, however, the neurosurgeon performs an **intraoperative cortical mapping**, in which the term *mapping* refers to identifying the brain areas responsible for speech and language during open brain surgery (Chang et al., 2015). It is performed to surgically remove the damaged cortical areas that trigger intractable and life-threatening seizure disorders that do not respond well to medications. During this surgery, the surgeon's goal is to spare the speech-language areas, hence the need to map them first. Penfield and Roberts (1959) pioneered this technique. It is now well established, experimental, and in vivo (performed on a living person, as against the postmortem). There is no substitute for the surgical method of finding a person's speech and language areas. In a given person, "the location of essential language sites can be extremely variable and nearly impossible to predict preoperatively," (Chang et al., 2015, p. 253). Furthermore, brain lesions force a reorganization of language within the hemisphere. Even the contralateral region may take over the function. Therefore, a general knowledge of language-specific areas in brain-healthy individuals is not precise enough to spare critical regions during surgery. Brain scanning techniques, including the functional magnetic

resonance imaging, are not as effective as the intraoperative cortical mapping in determining cortical speech and language areas (Chang et al., 2015).

The method requires only local anesthesia because the brain tissue has no pain sensors. During this **awake craniotomy**, the patient is fully conscious during the operation and able to talk and respond in various ways. The neurosurgeon cuts open a portion of the skull and exposes the cortex. Placing an array of electrodes directly on the surface of the cortex, the surgeon electrically stimulates the relevant brain regions with a handheld probe. The person is asked to perform various verbal tasks including naming the pictures shown on a computer monitor, counting, repeating words and sentences, answering questions, reading, writing, and other tasks of interest (Chang et al., 2015).

Electrical stimulation of motor areas of the brain elicits motor responses. The same stimulation of language areas does not elicit speech, however, but disrupts or confuses ongoing speech. For example, while the person is trying to name an object shown on the monitor, stimulation of the posterior inferior frontal gyrus and the posterior temporal and inferior parietal regions may prevent naming. Such speech disturbances are temporary. Incidentally, this finding contradicts the traditional knowledge that the temporal lobe is not concerned with speech production, only the frontal lobe is. Production and comprehension are integrated.

Electroencephalography

Electroencephalography is a time-honored method of studying cerebral activity and pathologies by recording the electrical potential (brain waves) of the cerebral cortex. The graphic representation of the electrical activity of the brain is called an electroencephalogram (EEG). EEG is a standard clinical diagnostic method of the neurologist. By using multiple-channel machines and placing multiple electrodes on the scalp, brain waves may be simultaneously recorded from different lobes of the cortex. Differences in the recorded electrical activity patterns within and across the lobes and across the corresponding areas of the two hemispheres may be studied. Any abnormal

electrical activity suggesting underlying cortical pathology may be identified (Meechan et al., 2021).

EEG is especially useful in diagnosing epilepsy as the method can detect abnormal discharges even when the patient is not experiencing seizures. Any focal lesion also is suggested by abnormal EEG patterns surrounding the site of lesion. EEG findings, when combined with other measures and clinical evaluation, may be helpful in supporting diagnosis of herpes simplex encephalitis or such forms of dementia as Creutzfeldt-Jakob disease.

Magnetoencephalography

Magnetoencephalography (MEG) is an electrophysiological method that can help assess the entire brain. MEG produces images of higher spatial and temporal resolution than EEG and functional magnetic resonance imaging (Bowyer et al., 2020; Meechan et al., 2021; Proudfoot et al., 2014). MEG records language-evoked magnetic fields when auditory or visual verbal stimuli are presented.

MEG is used mainly to study lateralization of language to the (usually) left hemisphere, rather than identifying specific language areas. Neurological diseases such as epilepsy, tumor, or stroke may shift language to the right hemisphere or to nontraditional areas in the left hemisphere. Therefore, it is often necessary to first determine the language-dominant area before performing neurosurgery for brain diseases. MEG is performed for this purpose. A Wada test may validate the MEG-determined language laterality (Bowyer et al., 2020). Unlike MEG, Wada, however, is an invasive test. A chemical anesthetic such as sodium amobarbital is injected into the internal carotid artery. Soon, one of the arms (typically the right) is anesthetized, suggesting the motor and speech motor control by the contralateral hemisphere (typically the left).

In the MEG procedure, language stimulation evokes peaks of activity in multiple language areas, including supramarginal gyrus, angular gyrus, and inferior frontal gyrus. MEG has documented language comprehension in the temporal lobe (including Wernicke's area) and language production in the

frontal (including Broca's area) and basal temporal regions. In the final analysis, a *language laterality index* is calculated to specify the language-dominant hemisphere. The index may confirm left, right, or dual dominance. In some cases, dominant may be undetermined (Bowyer et al., 2020).

Computed Tomography

Computed tomography, also known as CT scan, is a noninvasive method of taking pictures of different sections of an organ. *Tomos* means sections and *tomography* means imaging the body sections. The method uses an x-ray scanning machine that rotates 360 degrees around the head (or other structures) to take images. On its monitor, a computer displays the processed images. Introduced for clinical applications in 1973, CT scan is an early method of visualizing the brain structures in living persons.

Tomographic scans show internal structures, lesions, tumors, and other neuropathologies. They are especially good at detecting recent ischemia and hemorrhages, suggesting immediate intervention options. Therefore, the method is one of the first applied to persons with stroke (Forkel & Catani, 2018). CT scans are better at showing stroke-induced pathology than that caused by tumors or trauma. Therefore, tomographic scans have been used to study the location and extent of lesions in various types of aphasia (Greenberg et al., 2021). CT scans are useful in detecting both focal and progressive pathologies. The method may fail to reveal small lesions, however. In routine clinical practice, localization of pathology may only be approximate.

Though not invasive, CT scanning exposes individuals to radiation. Injecting radiopaque material to a person's bloodstream improves the image contrast, but it also increases the chances of negative side effects (e.g., pain, nausea, kidney problems).

Magnetic Resonance Imaging

Magnetic resonance imaging (MRI) is a group of techniques to generate pictures of brain with the help of a powerful magnetic field that alters the electrical activity of the brain. Because water is a main content of brain tissue, the behavior of atomic nuclei of hydrogen in the brain can help create images of that tissue. This noninvasive method does not introduce radioactive material into the patient's body, and it does not use the x-ray. MRI produces images similar to those from CT scanning. To acquire the images, the person's head is placed in a strong magnetic field. The randomly spinning hydrogen atoms of the brain produce magnetic properties that are exploited in constructing a picture of the brain structures. Much like iron filings lining up and pointing in the direction of a magnet, the hydrogen atoms of the brain align themselves with the magnetic field.

An electromagnetic pulse is then introduced (the person hears a noise similar to that produced by a washing machine). This pulse of energy disturbs the alignment of the hydrogen atoms of the brain for a brief moment and then the atoms swing back to alignment as the electromagnetic pulse is withdrawn. While swinging back to alignment with the magnetic field, the atoms of specific areas produce particular electromagnetic signals at a certain radio frequency, called **resonant frequency** (hence the name, *magnetic resonance imaging*), similar to a faint echo. The computer detects, analyzes, and uses the resonant magnetic signals to construct an image of the structures (Goodman & Szaflarski, 2021).

MRI provides clearer images of body structures than the CT scan. It can detect small lesions missed by CT scans along with demyelinating and traumatic lesions. MRI is less prone to artifacts from bone tissue than are CT scans. The white and gray matter of the brain are better contrasted in MRI than in CT because the gray matter contains more water. CT scans are not especially productive during the first 48 hours of stroke, although they are still superior to MRI in detecting cerebral hemorrhage. Hematoma that is 3 to 4 days old is better revealed by MRI than CT scan, however. Although CT scans are very sensitive to changes in tissue density, the MRI is sensitive to varying chemical composition of tissue. MRI is informative of brain structures only, not their function (Bowyer et al., 2020).

The basic technique of magnetic resonance imaging has evolved into several specialized procedures, each designed for a specific purpose. All use the same scanner. A brief description of each type of imaging follows.

Functional magnetic resonance imaging (fMRI) helps assess the brain function, as against the structures the MRI shows. fMRI detects changes in cerebral blood flow, an index of metabolic rate. Different metabolic rates are associated with different activities (e.g., at rest versus listening to music). fMRI detects increased blood flow that is typically associated with cerebral activity in the regions that are activated (Goodman & Szaflarski, 2021; Greenberg et al., 2021). fMRI has been used to study normal as well as impaired cognitive functions. The method has been studied in documenting neuropathology, spontaneous recovery of language functions, and brain function changes following behavioral therapy in persons with aphasia and other communication disorders (Fridriksson et al., 2010; Schevenels et al., 2020).

Diffusion-weighted MRI constructs pictures of structures based on its detection of microscopic motion of water protons in the brain tissue. The method has been found to be clinically useful because it can detect cerebral ischemia in stroke patients soon after onset and with good clarity and specificity. It also can help distinguish stroke-induced cerebral edema from edema due to other factors.

Perfusion-weighted MRI produces images based on blood flow variations (as against water proton movements detected in diffusion-weighted MRI). Diffusion-weighted and perfusion-weighted images may be compared to analyze ischemic damage in stroke patients.

Magnetic resonance angiography (MRA) is a procedure in which the rate and velocity of blood supply to the selected cerebral structures are measured. Compared to the traditional cerebral angiography (described later), MRA is noninvasive, less risky, and more economical. It is especially useful in visualizing the carotid arteries and the fast proximal intracranial blood circulation. Occlusion of vessels may be detected with this procedure, although the image clarity is inferior compared to those of the traditional angiography.

Diffusion MRI fiber tractography is another MRI-based technique that helps visualize the subcortical white matter fiber pathways (tracts) that connect

the different parts of the brain. Study of such cortical and subcortical connecting fibers is known as **connectomics**. Previously, subcortical white matter connecting fibers have been studied only through postmortem. Diffusion tractography is the only method to study deep-brain connecting fibers in vivo (Campbell & Pike, 2014). The method has proven useful in studying language in healthy and diseased brains (Jeurissen et al., 2017). Fiber tractography studies of the brain have more definitively identified the subcortical connecting fibers described in [Chapter 1](#). These fibers include the superior and inferior longitudinal, the middle longitudinal, and inferior frontal occipital and uncinate fasciculi that are critically important in language formulation, comprehension, and production (Chang et al., 2015). Such studies have supported the dual-stream hypothesis of language processing in the brain (Hickok, 2022), as described in [Chapter 3](#).

Voxel-based morphometry is yet another variation of structural MRI. This method allows for more precise localization of language functions. A **voxel** is an extremely small area of the brain (1×1 mm). Combined with high-resolution MRIs and analysis of precisely defined areas, the voxel-based morphometry has enhanced our understanding of brain lesions that affect language. The method has been used in lesion-symptom matching in persons with stroke and aphasia (Kim et al., 2021; Mirman & Thye, 2018). See [Chapter 3](#) for additional information on voxel-based symptom mapping in aphasia.

Positron Emission Tomography

A nuclear medical procedure, **positron emission tomography (PET)** is used to study brain activity through differences in metabolic rates of different areas of the brain. A *positron* is a positively charged particle having the same mass and magnitude of charge as the electron and constituting the antiparticle of the electron. The method helps detect differences in the rate of metabolic activity through differential glucose consumption, which indicates differences in the health of the tissue studied (Valotassiou et al., 2018).

In the PET procedure, positron-emitting radioactive isotope (carbon, nitrogen, or oxygen), also called a *tracer*, is injected into the person's bloodstream. The tracer gets concentrated in areas of greater metabolism (greater neural activity and blood flow). Such areas tend to be lesion or tumor sites in the brain. The PET scanner detects and amplifies the positrons (radiation) emitted by the isotopes. A computer analyzes the data to show areas of high or low metabolic activity in two- or three-dimensional pictures.

PET is good in detecting brain tumors as the tumor sites are associated with increased metabolism. PET also can reveal damage to areas that have normal blood flow but reduced metabolic activity due to that damage. Furthermore, PET also can show areas that are not working although no structural damage is seen. Therefore, PET scans can reveal tissue changes that are not yet evident on CT and MRI. A combination of PET and CT scan provides x-ray as well as scanned images. Hybrid PET/MRI procedures are also in use, providing high-contrast images. PET is capable of showing how focal brain damage may affect functions elsewhere in the brain, supporting the view that functions in the brain are not strictly localized.

Single Photon Emission Computed Tomography

Another nuclear medical procedure, **single-photon emission computed tomography (SPECT)** produces three-dimensional images of the brain by detecting the gamma rays the intravenously injected radioisotopes radiate. The method may be a follow-up to an MRI. Effective in detecting diffuse as well as focal brain damage, the results help diagnose cerebral tissue damage associated with epileptic seizures, strokes, neurodegenerative diseases, and traumatic brain injury. It can also detect tumors and cerebral infections (Valotassiou et al., 2018).

DAT-SPECT (also known as SPECT DaTscan) detects the dopamine levels in the brain to diagnose neurological diseases. The method combines DAT, a radioactive dopamine transporter with SPECT scanning. The radioactive material is injected into the patient. The diseased brain tissue absorbs DAT more than the healthy tissue. The SPECT procedure is then

performed to obtain the brain images. The method is helpful in diagnosing Parkinson's disease, a movement disorder for which no definitive imaging technique was available. The abnormal dopamine levels, such as those found in persons with Parkinson's disease, may be distinguished from similar diseases including drug-induced parkinsonism and essential tremor in which dopaminergic deficiencies are not evident (Palermo & Ceravolo, 2019).

Regional Cerebral Blood Flow

Regional cerebral blood flow (rCBF) is a PET- or SPECT-based technique to assess the amount of blood flow in different areas of the brain. An active cerebral region requires more blood. Thus, an estimate of the amount of blood flowing into a region indicates the presence of, or increase in, cerebral activity. Increased blood flow also is an indication of increased metabolic activity (Bhatnagar, 2017). In this method, a radioactive material, often an inert gas called *xenon-133*, is introduced to the patient's bloodstream, either by having the patient inhale the gas or by injecting it.

A gamma ray camera attached to a computer scans the brain for the radioactive material in the cerebral blood flow. A computer constructs the images of the regions of the brain and shows differential blood flow in different colors and hues.

Regional blood flow studies have shown that the right hemisphere is active in speech and language to a greater extent than previously suggested. The method has been valuable in assessing pathological changes in the brain associated with aphasia and various neurodegenerative diseases, including Alzheimer's disease (Peretti et al., 2019).

Cerebral Angiography

Cerebral angiography is a radiographic invasive procedure combined with the injection of radiopaque contrast material into selected arteries (Dowd, 2021). The x-rays do not penetrate radiopaque material; arteries that are infused with

this material appear white or light on the exposed x-ray film. A catheter is inserted into the femoral artery in the groin and guided into the carotid or vertebral artery. A radiopaque material is then injected into the catheterized artery.

A rapid series of x-rays is taken immediately to evaluate the health of the vascular system by examining the outline of major arteries on the film. The cerebral anterior, posterior, and middle arteries may be visualized.

The x-rayed radiopaque material shows variations in blood circulation that might suggest vascular occlusions. Arterial sections beyond a clot that blocks the blood flow will not show up, however, because the radiopaque material is blocked. The method also can help detect hemorrhages and aneurysm.

One limitation of cerebral angiography is that it does not outline the smaller arteries. Because it is an invasive procedure, it poses a risk of stroke.

Other Neurodiagnostic Techniques

There are several other neurodiagnostic techniques. Among them, some construct images from the sounds deflected from physiological activities. **B-mode carotid imaging** (also known as *echo arteriogram*) is a technique of that kind. It is used to assess the health of arteries in the neck by placing a high-frequency sound generator over the neck. The sound that is reflected back by the arteries is used to assess the health of those arteries. The technique can help detect such arterial pathologies as stenosis and ulceration.

Carotid phonoangiography is a method of assessing the health of the carotid arteries by the characteristics of the sound generated by the blood gushing through the arteries. Arterial stenosis, which is narrowing of the arteries, creates turbulence as the blood moves and this can be detected through carotid phonoangiography.

Doppler ultrasonography (also called ultrasound) is another sound-based technique. It measures the velocity of blood flow by exposing the head to high-frequency sound. Depending on the blood flow characteristics, the frequency of reflected sound changes. The computer analyzes such changes and

constructs an image of the blood flow, pressure, and velocity. Increased velocity of sound suggests a narrowing of the arteries.

Pathologies of Neurologically Based Communication Disorders

Like all disease, neurologically based communication disorders have many causes; most are chains of causes with remote and immediate elements of the chain interacting with each other. Some form of brain damage is the most immediate or even simultaneous cause of aphasia and communication impairments associated with both traumatic brain injury and right hemisphere disorder. A slow and insidious disease process is at work in most forms of irreversible dementia (Walker et al., 2019).

As an example, look at the chain of events depicted in the next section, going backward from the existing aphasia; the arrows suggest the sequence of events from the most remote (genetic predisposition) to the most recent (injury to the language structures in the brain).

An illustration of elements of a potential causal chain leading to aphasia:

Aphasia



Injury to language structures in the brain



Interrupted blood supply



Arteriosclerosis



High blood cholesterol



Poor eating habits, sedentary lifestyle



Genetic predisposition

The causal elements are only examples, showing one possible (not certain) sequence of events. The various elements within the chain interact with each other to produce the eventual health consequence. Other chains of events with different causes may also cause damage to the speech and language structures in the brain. Evidence of such damage is essential for a diagnosis of aphasia and other neurologically based communication disorders.

The brains of patients who have sustained damage, infection, or other insults show various kinds of neuropathological elements. The following kinds of cerebral changes may be found in varying degrees across individuals with aphasia and other disorders (Greenberg et al., 2021):

- **Edema.** A brain area that has recently suffered some kind of damage is typically soft and swollen. Swelling may extend from the gray to the white matter. The edema reaches its maximum level within the first 4 to 5 days of the onset of a stroke. Most deaths following a stroke are due to cerebral edema.
- **Cellular changes.** Some cortical cells in the affected area may die. Others may be shrunk.
- **Destruction of glial cells.** The glia, the special kinds of cells that support the neural cells, may be destroyed.
- **Necrosis of small blood vessels.** *Necrosis* is cell damage. Damage to small blood vessels is often present.
- **Ischemic penumbra.** A tissue region surrounding the major locus of infarction is called *ischemic penumbra*; improving the blood supply to this region is a major medical concern following an ischemic stroke.

Transient Ischemic Attacks

Symptoms of strokes last long enough to be noticed by the patient and the family members. There are transient (brief) “brain attacks,” however, which, although the patient notices, may not lead to swift action because such attacks do not produce lasting effects. **Transient ischemic attacks (TIAs)** are “ministrokes” that last a few seconds and the patient recovers without more permanent disability; however, they may be warning signs of more serious strokes that do produce lasting effects (Coutts, 2017).

It is important to recognize the signs of TIAs as urgent medical attention may be needed. Prompt treatment may reduce the chance of a more serious stroke. Patients and family members should suspect a TIA when the following symptoms are experienced:

- Sudden weakness, numbness, or paralysis in facial muscles, arm, or leg (often on one side of the body)
- Sudden impairment in understanding speech
- Slurred or garbled speech
- Sudden blindness or double vision
- Dizziness, impaired balance, or impaired consciousness

A person may have multiple TIAs that may produce serious cumulative effects. A full-fledged stroke also may follow a TIA. Therefore, it is important to seek immediate medical attention by calling 911 when symptoms of the first TIA are experienced.

Cerebrovascular Accidents (Strokes)

Cerebrovascular accidents, popularly known as strokes, are a frequent and immediate cause of aphasia; they may also be involved in right hemisphere disorder. A stroke has an acute onset, resulting in focal brain damage, caused

by impaired cerebral blood circulation (Barthels & Das, 2020). The resulting focal brain damage leads to an array of communication disorders.

Symptoms of a stroke are similar to those of TIAs, although some of the more serious symptoms associated with strokes are infrequent in TIAs (e.g., vomiting, seizures, and coma are uncommon in TIAs). A stroke is the likely problem if an individual experiences the following symptoms (Greenberg et al., 2021):

- Severe headache combined with other symptoms (more common in anterior ischemia)
- Altered consciousness (more common in posterior ischemia)
- Impaired speech production, comprehension, or both (in anterior ischemia)
- Impaired vision, often in one eye (more common in posterior ischemia)
- Dizziness, sudden falls, inability to stand or walk (in posterior ischemia — cerebellar hemorrhage)
- Paresis (weakness) or paralysis (more common in anterior ischemia)
- Impaired sensation, especially on one side of the body (more common in anterior ischemia)
- Vomiting (in hemorrhagic stroke)
- Seizures (mostly in hemorrhagic strokes affecting subcortical white matter)
- Coma (usually in hemorrhage in pons; often fatal)

Stroke symptoms last at least 24 hours. In a few cases, symptoms may last more than 24 hours and yet the patient may recover completely or nearly so. TIAs may precede strokes in many cases.

Etiology of Cerebrovascular Accidents

Various vascular disorders cause strokes that generally produce focal symptoms. The nature and extent of the symptoms depend on the site of the vascular region and that of brain tissue involved. Focal symptoms produced by trauma or metabolic disorders are usually not called *aphasia*, because aphasia refers to a series of events triggered by vascular pathology.

Based on their vascular pathology, strokes are classified as either *ischemic* or *hemorrhagic*. Ischemic strokes are more common than the hemorrhagic strokes. Ischemic strokes account for 87% of all strokes in the United States. The risk factors for strokes include sedentary lifestyle, unhealthy eating habits, smoking, hypercholesterolemia, hypertension, and diabetes. A combination of hypercholesterolemia and hypertension is a strong risk factor for strokes in younger persons in the age range of 35 and 44 years. Hypertension may be a more common risk factor in the older adults (Barthels & Das, 2020).

Ischemic Strokes

Ischemic strokes are caused by occlusive vascular disorders that block or interrupt arterial blood flow to a region of the brain resulting in an **infarction**, also called **necrosis**, which is death of neural tissue for lack of oxygen and glucose, caused by interrupted blood supply. There is a single effective drug called the tissue plasminogen activator (tPA) that dissolves the blockage and restores blood supply to the brain, but to be effective, it must be administered within 4.5 hours of the onset of symptoms (Barthels & Das, 2020).

Blood supply to the brain may be interrupted by either *cerebral thrombosis* or *cerebral embolism*.

Thrombosis is a vascular disease involving the formation of thrombus, which is a collection of blood materials that get entrapped with cell material, blocking blood circulation. In general terms, a **thrombus** is a special kind of blood clot. Thrombosis may be the cause of roughly two thirds of all ischemic strokes.

The formation of thrombi (plural of *thrombus*) are made possible by **atherosclerosis**, which is a slowly developing arterial disease process in which

the arteries are hardened and narrowed from an accumulation of lipids, other fatty particles, calcium deposits, and fibrous material that form atherosclerotic plaques. Atherosclerosis is associated with a variety of factors including high blood pressure, high cholesterol (especially the low-density lipoproteins), triglycerides, diabetes, free radicals, smoking, use of oral contraceptives, and so forth. Genetic predisposition also is suspected.

Atherosclerosis and the resulting thrombi are typically formed in large arteries (the internal carotid, middle cerebral, and basilar), small penetrating arteries, cerebral veins, and venous sinuses. Specific sites include the origins of the common carotid, middle cerebral, and vertebral carotid arteries; the basilar artery; vertebral artery just above its entry point to the skull; and the internal carotid artery just above the common carotid bifurcation. Generally, the points at which an artery divides and bends are more vulnerable to an accumulation of material leading to atherosclerosis and consequent thrombosis.

The accumulation of materials in the arteries thickens and hardens their walls. This results in narrowed or constricted arteries through which blood does not flow normally. Slow-flowing blood is vulnerable to thrombus (clot) formation. Thrombi remain at the site of their formation and occlude the artery. Consequently, the tissue down the stream may die for lack of oxygen.

Thrombotic strokes tend to occur when the person is asleep or engaged only in low levels of physical activity. Transient ischemic attacks may precede such strokes.

Embolism is another arterial disease in which a moving or traveling fragment of arterial debris blocks a small artery through which it cannot pass. Thus, an embolus (emboli, plural) is a traveling mass that may have been formed farther away from the place where it occluded a vessel. Thus, a traveling embolus contrasts with a stationary thrombus. [Figure 2–1](#) shows plaque buildup that eventually leads to embolism that occludes (blocks) the blood flow in an artery, leading to cerebral cell damage in an area supplied by the blocked artery.

An embolus may be a clump of tissue from a tumor or a diseased artery, moving atherosclerotic plaque, a mass of bacteria, a blood clot or a piece of clot that has broken loose from its point of origin, an air bubble, and so forth.

An embolus may have been a thrombus to begin with; when it breaks off from the site of its origin and begins to move, it is called an embolus, not a thrombus. Heart diseases like atrial fibrillation (heart palpitation) may lead to embolism because pooled blood in the heart can promote clot (embolus) formation. Emboli also may form at the site of cardiac surgery and then migrate to brain vessels to block one of them.

Emboli often occlude the middle cerebral artery and its branches, which supply 85% of hemispheric blood. The superior division of the middle cerebral artery supplies blood to anterior portions of the brain, including Broca's, motor, and sensory areas. The inferior division supplies Wernicke's area. Emboli also occlude blood supply to the posterior portions of the brain when they are formed in basilar and posterior cerebral arteries.

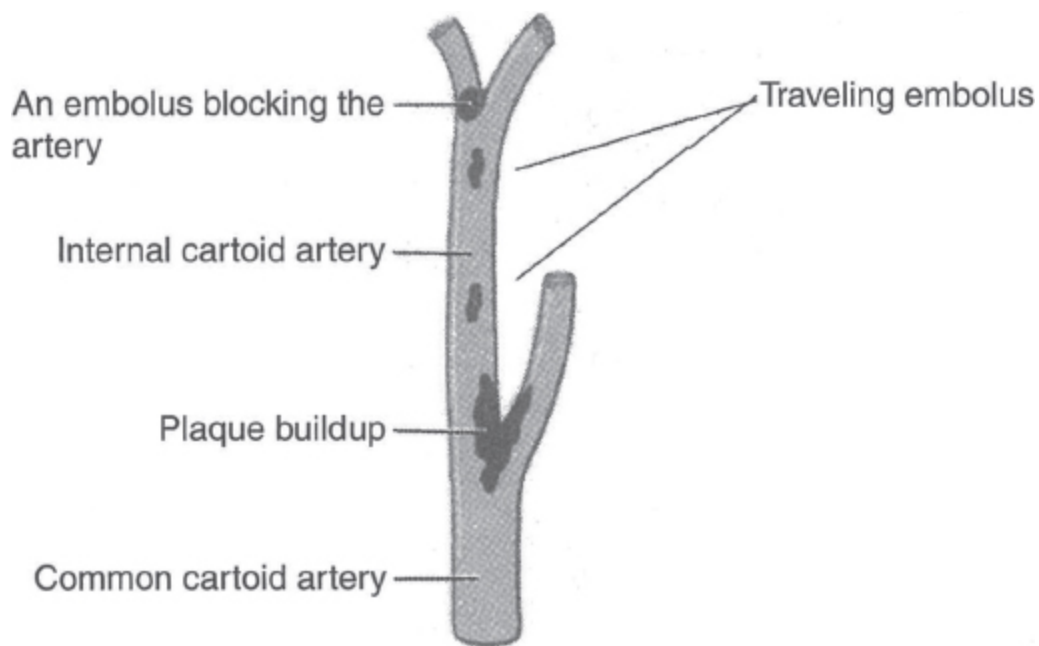


Figure 2-1. Plaque buildup in an artery leading to an embolus that eventually blocks an artery. Brain tissue ahead of the lodged embolus will be the area of infarction.

Embolic strokes tend to occur when the person is awake and active. Typically, the patient may have no warning signs, although in a few cases, headaches and seizures may precede such strokes.

Hemorrhagic Strokes

Hemorrhages cause the second most common variety of strokes. **Hemorrhagic strokes** are those that result from ruptured cerebral blood vessels that cause cerebral bleeding, as contrasted with ischemic strokes resulting from occlusion of blood vessels (Barthels & Das, 2020; Greenberg et al., 2021).

The common causes of blood vessel ruptures include weakened arterial walls, various malformations of the blood vessels, and high and fluctuating blood pressure. A less frequent cause is trauma to the blood vessel.

There are two main kinds of hemorrhagic strokes: intracerebral and extracerebral. [Table 2–1](#) shows their causes and characteristics.

An aneurysm is a balloon-like swelling of a weak and thin portion of an artery that eventually ruptures. A common site of an extracerebral aneurysm is the base of the brain and can involve the vertebral arteries, the basilar artery, or the circle of Willis. Anterior and middle cerebral arteries also are aneurysm sites. See [Figure 2–2](#) for an aneurysm that might eventually rupture and produce a hemorrhagic stroke.

Table 2–1. Causes and Characteristics of Intracerebral and Extracerebral Hemorrhages

Intracerebral Hemorrhage	Extracerebral Hemorrhage
Ruptures within the brain or brainstem cause <i>intracerebral hemorrhage</i>. Hypertension is the most common cause of blood vessel ruptures. Intracerebral hemorrhage is more common in small arteries deep within the brain structures,	Ruptures within the meninges are called <i>extracerebral hemorrhage</i>, and are classified into three types: Subarachnoid: The most common of the extracerebral hemorrhages that occur on the surface of the brain, brainstem, or cerebellum; ruptures occur beneath the arachnoid;

especially around the thalamus and basal ganglia.	frequently caused by aneurysms. Subdural: Rupture occurs beneath the dura matter; often caused by TBI. Epidural (Extradural): Ruptures occur above the dura, between the dura and the skull; often caused by TBI.
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Intracerebral hemorrhage in basal ganglia may cause subcortical aphasia, high mortality rate, and more permanent brain damage in survivors. Hemorrhaged blood forms clots and destroys brain cells. Strokes in such cases tend to have a sudden onset. Headache, vomiting, coma, stupor, paralysis, sensory loss, confusion, memory loss, and impaired communication are the frequent symptoms.

Patients who have had ischemic or hemorrhagic strokes tend to show different patterns of recovery, as summarized in [Table 2–2](#).

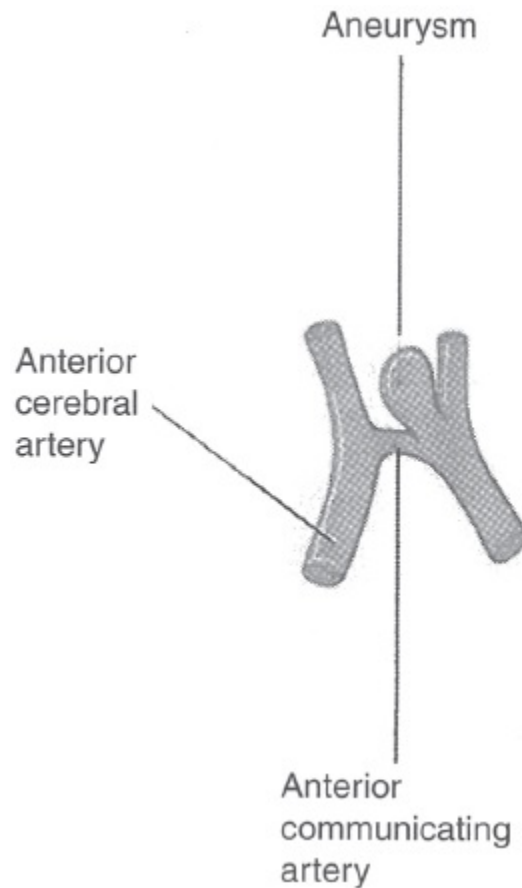


Figure 2-2. Aneurysm of an artery that might rupture and cause a hemorrhagic stroke.

Table 2-2. Different Patterns of Recovery from Typical Ischemic and Hemorrhagic Strokes

Ischemic	Hemorrhagic
Greater and sooner recovery	Little recovery in the first 4 to 8 weeks
Noticeable recovery in the first few weeks	More rapid recovery after 4 to 8 weeks
Maximum recovery in 3 months	A slowing down of recovery; stabilizing with greater residual

Brain Trauma

Traumatic brain injury (TBI), which is externally induced trauma to the brain, is another cause of neurologically based communication disorders. Automobile and other kinds of accidents, gunshot wounds, and blows to the head, among others, are the frequent causes of trauma to the head and the brain. TBI may cause a variety of communication problems, as described in [Chapter 8](#).

Intracranial Neoplasms

Intracranial neoplasms (tumors) are pathological growths within the cranial structures. They are a less frequent cause of aphasia than are strokes. **Tumors** are space-occupying lesions that cause swelling in the surrounding tissue and lead to increased intracranial pressure (Greenberg et al., 2021). Tumor epidemiology has stayed relatively stable over the decades (McNeill, 2016).

Tumors may be primary or secondary (metastatic), benign, or cancerous. Benign tumors are not cancerous and grow slowly. Malignant (cancerous) tumors grow uncontrollably because they form new blood vessels that continue to feed the tumors (McFaline-Figueroa & Lee, 2018).

Primary intracranial tumors originate in the brain. Often found in the cerebrum and the cerebellum, these tumors are more common in persons 25 to 50 years of age. The cause or causes of tumors are unknown, although the loss of a tumor-suppressing gene, an inappropriate expression of a cancerous gene, or a combination of both may be responsible. Heredity and former sites of injury correlate with the formation of such tumors (McNeill, 2016).

Malignant primary brain tumors include several varieties that differ in their degree of malignancy and their rate of growth. The term *glioma* is often used to include all forms of malignant primary brain tumors. Gliomas arise in

neuroglia (also called glia, meaning glue in Greek), which are nonneural supportive cells of the nervous system. Glia cells include astrocytes, oligodendrocytes, and microglia.

Astrocytomas are primary malignant tumors that arise from astrocytes. Their extent of malignancy may vary significantly, however. Astrocytomas are the most common form of glioma. The most malignant of glioma is **glioblastoma multiforme**, which is associated with a high death rate. **Oligodendrogliomas** arise from oligodendrocytes and are typically found in the adult frontal lobes.

Intracranial tumors may cause destruction of healthy tissue, swelling of nearby tissue, headaches, seizures, and a variety of initially focal and eventually generalized symptoms that include impaired vision, loss of sensation, vertigo, memory problems, behavior changes, and so forth.

Successful surgical treatment of tumors depends on their size, location, and time of diagnosis. Whether benign or malignant, a tumor located in an inaccessible and vital part of the brain (e.g., thalamus or brainstem) may be fatal.

Meningiomas, a variety of intracranial (and extracerebral) tumors, form within the meninges. Benign, slow growing, and generally localized, meningiomas tend to cause focal symptoms and are most effectively removed surgically.

Secondary (metastatic) intracranial tumors are those that have grown elsewhere in the body but have migrated into the brain. Such migration or spreading of cancer cells throughout the body is called **metastasis**. Most metastasized intracranial tumors are originally cancer cells that break off from their primary site elsewhere in the body and are carried by the bloodstream into the cranial space where they begin to grow. Most common secondary intracranial tumors come from cancer of the breast, the lungs, the pharynx, or the larynx. Notably, the primary intracranial tumor cells do not travel to other parts of the body. Metastatic tumors carry a high mortality rate. Death typically occurs within 2 to 3 months of diagnosis (McFaline-Figueroa & Lee, 2018).

Based on the degree of their malignancy (aggressiveness of their growth), tumors are graded as Grade I, II, III, or IV. Grade I tumors have distinct borders (circumscribed in structure) and grow slowly. Grade IV tumors grow most aggressively and signal the worst prognosis for the patient. Glioblastoma multiforme, described previously, is typically graded IV. Tumors that fall in between the least and the most aggressive are graded II or III.

Central Nervous System Infections

Central nervous system (CNS) infections can be life-threatening and tend to produce serious physical symptoms. Language disturbances they produce tend to be overlooked. Older adults and children and persons with compromised immune system are especially vulnerable to CNS infections. Both MRI and laboratory tests of the cerebrospinal fluid are done to diagnose them, although the former produces more immediate results.

Diagnosed promptly and accurately, CNS infections are treatable with antibiotics or antiviral medications (Saini et al., 2014). Major and relatively common CNS infections include (1) bacterial, (2) viral, (3) fungal, and (4) parasitic (Saini, 2014; Solomon, 2020). Poorly managed CNS infections can damage the brain permanently, leading to neurologically based communication disorders. Brain infections tend to cause meningitis and encephalitis. [Chapter 10](#) on dementia has additional information on cerebral infections, some of which cause reversible dementia.

Meningitis is inflammation of the meninges, the membrane covering the brain and the spinal cord. It is caused by infection of the cerebrospinal fluid. Viral infections are the most common causes, although bacterial, parasitic, and fungal infections also may cause meningitis. The common symptoms include headache, sudden high fever, nausea and vomiting, and neck pain. Additional complications may include seizures, confusion, and sensitivity to light. Untreated meningitis, especially the bacterial variety, may result in death or permanent brain damage affecting speech and language skills.

Encephalitis is the inflammation of the brain. The most common cause is a viral infection. It can also be an autoimmune response. Symptoms include

mild or severe flu-like symptoms. In severe cases, seizures, confused thinking (affecting speech and language), memory problems, hallucination, heightened sensitivity to light, muscle weakness or paralysis, joint pain, impaired movement, coma, and vision and hearing problems may be evident.

Bacterial infections include the pneumococcus (streptococcus), meningococcus, haemophilus, and Listeria (*Listeria monocytogenes*), neurosyphilis, neurobrucellosis (bacteria in uncooked meat), and CNS tuberculosis, among others (Saini et al., 2014). Several vaccinations are available to prevent bacterial infection of the CNS. Bacterial infections may cause, among other problems, cerebritis (brain inflammation), cerebral abscess (a confined pocket of pus), ventriculitis (inflamed lining of the cerebral ventricles), and meningitis. Neurosyphilis may cause strokes, and haemophilus and streptococcus may cause meningitis.

Viral infections of the CNS often cause encephalitis, but meningitis is also possible. Resulting brain damage may be associated with communication impairment. Common viral infections include rabies, AIDS, herpes simplex virus, mumps, measles, enteroviruses, mosquito-borne viruses (causing dengue, chikungunya, or Japanese encephalitis), Epstein-Barr virus of the herpes family, West Nile virus, and so forth. Herpes simplex encephalitis is a frequent viral infection that damages the brain. As the patients recover, severe amnesia and generalized dementia may be more apparent than the naming problems they do experience. AIDS is associated with an increased incidence of transient ischemic attacks, ischemic strokes, and, in some cases, dementia.

Fungal infections of the CNS include yeast that causes meningitis, pseudohyphae that causes focal infarcts of brain tissue, and the true hyphae that causes strokes (Solomon, 2020). Fungal infections are known to cause intracerebral and subarachnoid hemorrhage. Cerebritis and cerebral abscess have also been noted. Some fungal infections lead to the formation of thrombosis and aneurysm, posing stroke risks (Saini et al., 2014).

Parasitic infections of the CNS include single-celled protozoans, free-living amoebas, and microscopic nematodes (roundworms), cestodes (tapeworms), and trematodes (flukes). Humans typically get infected by consuming solid and liquid food contaminated with parasitic eggs. These

infections cause a variety of problems, including focal neurological deficits, encephalitis, seizures, and other effects (Saini et al., 2014; Solomon, 2020).

Other Factors

Several other factors may be related to brain pathology and communication disorders. For instance, cerebral toxemia may result from drug overdose, drug interactions, and heavy metal (e.g., lead and mercury) poisoning. Cocaine addictions cause strokes in persons who are younger than 35 years of age. Use of cocaine hydrochloride most frequently causes hemorrhagic strokes, although ischemic strokes are also possible. Crack cocaine (alkaloidal cocaine) typically leads to ischemic strokes, although hemorrhagic strokes also may occur. Amphetamines usually cause ischemic strokes (Greenberg et al., 2021).

Epilepsy may be associated with aphasia or similar focal language impairment. Such impairments may more often be observed soon after a seizure attack. Language impairments associated with epilepsy tend to be transitory.

CLASS NOTES

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PART II

APHASIA



3

APHASIA: PREVALENCE, DEFINITION, AND CLASSIFICATION

Chapter Outline

- Prevalence of Aphasia and Associated Diseases
- Definition of Aphasia
- Limitations of the Classic Aphasia Classification
- Alternative Aphasia Classifications
- General Evaluation of Aphasia Classification
- Overview of Aphasia Symptoms
- Nonfluent Aphasias
- Fluent Aphasias
- References

Learning Objectives

After reading the chapter, the reader will:

- Summarize the data on the prevalence of aphasia and related diseases
- State different definitions of aphasia

- Give an overview of different approaches to aphasia classification
- Evaluate the different aphasia classifications
- Describe the major language deficits associated with varieties of aphasia
- Describe the neuropathologies and language and speech characteristics of the four varieties of nonfluent aphasia
- Describe the neuropathologies and language and speech characteristics of the four varieties of fluent aphasia
- Distinguish the different varieties by their critical diagnostic features

PRINTED NOTES

Systematic scientific study of aphasia began in the mid-1800s. Paul Broca (1824–1880), a French neurosurgeon and physical anthropologist, was the first to offer clinical and pathological evidence relating the frontal lobe and left brain to nonfluent, agrammatic, telegraphic speech but mostly intact language comprehension (Broca, 1861/1960). This disorder came to be known as *Broca's aphasia*. Broca has been credited with the famous statement, “We speak with the left hemisphere” (Benton & Anderson, 1998).

Another classic contributor was Carl Wernicke (1848–1905), a German neuropsychiatrist. He described a type of aphasia that contrasted with Broca's aphasia. His patients spoke fluently and grammatically, but their speech did not make much sense. They had severe problems in understanding spoken language. Wernicke (1874/1977) had concluded that a lesion in the posterior portion of the left superior temporal gyrus caused the symptoms he had observed. This portion of the brain came to be known as *Wernicke's area*, and the type of language problem he described was accorded the name *Wernicke's aphasia*.

Ludwig Lichtheim (1845–1928), another German neurologist and a pioneer, built an aphasia classification system based on the works of Broca and Wernicke (Ueno et al., 2011). Lichtheim also proposed a more comprehensive analysis of language impairments associated with aphasia, including spontaneous speech, repetition, spontaneous and writing to dictation, and comprehension of speech as well as orally read materials (Krestel, 2013). Eventually, the classic Broca-Wernicke-Lichtheim model came to dominate aphasia classification. Neurologist Norman Geschwind (1926–1984) Americanized the aphasia classification. Consequently, the classic approach is also known as the Broca-Wernicke-Lichtheim-Geschwind model of aphasia and its classification. Still commonly used, the classic classification of aphasia includes two broad categories of fluent and nonfluent aphasias. There are several subtypes under each.

The classic model has come under severe criticism in recent decades. Alternative approaches to understanding aphasia have been advocated. We will review data on prevalence and definitions of aphasia before we return to the classic and newer models of aphasia.

Prevalence of Aphasia and Associated Diseases

Various neurological diseases and pathologies determine the prevalence of aphasia. A **cerebrovascular accident** (CVA), popularly known as a **stroke**, is frequently an immediate cause of aphasia. Because heart diseases and strokes are closely related, they produce a combined effect on the prevalence of both the conditions.

Published health statistics have a lag of 3 to 4 years. For the latest figures, the reader should consult such sources as the National Center for the Health Statistics, Centers for Disease Control and Prevention (CDC; <https://www.cdc.gov/stroke/facts.htm>), and the American Heart Association for the latest figures available. On these sites, 2020–2021 updates suggest the following figures, although the exact numbers on various sources may differ somewhat:

- More than 795,000 persons in the United States have a stroke every year; that is a stroke every 40 seconds; of those, 610,000 are first-time strokes and 185,000 are second or subsequent strokes.
- Historically, strokes were the third leading cause of death in the United States, after coronary heart disease (#1) and cancer (#2), but in 2020–2021, COVID-19, with more than 350,000 deaths, and accidents at 200,955 deaths surpassed deaths due to stroke (160,000); combined, heart diseases and strokes are still the number 1 cause of death; someone dies of stroke every 4 minutes.
- In 2019, one in every six deaths from cardiovascular disease was due to stroke.
- Mortality rates differ across the types of strokes; 8% to 12% of ischemic (blockage of blood) strokes and 37% to 38% of

hemorrhagic (bleeding in the brain) strokes result in death.

- Stroke is a leading cause of disability in the United States. In excess of 300,000 persons who suffer strokes are permanently disabled.
- Incidence of strokes is low in children 18 and under; it is estimated that 1 to 2 in 100,000 may experience a stroke; more boys than girls have strokes; as in adults, ischemic strokes are more common than hemorrhagic strokes in children.
- After age 55, the incidence of strokes increases rapidly during each decade of life; about two thirds of all strokes are associated with age 65 and older; between the ages of 45 and 54, 1.2% of men and 2.1% of women have a stroke; but at age 75+, 12% of men and 11.5% of women have a stroke.
- At comparable age levels, men's stroke incidence is 1.25 times higher than women's; however, because of increased longevity, annually, about 40,000 more women than men have strokes; also, of every five deaths due to strokes, two are men and three are women.
- About 1 million people (i.e., 1 in 250 individuals) in the United States have aphasia.
- Annually, about 80,000 new cases of aphasia are reported in the United States.
- About 15% of the survivors of one or more strokes need institutional care; stroke patients constitute a large number of older people admitted to nursing homes.
- Of all strokes, 87% are ischemic, 9% are intracerebral hemorrhagic, and 3% are subarachnoid hemorrhagic.
- Broca's aphasia is more common in younger persons and Wernicke's aphasia in older persons.

Aphasia and Associated Diseases in Ethnocultural Groups

Payne (2014) has reviewed variations in health status that affect strokes and cardiovascular diseases in the general population and in major ethnocultural groups. The previously cited sources, including the National Center for the Health Statistics, provide updated demographic data.

- The age-adjusted stroke incidence rates per 100,000 persons is 167 for White males, 138 for White females, 323 for African American males, and 260 for African American females.
- African American males have the highest death rates due to strokes; Hispanics and American Indians/Alaska Natives have the lowest.
- African American women tend to have strokes at an earlier age than White women.
- Women with aphasia from all ethnocultural groups combined outnumber all men with aphasia.
- Ischemic attacks, transient attacks, and extracerebral strokes are more common in Whites than in other ethnic groups.
- South Asians (people from the Indian subcontinent) living in the United States have a higher incidence of strokes (and heart diseases) than Whites.
- Whites have a higher prevalence of heart diseases than African Americans or Hispanics.
- Compared to Whites, Hispanics and African Americans tend to have strokes at earlier ages.
- In the age range of 44 to 55 years, more African Americans than Whites die of strokes. Black and Hispanic younger adults in the age range of 20 to 44 years also have a higher incidence of strokes than Whites of similar age.
- African Americans, Asians, Native Americans, and Hispanics are more prone to intracerebral hemorrhagic strokes than Whites; these strokes are associated with higher mortality rates.
- Mortality rate from strokes in Hispanics after age 65 is lower than that in Whites; this is attributed partly to Hispanic people's typically lower blood pressure.

- Risk factors for strokes vary somewhat among the different ethnic groups but include high blood pressure, diabetes, smoking, high cholesterol levels, obesity, poor diet, malnutrition, alcoholism, high sodium, and lack of exercise.
- The first strokes in African Americans produce more severe effects and greater disability than they do in Whites, and they need more recovery time, mostly due to lack of access to quality health care.

White men tend to be oversampled in epidemiological studies and women and ethnoculturally diverse groups are typically undersampled. Therefore, all generalizations about prevalence of aphasia and related diseases in ethnocultural groups must be treated with caution.

Definitions of Aphasia

Aphasia is a language disorder in adults attributable to recent cerebrovascular events and documented cerebral pathology or trauma. Language production, speech comprehension, gesturing, reading, and writing may all be impaired to varying extents. These communication impairments may result in disrupted interpersonal relationships, limited social participation, social isolation in extreme cases, and occupational difficulties in working individuals. A person's aphasia affects all members of the family.

Disrupted language learning in childhood due to unfavorable neurological events (such as cerebral palsy) is not aphasia. Similarly, as a disorder of language formulation and expression, aphasia is neither apraxia of speech nor dysarthria — two varieties of motor speech disorders.

Definitions of aphasia vary for different reasons. One reason is the dissatisfaction with its unreliable classification. For example, rejecting the idea that there exist different types of aphasia, Darley (1982) had defined it as “an impairment, as a result of brain damage, of the capacity for interpretation and formulation of language symbols” (p. 42) and had avoided the practice of classifying it into different varieties. Benson and Ardila (1996), who believe that aphasia is an overly classified disorder, defined it simply as “the loss of or

impairment of language caused by brain damage” (p. 3). A limitation of such overly broad or generic definitions is that they do not distinguish aphasia from the speech-language effects of traumatic brain injury, right hemisphere disorder, and dementia.

Other researchers emphasize the social consequences of aphasia. The disorder has broader ramifications for the individual and his or her family members, professional caregivers, and social acquaintances. Consequently, Simmons-Mackie (2001) defined aphasia as an “impairment due to brain damage in the formulation and reception of language, often associated with diminished participation in life events and reduced fulfillment of desired social roles” (p. 248). However, aphasia is not unique in producing social consequences; all disorders of communication affect social participation. It is important to consider the effects of a disorder, but the effects do not define the disorder. Nonetheless, an emphasis on social aspects of aphasia broadens assessment and treatment concerns and widens the scope of treatment to include social situations, family members, professional caregivers, and so forth.

Limitations of the Classic Aphasia Classification

As noted previously, the classic Broca-Wernicke-Lichtheim-Geschwind model that classified aphasia into fluent and nonfluent varieties was based on the neuroanatomy and brain function known at that time (Monroy-Sosa et al., 2021). The classic model, especially as Geschwind revised it, is a hypothesis of how the brain normally mediates language. Geschwind (1965, 1970) had recognized only the arcuate fasciculus as the bundle of fibers connecting Broca’s and Wernicke’s areas. Wernicke, however, had recognized uncinate fasciculus as another pathway that connects frontotemporal regions. To simplify the model, Wernicke’s area recognizes the meaning of the words heard, whereas Broca’s area formulates articulated speech (Nasios et al., 2019).

In the late 20th century, new information began to emerge on newer cerebral networks, subcortical white matter, and distant brain regions that are involved in typical and impaired language functions following brain diseases. It has been known for decades that the classic classification of aphasia is

unreliable and inaccurate. However, criticisms of the traditional diagnostic categories and their advocated neuroanatomical bases have intensified in the past two to three decades. Objections to the traditional classification of aphasia include the following:

- Anatomical sites thought to be exclusive to general or particular language functions may not be so at all. Broca's aphasia may *need* damage to subcortical white matter. High-resolution imaging studies of Broca's original patients' preserved brains showed extensive subcortical damage that Broca had missed (Dronkers et al., 2007). Broca's aphasia may *not* need damage to Broca's area at all as the symptoms may be observed in individuals with intact Broca's area. Damage limited to Broca's aphasia may produce transient mutism and subsequent apraxia of speech, not aphasia (Ardila et al., 2016; Trupe et al., 2013). Chronic nonfluent aphasia may involve damage to both Broca's and Wernicke's areas (Hickok, 2022). Broca's area may contain one subregion specific to language and another for domain-general tasks (Fedorenko et al., 2012). The motor cortex thought to be involved only in speech production reacts when speech is heard; its activation during listening is similar to the acoustic feature-based activation in the auditory cortex (Cheung et al., 2016). Cerebral localization of Wernicke's area has been even more problematic. Some research suggests that no single area in the brain mediates speech comprehension (Mesulam et al., 2015).
- Boundaries of the anatomical language sites are not as clear as the classic model suggests. Trembley and Dick (2016) have pointed out inconsistent definitions of Broca's and Wernicke's areas. The authors have argued that the classic model excessively emphasizes the cortical structures and underappreciates the contributions of subcortical regions, subthalamic nucleus, and different connecting fibers. They consider the assumption that only the arcuate fasciculus supports language function obsolete.
- The classic model underemphasizes certain other neurological consequences of strokes. Most people with aphasia have

hypometabolism in widespread cerebral regions that could contribute to the initial symptoms (Shin et al., 2020).

- The anatomical lesion sites of each aphasia subtype may be ambiguous, vary across individuals and for the same set of language symptoms, and may cross the presumed anatomic boundaries.
- More than a quarter of persons with obvious aphasia are unclassifiable, and in a majority of persons (63.5%), the traditional lesion-to-syndrome correspondence may not be established (Kasselimis et al., 2017).
- Classical diagnostic categories are unstable; intraindividual diagnostic categories change over time; poststroke initial diagnosis may not hold 1 year later; nonfluent aphasia may evolve into fluent aphasia (but not vice versa); other diagnostic categories may change to a chronic anomia (Pedersen et al., 2004). When persons recover from aphasia, different symptoms recover to different extents (Wilson et al., 2019).
- No two individuals with the same type of aphasia have the same pattern of language deficits. Symptoms and the sites of lesions vary for the same type of aphasia (Fridriksson et al., 2018).

Newer neurodiagnostic techniques and language mapping studies of neurosurgical patients have greatly advanced our understanding of brain-language relationships. Therefore, in the historical context, it is not at all surprising that the classic aphasia classification has come under criticism. It would be surprising if it had not.

General Validity of the Classic Classification of Aphasia

In spite of the problems described, the classic model of aphasia continues to be used in clinical practice. Some attribute this to an existing gap in research and clinical practice (Kasselimis et al., 2017). All kinds of clinical practices lag behind research. Nonetheless, clinicians may find the classification useful for several reasons, including the following:

- While it may get specific details wrong, the classic model has broad general validity. There is enough evidence to assert that generally and broadly, lesions in the frontal cortex cause agrammatic nonfluent aphasia, lesions in the temporal cortex cause fluent but meaningless language, and lesions in the arcuate fasciculus cause conduction aphasia (Nasios et al., 2019). For the clinicians, the possibility that each of these aphasia types may be associated with cerebral damage elsewhere does not diminish the broad significance of the classic model.
- With their newer analytical tools, Matchin and Hickok (2020), two notable critics of the classic model, nonetheless confirmed that the posterior middle temporal gyrus and the posterior superior temporal gyrus are involved in language comprehension and that the inferior frontal gyrus (the anterior part of Broca's area) is involved in speech and language production. Sine-wave speech intelligibility studies have confirmed that temporal cortex is indeed involved in speech comprehension, although the frontal cortex also may be involved (Khoshkhoo et al., 2018).
- Newer models have their own theoretical and clinical limitations, as noted in the next section. They have not solved all the aphasia assessment and diagnostic problems. They have not yet penetrated clinical practice, possibly because of those limitations. It is also likely that the training most clinicians receive in assessing and treating aphasia is based on the traditional model.

Alternative Aphasia Classifications

There are several newer models of neurobiology of language, neuroanatomy of language, and classification of aphasic language disorders. Investigators of these newer models believe that the classic model is overly simplistic, inaccurate, and inadequate to classify and explain aphasia (Ardila, 2021; Fridriksson et al., 2015; Fridriksson et al., 2018; Hickok, 2022; Nasios et al., 2019; Tremblay & Dick, 2016). The most researched of the newer models is the *dual-stream*

approach to aphasia classification, which is described in this section. I briefly mention two other approaches.

The origin of the dual-stream model can be traced back to Wernicke, who had hypothesized the existence of two pathways: one for mediating language comprehension and the other for production (Bhaya-Grossman & Chang, 2021; Chang et al., 2015; Hickok & Poeppel, 2004; Nasios et al., 2019; Trembly & Dick, 2016). Wernicke had emphatically stated that a single center, such as Broca's area, could not control language in the brain and that the fibers that connect the different areas play a major role.

The Dual-Stream Model

The **dual-stream model**, as the name implies, includes two speech-language pathways in the brain. *Dual streams* are two interacting neural pathways or networks that connect different cerebral regions. The dorsal stream and the ventral stream are the two pathways. In general, the dorsal stream is concerned with speech production and the ventral with speech comprehension (Fridriksson et al., 2018; Hickok, 2022; Hickok & Poeppel, 2004). Several studies, using functional imaging, lesion site mapping, and intraoperative cortical mapping methods have produced generally supportive evidence for the dual-stream model (Chang et al., 2015; Fridriksson et al., 2016; Hickok, 2022).

The **dorsal stream** is unilateral, left-lateralized. The subcortical arcuate fasciculus (AF) and the superior longitudinal fasciculus (SLF) are major pathways of the dorsal stream. Some researchers consider the AF and SLF the same; others consider them different (Dick & Tremblay, 2012). Historically, these two bundles were thought to connect Broca's area with Wernicke's area, but there is no agreement among contemporary researchers who have tried to divide and subdivide AF and SLF (Dick & Tremblay, 2012). Subcomponents of the SLF make frontal-parietal and temporal-parietal connections (Chang et al., 2015). Areas of the superior temporal gyrus, posterior-inferior frontal gyrus (including pars opercularis and pars triangularis), and the lateral premotor cortex are parts of the dorsal stream.

The dorsal stream mediates phonological aspects and speech production. Damage to this stream causes speech production problems. The more severe the aphasia, the more extensive the damage to the dorsal stream. The dorsal stream also controls self-feedback and online corrections of speech production errors (Fridriksson et al., 2016).

The bilateral **ventral stream** consists of a large number of fibers including the extreme capsule and the uncinate, the middle longitudinal, the inferior longitudinal, and the inferior fronto-occipital fasciculi. Located between the claustrum and the insula, the extreme capsule is either a place for the uncinate and the inferior fronto-occipital fibers to pass through (Shekari et al., 2021) or a bundle of long association fibers (Dick & Tremblay, 2012). The uncinate connects anterior temporal and inferior frontal areas. The middle longitudinal fasciculus connects the inferior parietal and temporal regions. Both the inferior longitudinal and the inferior fronto-occipital fasciculi are thought to connect occipital cortices to the anterior temporal and inferior frontal cortices. The ventral stream extends from the posterior middle and inferior temporal gyrus to the anterior middle temporal gyrus.

The ventral stream predominantly mediates auditory speech-language comprehension (perception of meaning), also described as *semantic-lexical processing*. Damage to this stream causes difficulties in understanding spoken speech. Severity of the difficulties depends on the extent of damage.

The two systems are asymmetric in their functions. In speech perception, only the ventral system is involved, but in speech production, both the ventral and dorsal systems are involved (Hickok, 2022).

The dorsal and ventral systems interact with each other. Because the two systems interactively mediate expressions of grammatically complex utterances, similar language disruptions are associated with lesions in disparate locations. Lesions anywhere in the brain disrupt language functions because of the coordinated manner in which the two streams work (Fridriksson et al., 2018).

The original Hickok and Poeppel (2004) dual-stream model has been extended to better account for syntactic processing in the brain (Matchin & Hickok, 2020). Accordingly, the middle temporal gyrus mediates both

receptive and expressive syntax, whereas the more anterior portion of Broca's area mediates syntactic expression to a greater extent.

Evaluation of the Dual-Stream Model

- The dual-stream research has greatly advanced our understanding of complex pathways that interconnect the various brain regions. This line of research is directly relevant to aphasia symptom complex and classification.
- Investigators of the newer models are united in opposing the classic model of aphasia but are divided about a model that should replace it.
- The origins, courses, terminations, and the extent of the dual pathways are controversial. The number of pathways within each stream has been debated extensively. The anatomy of the ventral stream is more controversial than that of the dorsal stream.
- Although there is consensus that the dorsal stream is associated with phonological/speech production skills and the ventral stream is associated with semantic/meaning/speech comprehension skills, there is no consensus on the typical speech-language functions attributed to the specific subpathways whose numbers and connections vary across reported studies.
- Clinical data on language disturbances following stroke and other neurological disturbances also are inconsistent across studies.
- Dual-stream research has not resolved some of the long-standing controversies about the lesion sites of certain types of aphasia. For instance, the lesion sites of conduction aphasia still remain unclear or controversial (Dick & Tremblay, 2012). And the aphasia lesion sites and the symptoms they produce have remained as varied and controversial as ever.
- The dual-stream model does not account for the role the cerebellum, basal ganglia, and thalamus play in mediating speech-language skills (Hickok & Poeppel, 2004; Nasios et al., 2019).

Intraoperative Language Mapping

Historically, patients with epilepsy undergoing brain surgery while awake and verbally responding to electrical stimulation of brain regions have supplied some of the earliest data on cortical language functions (Penfield & Roberts, 1959). Since then, intraoperative mapping of language cortex has advanced to a great extent and has supplied a vast amount of data, a good portion of which has been generated in recent decades by the neurosurgeon Chang and his associates (Chang et al., 2015). Chang and his team of researchers have targeted mapping of progressively more complex language structures. In one of their studies, Chang et al. (2018) have demonstrated that at least nine sites within the inferior frontal gyrus, mostly pars triangularis and opercularis (Broca's area), mediate the production of complex syntactic constructions. The study also demonstrated, like other cortical electrical stimulation studies, that individual differences are significant. The authors identified syntactic encoding sites in only 50% of the participants. Bhaya-Grossman and Chang (2021) and Yi and associates (2019) demonstrated that the superior temporal gyrus detects acoustic-phonetic features and promotes phonological processing. Consistent with findings of other investigators, Rolston and Chang (2018) have demonstrated that language areas have greater local network connectivity than the nonlanguage networks, potentially explaining the varied and sometimes extensive aphasic symptoms associated with lesions in different parts of the brain.

Most data driven and least speculative, Chang and associates are unique among neurolinguistic researchers in that they consider contextual, experiential, and behavioral learning effects on the eventual establishment of brain regions that mediate and modulate speech-language functions. In one of the studies, Bao et al. (2004) demonstrated that operant conditioning (training) improves temporal processing in the brain. A finding of importance is that electrical stimulation may not produce the same results across individuals and may be unstable within the same individuals. Prior stimulation in an individual may either facilitate or extinguish subsequent stimulations (Breshears et al., 2015).

Chang and associates assert that speech is not a purely linear feedforward process in which phonetic, phonemic, and lexical elements are sequentially identified or programmed. Instead, speech requires integration of knowledge generated from multiple sources, most of which is learned. The authors have demonstrated that sensory processing in the brain is adapted to statistics of behaviorally relevant stimuli (Hullett et al., 2016). Most neurolinguistic researchers treat brain as an autonomous structure unaffected by environmental variables, except for mentioning nonspecific variables like “speech heard,” “auditory input,” and so forth. Chang and associates have generally supported the dual-stream model of speech and language representation in the brain (Chang et al., 2015).

Evaluation of Intraoperative Language Mapping

- The intraoperative method has greatly expanded the database on cerebral language processing. Research has moved from simple phoneme and word level to the levels of comprehension and production of syntactically complex sentences.
- The method has confirmed that language areas are more extensively connected than nonlanguage areas.
- Combined with animal studies, the method has demonstrated the effects of learning and training on the relationship between brain and language.
- The intraoperative method generally recruits persons with epilepsy who are candidates for neurosurgery. The method is not designed for evaluating aphasic disorders.
- While the method generates data on cerebral language sites, it is not clear whether the results of intraoperative data may be generalized to persons with stroke.

Lesion-Symptom Mapping

Also known as lesion-behavior mapping, this approach uses neurodiagnostic methods that produce images of greater resolution (e.g., computed tomography and fMRI coupled with voxel technology) to relate aphasic language impairments to lesion site and size (Forkel & Catani, 2018; Landrigan et al., 2021; Matchin et al., 2022; Mirman & Thye, 2018; Sperber & Karnath, 2018). High-resolution images of the brain make it possible to locate lesions in narrowly defined brain patches called *voxels* that may be as small as 1×1 mm. Lesions in specific voxels may be matched with observed symptoms in persons with stroke or other brain diseases.

Lesion-symptom matching began with Broca and Wernicke in the 19th century and has continued ever since. While the approach is not new, the technical sophistications of imaging procedures and statistically correlating brain lesions to their behavioral consequences are new and evolving. Aphasic symptoms may be grouped or combined into composite speech-language measures. Statistical analysis is made to find comparable patterns across diagnostic groups. Several investigators claim that separate brain regions may be identified for three cognitive systems that underlie language and its neurologically based impairments: phonological, semantic, and fluency. Research on a fourth system, impairments in executive tasks in persons with aphasia, has not produced consistent results.

Mirman and Thye (2018) reviewed multiple lesion-symptom matching studies. The following summary is based on their review.

Phonological *comprehension* impairments are assigned to the posterior to anterior superior temporal lobe, part of the ventral pathway. Phonological *production* impairments are assigned primarily to inferior parietal and frontal regions, part of the dorsal pathway. Integrated (not dissociated) speech perception and production problems suggest damage to the superior temporal sulcus.

Semantic impairments (e.g., matching words or sentence to pictures) tend to be associated with lesions in the anterior temporal lobe, including the middle and superior temporal gyri. The anterior temporal lobe is thought to integrate different kinds of semantic information (recognition and comprehension of sounds, syllables, and words). Semantic disruptions also

may be associated with such subcortical white pathways as the inferior fronto-occipital and uncinate fasciculus.

Fluency impairments (shorter or incomplete sentences, hesitant speech) may be associated with lesions in the insula and inferior frontal and precentral regions. Lesions in the middle frontal gyrus and frontal white matter (including the anterior portions of the arcuate fasciculus) also may impair fluency.

Evaluation of Lesion-Symptom Mapping

- This comprehensive and data-driven method has failed to produce reliable and valid symptom-lesion matching. Kasselimis et al. (2017) could not match symptoms with lesions in 63.5% of their study participants.
- Theoretically, symptom grouping may help reduce the unmanageable diversity of aphasic language impairments, but the clinicians who need to measure each individual deficit in a patient may not find the method useful.
- Just as with other approaches, symptom-lesion mapping ignores individual variability in symptoms.
- A significant limitation of the approach is that it does not address word retrieval problems or grammatical (syntactic) impairments.
- The clinical relevance of lesion-symptom mapping is not established (Mirman & Thye, 2018).

General Evaluation of Aphasia Classification

Aphasia classification continues to be problematic. The classic model's limitations are well known and undisputed. The lesion sites for different types of aphasia vary across individuals.

Researchers criticize the clinicians for using an outdated and inaccurate model of aphasia (Charidimou et al., 2014; Hickok, 2022; Hickok & Poeppel,

2004; Nasios et al., 2019). Clinicians and researchers perform different tasks, however. Researchers analyze group data. Clinicians analyze an individual's aphasic symptoms and take note of symptoms that may not be consistent with the client's assigned diagnostic category or known lesion sites. They then proceed to design an intervention to remediate the deficits based on behavioral deficits, not brain lesions (Fabian et al., 2020; Fridriksson & Hillis, 2021). Lesion sites alone do not dictate treatment targets or procedures. For the clinicians, the rest is theoretical.

Advocates of the newer models admit that their models are not yet ready for clinical application (Mirman & Thye, 2018). Model builders hope to find consistent patterns in symptoms and correlations with lesions. Criticizing the results of single-subject or small sample group design studies, researchers suggest that the larger the sample, the greater the uniformity of data that can support a theory or a symptom classification (Charidimou et al., 2014). However, the larger the sample, the greater the variability and smaller the chances of finding that sought-after uniformity. Pooled data cannot be uniform when unique individuals contribute to that pool. Statistical analyses of large group data do not create uniform data; they only mask individual differences. Unique individual data in persons with aphasia are paramount to clinicians, but they are “noise” to group researchers who have an elusive goal of data uniformity (Halai et al., 2017). To achieve data uniformity, large group researchers do an “initial data cleaning” (Landrigan et al., 2021, p. 1379) by eliminating troublesome individual data and then by generating helpful data by statistics (not by observing individuals).

Data that vary across individuals *are the data*. Differences in individual brain anatomy; the various types, numbers, and extents of vascular diseases that lead to strokes; extensive and diffuse connecting pathways in the brain that may be differentially affected by brain diseases; still unknown micro- and macrostructures of the brain; the premorbid health of the stroke patients; premorbid personal variables (e.g., age, sex, education, ethnicity, socioeconomic status); different postonset times at which measures are obtained; poor agreement among standardized deficit measuring instruments; different recovery patterns in different stroke patients; and likely other unknown factors create highly variable symptoms in stroke patients (Charidimou et al., 2014; Landrigan et al., 2021).

In spite of numerous newer efforts, a new classification of aphasia that is comprehensive and clinically applicable has not emerged. A survey of aphasiologists showed that a majority still use the classic model, if only because they do not agree on a better model (Trembley & Dick, 2016).

Aphasia can be classified only into broad and gross categories, with known limitations. A diagnosis may be made based on dominant symptoms. Within a diagnostic category, atypical symptoms are expected, assessed, and treated. Heeding Darley (1982), who wrote, “Little is to be gained clinically by splitting aphasia into subgroups” (p. 42), most clinicians use the classification only as a rough guideline. And as Kasselimis et al. (2017) have suggested, aphasia symptom classification research may have to begin “on a blank slate” (p. 65).

Overview of Aphasia Symptoms

Most individuals with aphasia exhibit a set of common symptoms. The severity of the same symptoms found across types of aphasia may vary, however. Most aphasia types are distinguished by only a few unique features. Therefore, an understanding of common as well as unique aphasia symptoms is essential for differential diagnosis.

Paraphasia

A central feature of aphasia, **paraphasia** is word or sound substitution error. Aware of their errors, persons try to correct them. Paraphasia may be **verbal** (less appropriate words substituted for the more appropriate ones), **semantic** (such semantically related word substitutions as *son* for *daughter*), **random** (substitution of words unrelated to the target word, for instance, *window* for *banana*), **neologistic** (substitution of meaningless, invented words), or **phonemic** (substitution of one sound for another, for example, *loman* for *woman*).

Although more common in Wernicke's aphasia, paraphasia is present to some extent in almost all cases of aphasia. Therefore, it is not useful in distinguishing the different types of aphasia.

Impaired Fluency

An aspect of language production, fluency is relatively effortless, smooth, and flowing speech without excessive interruptions, often described as dysfluencies. Traditionally, a much researched disorder of fluency is stuttering, characterized by a higher-than-normal frequency of dysfluencies in speech. In the study of aphasia, the terms fluency and fluency impairment have a different meaning.

Aphasiologists define fluency as speech that approximates the normal rate, typical word output, length of sentences, and the melodic contour. Persons who produce five or more connected words may be judged fluent. When word output is less than 50 words per minute in conversation, fluency is significantly impaired (Benson & Ardila, 1996). Based on the fluency dimension, two major types of aphasia (fluent and nonfluent) have been identified.

Impaired Auditory Comprehension

Difficulty understanding spoken language is another general characteristic of aphasia. The degree of impairment across individuals may vary from mild to severe.

Moderate-to-severe impairment in speech comprehension is likely in persons with global, Wernicke's, and transcortical sensory aphasia. Mild-to-moderate impairment is seen in persons with Broca's and transcortical motor aphasia. Mild comprehension problems may be evident in persons with conduction and anomic aphasia. Auditory comprehension problems do not reliably distinguish the fluent from the nonfluent. Those problems are present in all types to varying extents.

Repetition

Repetition in aphasia literature is an individual's imitation of single words, phrases, and sentences a clinician models. (In stuttering, repetition is the reiteration of one's own productions.) Imitation is impaired to varying extents in different types of aphasia. Impaired repetition is the dominant symptom of conduction aphasia, a fluent variety. To the contrary, relatively intact repetition skills are a distinguishing feature of transcortical motor aphasia, a nonfluent variety.

Repetition errors are more frequent or more severe in individuals with Wernicke's aphasia, who also have significant impairments in auditory comprehension of speech. Nonetheless, persons with transcortical sensory aphasia, who also have relatively poor auditory comprehension, may have good repetition skills.

Similarly, some persons with good auditory comprehension may exhibit poor repetition skills. For instance, individuals with Broca's aphasia cannot repeat because of impaired word production, not comprehension. Hence, persons with different types of aphasia may demonstrate the same problem for different reasons.

Anomia

Anomia is difficulty in naming or finding correct words during verbal expression. Anomia is not particularly useful in making differential diagnosis because it is found in almost all types of aphasia. Naming problems tend to persist in persons who have recovered from aphasia. *Anomic aphasia* is a separate disorder, however.

Confrontation naming is naming in response to a verbal demand (e.g., "What is this?"). The problem may also become evident in word fluency tasks, in which the person is asked to name as many items of a certain category as possible (e.g., "Name as many California cities as you can"). Confrontation naming may be easier than word fluency tasks for some persons.

Other factors that influence naming include frequency with which words occur in language and the semantic category to which the words belong. Low-frequency words are more difficult to recall; high-frequency words are easier. Certain kinds of words, for instance, names of animals or vegetables, may be especially difficult for some individuals.

Pronoun Production Errors

Misuse or failure to use pronouns is an aphasic problem regardless of the age, postonset duration, language spoken, and types of aphasia (Arslan et al., 2021). Nominative or personal pronouns (e.g., *she*, *he*) are more often correctly produced than the accusative pronouns (e.g., *her*, *him*). Production of reflexive pronouns (e.g., *herself*, *himself*, *themselves*) may be relatively better preserved. Pronoun production or comprehension may be more seriously impaired in more complex (longer) as opposed to simpler (shorter) sentences. Pronoun comprehension deficits, production deficits, or both may be variable across persons with the same type of aphasia (Arslan et al., 2021).

Writing Problems

Writing problems associated with cerebral lesions in the adult are called **agraphia**. Most if not all individuals with aphasia have some level of impairment in their writing skills. Hence, these problems do not distinguish different types of aphasia.

Agraphia associated with aphasia includes a variety of writing problems, which generally parallel the speech-language problems. Agrammatic, effortful, and sparse speech of Broca's aphasia is reflected in agrammatic, effortful, and sparse writing. Similarly, fluent but meaningless speech of persons with Wernicke's aphasia may be associated with free, copious, but meaningless writing. Collectively, persons with aphasia may write with poor letter and word formation. Spelling errors are common. They may reverse, confuse, or substitute letters. Errors tend to be more in writing less frequently used words.

Efforts at self-correction may be numerous but unsuccessful. Some words they write may be nonsensical (neological, invented). In the most severely impaired persons, writing may be totally unreadable.

Reading Problems

Reading problems also vary across individuals with aphasia. Some individuals may orally or silently read printed passages but may understand little or nothing of what they read.

The terms *alexia* and *dyslexia* may be used interchangeably to refer to reading problems. Medical specialists, however, limit the term **dyslexia** to the reading problems found in children who are still learning to read; it is a learning disorder, not an impairment of learned reading skills (<https://www.mayoclinic.org/diseases-conditions/dyslexia/symptoms-causes/syc-20353552>). On the other hand, **alexia** is defined as “the loss of ability to comprehend written language” (Benson & Ardila, 1996, p. 97) following brain injury, found in persons with good premorbid reading skills.

Reading and writing problems often coexist in aphasia, although rare cases of pure alexia with only reading difficulty have been reported. Individuals with severe aphasia may exhibit equally severe reading problems. Some may not recognize the printed word at all. Many struggle to read and read aloud with many errors. Individuals may read sentences word-by-word. A fairly common difficulty is a failure to understand what is read.

Gestures

Understanding or using gestures and pantomime in communication may be disturbed in individuals with aphasia. The disturbance is not entirely a function of any motor deficit (e.g., paresis or paralysis of hands) that may exist in a person with aphasia.

Individuals with aphasia may have difficulty imitating specific gestures (such as those of the American Sign Language). Sign language users with aphasia may exhibit difficulties in their sign expression that parallel verbal deficits found in oral language users who have aphasia.

Changing Symptom Complex

Aphasia symptoms change over time. Clinicians need to consider this dynamic nature of aphasia in assessment and treatment. Aphasia at onset is usually more severe; improvement over time is common. Pedersen et al. (2004) reported that among 270 persons at the acute stage, the diagnosis was 32% global, 25% anomic, 16% Wernicke's, 12% Broca's, and 7% transcortical sensory aphasia. One year later, they were only 7% global, 13% Broca's, 5% Wernicke's, 0% transcortical sensory, 6% conduction, and 29% anomic.

All types of aphasia change to a less severe form during the first year. Nonfluent aphasia may change to fluent. Global aphasia may sound more like Wernicke's and Broca's may be predominantly anomic. Fluent aphasia, however, may become less severe but may not become nonfluent. Persons with global aphasia at onset show the greatest improvement over 12 months. Other kinds of aphasia may turn into anomia, with a persistent naming difficulty (Pedersen et al., 2004).

Nonfluent Aphasias

Four varieties of aphasia are classified as *nonfluent*: (1) Broca's aphasia, (2) transcortical motor aphasia, (3) mixed transcortical motor aphasia, and (4) global aphasia. Nonfluent speech associated with these four types is characterized by the following:

- Excessively slow speech rate (fewer than 50 wpm)
- Excessive speaking effort, struggle, facial grimaces, and hand gestures
- Limited phrase length, speech limited in both length and complexity

- Abnormal prosody, lack of intonation and rhythm, and monotonous or choppy speech
- Vastly reduced amount of speech
- Difficulty initiating speech
- Excessive use of content words (nouns and verbs) and omission of function words (grammatical morphemes)

Generally, fluency is impaired in individuals who sustain lesions in the anterior parts of the brain. Those with lesions in the posterior parts of the brain retain fluency the best.

Broca's Aphasia

The main language features of *Broca's aphasia* include nonfluent, effortful, agrammatic, and generally meaningful language. Naming and word repetition may be impaired. Auditory comprehension may be better than in individuals with Wernicke's aphasia.

Neuroanatomical Bases and Features of Broca's Aphasia

The posterior-inferior (third) frontal gyrus of the left hemisphere is known as Broca's area (Brodmann's area 44, which may extend to parts of area 45). It is supplied by the upper division of the middle cerebral artery. Broca's aphasia is a controversial diagnosis because of its varied symptom complex. Lesion location is typically ambiguous (Ardila et al., 2016; Charidimou et al., 2014; Fridriksson et al., 2015; Gajardo-Vidal et al., 2021; Kasselimis et al., 2017; Kim et al., 2019; Ochfeld et al., 2010; Tremblay & Dick, 2016). Generally, research evidence suggests that damage to the postcentral gyrus and the supramarginal gyrus may cause phonetic errors, damage to inferior and middle frontal gyri and underlying white matter may affect sentence production, and damage to postcentral and supramarginal gyri (parietal regions) may impair phonetic-articulatory planning and execution (Mirman et al., 2019).

Figure 3–1 shows potential primary and secondary cortical areas that may be involved in Broca’s aphasia.

Nonetheless, there is conflicting evidence about Broca’s aphasia:

- Imaging studies of Broca’s original patient have revealed extensive cortical as well as subcortical damage beyond Broca’s area. Broca’s aphasia typically involves more extensive and deeper cortical (white matter) damage.
- Areas other than Broca’s may be involved in Broca’s aphasia (e.g., the lower portion of the motor strip, areas anterior and inferior to area 44).

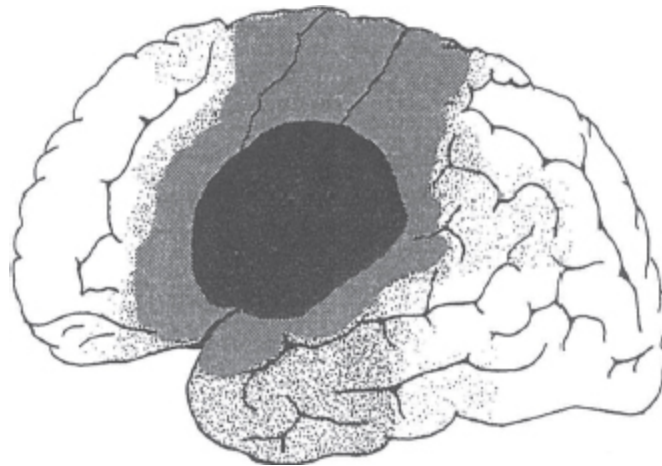


Figure 3–1. Potential neuroanatomical sites of Broca’s aphasia. The darker area is the primary site of lesion in most cases; lighter shading shows other areas that may be involved.

- Chronic Broca’s aphasia is associated with damage to both Broca’s and Wernicke’s areas (Fridriksson et al., 2015).
- Instead of aphasia, damage limited to Broca’s area may produce transient mutism, subsequent mild apraxia, and transcortical motor aphasia, another nonfluent variety.
- Involvement of portions of frontal, temporal, and parietal regions may produce symptoms often grouped under Broca’s aphasia.

- Individuals with Broca's aphasia but without damage to Broca's area have been observed.
- Lesions alone do not explain the observed language deficits. Hypometabolism and reduced capacity of unaffected areas also may contribute to functional deficits.

Language Disorders of Broca's Aphasia

Typical language disorders associated with damage to Broca's area are more likely to be observed in the acute stage, rather than in the chronic state (Ochfeld et al., 2010). Some persons who sound like they have Broca's aphasia may have had global aphasia at the onset of stroke and aphasia. The following language features characterize Broca's aphasia (Ardila et al., 2016; Benson & Ardila, 1996; Hegde & Freed, 2022):

- **Nonfluent and effortful speech.** The speech is filled with many pauses, interjected sounds (e.g., "uh"), revisions, sound and syllable prolongations, and repetitions.
- **Agrammatic, limited speech.** Telegraphic, agrammatic speech is limited to nouns and verbs (content words) and excludes articles, conjunctions, auxiliary verbs, copulas, prepositions, and inflections (function words or grammatic morphemes). Speech may be limited to just a few words and short sentences.
- **Slow rate with uneven flow.** The frequent pauses disrupt the flow of speech and create an unnatural rhythm, prosody, and monotonous speech.
- **Impaired repetition of words and sentences.** Significant difficulty imitating (repeating) modeled utterances; omission of grammatical elements while repeating (e.g., when asked to repeat "the boy is walking," the individual may repeat "boy walk").
- **Impaired confrontation naming.** Naming an object, a picture, or a person when asked (e.g., "What is this?") is especially difficult.
- **Circumlocution.** Persons may use imprecise words or beat around the bush as they have difficulty producing the precise words in

running speech.

- **Some impairment in auditory comprehension.** Pointing to objects named in sequence, distinguishing between relational words (e.g., bigger/smaller), and understanding meanings implied in word order may all be difficult. For instance, the client may fail to distinguish the meaning of such phrases as *my sister's husband* versus *my husband's sister* even though they understand the meaning of individual words involved.
- **Generally poor oral reading.** The oral reading is effortful and nonfluent with limited comprehension of what is read.
- **Writing problems.** Writing tends to be slow, laborious, agrammatic, and with many spelling errors and omitted or poorly formed letters.

Associated Speech Disorders of Broca's Aphasia

Broca's aphasia may often coexist with motor speech disorders because of extensive brain damage, especially in the vicinity of motor speech areas (Hegde & Freed, 2022):

- **Apraxia of speech.** This motor speech disorder includes difficulty in initiating articulation, effortful articulation, groping articulatory movements, and articulatory inconsistency in the absence of speech musculature weakness or paralysis.
- **Dysarthria.** Another motor speech disorder, dysarthria is caused by muscular weakness that impairs respiratory, phonatory, articulatory, resonance, and prosodic features of speech, resulting in reduced speech intelligibility. Dysarthria may be mild but, in combination with apraxia of speech, intelligibility may be significantly affected.

Other Neurobehavioral Problems of Broca's Aphasia

Because of the involvement of the motor cortex, persons with Broca's aphasia may exhibit:

- **Contralateral hemiplegia** (paralysis of one side of the body), hemiparesis (muscular weakness on one side), and weakness of facial muscles. Persons may use a wheelchair, a walker, or a cane. These motor problems are due to the likely damage of the descending pyramidal tracts.
- **Depression.** Being aware of their deteriorated communication skills, persons with Broca's aphasia may be depressed and react emotionally when they fail on assessment tasks (Laures-Gore et al., 2020). Called *catastrophic reactions*, persons may weep and refuse to continue or cooperate during testing and assessment.

In spite of their language difficulties, individuals with Broca's aphasia may communicate more effectively than the more fluent individuals with Wernicke's aphasia. Even with their halting speech, the former may succeed in communicating their basic needs because their speech is not filled with jargon and irrelevant utterances that characterize Wernicke's aphasia. Individuals with Broca's aphasia try to self-correct their mistakes and give relevant responses. They are socially appropriate, alert, and cooperative during assessment.

Transcortical Motor Aphasia

Transcortical motor aphasia (TMA) is another variety of nonfluent aphasia in which repetition skills are well preserved. Otherwise, speech-language impairments are similar to those found in Broca's aphasia. Both Wernicke and Lichtheim suggested the term transcortical motor aphasia, which has gained acceptance. TMA occurs much less frequently than Broca's aphasia.

Neuroanatomical Bases and Features of TMA

The lesions causing TMA are usually outside Broca's area. Most often, the anterior superior frontal lobe is involved. The lesions often are found in deep portions of the left frontal lobe or below or above Broca's area (Bhatnagar,

2017; Hegde & Freed, 2022). The most common neuroanatomical damage sites of TMA are shown in [Figure 3–2](#).

Other observations about the neuroanatomical basis of TMA include the following:

- Lesions often affect association pathways that connect perisylvian regions with other cerebral regions.
- Supplemental motor areas may be involved. Lesions may separate the supplemental motor cortex from Broca's area.
- The areas supplied by the anterior cerebral artery and the anterior branch of the middle cerebral artery are commonly affected, and the watershed region between the middle cerebral and anterior arteries may be involved.
- Head trauma with frontal lobe damage, tumors, herpetic encephalitis, and progressive neurological diseases affecting the frontal lobe also may cause TMA.
- Motor disorders — including rigidity of upper extremity, akinesia (absence or poverty of movement), and bradykinesia (slowness of movement) — may be seen to varying extents, especially soon after the onset of aphasia. Hemiparesis (leg more involved than the arms) and buccofacial apraxia may be seen in a few cases.
- Apathy or behavioral withdrawal may be evident. Most individuals show little or no interest in verbal communication.

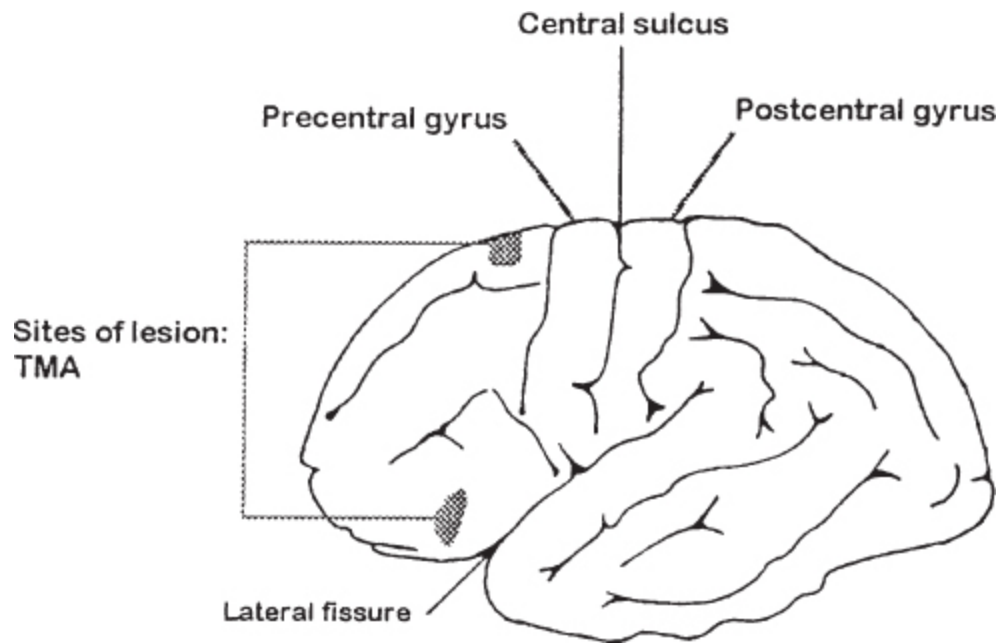


Figure 3-2. Two common neuroanatomical lesion sites of transcortical motor aphasia.

Language Disorders of TMA

Individuals with TMA exhibit the following language characteristics (Benson & Ardila, 1996; Hegde & Freed, 2022):

- **Muteness.** Speechlessness soon after the onset, possibly due to akinesia; subsequently, marked difficulty in initiating speech.
- **Echolalia and perseverative speech.** As muteness fades, echolalia (repetition of heard speech) and perseveration (multiple repetitions of one's own utterances) may emerge.
- **Intact repetition skill.** The unique and diagnostic feature of TMA is a remarkably preserved repetition skill. An individual who may give only a single-word response after much coaxing may correctly repeat long and complex sentences that the clinician models for imitation. This is in contrast to people with Broca's aphasia whose repetition skills are seriously impaired.
- **Reduced spontaneous speech.** Lack of interest in verbal communication and a disinterest in spontaneous speech are

common.

- **Nonfluent, paraphasic, agrammatic, and telegraphic speech.** Recovered speech is similar to that found in Broca's aphasia, with missing complex and precise syntactic structures. Single words that need to be coaxed and unfinished sentences characterize the speech.
- **Naming problems.** Confrontation naming problems may improve with various cues.
- **Relatively intact serial speech.** Even though assistance may be needed to initiate it, counting numbers and reciting days and months may be unimpaired.
- **Intact knowledge of grammaticality and meaningfulness.** Individuals may correct a grammatically incorrect model the clinician provides and repeat it in its correct form (e.g., "These is books" may be corrected and repeated as "These are books"). Individuals may refuse to repeat nonsense syllables.
- **Limited word fluency.** Difficulty generating word lists (e.g., production of as many names of flowers as possible) and mixing words from different categories may be evident.
- **Stuttering** or a high frequency of disfluencies in some persons.
- **Use of motor prompts to initiate speech.** Individual may clap, vigorously nod, or wring hands to initiate speech.
- **Better comprehension than production.** Comprehension is generally good for simple conversation but may be impaired for complex speech.
- **Reading problems.** Reading aloud may be slow and difficult to maintain; reading comprehension may be good for simple material and impaired for complex passages. Both are better preserved than either speaking skills or writing skills.
- **Writing problems.** Disinterest in writing; when coaxed to write, the individual may write little and with poor spelling and large and ill-formed letters.
- **Apraxia of speech.** Some individuals may exhibit this motor speech disorder.

- **Sensory or visual-field loss and neglect.** Right limb neglect may occur in some cases.

Across individuals, the similarity between TMA and Broca's aphasia can vary; the closer the lesion to the Broca's area, the greater the resemblance between the two types. In essence, spared repetition skills most clearly distinguish TMA from Broca's aphasia.

Mixed Transcortical Aphasia

A rare variety, mixed transcortical aphasia (MTA) includes features of both fluent and nonfluent varieties (Benson & Ardila, 1996; Bhatnagar, 2017; Hegde & Freed, 2022). It combines symptoms of TMA and transcortical sensory aphasia, a fluent form described in a later section. Both production and comprehension of language are impaired. MTA is similar to global aphasia in that the language impairment is extensive and often severe. As in TMA, speech repetition skills are retained in MTA, a feature that distinguishes it from global aphasia.

Neuroanatomical Bases and Features of MTA

The damage most often associated with MTA is found in the cerebral border zone or the watershed area, between the areas supplied by the middle cerebral artery on the one hand and the anterior and posterior cerebral arteries on the other. Typically, Broca's and Wernicke's areas may be spared but isolated from the rest of the brain (Benson & Ardila, 1996; Hegde & Freed, 2022). The arcuate fasciculus may also be unaffected. Symptoms and features include:

- Hypoxia of various origins including carbon monoxide poisoning, reduced oxygen supply to tissue in spite of normal blood supply, anoxia (total lack of oxygen supply), chronic hypoperfusion (reduced blood flow), acute carotid occlusion or stenosis, and sudden cardiac arrest have all been suggested or documented in individual cases.

- Cerebral shock that results in severe cerebral swelling (edema) may be associated with MTA.
- Multiple embolic strokes, especially those involving the peripheral branches of the middle cerebral artery, can cause MTA.
- Neurological features of individuals with MTA include bilateral upper motor neuron paralysis (spastic type of paralysis affecting volitional muscle movements), severe spastic quadriparesis (weakness in all four limbs, suggesting bilateral brain damage), visual field defects, and weakness in hip and shoulder muscles.

Language Disorders of MTA

MTA resembles global aphasia, with profoundly impaired communication skills except for repetition skills. Language production and comprehension are both impaired, often severely (Benson & Ardila, 1996; Bhatnagar, 2017; Hegde & Freed, 2022). Persons with MTA are characterized by:

- **Severely limited spontaneous speech.** Some individuals may have no spontaneous speech.
- **Echolalia.** Speech may be limited entirely to echolalia, a parrot-like imitation of what others say, although repetition may be limited to three or four words. Nonsense syllables or foreign language words may also be repeated.
- **Repetition combined with sentence completion.** If an examiner says the initial part of a familiar statement or song, the person may immediately repeat it and then go on to complete it.
- **Severely impaired fluency and marked naming difficulty.** Neologisms may be present, but confrontation naming skill may be absent.
- **Severely impaired auditory comprehension.** Individuals may not comprehend even simple conversation (a feature of fluent aphasia).
- **Mostly unimpaired automatic speech.** Once initiated and uninterrupted, the months in a year or a number series may be recited. Once interrupted, the individual may not reinitiate, however.

- **Normal articulation.** Although many aspects of language are impaired, speech sound production skill may be relatively intact.
- **Severe reading and writing deficits.** Oral reading, reading comprehension, and writing may be severely, often totally, impaired.

Prognosis is thought to be poor for persons with MTA. Definitive statements are hard to make because only a few cases have been studied.

Global Aphasia

Global aphasia is the most severe form of aphasia that has a generalized (global) effect on communication skills (Bhatnagar, 2017; Hegde & Freed, 2022; Sierpowska et al., 2020). Affecting both verbal and nonverbal communication, the disorder spares no particular skill.

Neuroanatomical Bases and Features of Global Aphasia

The entire perisylvian region is affected in global aphasia. In addition to both Broca's and Wernicke's areas, damage may extend deeper into the basal ganglia, internal capsule, and thalamus (Bhatnagar, 2017). In exceptional cases, the damage may be restricted to lesions in the anterior region, the posterior region, or subcortical structures.

Affected regions are supplied by the middle cerebral artery, the largest branch of the internal carotid artery. If occlusions occur before it branches out to supply the various areas of the brain, effects are serious. Lesions in the frontal, temporal (especially the posterior temporal gyrus), and parietal lobes cause significant impairments. [Figure 3–3](#) shows the areas that are commonly affected in global aphasia.

Additional features of global aphasia include the following:

- Strong neurological symptoms include right hemiparesis or hemiplegia and right-sided sensory loss.
- Apraxia and apraxia of speech.

- Hemineglect, a tendency to neglect one side of his or her body, especially the left side, is more common in right hemisphere damage.

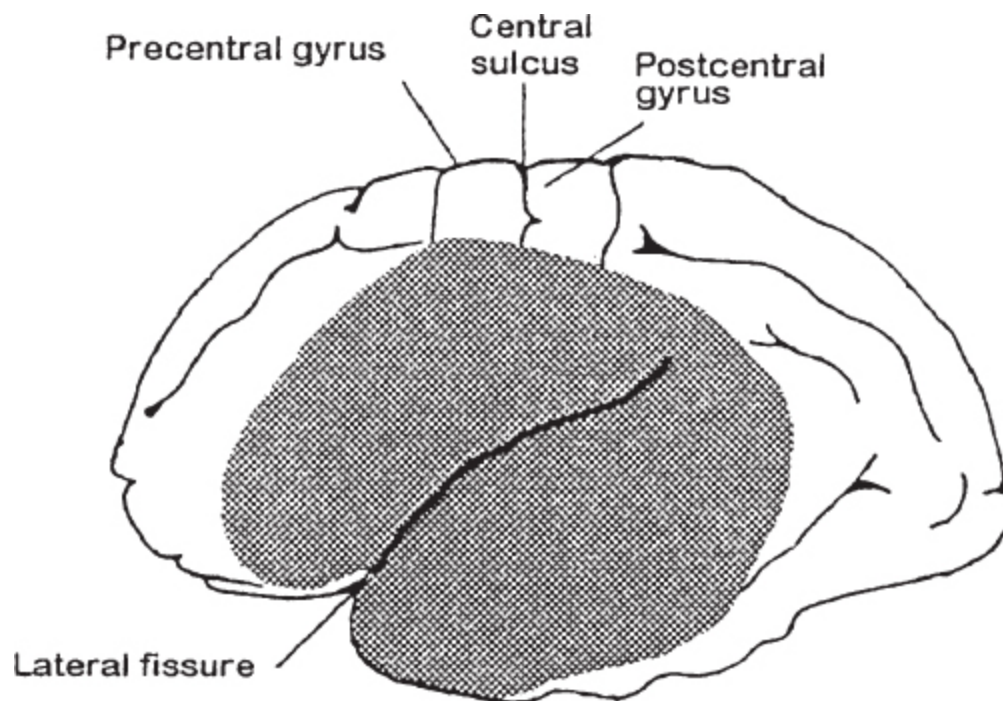


Figure 3-3. Neuroanatomical basis of global aphasia. The entire language zone surrounding the sylvian fissure (*shaded area*) is involved in global aphasia.

Language Disorders of Global Aphasia

Extremely limited language skills distinguish global aphasia from other forms (Hegde & Freed, 2022):

- **Globally impaired communication skills.** All verbal and nonverbal communication skills may be profoundly impaired. Language may be limited to a few recognizable or unrecognizable words, exclamations, and serial utterances. Repetitions of such consonant-vowel combinations as *do-do-do* or *ma-ma-ma* may be heard.
- **Severely reduced fluency.** This results in equally restricted speech output.
- **Impaired repetition.** Even simple words may not be repeated.

- **Impaired naming.** Severity of this difficulty reflects the extremely limited speech.
- **Impaired auditory comprehension.** Comprehension may be limited to single words, although it may be better than what expression.
- **Impaired reading and writing skills.** These problems parallel their severe language deficits.
- **Symptoms may change.** Global aphasia at the onset may evolve into Broca's, Wernicke's, anomic, or conduction aphasia.
- **Prognosis is generally poor.** Nonetheless, communication skills may improve with treatment.

Among the four varieties of nonfluent aphasia, clinicians probably treat more individuals with Broca's aphasia than any other type. The next section describes the four types of fluent aphasia.

Fluent Aphasia

Symptoms across varieties of fluent aphasia vary, but a common dominant element is relatively well-preserved fluency. Speech is abundant, with verbal output in the range of 100 to 200 words per minute. Individuals produce their speech with ease, good articulation, typical prosody, and a normal phrase length of five to eight words per phrase (Hegde & Freed, 2022; Hickok & Poeppel, 2004).

Fluent does not imply typical language, however. Individuals omit meaningful and significant words and go on creating meaningless words. In spite of their flowing and copious speech, persons are less effective in their communication than those with nonfluent aphasia.

Wernicke's Aphasia

Carl Wernicke, the German neuropsychiatrist, first described a form of aphasia in 1874 and called it *sensory aphasia*. *Receptive aphasia* is still in use. Wernicke's aphasia is clinically well established as it is one of the early aphasia syndromes to be described. It is less controversial than Broca's aphasia.

Wernicke's aphasia is characterized by fluent and sometimes excessive verbal expressions that are grammatically intact but are full of paraphasias and neologisms that may render the utterances unintelligible. There is a significant auditory comprehension deficit. This combination of fluent and jargon-filled speech and poor speech comprehension help distinguish Wernicke's aphasia from other varieties.

Neuroanatomical Bases and Features of Wernicke's Aphasia

Wernicke had localized the damage in the superior temporal gyrus in the dominant hemisphere. Subsequent studies and brain imaging techniques have confirmed that, in most individuals, the lesion is found in the posterior portion of the superior temporal gyrus in the left hemisphere, now known as Wernicke's areas (Alyahya et al., 2020; Hickok & Poeppel, 2004).

The lesion may extend to the second temporal gyrus, the surrounding parietal region, the angular gyrus, and the supramarginal gyrus. In addition, subcortical damage that impairs connections to the temporal cortex may produce Wernicke's aphasia (Alyahya et al., 2020; Benson & Ardila, 1996; Fridriksson et al., 2018; Hickok & Poeppel, 2004; Lwi et al., 2021). [Figure 3–4](#) shows the primary and secondary areas of lesion often found in individuals with Wernicke's aphasia.

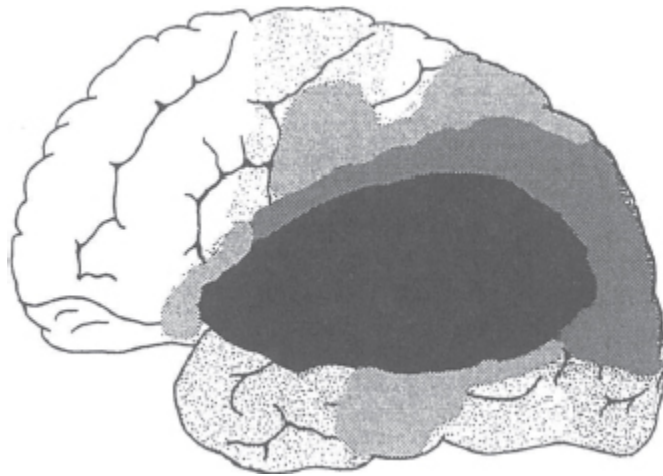


Figure 3-4. Neuroanatomical basis of Wernicke's aphasia. The darker area shows the primary site of lesion found in many cases; lighter shades show other areas that may be involved.

Embolic or thrombotic cerebrovascular accidents and intracranial hemorrhage in the posterior temporal lobe, trauma, or a tumor in the same area are among the more common causes of damage to Wernicke's and surrounding areas. The sites of lesion associated with Wernicke's area are supplied by the posterior branches of the left middle cerebral artery. Untreated ear infections also may cause damage to Wernicke's area.

In contrast to persons with Broca's aphasia, those with Wernicke's aphasia do not exhibit such obvious neurological disabilities as paresis or paralysis.

Other Behavioral Features of Wernicke's Aphasia

Persons with Wernicke's aphasia may also exhibit the following behavioral (and psychiatric) symptoms:

- **Confusion.** Jargon-filled, irrelevant-sounding, but somewhat incessant speech may give the impression of confusion.
- **Lack of awareness of their disability.** Individuals with Wernicke's aphasia do not seem to appreciate their communicative disorder and its effects on the listener.
- **Lack of frustration in failed communication.** Persons with Wernicke's aphasia are not as frustrated as those with Broca's aphasia,

possibly because of deficit unawareness.

- **Psychiatric symptoms.** Persons may be paranoid, homicidal, suicidal, or depressed. They may accuse others of speaking in a code they cannot decipher.

Because of confused and irrelevant speech, paranoia, and depression, combined with normal or near-normal neurological findings, Wernicke's aphasia may be misdiagnosed as a psychiatric disorder (e.g., schizophrenia).

Language Disorders of Wernicke's Aphasia

Fluent but jargon-filled speech and impaired speech comprehension are the hallmarks of this syndrome (Alyahya et al., 2020; Benson & Ardila, 1996; Lwi et al., 2021).

- **Normal or even hyper-speed fluency.** Prolific and excessive word output is typical, with the addition of extra syllables to some of the words produced. Produced effortlessly, the speech is so incessant that others may seek to stop it.
- **Rapid speech.** Fast rate of speech, as though there is so much to say in a short duration.
- **Normal prosodic features.** Rhythm, intonation, and syllable stress of meaningless utterances sound normal.
- **Good articulation.** Symptoms of apraxia of speech and dysarthria are typically absent.
- **Normal phrase lengths.** Phrase length is generally within the normal range (five to eight words), but some sentences may be excessively long.
- **Generally intact grammatical forms.** Grammar may appear normal, although the language may be characterized by para-grammatism (excessive use of grammatical morphemes). Goodglass (1993) characterized the language production of Wernicke's aphasia as displaying "pseudogrammatical sentences" (p. 211).

- **Severe word-finding problems.** Failing to name objects shown, a semantic substitution, or an incomprehensible neologistic jargon may emerge. Description, instead of naming, is common (circumlocution). There may be slightly better success in pointing to objects when asked, especially when choices are limited.
- **Empty speech.** Speech filled with semantic and literal paraphasia, extraneous syllables, and meaningless word creations communicates little. Grammatical words may be retained, but the content words may be omitted or substituted with neologistic words. Such general words as *this*, *that*, *stuff*, and *thing* may characterize the speech.
- **Poor auditory comprehension of speech.** A dominant feature of Wernicke's aphasia, this deficit may show different patterns across individuals. Some may comprehend only certain elements of spoken speech, whereas others may understand nothing. Names of common objects may be difficult to understand, but they tend to have greater difficulty with sentences. Distinguishing spoken words that contain minimally different phonemes (e.g., the /p/ and /b/ in *pat* and *bat*) may be problematic.
- **Impaired conversational skills.** This may be due to poor auditory comprehension; background noise makes it worse. An apparent good comprehension of conversation on a topic is short-lived and may cease when a new topic is introduced, requiring an extended time to establish new conversational topics. Turn taking may be grossly impaired in cases of severe involvement; individuals may fail to yield to their conversational partners.
- **Impaired repetition skill.** The degree of impairment may correspond to the degree of auditory comprehension deficit. Individuals may not repeat what they do not understand and may repeat some of what they do.
- **Impaired reading comprehension.** Individuals may fail to recognize sounds associated with written words, understand the meanings of printed words, match alphabets, or recognize the names of the alphabets.

- **Impaired writing skills.** Individuals may write a lot, write freely with their dominant hand, write in cursive style, but what is written tends to be meaningless. As copious as their speech, writing may be full of misspellings and nonexistent words. They may be unaware of their writing problems.

The extent and the nature of auditory comprehension deficits are not well understood. These deficits are difficult to measure. Better auditory comprehension is related to greater and earlier recovery of language functions, which is true in all individuals with aphasia. Wernicke's aphasia may evolve into anomic aphasia.

Transcortical Sensory Aphasia

Wernicke first described transcortical sensory aphasia (TSA) in 1881. Four years later, in 1885, Licht-heim described the same syndrome.

Most early descriptions of TSA highlighted a tendency to echo and imitate heard speech. Because of their similarities, TSA and Wernicke's aphasia may be confused with each other. The distinguishing feature, however, is that repetition is intact in TSA but impaired in Wernicke's aphasia.

We noted earlier that the nonfluent transcortical motor aphasia (TMA) is also characterized by intact repetition skills. Thus, TSA is a fluent counterpart of the nonfluent TMA.

Neuroanatomical Bases and Features of TSA

The most frequent cause of TSA is a lesion in the temporoparietal region. Damage to the posterior portion of the middle temporal gyrus is typical. The lateral aspects of the occipital lobe, the angular gyrus, and visual and auditory association cortex also may be involved. Wernicke's area, Broca's area, and arcuate fasciculus are typically intact (Benson & Ardila, 1996; Hickok & Poeppel, 2004).

The cortical regions of the dominant hemisphere that are typically damaged in TSA are in the watershed areas of the middle cerebral artery. In addition to ischemic lesions, head trauma also can cause TSA, but the symptoms are transient. The most common site of TSA is identified in [Figure 3-5](#).

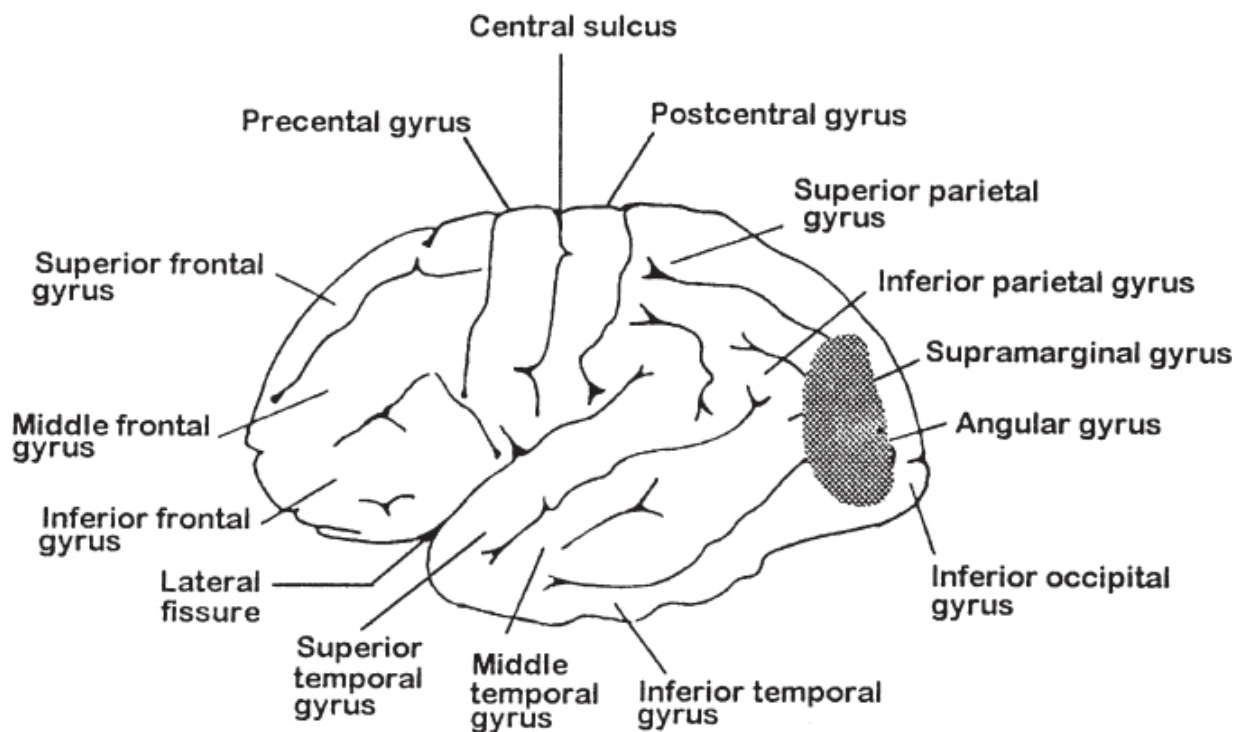


Figure 3-5. Neuroanatomical basis of transcortical sensory aphasia (TSA). The shaded area is the most common site of lesion.

The general and neurological characteristics of TSA may change over time. Related deficits associated with TSA include:

- Hemiparesis at onset of TSA. Because of good recovery, there may be no significant physical symptoms in later stages.
- TSA may be associated with Gertsmann syndrome, finger agnosia, agraphia, confusion of laterality, and acalculia due to occipital and angular gyrus lesions.
- Unilateral inattention (neglect of one side of the body).

- Sensory loss may or may not be present; visual loss is reported in some.

Language Disorders of TSA

TSA is characterized by fluent, well-articulated, paraphasic, somewhat echolalic, empty speech in the context of poor auditory comprehension (Benson & Ardila, 1996; Hegde & Freed, 2022; Hickok & Poeppel, 2004).

- **Fluent speech.** Fluency in TSA is similar to that in Wernicke's aphasia. Good articulation, normal phrase length, acceptable intonation, and stress patterns are evident. In essence, the phonemic, phonetic, and prosodic features are intact.
- **Generally good syntactic skills.** The agrammatic speech of Broca's aphasia is absent, and syntactic structures are comparable to those found in Wernicke's aphasia.
- **Paraphasias.** Both semantic and neologistic paraphasia may be present.
- **Empty speech.** Communication is poor because of paraphasia and neologism.
- **Impaired naming.** Striking features include failure to point to objects shown, poor confrontation naming, a tendency to describe instead of name objects, and word-retrieval difficulty in conversation. Neologism and nonspecific words at the expense of meaningful communication may be equally striking.
- **Good repetition skills.** Unlike those with Wernicke's aphasia, individuals with TSA can repeat words and phrases modeled for them, although without comprehending the meaning of what they repeat.
- **Echolalic behavior.** Absent in Wernicke's aphasia, repetition of words and phrases heard is a significant aspect of TSA. Clinician's words and phrases may be incorporated into one's own expressions with no regard for meaning. Echolalia may extend to grammatically

incorrect forms, nonsense syllables, and words from foreign languages, a feature not found in TMA.

- **Impaired auditory comprehension.** Many fail to understand simple requests to point to objects shown, follow motor commands, or answer yes/no questions.
- **Once initiated, normal automatic speech.** Once initiated with help, serial tasks (e.g., counting or reciting the days of the week) may be completed.
- **Completion of poems and sentences.** When a clinician starts a sentence or a poem, the individual may promptly complete them, a puzzling aspect of their otherwise poor comprehension of spoken language.
- **Poor reading comprehension.** Comprehension of what is read may be limited or absent. Oral reading, however, may be normal, albeit with some word substitutions.
- **Writing problems.** The writing in TSA resembles that in Wernicke's aphasia with problems paralleling expressive language problems.

In essence, a puzzling echolalia and repetition in the presence of impaired auditory comprehension distinguishes TSA from Wernicke's aphasia. In most other respects, the two types of aphasia are more similar than different. In fact, in some individuals, an initial Wernicke's aphasia may evolve into a chronic form of TSA. And in others, an initial TSA may evolve into anomia from which some may recover fully.

Conduction Aphasia

A type of fluent aphasia, conduction aphasia, is both rare and controversial (Benson & Ardila, 1996; Hegde & Freed, 2022; Hickok & Poeppel, 2004). Wernicke first postulated the existence of conduction aphasia. Individuals with conduction aphasia are mostly similar to those with Wernicke's aphasia. A notable exception is that auditory comprehension is near normal in conduction aphasia.

Neuroanatomical Bases and Features of Conduction Aphasia

The neuroanatomical basis of conduction aphasia is more controversial than it is for most other varieties. Lesions in the left posterior temporal-parietal boundary are often cited as the cause of conduction aphasia. Either the superior temporal gyrus, the supramarginal gyrus, or both may be affected (Benson & Ardila, 1996; Hegde & Freed, 2022; Hickok & Poeppel, 2004). Wernicke (1874/1977) had hypothesized that lesions in the subcortical white matter pathway, particularly the arcuate fasciculus, disconnect the sensory language area (later named Wernicke's area) from Broca's motor language area and cause conduction aphasia. Recent research suggests that instead of in the subcortical arcuate fasciculus, lesions in cortical areas cause conduction aphasia (see Hickok & Poeppel, 2004, for a review of studies). More anteriorly located lesions cause aphasia that is relatively nonfluent but with better auditory comprehension. More posterior lesions may produce aphasia that is more fluent but with poor auditory comprehension.

A frequently observed site of lesion in conduction aphasia is depicted in [Figure 3–6](#).

Varied neurological and other symptoms characterize conduction aphasia. Some individuals may present no neurological symptoms, whereas others may present mild to severe paresis of the right side of the face and right upper extremity from which good recovery may be expected.

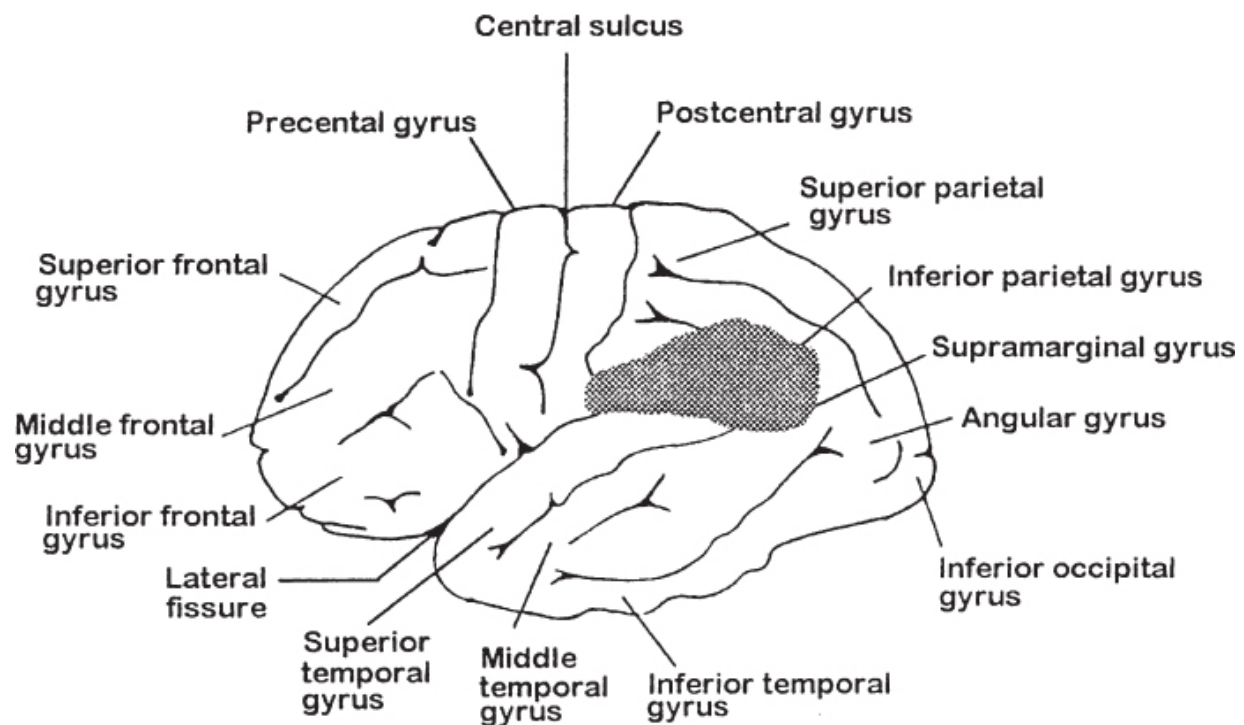


Figure 3-6. Neuroanatomical basis of conduction aphasia. The shaded area is the most common site of lesion.

Language Disorders in Conduction Aphasia

Impaired repetition and fluent and paraphasic speech with naming difficulties characterize conduction aphasia (Benson & Ardila, 1996; Hegde & Freed, 2022; Hickok & Poeppel, 2004). The main features of conduction aphasia include the following:

- **Impaired repetition.** Disproportionately impaired repetition with good auditory comprehension is a key diagnostic feature of conduction aphasia. In Broca's and Wernicke's aphasia, impairments in repetition and other language skills are proportionate; in conduction aphasia, repetition impairment is disproportionate to other impairments. Attempts at repeating modeled productions may include added or deleted phonemes (phonemic paraphasia). Repetition of longer words, phrases, sentences, function words, and unfamiliar phrases may be the most difficult. Nonetheless, the

individuals may use the words in spontaneous speech that they cannot repeat.

- **Varied speech fluency.** Some individuals are more fluent than others, but generally, even the more fluent persons are less fluent than those with Wernicke's aphasia. Hesitations and self-corrections interrupt fluency.
- **Paraphasic speech.** Literal paraphasia is more common than semantic paraphasias and neologisms.
- **Marked word-finding problems.** Content words may be more problematic than function words, resulting in empty speech.
- **Recognition of errors.** Individuals tend to make efforts at correcting their speech errors but typically are no more successful than those with Broca's aphasia.
- **Acceptable syntactic and prosodic features.** These normal features distinguish conduction aphasia from Broca's aphasia.
- **Mostly normal speech articulation.** However, there may be literal paraphasia (sound substitutions), more prominent in repetition tasks. Simpler phonemes may be substituted for more difficult ones. Phoneme deletions are common, but phoneme additions are rare.
- **Naming problems.** Generally mild to severe, but some individuals simply fail to name; most produce a plethora of literal paraphasias. Interestingly, those who cannot name objects may easily point to them when named.
- **Normal or near-normal auditory comprehension.** Comprehension may be adequate for typical conversation; if not, the problem may be limited to grammatically more complex structures.
- **Variable reading problems.** Reading aloud is difficult and may be filled with paraphasias. Failure to read short, printed material is not uncommon. Nonetheless, the individuals can silently read and comprehend such long and complex material as a novel or a scientific book.
- **Writing problems.** Individuals can write a few words with well-formed letters; spelling errors, letter omissions, reversals, and

substitutions may be common in these and in extended writing.

- **Buccofacial apraxia.** Difficulty in performing buccofacial movements when requested may be present in most cases. Some may fail to perform limb movements upon request.

In essence, a disproportionate impairment in repetition, fluent speech but less fluent than in Wernicke's aphasia, phonemic paraphasias, and good comprehension distinguishing conduction aphasia.

Anomic Aphasia

Anomic aphasia is the most controversial of all aphasia types. Its existence as a reliable syndrome of aphasia is doubted because of its highly variable lesion sites (Fridriksson et al., 2018).

Anomia is naming difficulty, a singular symptom of almost all types of aphasia and a residual symptom in many persons who have recovered from aphasia. **Anomic aphasia** is a variety of aphasia whose overriding feature is a persistent and severe naming problem with relatively intact language skills. Therefore, it is necessary to distinguish anomia as a symptom from anomia as a syndrome.

As a symptom, anomia is found not only in most varieties of aphasia but also in right hemisphere disorder (see Chapters 6 and 7), several forms of dementia (see Chapters 10 and 11), encephalitis, increased intracranial pressure, subarachnoid hemorrhage, concussion, and in all cases with generalized (diffuse) brain damage (see Chapters 8 and 9 on traumatic brain injury).

Neuroanatomical Bases and Features of Anomic Aphasia

Lesions that cause anomic aphasia are generally varied; in some cases, the loci may not be identified with any degree of certainty. Damage to multiple sites and cortical networks may be associated with anomic aphasia (Fridriksson et al., 2018). Some frequently cited sites of lesions include the angular gyrus and

the second temporal gyrus (Helm-Estabrooks & Albert, 2004). The juncture of temporoparietal lobes also may be involved. The damaged cerebral regions shown in [Figure 3–7](#) should be considered potential and incomplete.

Language Disorders in Anomic Aphasia

The most significant positive sign of anomic aphasia is a persistent naming problem. Fluent, if somewhat empty, speech may be another positive sign. Other signs of language impairment seen in other types of aphasia are either negative or somewhat variable (Helm-Estabrooks & Albert, 2004).

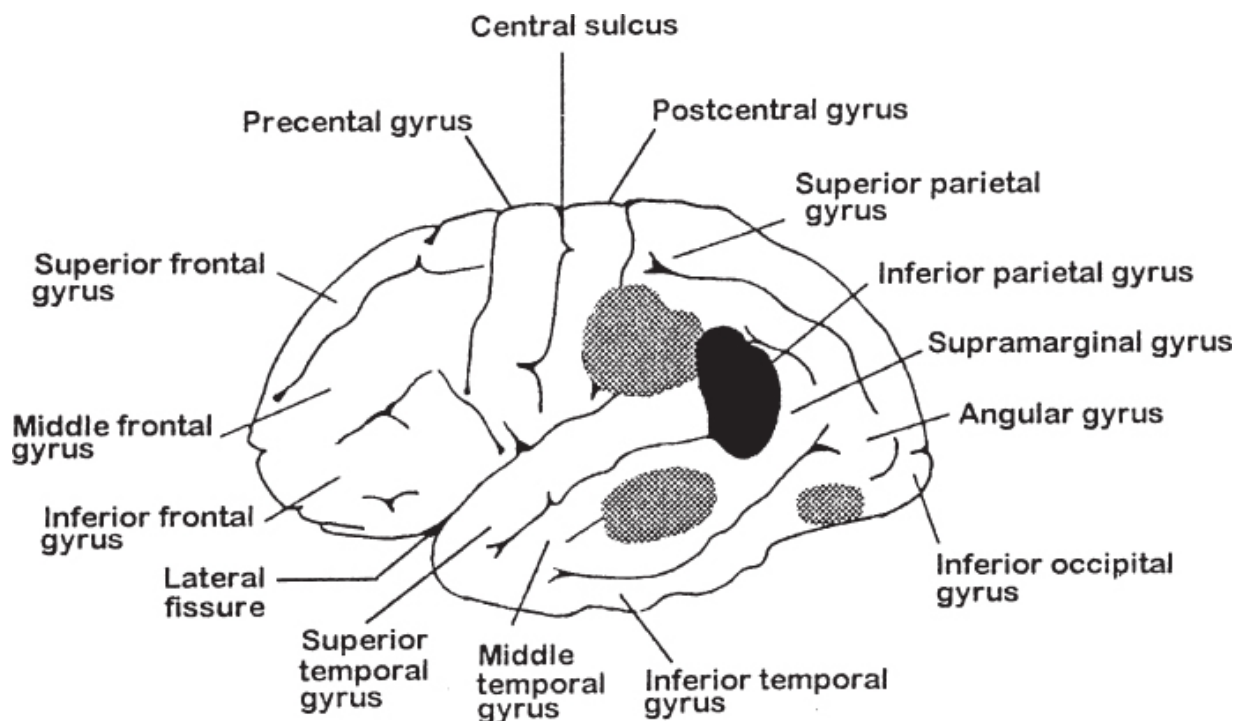


Figure 3–7. Potential neuroanatomical bases of anomic aphasia. The dark and the lighter-shaded areas are among the multiple possible sites of lesion.

Individuals with anomic aphasia tend to exhibit the following:

- **Word-finding difficulty.** To diagnose anomic aphasia, this difficulty should be debilitating, be pervasive, and exceed all other communication problems.

- **Unimpaired pointing.** The person who cannot name may easily point to the named objects, suggesting good comprehension of spoken names.
- **Fluent speech.** Often so described, but this is not fully accurate. Pervasive naming problems cause pauses, excessive repetitions, circumlocution, and nonspecific word productions (e.g., *this*, *that*, and *thing*) that result in empty speech.
- **Normal syntax.** Sentence structures may be normal or nearly so.
- **Good auditory comprehension.** Any problems that exist may be subtle.
- **Intact repetition.** Modeled words, phrases, and sentences may be accurately imitated.
- **Good articulation.** Speech sound production problems are not a feature.
- **Normal or near-normal oral reading and writing.** Individuals can write well and understand what they write or read.
- **Symptom variability.** Consistent across the individuals is the naming difficulty. The existence and the extent of other problems may vary due to the varied lesion sites. For instance, phonemic cues may evoke a naming response if the lesions are in the more frontal areas. Repeating a word endlessly with no comprehension of its meaning is associated with lesions in the angular gyrus.

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Chapter Outline

- Overview of Assessment
- Screening for Aphasia
- Diagnostic Assessment of Aphasia
- Assessment of Functional Communication and Quality of Life
- Assessment of Bilingual and Ethnoculturally Varied Individuals
- Diagnostic Assessment Summary
- Differential Diagnosis
- References

Learning Objectives

After reading the chapter, the reader will:

- Describe why aphasia is screened for and assessed
- Give an overview of aphasia screening and assessment tools
- Outline a basic aphasia assessment plan
- Describe how clients of varied ethnocultural backgrounds may be evaluated
- Summarize how aphasia may be distinguished from other, potentially confusing, disorders

PRINTED NOTES

Assessment of communication skills is made in the context of the person's current health status. The rate of recovery from the physical problems the recent neurological events created for the person will affect communication assessment. What the client and the family members expect from treatment and rehabilitation also affects a speech-language pathologist's (SLP's) assessment.

The main objectives of aphasia assessment are to (1) determine the current communication deficits and needs, (2) make a diagnosis of aphasia, (3) suggest the type of aphasia if justified, (4) counsel the client and the family members about the needed clinical services, (5) describe what might be expected if treatment is offered or withheld, (6) make tentative treatment plans, and (7) refer the person to other medical and rehabilitation specialists if warranted.

Overview of Assessment

To accomplish the objectives of assessment, the clinician uses both the standard protocols (e.g., the case history and interview) and procedures that are unique to aphasia (Hegde & Freed, 2022). In all cases, the clinician needs to take the following steps:

- **Obtain a case history.** Information on the person's education, occupation, and health history may be obtained from family members as well as the client. Details on strokes and other diseases, their immediate consequences, medical management, and the client's current health and speech-language status will be documented in the case history.
- **Conduct an interview.** The client, the family members, and the caregivers may all be interviewed to understand the events leading to

aphasia, perspectives of the caregivers and family members on the client's current situation, and the expectations from speech-language clinical services.

- **Screen the client's hearing.** A routine hearing screening is part of assessment to judge whether an audiological assessment is needed.
- **Conduct an orofacial examination.** The clinician may use a standard protocol and include diadochokinetic rates.
- **Administer standardized tests.** Ethnoculturally appropriate tools may be used.
- **Assess speech and language skills.** As described in subsequent sections, speech comprehension and production, functional communication skills, reading and writing skills, and the quality of life will be assessed.
- **Analyze and integrate the assessment results.** Information gathered through case history, interviews, formal and informal assessment procedures, direct observations, and information supplied by medical and other professionals may be analyzed and integrated to give a comprehensive picture of the person's strengths and limitations.
- **Make a differential diagnosis.** The clinician may identify the particular type of aphasia and distinguish it from such other disorders as dementia, communication disorder associated with traumatic brain injury, or right hemisphere disorder.
- **Counsel the client and the family members.** In a postassessment counseling, the clinician may share the results with the client, the family members, and the caregivers. A diagnosis and prognosis may be suggested. The clinician will answer questions the client and the family may have. A need for treatment may be suggested.

Screening for Aphasia

Before a diagnostic assessment is attempted, most individuals admitted to a hospital following a cerebrovascular accident (stroke) are screened for aphasic communication deficits. **Screening** is a brief procedure that gives an initial

impression of a disorder and suggests a need for in-depth assessment at a suitable time. Most individuals who have had a stroke or exhibit other signs of cerebral injury are screened at their hospital bedside.

Experienced SLPs may have their own informal but uniform means of screening for aphasia. A few minutes of conversation; some common objects to name or point to; questions that evoke brief descriptions; a request to recite numbers and days of the week; or repetition of words, phrases, and sentences may give an expert clinician sufficient screening information to determine a need for detailed assessment (Davis, 2000).

Clinicians may also use one of the standardized screening tests listed in [Table 4–1](#).

Results of a quick screen may suggest that there is a need for diagnostic evaluation, but that will have to wait. Most persons in their postonset stage in hospitals may need to be physically stabilized before they may be fully assessed.

Diagnostic Assessment of Aphasia

Assessment is about finding out the pattern of deficits a person with aphasia exhibits so that treatment may target those deficits. Diagnosis is a waystation to effective treatment. Therefore, a significant variable to consider is what the clients and families wish to see improved (as well as assessed). Fucetola and Connor (2015) reported that family members rate expressive language skills as more important than other skills that most standardized tests assess (e.g., word fluency and picture description). Conversational speech is especially important for family members. Another study has shown that people with aphasia want social engagement and work activities (Worrall et al., 2010). Therefore, as addressed in a later section, the SLP may place a proper emphasis on assessing functional communication, social participation, and quality of life.

Two considerations in assessment will determine the scope and procedures of assessment. First, not all skills need to be assessed in all clients, especially not initially, unless the skill is of diagnostic significance and an obvious treatment

target. For instance, in some cases, an initial assessment of reading and writing may be less important than functional verbal skills.

Second, not all skills need to be assessed by standardized instrument. Normative comparison is not that critical in assessing older persons with disrupted communication skills that were present prior to the neurological disease or trauma. Client-specific, functional skill-oriented, and criterion-referenced procedures will help tailor a treatment program specific and useful to the individual. Treatment results themselves may suggest a need for additional assessment. Subtle auditory comprehension problems, noted but not fully assessed earlier, may later become an important assessment target.

Table 4-1. Screening Tests for Aphasia

Test Name	Skills Screened
Aphasia Language Performance Scales (ALPS) (Keenan & Brassell, 1975)	Listening, talking, reading, and writing (also available in Spanish)
Aphasia Screening Test, Second Edition (Reitan, 1991)	Naming, copying, drawing, reading, repetition, following commands, simple arithmetic
Aphasia Rapid Test (Azuar et al., 2013)	Following orders, word and sentence repetition, object naming, fluency
Acute Aphasia Screening Protocol (AASP) (Crary et al., 1989)	Attention, language comprehension and expression
Aphasia Screening Test, Second Edition (Whurr, 1996)	Comprehension, oral and written language production, calculation

Bedside Evaluation Screening Test, Second Edition (Fitch-West et al., 1998)	Expression, object naming and description, repetition, comprehension, reading
Western Aphasia Battery (Kertesz, 2007)	Fluency, repetition, naming, reading, writing, drawing, comprehension through its bedside form
Boston Diagnostic Aphasia Examination–Short Form (BADE) (Goodglass et al., 2000)	Language comprehension, expression, repetition, reading, writing
Frenchay Aphasia Screening Test, Second Edition (Enderby et al., 2012)	Language comprehension and expression, reading, writing
Language Screening Test (LAST) (Flamand-Roze et al., 2011)	Language comprehension and expression
Mississippi Aphasia Screening Test (Nakase-Thompson, 2004)	Naming, automatic speech, repetition, object recognition, verbal instructions, reading instructions, fluency, writing
Multilingual Aphasia Examination, Third Edition (Benton et al., 1994)	Naming, repetition, fluency, aural comprehension, reading, writing
Quick Assessment for Aphasia (Tanner & Culbertson, 1999)	Naming, answering, questions, conversation
Sheffield Screening Test for Acquired Language Disorders (Syder et al., 1993)	Language comprehension and expression

Short-Form Philadelphia Naming Test (Walker & Schwartz, 2012)	Naming skills (words and pictures)
Sklar Aphasia Scale (Sklar, 1983)	Language comprehension, graphic sign recognition, naming, writing

Take a Detailed Case History

A detailed case history helps understand the client's current problems and premorbid communication and intellectual skills. Case history combined with a good interview is essential to understand the cultural and verbal background of the client. The SLP may gather details on the following:

- **Client's biography.** Biographical information on the client's age, education, occupation, ethnocultural background, oral and written language skills, and hobbies and interests will be useful in understanding the changes in the client that are brought about by the stroke.
- **Family constellation and medical resource access.** Information on the current living arrangements, family members and others who live with the client, and the family's access to clinical services will be useful in understanding the person, family support for treatment, and the health care access problems that need to be addressed in treatment planning.
- **Medical history and data.** Information on the client's health history, the potential causes and consequences of aphasia, associated diseases, current physical condition, any psychiatric complications (e.g., depression), current medications, and physical limitations or disabilities will help make a comprehensive assessment and set realistic treatment goals.
- **Behavioral observations.** Important diagnostic and treatment implications may emerge from careful observations of the client's

communication and general behavior. Changes in a client's behavior over time may suggest diagnosis and treatment targets that are different from the ones initially found appropriate.

Administer a Standardized Test

Standardized diagnostic tests give a format to assess various skills of interest. They provide standard procedures for scoring and interpreting the client's responses.

It is the responsibility of the clinician to select tests that are reliable, valid, and appropriate for the individual's ethnocultural background. Many standardized tests fall short of acceptable reliability and validity (Wallace et al., 2020). Conducting a systematic review of standardized tests of aphasia, Rohde et al. (2018) reported multiple problems with many of the commonly used tests. For instance, during standardization, (1) several tests did not compare the performance of persons with aphasia with that of healthy comparable persons; (2) some included nonstroke patients, but the etiology of their disease or disorder was unspecified; (3) some included, along with persons with aphasia, those with traumatic brain injury or right hemisphere disorder; (4) many did not collect validity data; and (5) some did not assess diagnostic accuracy, sensitivity, and specificity (Rohde et al., 2018).

Several other problems render many standardized tests irrelevant for assessing aphasia in certain groups of persons. Some standardization samples may not have included sufficient numbers of women, individuals belonging to minority groups or bilinguals, people from lower socioeconomic levels, and those from rural settings.

Some assessment tasks may be less informative. For instance, naming difficulty (word retrieval), one of the most debilitating problems found in many persons with aphasia, is typically assessed by asking the person to name pictures or objects shown. But there is evidence that individuals with aphasia perform more poorly on single-word naming tasks than on such tasks as picture description or connected speech (see Boyle, 2014, for a review of studies). During assessment, it may be more useful to take extended

conversational samples that include description of events, narration of stories, and discourse on a topic of interest. Alyahya et al. (2020) reported that storytelling narratives of persons with aphasia contained a higher proportion of content words and a greater variety of words compared to picture and procedural descriptions (e.g., *how do you make a sandwich?*). Such tasks may be client specific and nonstandardized because the scores will not be compared against any norms. In any case, the need to compare an individual's performance to norms is minimal in cases of older adults with aphasia and other communication disorders related to brain diseases and injury.

Considering both the advantages and limitations of standardized tests, clinicians can select one or more of the standardized aphasia diagnostic tests listed in [Table 4–2](#).

By reviewing the test manuals, the clinicians may note that:

- Tests have varying degrees of reliability and validity.
- Some tests are heavy on linguistic skills and others are more functionally oriented.
- Some tests help make a differential diagnosis of types of aphasias, but others do not.
- Some tests help grade the severity of the disability.
- Some have close to 50 subtests and take several hours to administer.
- Some tests are comprehensive and help assess most of the aphasic impairments.
- Several tests are specific to a single skill or impairment (e.g., auditory comprehension, naming, or reading and writing).

Generally, several procedures are needed because no single measure can assess the multiple aphasic deficits. Standardized diagnostic tests, clinician-prepared procedures, functional communication and quality of life evaluation, complex language measures, rating scales, and clinical judgments are all important. Each kind of assessment helps evaluate a dimension of communication that the other might miss or underevaluate. For instance, much useful functional communication assessment may not help make a diagnosis or a differential diagnosis of a particular type of aphasia. The

traditional standardized tests may not address functional communication skills, social participation, or quality of life. Therefore, assessment needs to use multiple instruments and measures.

Table 4-2. Standardized Tests of Aphasia

Test Name	Skills Measured
Amsterdam-Nijmegen Everyday Language Test (Blomert et al., 1994)	Understandability and intelligibility of verbal expressions; changes in communication over time
Aphasia Diagnostic Profile (Helm-Estabrooks, 1992)	Speaking, comprehension, reading, writing, and gestures; severity, classification, and behavior profiles
Boston Assessment of Severe Aphasia (Helm-Estabrooks et al., 1989)	Comprehension and expression, apraxia, gesture recognition and expression, reading, and writing
Boston Diagnostic Aphasia Examination, Third Edition (Goodglass et al., 2000)	Comprehension, conversation, expository speech, expression, repetition, reading, writing
Comprehensive Aphasia Test, Second Edition (Swinburn et al., 2022)	Cognition, language, and disability; measures the effects of aphasia on the individual's life; does not promote aphasia type classification
Discourse Comprehension Test, Second Edition (Brookshire &	Listening and reading comprehension, stated and

Nicholas, 1997)	implied meanings of stories
Neurosensory Center Comprehensive Examination for Aphasia (Spreen & Benton, 1977)	Naming, repetition, fluency, object description, memory, reading, writing
Philadelphia Naming Test (Roach et al., 1996)	Object naming
Porch Index of Communicative Ability, Fourth Edition (Porch, 2001)	Gestural, verbal, and graphic skills
Progressive Aphasia Severity Scale (Sapolsky et al., 2014)	Articulation, fluency, comprehension, repetition, reading, writing, and functional communication in primary progressive aphasia
Quick Aphasia Battery (Wilson et al., 2018)	Connected speech production, word and sentence comprehension, naming, repetition, reading
Reading Comprehension Battery for Aphasia-2 (LaPointe & Horner, 1998)	Silent and oral reading comprehension
Revised Token Test (McNeill & Prescott, 1978)	Auditory comprehension of verbal commands
SOAP Test of Syntactic Complexity (Love & Oster, 2002)	Comprehension of active, passive, subject-relative, and objective-relative types of sentences

Western Aphasia Battery– Revised (Kertesz, 2007)	Fluency, comprehension, repetition, naming, word finding, reading, writing, drawing
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As the brief descriptions of standardized aphasia tests offered in [Table 4–2](#) make it clear, the clinician who has administered and scored a test will have a basic understanding of the client’s speech comprehension, speech sound errors of coexisting dysarthria or apraxia of speech, basic expressive speech, word and sentence repetition, picture or object (confrontation) naming, picture description, speech fluency, simple conversation, agrammatism, word-finding difficulties, rate of speech, meaningful or meaningless speech, reading and writing deficits, counting and reciting, and so forth. To make a thorough assessment of a person with aphasia, however, it is essential to assess complex language skills in sufficient detail.

Assess Complex Language Skills

Complex language is connected, continuous, and extended speech-language production. In aphasia assessment, multiple parameters of complex language are important: interview, language sampling, conversation, narration, and discourse. An independent property of complex language, fluency may be assessed through those parameters of complex language.

Clinical interview and language sampling are basic and noncontroversial procedures of evoking connected and naturalistic speech-language skills. Conversation, narration, and discourse, on the other hand, are well defined and distinguished in child language (Nippold, 2022), but in aphasia assessment and research, their definitions are imprecise and distinctions among them are unclear (Prins & Bastiaanse, 2004). Evoked complex and continuous speech may be called *narrative* or *discourse* and sometimes both. For instance, Kim and Wright (2020) referred to telling a story after looking at the pages of a wordless picture book as *narrative discourse*. Others call picture book

description *narrative language production* (e.g., Mack et al., 2021). Some investigators consider narratives as a type of discourse and describe narrative, expository, and procedural discourse as subtypes of *monologue discourse*. Storytelling also can be described as *narrative discourse* (Leaman & Edmonds, 2021). Still others consider picture description, story narration, and interview as forms of connected speech or language samples (Boschi et al., 2017). Some describe all kinds of language production beyond the level of the sentence as *discourse* (e.g., Stark et al., 2021). Accordingly, two consecutively produced sentences qualify for discourse (Bryant et al., 2017), but it is doubtful if any clinicians would agree with that. Most everyday speech is beyond a single sentence, but it is not necessarily discourse. For instance, “I’m hungry. Do you have something to eat?” has two sentences but it is not discourse. Still others have defined *language use* as discourse (Bryant et al., 2017), but not all language use is discourse, and nondiscourse language (e.g., narration and everyday speech) is also “use” of language. Some of these definitions and descriptions are overly restricted, others are overly inclusive, and most are inaccurate or too vague to be useful in assessment or treatment outcome measurement. Purely linguistic structural definitions (such as the *production of two sentences*) do not capture the functional essence of narratives, discourse, or any type of simple or complex speech.

Complexity of language productions may vary across children and adults, but it is useful to recognize distinctions among interviews, language samples, narration, and discourse. They have procedural differences and may produce differences in the quantity, complexity, and variety of language evoked. An umbrella term, *complex language*, serves the purpose and includes the clinical interview, language sampling, conversation, narration, and discourse.

Interviews are question-based conversational exchanges in which one person mostly asks questions and the other person responds. The interviewer talks less than the interviewee. The interviewer seeks information, and the interviewee supplies his or her own biographical information, especially health- and communication-related information, other kinds of views and opinions, and relevant experiences. The roles of the interviewer and the interviewee are unequal.

Conversations are verbal exchanges between two (or more) people who typically play roughly equal roles in seeking and supplying information, although one person may dominate on occasions. Brief segments of narration or discourse may be embedded in conversations, but those segments should be distinguished from each other and from conversation. The give-and-take exchanges between two or more conversational partners are fewer in narratives and discourses. The latter two are more like monologues.

A **narration** is storytelling. During storytelling, the conversational partner is only minimally interactive. Stories may be objective or personal experiences. Stories may include an element of amusement, entertainment, mystery, surprise, or delight, none of which is essential to discourse.

Discourse is a short or long lecture on a general and public topic, a social or political issue, or a common societal problem. Discourse is less personal than some types of narratives (stories). Because a discourse is more like a monologue than conversation, the communication partner in a discourse may be a silent listener for the most part. In evoking a discourse from a person with a communication impairment, such as aphasia, the clinician needs to periodically prompt and request more information. Therefore, clinical discourse is slightly more *managed* than the typical discourse.

Procedural discourse is a subvariety in which a person describes how to perform certain actions (e.g., how to lace a shoe or how to book a hotel room). In assessing individuals with brain injury or diseases, procedural discourse may be recorded as a part of assessing daily living skills that may be negatively affected.

Assessment and treatment research suggests that traditional tests of aphasia are insensitive to impairments in social conversation and that narration and discourse better assess skills that standardized tests fail to capture (Alyahya et al., 2020; Armstrong, 2010; Bryant et al., 2016). Nonetheless, complex language sampling is not routine in clinical practice. Recording, transcribing, scoring, and analyzing language samples, narratives, and discourses are extremely time-consuming. Many clinicians are not trained adequately, especially in scoring and analyzing complex language samples. As an international survey of aphasia clinicians has revealed (Stark et al., 2021), the problem is further compounded for the clinician because there is no single,

uniform, objective, standardized, and practical method of scoring complex language samples, such as storytelling or discourse. For the most part, complex language collection and analysis are a research enterprise with little agreement on terms and methods, as summarized earlier. For the same reason, treatment researchers often do not include complex language in their outcome assessment. Surveys have suggested that most clinicians entertain their own and highly varied conceptualization of complex speech forms and listen to their clients as they produce connected speech and make online clinical judgments about deficiencies. Some surveys report as low as 16% of responding clinicians who record, transcribe, and formally analyze complex language samples (Bryant et al., 2017; Cruice et al., 2020), and others report a higher number of 78% (Stark et al., 2021). The higher number may be due to participant selection and consent bias. Stark et al. (2021) recruited only those clinicians who did discourse analysis. Therefore, their data cannot be generalized to unselected SLPs who routinely assess aphasia.

In the following sections, assessment of conversational speech samples, story narration, discourse, and speech fluency are described. These assessments are needed to go beyond the results of standardized tests to obtain additional details needed for effective treatment planning.

Record a Conversational Speech Sample

The initial interview designed to obtain the case history and related information is also a client's conversational speech sample. Clinicians typically ask questions designed to obtain biographic information, details on past and current health status, the work the client has done in the past or still doing, hobbies and leisure activities the person enjoys, information on current family constellation, and so forth. During the interview, the SLP may take note of any auditory comprehension problems that become evident. If necessary, another conversation may be recorded for analysis.

The recorded conversational speech sample may be analyzed for:

- Number of words in typical sentences
- Sentence varieties

- Grammaticality of sentences (grammatically incorrect and incomplete utterances)
- Production of grammatical morphemes (e.g., plurals, possessives, past-tense inflections)
- Speech fluency
- Word-finding problems (hesitations, circumlocution, nonspecific words replacing specific words)
- Flow of speech (smooth, labored)
- Rate of speech (slow, normal, excessively fast, uneven)
- Speech sound production problems (associated dysarthria or apraxia of speech)
- Conversational skills (turn taking, topic initiation and maintenance, conversational repair strategies, eye contact)
- Prosodic features (rhythm, intonation, loudness of speech)

The content and extent of conversational speech samples may be limited in persons with Broca's aphasia. The *clinician may have to limit them* in case of Wernicke's aphasia because of excessive and rambling speech with little or no meaning. Conversational speech may be difficult to record at all in case of global aphasia. Such variations of conversational speech are diagnostic, however.

Record a Story Narrative Sample

Narratives are an excellent addition to conversational speech samples. A narrative may be defined as **storytelling**. Typically, the client is requested to tell a story that may come from personal life, travel, or work experiences. It may be a story the person has heard or a synopsis of a novel read. Alternatively, clinicians may tell or read aloud a brief story and ask the person with aphasia to **retell** it. In aphasia clinical assessment and research, however, storytelling is often evoked by wordless picture books. The client is asked to look at sequenced pictures without words in a book, set it aside, and tell the story the pictures depicted. Some standardized tests include a picture description task (e.g., the Cookie Theft picture in the Boston Diagnostic Examination;

Goodglass et al., 2000). However, picture description alone will not generate a sufficient amount of connected and complex speech for analysis (Bryant et al., 2016). Storytelling and discourse on a given topic are more productive. Brookshire and Nicholas (1994) recommended a minimum of 300 to 400 words for connected speech analysis.

Although more like a monologue than conversation, a story narration may involve some exchange between the clinician and the client. The clinician begins a conversation and requests a narration. During the narration, the clinician may prompt or ask a few questions requesting details or clarifications. Narratives are still more naturalistic than responses to standardized test items.

Storytelling and story retelling help assess a variety of communication skills, including:

- Story introduction (giving the story background or starting abruptly)
- Event sequences (correct or confused sequences)
- Inclusion of irrelevant events (an estimation of the number of irrelevant elements)
- Omission of critical story elements (an estimation of missing elements)
- Descriptions of perspectives of other persons (objectivity in describing the views and feelings of different story characters)
- Comments on emotions and thoughts of other people and the narrator (the narrator's perspective on the experiences of the story characters)
- Story conclusion (ending the story properly or abruptly)
- Lesson from the story (the meaning of the story)

In addition, the narratives are also useful for analyzing fluency and dysfluencies that suggest word-finding problems. Sentence length, sentence types, concrete and abstract vocabulary, and word variety may also be analyzed. Story retelling may reveal memory problems more readily than narration of personal stories. Omission of story elements that suggest impaired immediate recall of events becomes obvious to the clinician who has told the story.

Record a Discourse Sample

Discourse-based treatment to improve global communication, as well as specific word retrieval problems, has been a fruitful area of research in aphasiology (Armstrong, 2010; Boyle, 2011; Bryant et al., 2016; Milman, 2016; Stark et al., 2021). Obviously, such a treatment approach requires a baseline assessment of discourse skills.

Discourse sampling also starts with a conversation in which the client may be requested to talk about an issue or a topic (e.g., global warming, health care system, or a pandemic). It may be better to let the clients suggest topics they would like to talk about. The clinician may request them to include opposing views, cite evidence, evaluate critically, and express personal views and offer support for them.

Being the most formal of the three naturalistic samples, discourse — to the extent the client is capable of producing — is likely to result in more complex and varied sentence forms. Analysis of discourse samples may include the following:

- Introduction to the topic of discourse (stated the background or started abruptly)
- Grammaticality (correct and complete utterances)
- Correct use of grammatical morphemes (e.g., present progressive, copula, auxiliary)
- Complexity of language (e.g., longer and complex sentences)
- Production of words relevant to the topic (production of needed technical terms)
- Organization (good subtopic transition, information easy to follow)
- Flow of information (systematic and coherent movement of arguments)
- Viewpoints (evidence for and against viewpoints and arguments)
- Logical and critical thinking (evaluation of arguments and evidence)
- Semantic clarity (the degree to which the utterance meanings are clear)

- Adequacy of the presentation (sample length, number of words produced, totality of the information presented)
- Concluding remarks (abrupt or proper conclusion)

The suggested analysis is mostly accomplished through clinical judgments. As the Stark et al. (2021) survey reported, clinicians have emphasized the need to develop a standard, objective, and clinically practical method of analyzing discourse samples. Nonetheless, clinical judgments of speech-language characteristics and disorders, with or without a formal rating scale, are standard practice in speech-language pathology. Casilio et al. (2019) have demonstrated that with sufficient training, clinicians can reliably rate aphasic speech characteristics. Those who wish to conduct linguistic analyses of complex speech may consult other sources (Brookshire & Nicholas, 1994, 1997; Bryant et al., 2016).

Assess Speech Fluency

As noted before, fluency helps distinguish the two main varieties of aphasia, Broca's and Wernicke's. Most standardized tests include a measure of fluency. However, additional measures of fluency in complex and extended speech samples are needed.

Standardized aphasia tests and specific fluency tests measure three kinds of verbal (speech) fluency: category (animal), action, and phonemic (Faroqi-Shaw & Milman, 2018; Scheffel et al., 2021).

- **Category or animal fluency.** The person with aphasia is asked to name as many animals as possible. Category fluency may also include such other tasks as naming furniture or clothing items.
- **Action fluency.** The person is instructed to name as many actions people perform as possible (e.g., *eat, sleep, cook*). Word variants (e.g., *eating, eaten*) are not accepted; single words are required (e.g., just "*eat*").
- **Phonemic or letter fluency.** The person is requested to say as many words as possible that begin with a given letter or speech sound,

excluding proper names (phonemes F, A, and S are commonly used).

All three tasks are timed for 1 minute. Category and action fluency are considered *semantic*, contrasted with the *phonemic*. Both the Boston Diagnostic Aphasia Examination, Third Edition (Goodglass et al., 2001) and the Western Aphasia Battery, Revised (Kertesz, 2007) assess category fluency.

Persons with aphasia, including those with mild aphasia, score lower than those without aphasia on all measures of word fluency. Fewer words and longer reaction times between words and categories characterize aphasic speech (Bose et al., 2017). Semantic fluency measures are better correlated with other language measures than the phonemic fluency. Action fluency may be better preserved than the other two kinds. The words persons with aphasia produce are shorter, acquired earlier, and more frequent in the language (Faroqi-Shaw & Milman, 2018). For more elaborate scoring procedures and normative data (Canadian), see Troyer (2000).

All three verbal fluency tasks are limited to timed word generation, not fluency in continuous speech. Fluency in everyday communication, formal narration, and discourse, however, is more functional to the client than naming animals and actions and generating phonemically prompted words. Typical communication rarely involves prompted word responses. Furthermore, aphasia treatment implications of word fluency are yet to be explored. It appears like a nonfunctional target for treatment.

Naturalistic fluency also better promotes communication partnership than category word naming. The emphasis, therefore, should be on an analysis of fluency in more naturalistic as well as challenging tasks (conversation, narration, and discourse).

Clinicians typically listen to connected speech samples and rate fluency. In a survey of how clinicians judge fluency of aphasic speech, Gordon and Clough (2022) found that most clinicians rate fluency based on the rate of speech, utterance length, and grammatical complexity. Lexical variety and intonational contour of speech may also be considered. These judgments are consistent with the typical conceptualization of fluency as the production of grammatically complex and relatively longer speech spoken at reasonably fast rate with appropriate prosodic variations. Another measure of significance is

the percentage of dysfluent words produced (repetitions of sounds and words, sound prolongations, interjections, pauses, and revisions). In the context of aphasia assessment, it is essential to judge the *meaningfulness* of the client's speech fluency to distinguish the meaningless hyperfluency of Wernicke's aphasia from nonfluent Broca's aphasia (Hegde & Freed, 2022).

There are three ways of rating fluency; two are discrete, and one is continuous:

- **Discrete verbal rating scales.** These include commonly used verbal categories such as *mild*, *moderate*, and *severe* impairment in speech fluency.
- **Discrete numerical rating scales.** These are commonly used 5-point rating scales on which 1 equals minimal fluency and 5 equals maximum or hyperfluency.
- **Continuous visual analog scales (VAS).** This is a 10-cm line drawn on a piece of paper, the beginning and end points of which may be verbally specified as the two extremes of the variable being rated (e.g., *minimal fluency* to *hyperfluency*). The intermediated points are left unmarked to suggest the continuous nature of the variable. The rating score on the line varies from 0 to 100. The clinician listens to a speech sample and makes a mark on the line to suggest more or less of the variable being measured. In a digital version of the VAS, the clinician may click a point on the line to rate fluency. The distance from the left end of the scale to the point marked is measured in centimeters and converted to a numerical score between 0 and 100 to rate the degree of fluency. Even though it generates a number, it is not evident that it is superior to other kinds of ratings that are simpler to use.

Clinicians also may use the modifiable Fluency Assessment Protocol given in Hegde and Freed (2022).

Assess Syntactic and Morphologic Productions

In addition to the results of standardized tests of language skills, the clinician may use the recorded conversational, narrative, and discourse speech samples to evaluate the syntactic and morphological aspects of the person's language production. If the test and the recorded conversational speech did not sample certain language features of interest, the clinician may design client-specific informal procedures to assess them. For instance, the person may be asked to name plural objects to evoke plural morphemes and evoke possessive morphemes by showing objects in the room or pictures and asking such questions as "Whose hat is it?" or "Whose tail is it?" and so forth. Clinicians may also use the Connected Speech and Grammatical Skills Assessment Protocol given in Hegde and Freed (2022). The protocol may be modified to suit individual clients.

Complex speech samples and the clinician-designed procedures help assess agrammatism and other deficits to better understand aphasic deficits, including:

- Omission of grammatical morphemes (plurals, possessives, tense markers, copula, auxiliary, conjunctions, prepositions, and so forth)
- Production of words and phrases instead of sentences
- Telegraphic speech
- Production of mostly nouns and verbs
- Presence of incomplete or grammatically wrong sentences
- Limited variety of sentence types (e.g., production of simple, active, declarative sentences; *wh*-questions, passive sentences, and so forth)
- Generally short and simple utterances

Obviously, the clinician will take note of generally acceptable grammatical features of a person's language production. The results of analysis will be useful in diagnosing agrammatism of Broca's aphasia and distinguishing it from Wernicke's aphasia with good grammatical skills.

Assess Comprehension of Spoken Language

Auditory comprehension of spoken language is totally lost or totally intact in only a few, if any, persons with aphasia. A hearing screening should precede auditory comprehension assessment, and failure should result in a referral to an audiologist.

Most aphasia test batteries include a measure of speech comprehension. Tests dedicated to this skill (e.g., the Token Test) also are available. In most cases, the outcome of a test may be supplemented with observations made during the interview, language sampling, narration, and discourse. Clinicians may also use the Auditory Comprehension Assessment Protocol given in Hegde and Freed (2022). The protocol may be individualized.

Comprehension problems that are at least moderate will become apparent during interview and complex language sampling. During assessment, the SLP may take note of any indication of speech comprehension problems. Irrelevant answers or requests to repeat what was just said are good indications of impaired comprehension.

Speech-language comprehension is assessed only through correct verbal or nonverbal responses, as comprehension is an inferred entity. The following kinds of tasks will help assess language comprehension:

- Responses to simple, single-element and complex, multielement commands or directions, using realistic color pictures or objects in the room (e.g., “point to a chair,” a simple command, and “pick up the pencil, book, and cup and put them in the box,” a complex command)
- Pointing to the correct picture in a field of three pictures on a page or objects on the table (e.g., “point to the fork” when it is displayed with two other objects)
- Responses to *yes-no* questions (e.g., “do you live in _____?” or “are you standing right now?”)
- Retelling a story (e.g., telling a brief story to the client and asking for a retell)
- Responses to questions about a story told to the client (e.g., “who are the three characters in the story?” and “what happened at the end?”)

Assess Reading Skills

The depth of assessment of reading skills will depend on the client's premorbid literacy skills, education, work history, the current level of those skills, and the need for regaining them. If there is a need to treat those skills, the clinician should assess them in some detail.

Several standardized aphasia diagnostic tests listed in [Table 4–2](#) include reading subtests. Beyond the tests, the clinician can devise individualized reading tasks to assess the skills. The clinician also may use the Reading, Writing, and Calculation Assessment Protocol given in Hegde and Freed (2022).

In assessing the reading skills, the clinician should select texts that are appropriate to the client's age, education, and ethnocultural background. For all clients, stories, instead of pedantic texts, may be especially useful. The client's interest may be another variable to consider. The judged degree of severity of language deficits also may be considered. It may be inappropriate to ask a college-educated person with obviously minimal aphasic deficits to point to printed letters of the alphabet or ask to match pictures to words. In such case, it is more appropriate to start with more complex skills and go down the complexity levels. The following tasks may be useful:

- **Read the text silently.** Ask questions about the text and score the responses as correct or incorrect to assess reading comprehension.
- **Read the text aloud.** Take note of the oral reading problems (e.g., omitted or substituted words or sentences; missed, substituted, or distorted phonemes; slow or labored reading; prosodic impairments). Make clinical judgments or use a rating scale.
- **Read printed words, phrases, and sentences.** Evaluate both production and comprehension with these tasks, but consider omitting them if the performance was acceptable on more complex tasks.

Assess Writing Skills

The variables specified for judging the need for reading assessment also apply to writing assessment. Premorbid writing skills may be more objectively determined than the premorbid reading skills, especially if family members supply a sample of the person's premorbid writing. The SLP then may compare it with the sample recorded during assessment.

Most standardized tests of aphasia include a writing subtest. Clinicians also may use the modifiable Reading, Writing, and Calculation Assessment Protocol given in Hegde and Freed (2022). In addition to reading and writing skills, this protocol allows for an informal assessment of basic arithmetic skills such as adding and subtraction, counting money, and so forth.

In assessing writing skills, the clinician may ask the client to:

- **Write a brief story.** The clinician may show a wordless picture book that tells a story and ask the person to write the story.
- **Write a biographical paragraph.** The person may write his or her own name, address, phone number, occupation, names of family members, hobbies and interests, and so forth.
- **Write to dictation.** The clinician may dictate a paragraph from a printed book.
- **Copy text.** The clinician may show a brief paragraph of printed material and ask the person to copy it.
- **Write the names of pictured items.** Show pictured common items and ask the person to write the names.
- **Write the letters of the alphabet to dictation.** Select a few letters of the alphabet and dictate them.

The writing sample may reveal failure to use the upper- and lowercase letters, poor letter formation, illegible letters, failure to produce script or printed letters, poor spacing between letters, errors in word arrangement, wrong usage of morphological features, omission of words and phrases, telegraphic writing, agrammatic sentences, and so forth. Generally, the writing problems of persons with aphasia parallel their oral communication problems.

Assess Motor Speech Disorders

Because aphasia may coexist with motor speech disorders, it is important to assess all clients with aphasia for apraxia of speech and dysarthria. Speech production problems, more commonly a symptom of Broca's aphasia than Wernicke's, may help distinguish the two types.

Clinicians may initially make a judgment about the presence of apraxia and dysarthria. In most cases, a thorough orofacial examination (including alternate motion rates and sequenced motion rates) and an analysis of conversational speech may be sufficient to judge speech intelligibility and acceptability of articulatory performance. The presence of a significant motor speech disorder, however, may warrant a more detailed assessment, which may include standardized test administration. Clinicians may consult sources on the diagnosis of motor speech disorders for details (Freed, 2020) and use the multiple protocols given in Hegde and Freed (2022) to assess apraxia of speech and dysarthria.

Assessment of Functional Communication and Quality of Life

Functional communication refers to speech and language skills that fulfill the needs of everyday living. Persons who can express their basic needs in any mode (e.g., verbal or gestural), seek needed information, respond to everyday questions, participate in simple family conversations, read newspapers, and write notes or shopping lists may be said to have functional communication skills. Functional communication is social and interactive in home and other naturalistic settings. Pure linguistic analysis of speech and language does not consider this social and interactive nature of communication (Doedens & Meteyard, 2020). Any attempt that is successful in affecting the behavior of listeners is *functional* and acceptable.

Quality of life is a subjective experience. The World Health Organization defines quality of life as “an individual's perception of their position in life in the context of the culture and value systems in which they live and in relation

to their goals, expectations, standards and concerns. It is a broad-ranging concept affected in complex ways by the person's physical health, psychological state, level of independence, social relationships, and their relationships to salient features of their environment" (WHOQOL Group, 1995, p. 1405). This conceptual definition is too broad to be operational and useful in assessment. Because it is assessed by evoking responses to specific questions, **quality of life** may be defined as self-reported statements on the degree of satisfaction with one's own health, well-being, and communication skills that fulfill the needs of everyday living. Such a practical definition prompts specific questions about the various aspects of quality of life.

Functional communication skills and quality of life are interrelated. Individuals with functional communication skills generally enjoy a better quality of life than those who lack such skills. Aphasia, especially in its severe form, affects the daily living activities, creates barriers to more fulfilling social participation, and reduces general satisfaction with life. A person who cannot effectively seek help for getting a drink of water or for going to bathroom is likely to not only get frustrated but also feel anxious.

Most traditional standardized tests emphasize sentence production, fluency, word repetition, naming, and such other formal verbal (linguistic) skills. Such tests do not sample everyday communication or quality of life. Formal stimulus presentation under a structured situation to evoke limited responses may not resemble everyday communication situations in which the environment is more variable, rich in contextual cues, and supported by gestures and facial expressions. Communication in everyday situations may be more or less efficient than what can be gleaned from standardized test results. Furthermore, traditional standardized aphasia diagnostic tests have not included measures of the *effects of aphasia* on the individual; this is the matter of quality of life. Tools specifically designed to assess functional communication and quality of life listed in [Table 4–3](#) overcome some of these limitations of traditional standardized tests of aphasia.

The instruments listed in [Table 4–3](#) differ somewhat in their emphasis and target skills. For instance, the Inpatient Functional Communication Interview is conducted in the hospital settings, but others may be administered at speech and hearing clinics or homes. Some emphasize relationships and social

interactions (ASHA QCL; Paul et al., 2004) and others may target daily living skills (Baines et al., 1999; Holland et al., 1998). Some instruments are designed for tracking changes in skills over time (Lomas et al., 1989). Clinicians may select the one that is most suitable for a given client or their assessment goals. Also, the instruments may vary in their effectiveness. For instance, Rangamani and Jadovski (2020) reported that in documenting changes in quality of life of persons with aphasia, the ASHA QCL (Paul et al., 2004) was more effective than the ASHA FACS (Frattali et al., 1995). For a critical review of available functional assessment tools, see Doedens and Meteyard (2020).

Table 4-3. Functional Communication and Quality of Life Assessment Tools

Test Name	Skills Assessed
Assessment of Language-Related Functional Activities (Baines et al., 1999)	Daily living skills including telling time, counting money, writing a check
Assessment for Living With Aphasia (Simmons-Mackie et al., 2014)	Effect of aphasia on everyday life, including communication (self-reported quality of life assessment)
Communication Activities of Daily Living, Third Edition (Holland et al., 2018)	Functional social communication, reading, writing, humor, metaphor, absurdity
Communicative Effectiveness Index (Lomas et al., 1989)	Change over time in functional verbal and nonverbal communication assessed by significant others

Functional Assessment of Communication Skills for Adults (ASHA FACS) (Frattali et al., 1995)	Social communication, communication of basic needs, reading, writing
Functional Communication Profile Revised (Kleinman, 2003)	Receptive and expressive language, fluency, pragmatic language skills
Inpatient Functional Communication Interview (O'Halloran et al., 2004)	Everyday communication skills in hospital settings
Quality of Communication Life Scale (ASHA QCL) (Paul et al., 2004)	Effects of communication impairment on relationships and social interactions
Quality of Life Questionnaire for Aphasics (Spaccavento et al., 2014)	Everyday functional communication, feelings, and autonomy in performing daily activities (e.g., shopping, money management)
Stroke and Aphasia Quality of Life Scale (Hilari et al., 2003)	Self-reported measures on mobility, work, language, thinking, self-care, and family and social roles
World Health Organization Quality of Life (WHOQOL)-BREF (World Health Organization, 2004)	Self-reported measures on overall and health-related quality of life, including mood, energy, capacity to work, support received from family and friends

Most functional communication and quality of life measures are questionnaires that may be filled by the individual with aphasia, a caregiver, a family member, or all three of them. The individuals themselves are the most important because it is their degree of satisfaction that matters. Typically, multiple sources and individuals may assist in collecting information. The clinician also may interview people who are in daily touch with the client and fill parts of the questionnaire or offer new information. Individuals' self-reports on their communication skills, the support they need, the degree of help they receive, their feelings and thoughts, and their own judgment about their quality of life add valuable dimension to assessment.

Functional communication assessment also may be indirect measures of persons' quality of life experiences. This assessment requires observation of a client's social interactions in everyday situations, including health care settings. Those with a higher level of social interactions may report better quality of life.

Questions related to quality of life usually fall into three main categories: a broader life quality in general (e.g., *Are you satisfied with your life in general?*), a narrower health-related life quality (e.g., *Are you satisfied with your health and well-being?*), and more specific communication-related life quality (e.g., *Are you satisfied with your everyday communication skills?*). Although SLPs are especially concerned with how communication disorders negatively affect the quality of life, other aspects are also important to consider because they all tend to interact with each other (Hilari et al., 2015).

Assessment of functional communication and quality of life is especially useful in documenting meaningful treatment outcome. In fact, documenting improvement in clients' quality of life may be just as important as documenting the main treatment effects on speech and language skills, including functional communication. Clients who regain their functional communication skills and begin to participate in social interactions will have achieved personally and socially meaningful treatment effects.

Assessment of Bilingual and Ethnoculturally Varied Individuals

Assessment of clients who are of minority ethnocultural background poses challenges to a clinician of any ethnocultural background. Clinicians need to understand the cultural and language background specific to the individual being assessed. Clinicians should avoid culturally biased tests. Only those standardized tests that have included a representative number of persons from the specific ethnocultural minority group the client belongs to should be selected. Several sources offer specific guidelines that the clinicians may consider. A few sources offer suggestions based on clinical experience and knowledge of different ethnocultural groups; all experts recognize the need for more careful research into assessment tools that produce reliable and valid information when used with specific ethnocultural individuals (Hegde & Freed, 2022; Payne, 2014).

In working with clients of varied ethnocultural background, clinicians may:

- **Adopt a client-specific approach.** They may design assessment procedures that are specific to a client's cultural and linguistic background.
- **Consider the test standardization sample.** Select tests that have been standardized on the relevant group (e.g., African Americans or Hispanic individuals).
- **Avoid culturally biased tests.** Excessive modification of test items to suit an individual's cultural and language background may invalidate the results.
- **Use a content-open assessment outline.** Consider an outline described in the previous section and make suitable modifications. Such outlines specify areas or skills to be assessed and give much freedom to develop stimulus materials and specific assessment items relevant to the client being assessed.
- **Avoid multicultural stereotypes.** Do not automatically assume that clients of a different ethnocultural background necessarily need unique procedures or that they do not have English proficiency. Case history and interview of the client and family will give some directions.

- **Consider potential barriers to health care.** Knowledge of, and financial accessibility to, health care and rehabilitation services are significant ethnocultural problems (Payne, 2014). Therefore, during the clinical interview, investigate the client's and the family members' knowledge of health care facilities in the community, need for additional services, availability of such services in the client's vicinity, and potential support sources.
- **Investigate the speech-language service affordability.** Treatment for aphasia generally requires multiple sessions extended over weeks or months. Consider whether the individual and the family can afford the recommended services, whether the family has transportation to the clinic, and whether a family member can accompany the client to the clinic. Keep a list of support services available in the community and suggest them to the person and the family.
- **Informally evaluate motivation for treatment.** Explore the client's and the family members' beliefs about health, disease, disability, disorders, and treatability of clinical conditions that may affect motivation to seek and sustain services.
- **Seek consultation.** If available, consult with an SLP who belongs to the client's ethnocultural background.

Persons of varied ethnocultural backgrounds may be bilingual. If so, it is essential to determine the dominant language, weaker second language, and the language or languages in which the individual needs to be assessed. English-dominant bilingual persons may prefer to be assessed in English. If a non-English language is the first and dominant language, then the assessment needs to be done in the client's first language. To make such an assessment, a monolingual English-speaking clinician may need the help of an interpreter.

Among the few tests that are available to assess multilingual or bilingual individuals, the Multilingual Aphasia Examination–Spanish Version by Ray et al. (1994) is noteworthy. It is also available in English, French, German, and Italian. Repeated testing of the same client may be accomplished by alternate

forms. The test helps assess naming, repetition, fluency, spelling, auditory comprehension, reading, and writing.

Another useful test is the Bilingual Aphasia Test (BAT) by Paradis (1987). A unique test that allows evaluation in 58 languages with parallel forms, BAT is one of the most comprehensive multilingual tests available. The test can evaluate a client's relative performance in a primary and a secondary language. BAT helps assess phonological, morphological, syntactic, lexical, and semantic aspects of the languages. It also helps assess language use in auditory, visual, and oral modalities. The bilingual section of the test evaluates skills in pairs of languages.

Several well-established English tests listed in [Table 4–2](#) have been translated to other languages. For instance, the Boston Diagnostic Aphasia Examination, the Boston Naming Test, Comprehensive Aphasia Test, Communication Activities in Daily Living, and Western Aphasia Battery are available in various languages. A potential problem with translated tests is that culturally inappropriate test items may evoke invalid responses.

Diagnostic Assessment Summary

After completing the assessment, the clinician may summarize the major findings to make a diagnosis. The most important parameters of aphasia can be summarized in a table format that facilitates diagnostic conclusions related to the person assessed.

[Table 4–4](#), adapted from Hegde and Freed (2022), may be used as a summative checklist of major symptoms the assessment will have revealed. The clinician may judge the severity of each documented deficit on a 4-point scale, ranging from *none* (no deficit) to *severe*. At the end, the clinician may enter the diagnosis for the individual.

In most cases, the clinician may need to consider the dominant features of the different types of aphasia before making a final diagnosis. To facilitate this task, Appendix A lists the main diagnostic features of the eight major types of

aphasia. Features that are not associated with a particular type are also important in making this differential diagnosis.

Table 4-4. Aphasia Diagnostic Assessment Summary

Name: _____ Age: _____ DOB: _____				
Diagnosis: _____ Date: _____ Clinician: _____				
<i>Summary of observed symptoms that justifies the diagnosis. The severity as checked in the box.</i>				
	Degree of Impairment			
	None	Mild	Moderate	Severe
Agrammatism	☐	☐	☐	☐
Syntactic features	☐	☐	☐	☐
Grammatical morphemes	☐	☐	☐	☐
Sentence complexity	☐	☐	☐	☐
Sentence variety	☐	☐	☐	☐
Sentence/phrase length	☐	☐	☐	☐
Paraphasia	☐	☐	☐	☐
Anomia	☐	☐	☐	☐
Confrontation	☐	☐	☐	☐
Spontaneous	☐	☐	☐	☐
Repetition	☐	☐	☐	☐
Of self-productions	☐	☐	☐	☐
Echolalia	☐	☐	☐	☐
Speech sound production	☐	☐	☐	☐
Apraxia of speech	☐	☐	☐	☐
Dysarthria	☐	☐	☐	☐
Fluency	☐	☐	☐	☐
Dysfluencies	☐	☐	☐	☐
Meaningfulness of speech	☐	☐	☐	☐
Rate of speech	☐	☐	☐	☐
Storytelling	☐	☐	☐	☐
Story retelling	☐	☐	☐	☐
Discourse	☐	☐	☐	☐
Conversational skills	☐	☐	☐	☐
Prosodic features	☐	☐	☐	☐
Speech comprehension	☐	☐	☐	☐

Oral reading	☐	☐	☐	☐
Misreading of words	☐	☐	☐	☐
Omission of words	☐	☐	☐	☐
Reading comprehension	☐	☐	☐	☐
Simple material	☐	☐	☐	☐
Complex material	☐	☐	☐	☐
Writing				
Biographic writing	☐	☐	☐	☐
Writing to dictation	☐	☐	☐	☐
Text copying	☐	☐	☐	☐
Functional communication	☐	☐	☐	☐
Quality of life	☐	☐	☐	☐
Client's rating	☐	☐	☐	☐
Caregiver rating	☐	☐	☐	☐
Diagnosis:				
Clinician: _____ Date: _____				

Differential Diagnosis

In some cases, ambiguity in observed and assessed symptoms of a person may compel further differential diagnostic considerations. The clinician may carefully distinguish aphasia from other neurologically based communication problems, including dementia, right hemisphere disorder, traumatic brain injury, apraxia of speech, and dysarthria. These disorders share neuropathology to some extent, and some of those disorders may coexist either initially or in subsequent stages of aphasia, making it difficult to distinguish among them.

Factors that help make a differential diagnosis include the following:

- The case history and the results of medical (including neurological) examinations.
- Detailed, adequate, and reliable sampling of verbal and nonverbal communicative behaviors.

- Individual differences within and across types of aphasia and across communication disorders of neurological origin. (Individuals with the same diagnosis may exhibit varied symptoms.)
- Changes over time in symptoms due to changes in the underlying neurological conditions. (Repeated probes will help confirm or change the original diagnosis.)
- The observed pattern of multiple symptoms. (A single symptom is rarely diagnostic.)

Appendices A through F, at the end of this chapter, contrast different types of aphasia and aphasia with dementia, right hemisphere disorder, traumatic brain injury, apraxia of speech, and dysarthria.

CLASS NOTES

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APPENDIX A

Differential Diagnosis of Aphasia Types

Type	Look for
Broca's	Nonfluent, sparse, agrammatic, aprosodic, effortful, anomic, and slow speech with impaired repetition; concerned and better communication.
Transcortical motor	Intact repetition, echolalia, difficult speech initiation.
Mixed transcortical	Intact but nonfunctional repetition, severely impaired fluency, naming, grammar, and comprehension.
Global	Severely and globally impaired speech-language skills and limited comprehension.
Wernicke's	Hyperfluency, meaningless, neologistic, paraphasic speech, severely impaired speech comprehension; unconcerned and poor communication.
Transcortical sensory	Paraphasic, anomic, fluent, echolalic speech; impaired comprehension.
Conduction	Variable fluency, impaired repetition, paraphasic speech, naming problems.

Anomic	Persistent and pervasive word finding difficulty, interrupted nonfluent speech, pauses and circumlocution.
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APPENDIX B

Differential Diagnosis of Aphasia and Dementia

Aphasia	Dementia
Sudden onset	Slow onset
Damage in the left hemisphere	Bilateral brain damage
Focal brain lesions	Diffuse brain damage
Mood is usually appropriate, although depressed or frustrated at times	May be moody, withdrawn, agitated
Generally intact nonverbal intellectual skills	Mild to severely impaired intellectual skills
Typically intact memory skills	Impaired memory
Generally relevant, socially appropriate, and organized	Often irrelevant, socially inappropriate, and disorganized
Fluent or nonfluent	Fluent until dementia worsens
Generally stable, unless in the case	Generally progressive, unless in the

of multiple strokes	case of reversible dementia
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Note: Most forms of dementia are irreversible, chronic, and degenerative.

Caution: Aphasia and dementia may coexist. A person with aphasia may develop a neurological disease resulting in dementia (Alzheimer's disease). A person with dementia may have a stroke that results in aphasia.

APPENDIX C

Differential Diagnosis of Aphasia and Right Hemisphere Disorder

Lesions in the right hemisphere produce a pattern of communication and behavior deficits that share a few common symptoms with aphasia.

Aphasia	Right Hemisphere Disorder
Naming, fluency, auditory comprehension, reading, and writing problems	Only mild problems
No left-sided neglect	Left-sided neglect
No denial of illness	Denial of illness
Relevant speech	Irrelevant, excessive, rambling speech
Good expression of emotion and understanding of other people's emotions	Difficulty expressing emotions and appreciating other people's emotions
Oriented to space and surroundings	Disorientation to space and surroundings
Recognizes familiar faces	May not recognize familiar faces

May simplify drawings	Rotation and left-sided neglect
Some prosodic defect	More pronounced prosodic defect
Appropriate humor	Inappropriate humor
May retell the essence of a story	May retell nonessential, isolated details
May understand implied meanings of statements	Understands only literal meanings
Pragmatic impairments less striking	Pragmatic impairments more striking
Although limited in language skills, communication can often be good	Although possessing good language skills, communication can be very poor
Verbal deficits are dominant	Attentional and perceptual deficits are dominant

Note: Right hemisphere damage in those few individuals whose right hemisphere is dominant for language results in aphasia. The causes of right and left hemisphere damage are similar.

APPENDIX D

Differential Diagnosis of Aphasia and Traumatic Brain Injury

Aphasia is due to vascular and other neurological diseases, whereas traumatic brain injury (TBI) is due to external trauma to the head and the brain. Events that cause aphasia are intraindividual and those that cause TBI are environmental.

Aphasia	TBI
Often a history of stroke	A history of head and brain trauma
Obvious neurological impairments only in some (e.g., in Broca's aphasia)	Obvious neurological impairments in most if not all
Conscious, alert, oriented in most cases	Stupor, coma, confusion, disorientation
Dysphagia, uncommon	Dysphagia, more common
Absence of psychiatric (behavioral) disorders	Auditory hallucinations, confabulation, delusions, apathy, irritability, unreasonable behavior, and depression in some cases
Absence of memory and intellectual	Presence of memory and intellectual

impairments	impairments
Mostly relevant speech and language	Confused, imprecise, rambling speech and language
Impaired social (pragmatic) communication generally not prominent, except for Wernicke's aphasia	Impaired social communication
Dysarthria, if present, is an independent coexisting problem; generally good intelligibility	Dysarthria, often the spastic type; generally impaired speech intelligibility
Abstract reasoning (e.g., proverb interpretation) problems are subtle, if any	Abstract reasoning problems may be obvious

APPENDIX E

Differential Diagnosis of Aphasia and Apraxia of Speech

Apraxia of speech is a motor speech disorder in the absence of muscular weakness, sensory loss, or paralysis. Apraxia and Broca's aphasia may coexist.

Aphasia Without Apraxia of Speech	Apraxia of Speech Without Aphasia
Neurogenic language problem	Neurogenic speech problem
More often associated with temporal or temporoparietal lesions	More often associated with posterior, frontal, or insular lesions
Right hemiparesis infrequent	Right hemiparesis frequent
Agrammatism and paraphasia often present	Agrammatism and paraphasia are generally absent
Trial-and-error, groping articulatory movements are not significant	Trial-and-error, groping articulatory movements are significant
Misarticulations less variable, more consistent	Misarticulations more variable, more inconsistent
Phonologic problems of fluently aphasic clients may be unpredictable	Phonologic problems of apraxic clients are predictable and

and not resemble the target sounds/words	approximate target sounds/words
Articulatory errors unaffected by complexity of productions	Articulatory errors significantly affected by complexity of productions
Lack of attempts at self-correction, especially in fluent aphasia	Repeated and failed attempts at self-correction
Some impairment in auditory comprehension	Generally, no impairment in auditory comprehension
Prosodic problems not dominant	Prosodic problems dominant
Less obvious difficulty in initiating utterances	More obvious difficulty in initiating utterances
Omission of function words	Omission of function words not significant
Word-finding problems	No significant word-finding problems
Reading comprehension deficits	No reading comprehension deficits
Severe aphasia may mask apraxia	Does not mask aphasia
Limb or oral apraxia not dominant	Such an apraxia may be dominant

APPENDIX F

Differential Diagnosis of Aphasia and Dysarthria

Dysarthria is a group of motor speech disorders due to poor muscle control caused by damage to the central or peripheral nerves that supply muscles involved in speech production. Subcortical lesions that produce aphasia are likely to be associated with dysarthria.

Aphasia Without Dysarthria	Dysarthria Without Aphasia
Neurogenic language problem	Neurogenic speech problem
Main difficulty is language formulation, expression, and comprehension	Main difficulty is speech production
Lesions in the central nervous system (language-dominant hemisphere)	Lesions in the central nervous system, peripheral system, or both
Normal orofacial mechanism except in cases of facial paralysis	Weakness, paralysis, incoordination of orofacial muscles a significant factor
Language problems not due to muscle weakness	Speech problems due to muscle weakness

Significant word-finding problems	No significant word-finding problems
Impaired auditory comprehension	Intact auditory comprehension
No consistent misarticulations	Consistent misarticulations
Intelligibility and rate of speech not clearly related	Intelligibility clearly related to the rate of speech
No respiratory problems associated with speech production	Respiratory problems associated with speech production
Phonatory problems not significant	Phonatory problems may be significant
Resonance disorders not significant	Resonance disorders are significant
Prosodic disorders not dominant	Prosodic disorders may be dominant
Abnormal voice quality not significant	Abnormal voice quality may be significant
Abnormal stress not significant	Abnormal stress may be significant
Reading problems are often present	Reading problems are not typical
Writing problems are often present	Writing problems are not typical

Note: There are several types of dysarthria; all of the distinguishing features just listed do not apply to all types of dysarthria. Whether it is aphasia or dysarthria, individuals differ in their symptom complex.

Chapter Outline

- Behavioral Treatment of Aphasia Is Effective
- Generalization and Maintenance Are Still Problems
- Most Treatment Programs Are Partial
- Treatment of Auditory Comprehension Problems
- Treatment of Verbal Expression
- Social Approaches to Aphasia Rehabilitation
- Treatment of Reading and Writing Problems
- Group Treatment for People With Aphasia
- Virtual Reality Therapy
- Telerehabilitation for Aphasia
- Augmentative and Alternative Communication
- Medical Treatment of Persons With Aphasia
- References

Learning Objectives

After reading the chapter, the reader will:

- Specify the target skills and treatment procedures for the communication deficits in individuals with aphasia

- Describe and evaluate traditional, social, and group treatment approaches
- Give an overview of augmentative and alternative communication devices

PRINTED NOTES

Multiple professions are involved treating persons with aphasia. Medical treatments including drugs and electrical stimulation of the brain are available. In some cases, occupational and physical therapists may offer their services. In this chapter, our main concern is the various communication impairments that speech-language pathologists (SLPs) treat.

The treatment SLPs offer to persons with aphasia is behavioral (Fabian et al., 2020; Fridricksson & Hillis, 2021; Sheppard & Sebastian, 2021; Vitti & Hillis, 2021). In most treatment sessions, SLPs present such stimuli as pictures, objects, and verbal devices (e.g., questions, requests). The SLP then models the target skills (naming, repetition, sentence production) for the client to imitate. Simpler responses may be gradually shaped into more complex ones. Withdrawing modeling, the SLP may *prompt* or give other kinds of cues to evoke the target responses. The individual receives positive reinforcement (e.g., verbal praise) for correct responses and corrective feedback for incorrect responses. A report by Johnson et al. (2014) illustrates the concepts and procedures of behavioral science in treating persons with aphasia.

Behavioral Treatment of Aphasia Is Effective

Treatment research reviewers have reported that behavioral treatment of aphasia is generally effective, although there is room for improvement in both short- and long-term effects (Ali et al., 2021; Bunker et al., 2019; Fabian et al., 2020; Fridricksson & Hillis, 2021; Sheppard & Sebastian, 2021; Vitti & Hillis, 2021). Authors of a meta-analysis (Ali et al., 2021) of treatment research concluded that (a) individuals of all ages improve with treatment, although younger persons with aphasia may gain more than the older persons; (b) enrollment in treatment any time poststroke may result in improvement, but 1-month poststroke enrollment results in the greatest magnitude of

positive outcome; and (c) improvement in auditory comprehension, functional communication, and naming follows the trend described in (a) and (b). Positive effects are evident in all kinds of studies — single-subject experimental, case studies, group design studies, and randomized clinical trials. Persons with the most severe aphasia gain less than those with milder forms, but treatment is recommended for all persons with all types and severity of aphasia and for persons with any degree of chronicity and any age group. Generally, the greater the number of treatment sessions, the higher the gains in communication skills (Fridricksson & Hillis, 2021). Evidence on high-intensity treatment, however, is contradictory, needing additional research to make recommendations.

More intensive treatment, also known as **massed practice**, may not be *more* effective than the less intensive, also known as **distributed practice** (Fabian et al., 2020). In their randomized clinical trial, Stahl et al. (2018) found that intensive therapy of 4 hours daily did not result in greater improvement than a schedule of 2 hours daily. Others have reported that the typical 1- to 3-hour a week treatment is as good as more intensive treatment (see Knollman-Porter et al., 2018, for a review of studies). Reviewing studies on distributed practice in aphasia naming treatment, Middleton et al. (2020) concluded that distributed practice (shorter session durations, sessions spaced apart) is superior to massed practice (longer sessions, closely spaced sessions).

Making a systematic review of studies reporting maintenance of gains after intensive therapy, Menahemi-Falkov et al. (2021) concluded that group averages may show improvement, but only 22% of individuals treated maintained their gains. Conflicting evidence on intensive therapy may be due to average group measures with a large amount of individual variability. The Menahemi-Falkov et al. (2021) review points out how group averages of treatment effects and statistical analysis can be misleading.

Studies or reviews of certain specific treatment procedures involving both group and single-subject experimental designs also have concluded that treatment is better than no treatment (DeDe et al., 2019; Efstratiadou et al., 2018; Kiran & Johnson, 2008; Kiran et al., 2011; Milman, 2016; Simmons-Mackie et al., 2010, 2014). Single-subject experimental designs have generated much of the evidence for aphasia treatment (Freed et al., 2004; Kurland et al.,

2012; Rider et al., 2008; Thompson, 2014). In recent decades, group treatment research and randomized clinical trials (RCTs) have increased, although the RCT is riddled with methodological problems. In spite of the name, there is no random selection of participants. The larger the RCT, the greater the interparticipant variability and the more ambiguous the meaning of statistically analyzed results (Hegde, 2007).

A complicating factor in claiming effectiveness for treatment procedures in general — and those for aphasia in particular — is that individuals recover their skills to varying extents without professional help. Improved brain functioning may help recover lost or impaired speech-language skills. Neuroimaging studies of persons with early stage aphasia have shown that the left hemisphere language regions recover to varying extents across individuals (Wilson & Schneck, 2021). Wilson et al. (2018) evaluated language deficits in 21 persons with aphasia every 2 to 3 days on 15 days poststroke period. Most persons did not receive substantial speech-language therapy during the period of evaluation, but some received unspecified amounts of therapy. The authors found linear improvement in language skills over the 2-week period. Persons with more severe aphasia improved to a greater extent than those with less severe aphasia. Improvement in language production (word finding, grammatical construction, repetition, reading) was greater than that in comprehension. The results of most other studies on spontaneous recovery are seriously confounded with potential effects of language therapy.

To claim effectiveness for the treatment researched, natural recovery should be ruled out. Separating spontaneous improvement from treatment effects is challenging, but generally single-subject reversal and multiple baseline designs do this more efficiently than group designs of clinical trials (Hegde, 2007; Hegde & Salvatore, 2021).

With all the positive claims about aphasia treatment effects, a caveat is in order. Most studies on aphasia treatment effects measure improvement in terms of changes on standardized test scores rather than the frequency of specific skills (such as naming or ordering in a restaurant). There is no guarantee that an improved naming score on a standardized test translates to functional naming in everyday situations. Language samples in clinical and

naturalistic settings give a better idea of improvement than test scores, but such samples are often not obtained.

Generalization and Maintenance Are Still Problems

It is generally agreed that aphasia treatment is better than no treatment. Nonetheless, two major problems pose challenges to treatment researchers as well as practitioners.

The first problem is to demonstrate generalization of treatment effects to the client's everyday functional and social communication. If a treatment is effective in treatment research, but the skills do not generalize to the natural environment or to other modes of response (e.g., conversation), then the method is still effective, but there is more clinical work to be done to achieve the missing generalization. A tendency is to establish effectiveness of treatment in clinical sessions and measure frequently occurring generalizations in the same sessions, and then terminate the study. Generalization to natural settings and other forms of communication may not be monitored. When generalization of treatment effects to naturalistic communication is not demonstrated, the intervention lacks ecological (social) validity.

Special procedures are needed to have the clinically established skills generalize to everyday communication. It begins with the target behavior selection for treatment. Regardless of the initial target skills, treatment should end with functional, conversational speech. If not, generalization to social communication is unlikely. Group treatment conducted in more naturalistic settings may also promote generalization. Training spouses and other caregivers may be necessary to have the clinically established skills generalized to home and other settings.

The second problem challenging the clinicians is that of maintenance of clinically established skills. Very few treatment research studies document longterm maintenance of treatment gains. Maintenance requires procedural intervention, not just measurement. Trained family members who prompt and

reinforce communication skills at home promote both generalization and maintenance. Teaching self-monitoring skills, periodic follow-up, and one or two booster treatment sessions may further strengthen maintenance.

Most Treatment Programs Are Partial

Aphasia treatment programs are rarely comprehensive. There are separate treatment programs each targeting a specific deficit (e.g., auditory comprehension or naming). Auditory comprehension and picture naming claim a large share of treatment research. Functional and social communication approaches receive insufficient amount of research attention. Successful functional communication training needs to move on to more complex language training. There is no need for the typical dichotomy of the deficit remediation versus social communication and participation.

There are a few comprehensive aphasia treatment programs, but they are rarely implemented (Rose et al., 2013). They all tend to be intensive (at least 3 hours of daily therapy for up to 6 days a week) and are implemented in specially organized clinics, sometimes periodically offered, generally expensive, often time limited (2 weeks), and typically self-funded by the participants. Some programs are held at the U.S. Veterans Administration hospitals (VHA facilities) (Babbitt et al., 2015; Winans-Mitrik et al., 2014). However, as noted before, intensive therapy that exceeds an hour or two per day may not be beneficial due to participant fatigue (Stahl et al., 2018). Other features of intensive therapy may not be practical for both individuals with aphasia and their clinicians.

Milman (2016) describes an **Integrated Discourse Treatment for Aphasia (IDTA)** that combines word-retrieval teaching strategies with sentence processing and discourse training. The program may be implemented with typical intensity (e.g., three 1-hour sessions per week). Johnson et al. (2014) have reported another comprehensive program that moves from word retrieval to social communication with the caregivers' involvement in all treatment sessions and training to conduct skill practice sessions at home. This program, however, is more intensive (3.5 hours of daily training for 3 weeks).

Treatment of Auditory Comprehension Problems

Auditory comprehension problems in persons with aphasia are often treated but least researched. It has not been demonstrated that improved auditory comprehension causes improved language production, and it possibly is an unlikely hypothesis. In persons with Wernicke's aphasia, impaired language comprehension may be serious and cause production problems (e.g., irrelevant responses). A study by Johnson et al. (2014) on teaching expressive language skills reported the second largest positive change (after the verbal skills that were taught) in untreated auditory comprehension. A review of several treatment studies that targeted only production also noted improvement, often substantial, in untreated auditory comprehension (Rose et al., 2013). Therefore, clinicians may offer language production training first and then probe for auditory comprehension to see if it has improved. If it has, much training effort and time will have been saved; if it has not, treatment may target comprehension.

Lesions within and outside the posterior superior temporal lobe (PST) can cause auditory comprehension problems. A lesion in the PST suggests poor prognosis for auditory comprehension and slower recovery. A lesion outside the PST suggests better prognosis and faster recovery.

Sequence of Auditory Comprehension Training

Historically, a treatment program for aphasia that Schuell et al. (1964) developed placed an extraordinary emphasis on auditory comprehension problems, and by so doing, firmly established a tradition of treating those problems. There has been no systematic research, however, to demonstrate the effectiveness of Schuell et al.'s method. Limited recent research on auditory comprehension has primarily targeted picture identification (e.g., pointing to correct pictures the clinician names), an extremely limited treatment target (Knollman-Porter et al., 2018).

Typical auditory comprehension treatment consisted of repeated trials, frequent modeling, and positive reinforcement for correct nonverbal responses

(e.g., pointing to the correct pictures). The target skills are sequenced from simpler comprehension tasks to progressively more complex tasks:

- **Single-word comprehension.** The person is asked to point to specific stimulus items (e.g., body parts, objects, pictures of objects) and to point to action pictures (e.g., picture of a man walking).
- **Sentence comprehension.** Sentences may be questions, directions to follow, and verification of spoken sentences. Acceptable responses may be verbal (“yes” or “no”) or nonverbal (e.g., a head shake) responses. Such questions as “Is San Francisco in Nevada?” may be asked. Spoken directions may be simple (e.g., “Please put the ball in the box”) or more complex (e.g., “Put the ball in the box and pick up the pencil from the box”). In sentence verification, the client is asked whether a spoken sentence is true or false (e.g., “Is a plant bigger than a tree?”). The client also may match a picture to the spoken sentence (“Point to the woman who is riding the bicycle”).
- **Narrative and discourse comprehension.** The clinician may narrate a story, give a short discourse, and question the client about what is told, prompt and reinforce correct responses, and correct any wrong statements.
- **Comprehension of conversation.** The clinician may engage the individual in conversation and ask questions about personal matters, current events, prior or current occupation, and so forth. Obviously, at this level of training, the individual responds verbally.

Helm-Estabrooks et al.’s (2014) **Treatment for Wernicke’s Aphasia (TWA)** treats auditory comprehension deficits. Candidates should have intact single-word reading comprehension. In the initial step, the person selects the correct picture to match a printed word. Subsequently, the person reads the printed stimulus and selects the picture, says the word (without the printed stimulus word), and repeats it. Finally, the clinician says the word and asks the client to select the picture. See Helm-Estabrooks et al. (2014) for details on TWA as well as other approaches to treating auditory comprehension deficits.

Treatment of Verbal Expression

In typical aphasia therapy, SLPs may reinforce individuals with aphasia for any kind of effort at communication, including the production of words, phrases, sentences, gestures, mimes, drawing, writing, and informal as well as formal signs — alone or in combination (Ross et al., 2006; Simmons-Mackie, 2008). An alternative approach insists only on verbal communication (Kurland et al., 2016; Pulvermuller et al., 2001). A clinician may prefer one or the other approach or select one of these opposing approaches considered suitable for an individual client.

Constraint-Induced Aphasia Therapy

Insisting exclusively on verbal responses, the **constraint-induced aphasia therapy (CIAT)** excludes nonverbal communication (Pulvermuller et al., 2001). If the person were to use gestures and other nonverbal means (e.g., drawing and writing), the clinician will ask not to do it.

CIAT is based on the assumption that nonverbal means of communication get reinforced poststroke because it is easier than speech, and speech for them is effortful and hence nonreinforcing. The reduced speech output and the increased nonverbal communication skills are learned outcomes; therapy should aim at reducing them by withholding reinforcement for it while increasing verbal communication by reinforcing it. Several studies have reported improved verbal communication in individuals with aphasia who have been treated with CIAT. Naming and other forms of verbal expressions have been taught with the technique (Kurland et al., 2012; Kurland et al., 2016). CIAT is intensive; up to 4 hours of daily therapy over 2 to 3 weeks is typical. Most investigators and systematic reviewers of research have concluded that CIAT is effective in promoting naming, comprehension, oral language, and writing but not superior to other procedures (Wang et al., 2020; Woldag et al., 2017; Zhang et al., 2017). Therefore, clinicians may find CIAT impractical due to its demand on daily treatment time of several hours.

An enhanced protocol, called the CIAT II, has also produced favorable results in a pre-post case series (Johnson et al., 2014). This modified procedure includes speech repetition, activities of daily living phrase repetition, language card game, picture description, and role-play with caregivers to promote everyday verbal interactions. It also includes a generalization package to reinforce verbal communication at home or special care settings.

CIAT and the CIAT II are group therapies. Two to three clients and a clinician play a language card game. Paired (duplicate) picture cards showing action, colors, number, phonemically similar words, and nouns of both high and low frequency are distributed to the participants. They cannot see the cards on the table or the hands holding the cards due to a cardboard barrier raised in front and to the sides of each client. One client requests a card and the other gives the requested card. For example, a man might pick up one of his cards and ask, “Mrs. Bentley, do you have a card with one red shoe?” The woman is required to say, “Yes, I do have it” and hand it to the requester, if she does have the requested card. If she does not have the card, she is required to say, “No, Mr. Dudley, I don’t have it.” Request complexity is gradually increased to make them longer, grammatically correct, and so forth. No hand or facial gestures, writing, miming, mouthing, or any such nonverbal means are allowed. All verbal responses may be prompted, shaped into higher response complexity, and reinforced as found appropriate. For the procedural details, the clinician may read the Johnson et al. (2014) report.

Another modification of CIAT is called the **Intensive Language Action Therapy**, which accepts and reinforces nonverbal means of communication (gestures, drawing, etc.), provided such means help evoke verbal responses. Such a use of constraints is called *guidance by constraint*. A study comparing this procedure with the more traditional PACE, described later (Davis & Wilcox, 1981), showed that neither one is clearly superior to the other (Kurland et al., 2016).

Some clinicians believe that preventing nonverbal means of communication is not prudent. They point out that there is modality interaction (between the verbal and the nonverbal) and that this interaction is helpful in learning verbal communication. Advocates of such multimodality teaching, which has been common in functional communication strategies, see

disadvantage in restricting communication to verbal means (see Rose et al., 2013).

Treatment of Naming

A difficulty in finding the right words at the right time is a pervasive problem for people with aphasia. A significant improvement in naming helps them achieve much more fluent and effective social communication.

In most naming intervention procedures, a client may be asked to match pictures with printed or spoken words and subsequently name pictures or objects shown. Various kinds of cues may be provided to evoke a correct naming response. A meta-analysis of word-finding treatment concluded that most naming techniques are effective in increasing naming skills in the treatment sessions, but generalized production of untrained (new) words in conversational speech produced at home and other naturalistic settings remains problematic (Wisernburn & Mahoney, 2009).

Confrontation naming and cueing hierarchies. A frequently used procedure is to place a picture or an object in front of the person and ask, “What is this?” “Who is this?” and so forth. Words selected for this kind of teaching should be related to (1) the client’s interests and hobbies, (2) names of family members, (3) food and clothing items, and (4) other functional words used in daily living, including everyday conversation, narration, and discourse.

A variety of stimulus cues may be used to evoke naming responses (see Nickels, 2002, for a review of studies on varied forms of cues). Cues are essentially the clinician’s models of partial responses or some other aspect of the response. Cues, for example, may be (a) auditory (“It makes a mew sound”), (b) phonetic (“It starts with a p” [for pen]), (c) syllabic (“It starts with stoo” [for stool]), and (d) semantic, which is related to meaning (“You wear it on your head”). Additional cues involve an incomplete sentence (“These are two . . .”), description (“You bounce and kick it”), demonstration of action (e.g., writing or drinking), and rhyme (“It rhymes with “hog” to evoke “dog”).

Freed et al. (2004) demonstrated that personal cues selected by the client or suggested by family members are better than standard cues. For instance, when allowed to choose a cue for hammer, the individual may suggest a cue

that is personal: “I once hit my finger with it.” The authors reported that training with personalized cues helps maintain naming skills better than training with phonological cues.

Data on generalization of naming skills are limited. When shown untrained pictures and objects, generalized naming has been documented (Freed et al., 2004; Nickels, 2002). Reviewing several studies, Boyle (2011) concluded that clinically taught confrontation naming may not generalize to discourse and conversation. A meta-analysis came to the same conclusion (Wisenburn & Mahoney, 2009). Teaching conversation, narration, and discourse skills that include naming may produce both generalized naming and improved complex language skills.

Semantic feature analysis (SFA). This method of teaching naming has been researched in controlled single-subject experimental studies (Boyle & Coelho, 1995). Words belong to families because of shared meanings. Object names have properties (e.g., color, shape, texture, size, use, activity, etc.). When a word like *ball* is selected as a treatment target, it may be evoked by other discriminative stimuli related to all properties of a ball or all balls (e.g., round, red, small, big) and the activities that may be performed with or on it (kick, bounce, throw).

In using the SFA, the clinician places a target picture (e.g., a fork) on the table in the center of a frame. Three cards may be placed above the target picture, one next to the other. On these cards, words like Group, Description, and Function may be written (a single word on each card). Three more cards may be similarly arranged just below the target word, each card containing either Context, Special Features, or Other/Personal. The words on the cards above and below the target word suggest the different semantic features of the target word (Wambaugh, Mauszycki, et al., 2013). The clinician asks the client to name the target word (“What is this?”). The correct response (“fork”) may or may not be produced. The client then goes through the semantic features. The clinician may ask a question relevant for each feature: “What category does it belong to?” “Where can you find it?” “How would you describe it?” “What does it make you think of?” “What is it used for?” “What makes it special or different?” Questions may be rephrased depending on the features displayed. All correct responses are positively reinforced with verbal praise. If an error is

made, the clinician and the client may say the correct word in unison, but not necessarily in all studies.

Specific details of SFA treatment procedures have varied enormously across studies. Semantic features (e.g., action or function), types of pictures (line drawings or realistic photographs), presence or absence of modeling, method of handling the errors, and other aspects have varied across studies. Each investigator seems to modify the procedure in some small or significant respects. Consequently, the results also have been variable across and within participants in every single study.

SFA studies have been amply and favorably reviewed. Authors of two recent meta-analyses have concluded that the studies are of high quality and the method is effective in teaching naming. Generalization to untreated names is documented though variable across individuals. Persons with milder aphasia may show greater generalization (Efstratiadou et al., 2018; Quicke et al., 2019). Generalization of nouns to conversational speech and discourse is limited. In their original study, Boyle and Coelho (1995) had noted that to have naming skills generalized to conversation and discourse speech, training may have to occur in those forms of speech.

Teaching naming skills during narration, discourse, and conversation. Teaching naming skills may be a part of conversation and discourse intervention (see Boyle, 2011, for a review of studies). This is better than teaching naming skills in isolated tasks with a hope to see an improvement in discourse and conversation. At best, words taught in a prior naming treatment may be produced during conversation without any increase in lexical diversity (Rider et al., 2008).

As the individual is engaged in discourse, narration, or conversation, the clinician may provide phonetic, phonemic, orthographic, or gestural cues to promote better naming and thus more fluent speech. If the cues are promptly provided at the earliest sign of a hesitation in speech, the client's speech may be smoother than if the clinician were to wait until the speech stopped to give the cue. Studies of this kind that Boyle (2011) reviewed were characterized by innumerable procedural variations, limited or variable generalization from trained discourse to conversation (and vice versa), poor maintenance, and

many other problems of measurement and experimental control, suggesting a need for better studies.

Expansion of Verbal Expressions

Interventions to promote effective social communication begin with an effort to expand the verbal expressions in the clinic to include conversation, narration, and discourse. These expansions may be extended to naturalistic settings.

By developing client-specific treatment programs, clinicians shape and reinforce progressively longer utterances. The clinician (a) starts with what the client can say, perhaps a few words; (b) adds additional words to create phrases or simple sentences; and (c) expands simple sentences into longer, more complex, and varied sentence forms. Conversations, narrations, and discourses may be targeted in the final stage of treatment in the clinic. Eventually, clinicians evoke these skills in naturalistic conversation between the client and themselves and, finally, between the client and the caregivers, including family members.

To teach **expanded sentence productions**, the clinician may:

- Model sentences initially and have the client imitate them (e.g., while showing a picture, ask, “What is the man doing here? Say *he is chopping the wood;*” reinforce correct imitations).
- Ask questions to evoke functional sentences (e.g., show the picture and ask the same question but do not model the response; if necessary, prompt the response by saying “He is chopping . . .”).
- Fade the prompts and only ask the question to evoke the response.

To teach **connected speech and story narration** the clinician may:

- Show action-filled pictures that are of interest to the client and ask the person to describe them.
- Select functional words with the help of the client and family members and ask the individual to make different kinds of sentences

using those words. Model, prompt, and shape longer sentences.

- Tell short stories and ask questions about them or ask the client to retell the story.
- Show sequenced pictures that tell a story and ask the client to construct one; shape longer stories by prompting.
- Ask the person to describe how certain tasks are completed (e.g., “How do you make apple pies?” “How do you make a hotel reservation?” “How do you change a flat tire?”). Prompt and reinforce longer and more complete sentences.
- Ask the client to give a short speech on a topic of interest (discourse). Prompt and reinforce longer productions, details, arguments, and expression and evaluation of opinions.
- Loosen the structure of treatment sessions by engaging the individual in more naturalistic conversations.
- Evoke more naturalistic conversational speech by introducing topics for dialogue (e.g., “Tell me about your favorite TV shows.”). Get the family members and other caregivers involved in conversation. Offer comments, express preferences, request more details, and invite criticisms and comments. Train multiple topics.
- Prompt the individual to initiate a topic of conversation without suggesting topics. For example, ask, “What shall we talk about today?” or “Do you have something to tell me today?” Generously reinforce any lead the client takes.
- Recruit a family member, a caretaker, a volunteer, or an unfamiliar clinician to participate in as many conversational treatment sessions as possible. Train them to let the individual take a lead in choosing the topic of conversation and to naturally reinforce (e.g., with nod, smile, approval) the targeted conversational skills.

The clinician may use one of several published treatment programs to treat sentence structures, elaboration of speech and language skills, discourse, and functional communication. Brief descriptions of a few such programs follow.

Promoting Aphasics' Communicative Effectiveness (PACE) (Davis & Wilcox, 1981). In the treatment of aphasia, PACE is one of the first to emphasize *effective communication*. The method later came to be known as *functional communication* that deemphasizes linguistic accuracy. It lets the clinician train conversational exchanges between two persons. The clinician and the client exchange the roles of listener and speaker equally often. The client is free to use any mode of communication. A set of stimulus cards (words or pictures) is used in treatment. The one whose turn it is to talk picks a card, hides it from the view of the conversation partner, and talks about what is on it. The listener may guess the word or give other kinds of feedback to the talker. See Davis and Wilcox (1981) for procedural details. A study comparing PACE with constraint-induced intensive therapy reported that both resulted in improved naming skills, but the latter promoted better generalization (Kurland et al., 2016).

Sentence Production Program for Aphasia (SPPA) (Helm-Estabrooks & Nicholas, 2000; Helm-Estabrooks et al., 2014). The program is designed to teach the production of different types of sentences (verities of intransitive, imperative, *wh*-, declarative, comparative, and *yes-no* questions) to persons with agrammatic (Broca's) aphasia. Within a story completion format, the clinician tells a short story. First the person is asked to repeat a sentence and then to say it without a model. The method is highly structured with specific criterion-defined steps. To use the method, the clinicians should obtain a copy of the program (Helm-Estabrooks & Nicholas, 2000). Limited case study and experimental evidence suggests that treatment and generalization effects are positive (Fink et al., 1995; Helm-Estabrooks et al., 2014).

The Response Elaboration Training (RET). This method emphasizes effective communication while supporting the production of expanded sentences that contain progressively increased amount of information (Kearns, 1985). Presenting a stimulus picture, the clinician requests the client to "talk about anything" in a more naturalistic and elaborate manner, drawing upon personal experiences and world knowledge. Stimulus naming or describing is discouraged. The same picture is re-presented to evoke different kinds of responses. Such behavioral techniques as modeling, prompting, and reinforcement for responses that offer more information are a part of this procedure. Wambaugh, Nessler, et al. (2013) modified this procedure to apply

it to teach procedural discourse (e.g., “Tell me in detail how you would go about moving to a new house”) and personal recounts (“Please talk for 5 minutes about anything you like”). After reinforcing the client’s responses, the clinician would request more elaborate sentences or retelling of a procedural detail, which would also be reinforced. The method has received experimental support that was comprehensively reviewed by Wambaugh, Nessler, et al. (2013) and reviewed with a meta-analysis by Bunker et al. (2019). Generalization and maintenance have been good, although with some variability across treated individuals (Bunker et al., 2019).

Script training. Designed to improve everyday conversation in social contexts with relative fluency, script training encourages clients to generate a list of personally relevant topics (e.g., telling personal stories or holding conversations with family members). The clinician may help generate the written script, which may be a dialogue or a monologue (Bilda, 2011). The person is taught to read the scripts in treatment sessions and then to repeatedly read them at home (massed practice). Finally, the person is asked to use the scripts in social communication. Massed practice of the script is expected to promote the use of the script in naturalistic communication. Scripts may be videoed, and such video scenes may be practiced on computers. Bilda (2011) reported that the participants in her study improved their verbal skills, increased everyday communication, and experienced increased confidence. A prerecorded script software is available (AphasiaScripts, <https://www.sralab.org>).

Conversation therapy. The ultimate goal of aphasia therapy is to promote everyday conversational skills that are essential for social integration and participation. However, many approaches teach skills (such as naming or sentence production) in structured stimulus-response formats and measure generalization of trained skills to conversation, held typically with the clinician. Most may not move to group conversation. A distinguishing feature of **conversation therapy** is that the treatment begins at the level of conversation with the clinician, with two individuals who have aphasia (dyads), and in smaller or larger groups. The following summary is based on a review of studies by Simmons-Mackie et al. (2014).

This clinically worthwhile enterprise has produced a diverse set of procedures, outcome measures, and investigators that continuously modify both. It is similar to, if not identical with, the **social approaches to aphasia rehabilitation** (Simmons-Mackie, 2008), described in a later section. It also includes such procedures as **communication partner training** (Kagan et al., 2001; Simmons-Mackie et al., 2010), **group aphasia therapy** (Elman, 2007, 2016; Marshall, 1999), **life participation approach** (Elman, 2016; Simmons-Mackie, 2008), and several other related variations (discourse therapy, volunteer training, spousal training, interaction-focused treatment, conversation partner training, conversation coaching). A limitation is that these varied therapies are not fully operationally described, creating difficulty for the clinician to implement an integrated, single, well-defined approach with a clearly specified sequence of steps.

Therapy session may be individual or group. Individual sessions may involve the client and the clinician and, more often, another person. Group format involves other persons including individuals with aphasia. A clinician especially trained in the group format may facilitate the meeting. Directions, instructions, target behavior specifications, modeling, reinforcement, correction, and such other procedures may be more or less explicit in given treatment procedures. In most programs, persons with aphasia are encouraged to use any mode of communication that is effective, including gestures, facial expressions, gaze, writing, use of pictures, visual scenes, computers and tablets, written scripts, role-playing, and so forth. Measures of behavior change (progress) may include the total number of words, sentence length, sentence variety, fluency, turns taken, number of self-corrections, frequency of topic initiations, reduced frequency of interruptions, rating scales, interviews, self-assessment, and so forth.

Much of the evidence comes from single-subject experimental studies and qualitative or case studies. A few were randomized clinical trials. Studies have generally reported positive outcomes of conversation therapy (Simmons-Mackie et al., 2014). More recent randomized clinical trials on group conversation treatment, however, have produced mixed and conflicting results. Hoover et al. (2021) reported that conversational treatment had no significant effect on discourse measures; the control group participants improved as much as those in the treatment group. On the other hand, Hoover et al. (2020)

reported that conversational treatment produced mixed results, generally highly variable across individuals with severe aphasia. No firm conclusions were warranted. Another randomized control trial that DeDe et al. (2019) conducted on the effects of conversation treatment showed improvement in functional communication and discourse production in a large group (six to eight participants with aphasia) but not in a dyad (two participants).

In general, conversation therapies of all formats are too varied in target behavior selection, treatment procedures, and outcome measures. Effective, operationally defined, succinct, and unified treatment targets, procedures, and outcome measures are needed to facilitate routine clinical applications.

Social Approaches to Aphasia Rehabilitation

Social approaches emphasize functional communication, social participation, training family members and other caregivers, conversation in everyday situations, group therapy, and aphasia support groups. An ultimate goal is to reduce the social isolation, increase social participation, and enhance the quality of life of individuals with aphasia (Simmons-Mackie, 2008; Simmons-Mackie & Damico, 2009). Like other methods, the social approach is also a collection of somewhat different intervention strategies. Several of the components of the social approach (e.g., group therapy, conversational therapy) are covered in other sections. Therefore, this section is limited to other major features of this approach.

An important point to note is that functional or social communication does not replace specific skill-oriented intervention (e.g., fluency, naming, syntactic structures). By adding additional intervention targets, the social approach takes treatment beyond verbal skills training. Therefore, deficit training and the social approach are not two opposing or incompatible methods; both are essential elements of comprehensive rehabilitation of persons with aphasia.

Drawing from various sources on the social approach to managing aphasia (Kagan et al., 2001; Lyon, 1997; Rayner & Marshall, 2003; Simmons-Mackie, 2008), the clinician may:

- Let persons with aphasia and their family members suggest the overall goals of rehabilitation and the selection of treatment targets. Have the client, the family members, and the caretakers evaluate the progress and satisfaction with the program. Consider their suggestions for procedural improvements.
- Address the social needs of people with aphasia to reduce social isolation and to promote community activities (e.g., volunteering in a hospital, attending and speaking at support groups for people with strokes). The clinician may facilitate the rejuvenation of premorbid social behaviors (e.g., playing cards with friends or cooking in collaboration with the spouse) and help create new social behaviors (e.g., learning to paint with friends, tending to a garden, or going to movies and talking about them).
- Extend communication skill training to naturalistic settings where they are much more authentic, relevant, and natural. Whenever practical, move treatment out of the clinic sooner, and in some cases, actually begin in the natural setting. For instance, conversational therapy may be started in many naturalistic settings, including the client's home, a restaurant, or a shopping center.
- Reinforce communicatively effective productions, even if they are linguistically truncated or incorrect.
- Target the collaborative nature of communication. Train others who regularly interact with the person in conversational support and management. The clinician may train conversational partners to (1) speak slowly and in simple sentences, (2) pause between topics, (3) introduce new topics, (4) pay attention to signs of failure to comprehend by periodically asking questions, (5) rephrase statements and be redundant when not understood, (6) avoid indirect references and make direct references (e.g., say Tom instead of *he*), (7) supplement verbal productions with gestures and facial expressions, (8) use props (e.g., pictures), (9) talk about the objects in the room like pictures on the wall, and (10) repeat key words (Simmons-Mackie, 2008).

Evidence in support of a social approach to treating people with aphasia is being accumulated (see Simmons-Mackie, 2008), although much needs to be done. The approach has many attractive features. Some of the methods of the social approach, however, need to be described in operational terms. Globally described methods need to be cast in more practical-methodological terms so that clinicians can readily and easily use them. Previously described limitations of conversational therapy are also found in the social approach as the two share many treatment targets and procedures.

Treatment of Reading and Writing Problems

The first step in designing a reading and writing intervention is to interview the client and family members to assess the premorbid level of literacy and the current need for, and interest in, reading and writing.

Clinicians may consider one or more of the reading, writing, or the combined programs that are commercially available. For instance, the Copy and Recall Treatment (Beeson et al., 2003) program is limited to writing single words shown on printed cards with drawings. The Multiple Oral Reading (Beeson et al., 2005) program initially measures the oral reading rate (words per minute) and errors and then teaches oral reading. The individual is then asked to read the same passage while receiving corrective feedback from the clinician and practice reading at home.

To be useful, the clinician needs to teach reading and writing skills that are more complex than some of the standardized programs include. Also, controlled data on reading and writing treatment programs are limited. Clinicians should measure the frequency of all target responses in all sessions so the technique may be continued, modified, or abandoned.

In implementing a **reading intervention program**, the clinician may:

- Select such functional reading skills as reading letters, menus, checkbook, bank statements, medicine labels, phone book, calendar, maps, and product labels. Let the client and family members suggest useful targets.

- Start with comprehension of printed words and phrases read aloud on repeated trials. Use modeling and prompting to get the client started. Ask the client to state the meaning of words and phrases read.
- Construct sentences using the comprehended target words and phrases and arrange for repeated practice of sentences. Initially, have the individual read orally with the help of modeling and prompts. Later, let the person read silently and answer questions to verify comprehension. Provide positive reinforcement and corrective feedback.

Reading comprehension can be improved only by having the client read more, read progressively more challenging material, and get appropriate feedback from the clinician and from the caregivers.

In implementing a **writing intervention program**, the clinician may:

- First find out what the person can do, and design intervention that goes beyond those existing skills. For instance, pointing to a printed alphabet or writing isolated words needs to be taught only if such basic skills are absent.
- Select such functional writing skills as writing short notes and brief letters, signing names, making grocery lists, writing one's address, and filling out forms. Ask the client and the caregivers to help make a list of useful writing skills.
- Have the person start with copying letters or words printed on cards. Remove the printed stimulus and have the person write letters and words to dictation. If the person fails, reintroduce the printed stimulus.
- Move on to copying sentences and print sentences to dictation. Ask the person to read aloud the printed material and say what it means to check for comprehension.
- Design additional steps as necessary (e.g., have the individual write paragraphs, brief notes, longer letters, a story the person knows, text

messages on the phone, emails, writing on the computer with word processors).

- Ask the client to practice the targeted writing skills at home at each level of training; assign additional homework through spelling and writing practice books; train a caregiver to supervise and encourage the persons and to submit the writing samples for evaluation.

Family members or other caregivers may help evaluate the writing samples for their accuracy and persistent errors. The clinician may address those residual errors.

Group Treatment for People With Aphasia

We noted earlier that constraint-induced aphasia therapy and conversation therapy use the group format. For the most part, these groups are formed to teach the same or similar language skills that may be taught in individual therapy sessions. There is another kind of group therapy that has slightly different objectives, although they are very close to those of the social approach to aphasia.

Unlike constraint-induced and conversational therapies, some group therapy sessions are held to promote generalization of specific and discrete language skills (e.g., naming, use of grammatical morphemes, certain types of sentences) established in individual treatment sessions. When social interactions are seen as a mechanism to promote generalization, group therapy follows individual therapy where the skills are first established. Simultaneous use of individual and group formats may also offer advantages.

Speech and language treatment groups have been formed to offer more or less direct treatment to members of the group (Simmons-Mackie et al., 2007). Groups in which communication skills are managed directly tend to be more structured than those in which the language skills are addressed globally, hence indirectly. In directly managed programs, the clinician is in charge of running the group and leading the group activities. Specific language targets may be taught with techniques used in individual therapy, including clinician-prepared

stimuli, modeling, cueing, positive reinforcement for correct responses, and milder and indirect forms of corrective feedback for incorrect ones (e.g., “Maybe you meant to say . . . ” instead of “That is not correct.”).

The language skills targeted in groups tend to be more complex (e.g., different sentence types) and exhibited in group conversational setup. Narrative description of events, sharing personal experiences, discourse, topic initiation, topic maintenance, and turn taking, may be targeted (Helm-Estabrooks et al., 2014). The more advanced the language skills targeted in the group format, the more likely such group treatment will follow individual therapy in which the basic skills are first honed in.

The groups in which speech and language skills are targeted only indirectly tend to be less structured, with the clinician playing the role of a facilitator. Group members may assume leading or relatively active roles in the indirect treatment groups. Both the target skills and the treatment procedures used in indirect treatment groups tend to be less specific. General improvement in communication skills, opportunities to practice skills learned in individual sessions, deriving and offering emotional support, learning from others about living with aphasia, and giving and receiving information, among others, may be the main goals of the group activity.

The distinction between direct and indirect treatment groups may be one of degree. The distinction may be blurred in some cases. Some highly structured groups may target such advanced skills as conversational turn taking and topic maintenance, narration, discourse, and so forth. Similarly, some loosely structured groups may target specific language features. For instance, each member’s sentence productions may be monitored, and more accurate productions may be reinforced.

Measuring outcome of group therapy has been a challenge, especially when the group structure is loose, the target skills are global or subjective, and the groups are large. Observing all members accurately and recording their actions reliably can pose both methodological and logistic problems. Measuring such clinically worthy goals as *sharing information, offering and receiving emotional support, learning from others about living with aphasia*, and so forth is difficult. Treatment procedures directed toward such goals, too, tend to be somewhat vague. Therefore, research on group therapy should address the following: (a)

the effectiveness of specific and replicable treatment strategies to be used in groups, (b) communication targets for which the group format is superior, (c) operational specification of skills targeted, (d) methods of measurement that will provide objective and verifiable data on the effects of treatment, (e) the relative effects of different kinds of group formats, and (f) experimental research that seeks to establish the absolute and relative effectiveness of different types of groups.

Virtual Reality Therapy

In theory, virtual reality therapy offers some advantages. Persons with aphasia can practice communication skills at home, engage in online social group interactions, and get entertained with game-like apps that target group interactions. Several virtual reality programs for various kinds of medical and nonmedical rehabilitation are available.

One of the better researched is the EVA Park, a computer program that depicts such virtual locations as shops, restaurants, house, café, health center, bar and disco, and other everyday places that people visit (<https://evapark.city.ac.uk>). Each user is a personalized avatar who walks, talks, runs, flies, and explores. Avatars can type messages. Multiple avatars interact on screen and talk to each other. Research findings on this reality therapy are unimpressive. An earlier study showed improvement on such measures as fluency and naming scores, but the control group also showed improvement (Marshall et al., 2016). Other measures, including word finding, conversation, and communication confidence, reduced feelings of isolation failed to reach significance. A later study failed to show any significant benefits of the EVA virtual group therapy (Marshall et al., 2020). A meta-analysis of available studies also found the method mostly ineffective (Cao et al., 2021). Technical improvements in the apps and methodological improvements in treatment applications may produce more favorable results.

Telerehabilitation for Aphasia

The COVID-19 pandemic in 2020–2022 has resulted in cancellation of many face-to-face outpatient clinical contacts, in-class teaching, and office medical visits. For the first time, many adults experienced sessions of telepractice with their primary care doctors. Schoolchildren and college students listened to lectures delivered on Zoom computer video apps. SLPs did assessment and treatment online. Telepractice got a big boost.

In medical fields, telepractice in clinical professions began long before the COVID-19 pandemic to support and monitor patients after their discharge from the hospital. In assessing and treating persons with aphasia, however, telepractice has a shorter history and limited efficacy data (Cassiante et al., 2021).

There are two main varieties of telepractice. In the **synchronous telepractice**, the SLP meets the client online and conducts therapy. It is a two-way videoconferencing session. The SLP provides the treatment stimuli, models the target response, and gives positive reinforcement or corrective feedback — much like in the face-to-face therapy, except administered remotely. In the **asynchronous telepractice**, the client, who is not in direct visual contact with the therapist, practices therapy exercises generated by specialized computer apps (Cassiante et al., 2021).

High-quality research on the effectiveness of telepractice without face-to-face therapy is lacking. In their meta-analysis, Cassiante et al. (2021) found only five synchronous studies. Only a few studies have compared telepractice with face-to-face therapy. Generally, the studies report that telepractice is not inferior to face-to-face therapy. The results of the two methods of treatment have been comparable. Investigators and reviewers alike stress the need for additional research (Cassiante et al., 2021; Macoir et al., 2017).

Augmentative and Alternative Communication

Face-to-face clinician-administered therapy is described as **restorative**, designed to restore the impaired communication skills. Some individuals with aphasia may need another kind of therapy, often described as **compensatory**, which aims at compensating for the deficits that are not modifiable or to be

modified later. Augmentative and alternative communication (AAC) is a form of communication that is compensatory.

Augmentative communication helps enhance, expand, or augment the existing verbal communication skills of a person. Technically defined, **alternative communication** replaces the typical oral (verbal) communication, but functionally defined, it is also a supplement to residual oral communication skills. Augmentative and alternative are not categorically different; the difference is one of degree. Generally, however, AAC is geared toward functional communication and, unlike restorative therapy, does not concentrate on speech production or grammatical accuracy.

AAC and the more typical verbal skills are not mutually exclusive; they may coexist to support or expand the communication repertoire of a person with aphasia. This is the social participation approach in which any means of communication and any combination of them are welcome (Beukelman & Mirenda, 2013). Clinicians need not always wait to implement AAC until verbal communication training fails. AAC training may be integrated with verbal skill training. Nor should they limit AAC to expressing only the basic needs. It may be used to promote other kinds of language skills, including social interactions. AAC should not be limited to persons with the most severe aphasia. Persons with any severity level may benefit from AAC (Dietz et al., 2020).

Upon admission to a hospital following a stroke, functional AAC may often be needed. Communication through gestures, facial expressions, body language, picture or object pointing, writing, or a communication board in the hospital setting in the acute or subacute stage of a stroke are examples of such simple, much needed, and *unaided* forms of AAC. In this brief introduction to devices that provide opportunities for augmented or alternative communication, we will consider only the increasingly popular mobile technology devices (laptops, tablets, and smartphones), which are *aided* forms of AAC.

There are two barriers to an integration of AAC with traditional speech-language therapy. The first is the unfortunate but real digital divide. For many persons, unaffordability of technological devices and services, lack of access to high-speed Internet in their communities, and limited education and income

may be roadblocks (Kelly et al., 2016). However, if other factors are favorable, the clinician may seek support from social, government, and philanthropic organizations to procure a device such as a smartphone or a tablet for a person with aphasia. The second is the clinician's training and orientation to aphasia treatment. Most clinicians do not include AAC in their treatment plan and do not offer significant information to persons with aphasia on AAC, and only 2% of their clients may receive AAC (see Dietz et al., 2020, for a review of studies). Dietz et al. (2020) recommend that clinicians consider using AAC earlier in their treatment plan rather than later. They argue that there is no reason to fear that AAC might be more permanent and hamper the reestablishment of speech and language skills.

The demographics of people who have aphasia have been changing rapidly. An increasing number of individuals with aphasia have used computers, tablets, and smartphones premorbidly. Such persons may now be willing to enhance their communication skills (Brandenburg et al., 2013) with technological devices. Because of their greater need, people with moderate aphasia may benefit to a greater degree than those with mild aphasia. Such persons may also use the Internet and communicate effectively with email messages. Email networks of people with aphasia and their communication partners may be established to enhance social communication opportunities (Mahmud & Martens, 2016).

People with severe aphasia are excellent candidates for simpler devices. These individuals may learn, on their own, to gesture, point to things, and write out the words. Moreover, with professional training, they may be able to use more sophisticated electronic devices to communicate more extensively and effectively than without such devices. For example, a person with aphasia may use a small handheld computer and type messages that are displayed on its screen. Another person who cannot type well may learn to display messages already stored in the computer by pointing to a graphic icon or clicking to display a stored word, phrase, single sentence, or a group of sentences narrating an experience (Hux et al., 2008). The person may also use such nonelectronic (low technology) devices like a communication board on which words, messages, pictures, and symbols are posted. The individual may point to words in a certain order to generate sentences that the communication partner

“reads.” One might also use an organized nonverbal language system, such as the American Sign Language.

An increasing number of applications for desktop computers, laptops, tablets, and smartphones are being developed to help individuals with aphasia communicate more effectively than otherwise. Also available are dedicated AAC devices that come with preloaded communication apps. The following are a few examples:

- Constant Therapy is an iPad-based software program to improve naming, reading, writing, sentence planning, memory, attention, problem solving, and executive functions (<https://constanttherapy.com>). This program is approved by the Food and Drug Administration (FDA).
- Tactus Therapy Solutions offers multiple apps for different platforms that help generate verbal skills such as asking and answering questions, naming, comprehension, reading, and writing (<https://tactustherapy.com>).
- Lingraphica, SmallTalk, and TalkPath are iPad-based apps that offer various speech and language exercise that may be completed at home between therapy sessions. They may be programmed to include frequently used utterances. Lingraphica also offers a midsize tablet called TouchTalk.
- AphasiaScripts allows for personalization of output and supports conversation (<https://www.sralab.org/aphasiascripts>).
- Coglink is a personalizable app that helps write, send, and receive simple email messages (<https://www.personaltechnologies.com>).
- StepByStep is a computer software program to practice a variety of language skills including naming, matching, spelling, repetition, and so forth (<https://www.aphasia-software.com>). It helps produce words with visual, semantic, and phonemic cues.
- Sentactics is another computer software program designed to teach complex sentences through interactive virtual reality sessions. A person with aphasia meets the virtual clinician on the computer

screen to participate in sentence production and comprehension trials.

- SentenceShaper is a computer software with which the users can create sentences from word fragments and record them with their own voice (sentenceshaper.com). The users can replay them when needed. The app contains built-in workbooks to practice sentence creation.
- Visual Scene Display contains communication scenes on an electronic device. The scenes include real-life interactions, situations, places, and experiences. The scenes may be combined with texts to communicate.

There are many other apps for digital devices, and new programs are constantly being developed. See Choi et al. (2019) and Russo et al. (2017) for a review of high-technology AAC devices and research. Clinicians are encouraged to do a periodic web search to find the updated information on apps, devices, and their effects.

AAC devices designed for persons with aphasia have several limitations that the clinician needs to evaluate before recommending one to their client. Considering their individual client's needs and capabilities, SLPs may search for current, well-researched, easy-to-use, and affordable devices. Such a device may help increase treatment intensity (Godlove et al., 2019) by providing an opportunity to practice the skills at home. However, high-technology AAC devices are not uniformly effective across individuals; some may not benefit at all (see Russo et al., 2017, for the studies). The quality and ease of use of such applications vary greatly. Many applications may be promoted without adequately evaluating their ease of use and effectiveness. Generalization to related but untrained language structures is still a problem, although face-to-face clinical training also shares this limitation. Devices and apps generally fall short of the speed of everyday conversational speech. A combination of the traditional face-to-face restorative therapy and AAC use may produce the best possible outcome for most individuals. Persons with aphasia and their communication partners should be appraised of such limitations and possibilities before they purchase a unit.

Medical Treatment of Persons With Aphasia

A few medical treatment options are available for persons with aphasia. Most medical treatments are experimental with limited evidence. They need additional research and evidence to support standard practice. Major medical interventions being researched in treating persons with poststroke aphasia include drug treatment, transcranial magnetic stimulation, transcranial direct current stimulation, epidural cortical stimulation, photobiomodulation, and stem cell therapy (Barthels & Das, 2020; Duncan et al., 2020).

Drug Treatment

Drug treatment during the acute stage may save a life or prevent additional strokes; it is not meant to directly affect communication deficits. **Tissue plasminogen activator (tPA)** is the only FDA-approved thrombolytic drug that helps dissolve blood clots in persons with ischemic stroke. It should be administered within a narrow window of 4.5 hours after the onset of stroke; if administered later, the drug will actually cause additional brain damage. If that window is missed, the neurosurgeon may perform a thrombectomy to remove the clot. To prevent a likely second or multiple strokes, anticoagulants and drugs that lower blood pressure and cholesterol may be administered (Barthels & Das, 2020).

Drug treatment to improve communication skills has not produced reliable and substantial positive effects (Picano et al., 2021). The effects of drugs thought to improve attention or neurotransmission have been investigated in persons with chronic aphasia. Most of the studies have administered these drugs without controlling for the effects of speech-language therapy the persons received simultaneously. One drug, **bromocriptine**, has been studied in isolation (without language therapy), and the only notable effect was a potentially dangerous side effect (Picano et al., 2021). Drug therapy aimed at improved language is currently not a desirable option for persons with aphasia (Duncan et al., 2020).

Electrical Brain Stimulation

Different methods of directly stimulating the brain with noninvasive electrical currents have been investigated for their effects on communication in persons with aphasia. Methods include transcranial magnetic stimulation and transcortical direct current stimulation (Duncan et al., 2020).

Transcranial magnetic stimulation (TMS) involves placing an electromagnetic coil on the scalp to induce a magnetic field in the brain. The stimulation is repeated in the repetitive transcortical magnetic stimulation (rTMS). This method poses a risk of seizures in persons with focal brain lesions, such as those who had a stroke. A low-frequency stimulation (1 Hz) inhibits the brain and sometimes is applied to the right hemisphere to prevent its participation during language therapy. Higher-frequency stimulation is excitatory and may be applied to the language-dominant left hemisphere. A few studies have reported some improvement in such limited language skills as object naming (see Duncan et al., 2020, for a review of studies).

Transcortical direct current stimulation (tDCS) is a noninvasive procedure of stimulating the brain with weak electrical currents of 1 to 2 mA through a pair of surface electrodes placed on the scalp. The current modulates the cortical excitability for about 20 to 30 minutes. Excitatory anodal (positive) stimulation is applied to the left hemisphere and inhibitory cathodal (negative) stimulation is applied to the right hemisphere in behavioral language therapy sessions, although most studies have used computer-based therapy, not the typical clinician-administered variety. Broca's and Wernicke's areas in the left hemisphere may be specifically targeted for anodal stimulation. Investigators using the Melodic Intonation Therapy have applied excitatory stimulation to the right hemisphere because of its presumed participation in prosodic features. Generally, the effects have been modest and temporary (Duncan et al., 2020).

The effects reported in hundreds of studies of electrical stimulation of the brain have produced modest effects on limited language skills (e.g., picture naming or description and word repetition). Elsner et al. (2020) reviewed studies on tDCS and concluded that there was no improvement in functional communication. While a few other reviewers have drawn a slightly more

favorable conclusion about the positive effects of brain stimulation, all agree that the method cannot be recommended for routine clinical practice (Breining & Sebastian, 2020). The method is experimental and promising, justifying better-designed studies that measure functional as well as complex communication (such as conversation). Whether a combination of behavioral language therapy and brain stimulation would prove more effective than the behavioral treatment alone in promoting functional as well as more complex language skills is yet to be determined (Picano et al., 2021). A survey of SLPs found that 94% have concerns about using brain electrical stimulation during their speech-language therapy sessions because of lack of training, safety, cost, administrative approval, and potential problems with reimbursement (Keator et al., 2020).

CLASS NOTES

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PART III

RIGHT HEMISPHERE DISORDER



6

RIGHT HEMISPHERE DISORDER

Chapter Outline

- Hemispheric Asymmetry
- Right Hemisphere Functions
- Neuropathology of Right Hemisphere Disorder
- Overview of Right Hemisphere Disorder
- References

Learning Objectives

After reading the chapter, the reader will:

- Describe the anatomic differences between the right and the left hemispheres
- Summarize the functions of the right hemisphere
- Describe the neuropathologies of right hemisphere disorder
- Describe the neurological and behavioral symptoms of right hemisphere disorder
- Specify the various communication disorders of right hemisphere disorder

PRINTED NOTES

Since Broca's investigation of aphasia associated with the left hemisphere (LH) damage in 1861 and Wernicke's discovery of sensory aphasia in 1874, the research and clinical attention has largely been paid to the LH. Strokes and aphasias clarified the relationship between the left side of the brain and language. Functions of the right hemisphere (RH) and their impairments, on the other hand, remained poorly understood until the latter part of the 20th century.

Early investigators of RH damage found that the symptoms were different from those associated with the LH damage. More careful anatomic studies revealed structural differences between the hemispheres.

Hemispheric Asymmetry

The two halves of the brain are asymmetric in their size. Brain postmortems and structural as well as functional magnetic resonance imaging (MRI) studies have confirmed structural differences between the hemispheres even in fetal developmental stages. For instance, ultrasound scanning has shown that the LH in both male and female fetuses is slightly larger than the corresponding RH (Hering-Hanit et al., 2001).

Morphological differences are the greatest in certain areas surrounding the lateral sulcus (fissure of Sylvius) that mediate language. The lateral sulcus in the LH is slightly longer than that in the RH; the left sulcus extends farther posteriorly than the right. The planum temporale on the superior surface of the temporal lobe may be more than five times larger in some individuals. The planum temporale is a part of the superior surface of the superior temporal gyrus, located posterior to the primary auditory cortex. Also, the left temporal opercular region is more infolded than the corresponding region in the RH.

This is the region that is also known as Broca's area, the motor speech cortex. In addition, the posterior thalamus region in the LH is also larger than that in the RH. The various white matter connecting fibers described in [Chapter 1](#), including the cingulate bundles and arcuate fasciculus that are important for speech and language, are bulkier in the left hemisphere (Nolte, 2020; Shu et al., 2015; Webb, 2017).

Initially, it was thought that the hemispheric difference was due to the LH's specialization in language. However, the LH is larger in apes as well as mice. Language may have been lateralized to an existing larger structure rather than language causing the structure to enlarge. It is thought that the larger left hemisphere is a product of both genetic as well as environmental variables (Nolte, 2020; Ocklenburg, 2012).

As described in [Chapter 1](#), the two hemispheres are well connected. A bundle of fibers called the corpus callosum (also called *commissural fibers*) connects the two hemispheres. Containing more than 300 million axons, corpus callosum is the largest bundle of fibers in the human brain (Nolte, 2020). Most commissural fibers connect structures that are mirror images in the two hemispheres, although many originate in one area only to connect different (not corresponding) areas.

Right Hemisphere Functions

The workings of the RH remained largely unknown until the 1950s; it was even named the *silent hemisphere*. It was thought that the RH took over the functions of a damaged LH. There is some evidence that in persons with left hemisphere stroke, the right hemisphere may assume certain language functions. During the past few decades, research on the structure and functions of the RH has increased to give us a better picture of its functions (Blake, 2018; Blake et al., 2002; Hartikainen, 2021; Hegde & Freed, 2022; Nolte, 2020; Payne, 2014; Webb, 2017).

In individuals with intact RH and normal hemispheric connections, the RH seems to predominantly mediate the following:

- **Arousal and attention.** The RH helps maintain a dynamic relation between the individual and the surrounding environment. Subtypes:
 - **Arousal:** General readiness to respond to external stimuli (being “in touch” with the surroundings).
 - **Orienting:** Attending to a specific stimulus, event, or location.
 - **Vigilance:** Heightened attention to detect changes in stimuli.
 - **Selective attention:** Paying attention to specific stimuli while ignoring others.
 - **Sustained attention:** Attending to stimuli or tasks for longer durations.
- **Visual perception.** The RH is dominant for visual perception and its various components:
 - **Holistic, gestalt-like stimuli:** Grasping the meaning of a total stimulus complex (e.g., a picture), with little time spent analyzing its components.
 - **Geometric and spatial information:** Understanding and recognizing stimuli that include spatially organized shapes and figures (e.g., recognizing handwriting of a known person).
 - **Constructional activities:** Putting a puzzle together, assembling a model car, taking apart and reassembling appliances.
 - **Facial recognition:** Infants between the ages of 4 and 10 months show an RH advantage for recognizing their mother’s face.
 - **Body image:** Maintaining one’s own body image.
- **Emotional experience and expression.** The RH is dominant for emotional experiences:
 - **Posterior region of the brain:** RH is more active than the LH when individuals experience emotions.
 - **Negative emotional experiences:** Anger and fear are lateralized to the RH. Positive emotional experiences (e.g., happiness) are lateralized to the LH.

- **Perception of temporal order.** The RH is dominant for perceiving the temporal order or sequence of events.
- **Perception of musical harmony.** The RH may be dominant for perception of musical harmony.
- **Certain aspects of communication.** Although the LH is dominant for language, the RH specializes in mediating certain aspects of communication (Barnes et al., 2019, 2022; Bartolomeo, 2014; Bartolomeo et al., 2017; Duque et al., 2021; Gold et al., 2012; Minga et al., 2021, 2022; Parola et al., 2016; Sheppard et al., 2021; Weed & Fusaroli, 2020). Communication-related functions the RH mediates partly or wholly include:
 - Comprehension of discourse, implied meanings and inferences during conversations, alternative and ambiguous meanings, and humor and jokes,
 - Comprehension of emotional tone of voice and expressions.
 - Responding to indirect requests and subtle questions.
 - Production of prosodic features (right inferior frontal gyrus) and comprehension of prosodic variations (right posterior temporoparietal region).
 - Comprehension of contextual information implied in verbal expressions.
 - Comprehension of abstract concepts and nonliteral meanings of metaphors, idioms, images, similes, and similar expressions used in everyday language as well as literary writings (such as poetry).
 - Conversational (pragmatic) skills including turn taking, topic maintenance, social appropriateness of communication, and eye contact.
 - Oral reading.
 - Word finding, word fluency, and body part naming.

Descriptions of right brain functions should be taken with caution. Most studies have shown individual differences that do not conform to generalized

statements either about the absolute and normal functions of the RH or about the impairments following its damage.

Because of a lack of consistent and stable pattern of symptoms across individuals with RH injury, the use of the term *right hemisphere syndrome* has declined in favor of *right hemisphere disorder*. Individuals often present unique or unexpected features. Therefore, generality of research findings on hemispheric specialization is limited.

Neuropathology of Right Hemisphere Disorder

The same neuropathological factors that affect the LH also affect the RH. The consequences, however, are different (Palmerini & Bogousslavsky, 2012).

- **Cerebrovascular accidents.** Ischemic and hemorrhagic strokes are the most common cause of RH disorder (RHD). RH strokes are nearly as frequent as the LH strokes. Treatment of persons with RH strokes may be delayed because those with transient ischemic attacks, experiencing mild cognitive deficits, may not arrive at the hospital to receive prompt treatment. Consequently, persons with RH strokes have a poorer outcome than those with LH strokes.
- **Tumors.** Various kinds of brain tumors in the RH tend to cause more focal symptoms than the cerebrovascular accidents.
- **Traumatic brain injury.** This factor also tends to produce more focal symptoms than cerebrovascular accidents.
- **Various diseases and infections.** For example, Alzheimer's disease also may affect the functioning of the RH. Bacterial (e.g., meningitis) and viral infections (e.g., mumps, measles, and untreated syphilis) may infrequently cause RHD.

Typically, posterior lesions in RH do not produce motor disabilities (hemiparesis or hemiplegia). Such disabilities are more commonly associated with RHD involving the frontal lobe. This may be another reason why persons with RH stroke do not seek immediate medical attention.

Overview of Right Hemisphere Disorder

As in the cases with LH damage, the functional consequences of RH damage depend on the site, nature, and extent of damage. The effects are highly variable across individuals. Similar damages in comparable sites produce different results in different individuals. A greater number of symptoms are noted at the acute stage than at the postacute stages.

In their study of symptom patterns observed in 123 persons with right hemisphere disorder, Blake et al. (2002) reported that no single deficit was observed in all of the individuals. Attention deficits were the highest, observed in 67.5% of the individuals. Left neglect was observed in 65.9% of the individuals, the next highest. Pragmatic language problems were noted only in 16.7% and “linguistic” deficits in 26% of the individuals.

In the following sections, the varied symptoms of RHD are described under three main categories: (1) perceptual and attentional deficits, (2) affective (emotional) deficits, and (3) communication deficits. A few other observed problems are briefly mentioned.

Perceptual and Attentional Deficits

Perceptual and attentional deficits include left neglect, facial recognition deficits, constructional impairments, attentional visuospatial impairments, and forms of disorientation. Such deficits negatively affect communication skills.

Various deficits associated with RHD are interrelated. Neglect and inattention to external stimuli and facial recognition problems affect social communication. Lack of attention, for example, may be related to such conversational problems as frequent topic shifts. Wandering attention may make it difficult to concentrate on the speech of conversation partners. Consequently, responses may be irrelevant.

Unilateral Neglect

Neglect is reduced sensitivity to stimuli, reduced awareness of space, or absence of previously learned responses to stimuli in certain visual fields, often in one side of the visual field. Therefore, it is *unilateral neglect*, also known as *hemispatial neglect*. Damage to any lobe of the brain in any hemisphere may produce neglect (Bartolomeo et al., 2017). Therefore, neglect can be right or left dominant. LH damage can produce *right neglect*, but due to a theoretical bias that only the RH is responsible for body awareness, left neglect has not been previously studied in persons with LH disease or disorder. Subsequent studies have reported that neglect may be found almost equally often in LH and RH damage (Nurmi & Jehkonen, 2014). Compared to right neglect, left neglect tends to be more severe, consistent, and almost incredible.

Because of its dominance for spatial attention, RH can compensate for spatial attentional impairment due to LH damage. When the RH is damaged, however, the nondominant left (for spatial attention) cannot compensate for the deficit. Lesions in the RH's posterior parietal cortex are the most common cause of left neglect. Less commonly, lesions in the prefrontal cortex of the RH also may cause left neglect.

Left neglect is characterized by greatly reduced responsiveness to stimuli on the left side of the body and impaired awareness of left-sided space. Visual, auditory, and tactile modalities may all be affected, although visual neglect has been the most frequently tested. Persons with visual neglect also may have left visual field blindness, but this form of neglect may be found in persons with an intact visual field. Within a few weeks or months postonset, persons recover from neglect either partially or completely (Bartolomeo et al., 2017; Blake, 2018).

With varying degrees of severity across individuals, the following characterize left neglect:

- **A right focus.** There is a strong tendency for the stimuli on the right side to capture and hold the person's attention, as well as difficulty switching attention to the left-sided stimuli.
- **Failure to perceive left-sided tactile or perceptual stimuli.** This is difficulty perceiving touch or pin-prick stimuli applied to the left side of the body. When both the sides of the body are simultaneously

touched (or otherwise stimulated), the person may report sensation only on the right side. Some mildly impaired persons can report left tactile stimuli only when the left side is stimulated but not when both the sides are simultaneously stimulated.

- **Failure to copy the left side of a picture or a geometric design.** For example, the person may copy only the right side of a clock and crowd all the numbers into the right half of the circle.
- **Painting only the right half of faces or other images.** For example, an artist may paint only the right side of his or her own face on a self-portrait.
- **Attention limited to extreme right-centered stimuli.** For example, a person may point to only the objects or pictures found at the extreme right when several are placed to the right of the person. This implies that what is right is relative.
- **Descriptions limited to the right side of spaces.** Persons may describe only those houses on the right side of their street.
- **Bumping into things or persons on the left side.** A person in a wheelchair may keep hitting objects on the left side. The person may not realize that a left shirt sleeve is stuck in a doorknob.
- **Using only the right-sided objects.** For example, the person may use only right-side pockets or drawers.
- **Reduced or absent awareness of illness.** Also called *anosognosia*, the person with RHD may be unaware or only be partially aware of the personal problems and limitations. The person may deny the existence of a paralyzed left arm or leg or may insist that the paralyzed body parts belong to someone else or that they have been amputated decades ago. Sometimes such beliefs are described as *delusions*, though they may not be the typical psychiatric delusions. Some persons who admit to their problems may underestimate the extent or effects of such problems. Others may be indifferent to admitted problems or their consequences, a condition called *anosodiaphoria* (lack of emotional response to personal disabilities). Being unaware of their problems, a person with RHD may *confabulate* explanations for their deficits, actions, and failures. Blake

(2018) gives the example of a man who bumped into the hallway wall on his left side and knocked the pictures off; he explained that his wife hung too many pictures that he didn't like, so he hit them to make them fall.

- **Auditory neglect.** Although less common than visual neglect, some persons may neglect sound coming from the left side. They may not respond to the telephone ringing or people talking on the left side. In addition, posterior lesions can cause sound localization problems that are independent of auditory neglect.
- **Motor neglect.** The person may behave as though the left side is paralyzed, even when it is neurophysiologically normal. The person's left leg or arm may drag while the person moves or the person may fail to withdraw the left limb from painful stimuli. Left motor neglect may cause loss of balance. Movements of the left limb may be slow, sluggish, and weak.
- **Left neglect in reading.** The person may read aloud only the right half of each line and then complain that what is read is meaningless. The person also may omit the leftmost letters in words (e.g., read *pastime* as *time*).
- **Left neglect in writing.** Failure to give the left margin in writing is common. Other writing errors include an inconsistent left margin, extra space between letters, split words, and letter repetitions or omissions. Frontal lesions tend to cause letter repetitions, and postrolandic lesions tend to cause left margin errors and other spatial problems in writing.

Facial Recognition Deficits

Also known as *prosopagnosia*, facial recognition deficits are a part of the perceptual and attentional deficits associated with RHD. These deficits are seen in persons with posterior RH damage. Bilateral damage may produce more persistent deficits. General facial recognition problems include the following:

- **Difficulty recognizing familiar faces.** This is most apparent when the person is shown line drawings or photographs of familiar faces,

but the difficulty may extend to pictures or actual faces of familiar animals, faces of older and younger persons, and male and female individuals. The difficulty may be limited to faces only. Persons with RHD usually can recognize a face as soon as the person speaks, moves, and does something. Such physical characteristics as body size, hair color and style, and gait also may prompt correct facial recognition.

- **Problems naming the pictures of faces of famous persons.** The persons may find it difficult to name pictures of persons they normally would know.

A rare and an extreme consequence of prosopagnosia is known as *Capgras syndrome*. In this syndrome, persons with RHD may entertain a delusional belief that their friends and family members are not their real selves but imposters or doubles.

Attentional Deficit

Injury to the RH or LH tends to impair attention. Nearly 68% of persons with RHD are likely to show various forms of attentional deficits that create problems in communication (Bartolomeo, 2014; Blake et al., 2002). The specific problems include:

- **Impaired state of arousal.** As noted, arousal is the physiological and behavioral readiness to respond to stimuli. RHD may reduce arousal (hypo-responsive) or increase it (hyper-responsive).
- **Impaired sustained attention.** Attention may wander from task to task or from stimulus to stimulus. Maintaining attention on treatment activities may be difficult.
- **Impaired selective attention.** Paying attention to a particular stimulus embedded in an array of stimuli may be impaired; individuals may be easily distracted by unimportant or irrelevant stimuli.

Because of these attentional deficits, persons with RHD find it difficult to respond promptly, sustain their conversation, stick to conversational topics, and maintain attention to a speaker's messages. Their speech may be incoherent at times because of their lack of attention to what others say.

Disorientation

Persons with RHD may show a variety of orientation problems, possibly due to impaired visual attention to environmental cues. Some of the specific problems include:

- **Topographic disorientation.** Confused about space, persons may have difficulty finding their way in familiar surroundings (e.g., one's own home or room), understanding maps, and giving directions.
- **Geographic disorientation.** Confusion about broader geographic location, persons may believe that their hospital or house is located in a state or country different from the actual one.
- **Reduplicative paramnesia.** A rare condition in which the person believes in the existence of multiple and identical persons, places, and body parts. Brookshire's (2015) examples include a man who believed in the existence of two identical hospitals in the same city, another man who believed that he possessed two left legs, and a woman who thought that she had two identical husbands at home.

Persons with RHD who show disorientation of the kind specified can be distinguished from other persons (e.g., those with dementia) who also are disoriented. Persons with RHD may not be disoriented to time, space, and persons in general, although they may be confused about themselves. Unlike persons with dementia, those with RHD who are lost in their familiar surrounding can usually find their way back.

Visuoperceptual and Constructional Deficits

Visuospatial deficits have been noted mostly in drawings and constructional deficits in block constructions and similar tasks. Parietal or parieto-occipital

damage is most often associated with visuospatial and constructional impairments. Major deficits include:

- **Difficulty recognizing drawings.** Line-drawn pictures or incomplete drawings may not be recognized.
- **Difficulty recognizing distorted drawings.** Distorted drawings that show unusual size, dimension, or orientation may not be recognized.
- **Difficulty recognizing superimposed images.** Layered drawings may be difficult to recognize.
- **Problems constructing block designs.** Persons may not construct (copy) block designs by looking at printed designs.
- **Errors in drawing or copying geometric shapes.** Drawings and copies of actual or printed geometric shapes may be inaccurate.

Visuospatial impairments may be seen in persons with all kinds of brain damage, although it is most prominent in persons with RHD. The persons with LH involvement draw with difficulty but make fewer mistakes; their drawings bear greater resemblance to models; their drawings are better with models than without. On the other hand, persons with RHD draw hastily, make many mistakes, add unnecessary lines to correct mistakes, turn three-dimensional figures into two-dimensional, and do not show improvement when provided with models.

It should be noted, however, that persons with RHD do not have difficulty recognizing real objects or objects depicted fully and naturally.

Affective Deficits

Emotions are felt internal bodily states; affects are observable emotional behaviors. The limbic system — a horseshoe-shaped structure in the frontal, parietal, and temporal lobes, forming a circle over the brainstem — mediates most biological functions, including emotions, affect, thirst, hunger, and sexual arousal. Lesions in the right posterior superior temporal gyrus (part of the ventral stream; see [Chapter 1](#)) will impair identification of emotions expressed

in speech (Harciarek & Mankowska, 2021; Hartikainen, 2020; Sheppard et al., 2020).

Although the limbic system is responsible for the actual experience of emotions, the RH is largely responsible for mediating the expression of emotions and appreciation of emotions other people express. Electroencephalographic (EEG) studies have shown increased RH activity when persons experience strong emotions or recall emotionally charged experiences. Persons with LH damage who fail to complete difficult and stressful tasks tend to react emotionally (known as *catastrophic reactions* and includes crying and cursing). In a similar situation, persons with RHD tend to be emotionally indifferent. Unless the limbic system also is involved, the person with RHD can experience emotions normally but may have the following kinds of difficulty expressing them.

- **Emotional apathy and indifference.** Apathy extends to one's own disabilities. They may give confabulatory explanations of their disabilities in an unconcerned manner.
- **Difficulty understanding or recognizing emotions.** Understanding emotions other people express through their facial expressions and recognizing emotional expressions depicted in pictures that tell stories may be impaired.
- **Problem recognizing emotional tone of voice.** Understanding emotions speakers express through their prosodic variations may be difficult. This problem is further described in the next section on communication deficits.
- **Difficulty in emotional expressions.** Persons with RHD may have problems in expressing their own emotions both verbally and nonverbally.

Communication Disorders

In one study, basic expressive and receptive language impairments were found in about 26% of 123 cases with RHD (Blake et al., 2002). Communication

disorders are more common in individuals who are left-handed and then sustain right cortical injury, presumably because in those individuals, the RH is dominant for language. Not all left-handed individuals who sustain RH damage exhibit predominant language problems, however. This is because the LH is still dominant for language in many left-handed individuals.

Impaired communication associated with RHD is unlike that associated with LH damage. Also, it is important to note that certain language skills are relatively preserved in persons with RHD. Generally, the persons with RHD do not experience word-finding problems, although exceptions might be noted. The persons can define words or name objects described to them. If there is any difficulty in individual cases, it is likely to be mild and more pronounced in naming categories or collective nouns (e.g., the person may name roses, petunias, and marigolds but may not readily say *flowers*).

Paraphasias and circumlocution due to word-finding problems are also not significant in persons with RHD. Any circumlocution they may exhibit is due to their attentional problems that make it difficult for them to be precise and direct in their communication. On the other hand, the persons with LHD do not have such attentional and cognitive problems; their circumlocution is due to word-finding problems. For these and other reasons, the communication problems of persons with RHD are not described as *aphasia*.

For the most part, the persons with RHD speak in grammatically accurate sentences. The persons may have difficulty with abstract or metaphoric meaning, but not with the literal meaning of language they hear. Although the RH may be involved in processing single words and sentences, most persons with RHD understand the main meaning of single words spoken to them. On most tests of aphasia, persons with RHD exhibit normal comprehension of simple sentences. However, as described later, RHD affects comprehension of ambiguous sentences with multiple meanings or abstract and metaphoric speech (Gajardo-Vidal et al., 2018; Gold et al., 2012; Minga et al., 2021). A study has revealed that if the dorsal parts of the superior longitudinal fasciculus and the right inferior frontal sulcus are damaged, auditory sentence-to-picture matching may be impaired. LH damage, on the other hand, affects picture description (Gajardo-Vidal et al., 2018).

Notable communication disorders of persons with RHD include (1) impaired conversation, (2) impaired narration and discourse, (3) prosodic deficits, and (4) difficulty with abstract language.

Impaired Conversation

Most individuals with RHD may not have serious and persistent difficulties with basic and everyday conversation. The speech of individuals with RHD may sound “awkward or somehow not quite right” (Blake, 2018, p. 48), but the listener may not be able to say what exactly was wrong or abnormal with their conversational speech. Reviewing the research, Barnes et al. (2019) noted that the studies on conversational skills in persons with RHD are surprisingly few, speech sampling techniques have been variable and unsystematic, and the evidence for impairments is extremely limited. Except for the single study by Barnes et al. (2019) that investigated conversation between a man with RHD (due to stroke), his female spouse, and a female friend, there are no studies on conversation between persons with RHD and their familiar conversational partners. Results of Barnes et al. showed that the conversational pattern of the person with RHD was mostly appropriate. Nonetheless, difficulty in engaging and sustaining meaningful conversation may be a problem in some persons; Blake et al. (2002) reported pragmatic language problems in 16.3% of individuals with RHD. Other types of deficits associated with RHD at a higher frequency, including inattention, neglect, impaired memory and learning, disorganized reasoning, reduced awareness, disorientation, and hyper- and hyporesponsivity, may contribute to conversational difficulties.

Commonly reported disorders of conversation include the following:

- **Impaired turn taking.** Because of inattention and rambling speech, the person may not yield to their conversational partners or frequently interrupt the speaker.
- **Frequent topic shifts.** Topic changes during conversations may be frequent and irrelevant. The speech may be excessive, but topic maintenance may be poor.
- **Lack of eye contact.** While talking excessively or rambling, persons with RHD may fail to maintain eye contact with their conversational

partners.

- **Insensitivity to contexts of communication.** The persons may assume too much about what their listener(s) know about the topic of conversation (e.g., persons may talk about their friends or colleagues as though the clinician knew them). The persons also may introduce new topics when their partners indicate a closure to the interaction.

Those conversational problems do not give a complete picture of the significant communication impairments found in persons with RHD. To get a better and comprehensive clinical picture, we need to consider the more serious problems in narratives, discourse, and comprehension of abstract language.

Impaired Narration and Discourse

Literature on communication disorders in adults does not clearly distinguish everyday conversation, narration, and discourse. Conversation is often equated with discourse or is considered a part of discourse (Blake, 2018; Blake et al., 2013). Narration and discourse may not be separated. These are clearly different kinds of verbal behaviors, appropriately distinguished in the literature in child language (Nippold, 2016, 2021). Conversation, narration, and discourse are measured differently.

Conversation may include discourse and narration, but not all conversation is either narration or discourse. **Conversation** is verbal interaction between two or more persons in which all partners may spend roughly equal time talking and listening. **Narration** is storytelling or describing a personal experience for an extended period of time while others mostly listen. **Discourse** is an extended talk on an objective subject or topic. The listeners are more like the audience in a lecture hall. In assessing narration and discourse, the clinician may interact minimally with the client to prompt or indirectly direct the course of storytelling or a brief lecture. Similar kinds of analysis, however, may be performed on narratives and discourses. Together, they reveal problems that the typical conversation may not.

Narrative and discourse skills in persons with RHD are impaired to a greater extent than their syntactic and morphological skills. Such an impairment may be the most significant aspect of communicative deficits of persons with RHD. They have difficulty narrating personal experiences or stories, describing pictures, or giving a brief expository speech in a coherent, well-organized, and precise manner. Most of the research on the topic has not sampled real storytelling or discourse on a given topic. Instead, the studies have sampled responses to single or sequenced story pictures (Minga et al., 2021). Possibly, a more naturalistic storytelling or discourse may reveal an even greater degree and number of deficits than reported. Commonly reported difficulties include:

- **Problems distinguishing the significant from the trivial.** Picture narratives and discourses of persons with RHD may be filled with trivial, tangential, irrelevant, and inconsequential information. Unimportant aspects of story picture shown may draw excessive attention and elaboration.
- **Lack of cohesion.** Within and across sentences that narrate, describe, or discuss something, there may be no connection or smooth transition. Incoherent narratives and discourses combine with irrelevant and tangential information, making them difficult to comprehend.
- **Lack of coherence.** Topical relevance within a group of utterances or sentences suggests coherence. Coherence is lost if a group of spoken sentences (akin to a paragraph in a printed material) refers to different topics instead of the same. Persons with RHD tend to mix topics within groups of utterances that are expected to be topically unified.
- **Poor narrative structures.** A good storytelling or a discourse has a clearly identified beginning, middle, and end. When narrative events flow in a predictable direction and in a proper chronological sequence, listeners find it easier to understand. Persons with RHD may mix up the chronological sequences. Their narratives and discourses may not have clear beginnings and end points.

- **Limited meaningful information.** Some persons may give excessively restricted and limited description of pictures or events. Others may say more, but the information may not be meaningful or significant.
- **Inappropriate speech.** The words and phrases of narratives and discourses may be inappropriate to the topic, context, and the listener.
- **Premature and incorrect inferences.** While describing a story picture, the person with RHD may make wrong and premature inferences about the characters, motivations, or background information.

Prosodic Deficits

Prosody of speech refers to the stress patterns, intonation, rhythm, and melodious qualities of speech that convey meaning that is not often conveyed by the words and phrases; sometimes such features may convey the opposite of what the words generally mean. For example, “Sure, you are invited to my party” may be said with such prosodic features to mean please stay away. Speech with adequate prosodic features is variable in intensity, pitch, stress, vocal quality, duration of syllables, and so forth. In their meta-analysis of acoustic features of prosody in persons with RHD, Weed and Fusaroli (2020) found reduced fundamental frequency variation. With diminished prosodic features, the speech may sound monotonous and fail to provoke interest. Lacking in pitch and intensity variations, aprosodic speech fails to convey subtle, important, or abstract meanings. Connotations and emotional tone of experiences being described may be lost. Therefore, prosodic deficits and affective deficits are interrelated.

When neurologically unimpaired individuals listen to speech with emotional prosody, the RH shows greater activation than the LH, confirming the importance of the hemisphere in mediating prosody. The RH damage may impair both production of normal prosodic features and comprehension of meanings others express through prosodic variations (Hartikainen, 2020; Sheppard et al., 2020). Correlating speech prosodic problems to brain lesion

sites, Sheppard et al. (2021) reported that subcortical structures, particularly the caudate, are especially involved in processing prosodic information. Recent evidence also suggests that the inferior frontal regions in the RH mediate emotional prosodic *expression* and the right temporoparietal regions mediate emotional prosody *recognition* (see Sheppard et al., 2021, for a review of studies).

Although the speech of depressed people shows reduced prosodic features, prosodic deficits of persons with RHD are not due to depression. Similarly, although persons with dysarthria (motor speech disorder) may show prosodic deficits, such deficits in persons with RHD may not be due to motor disturbances. Finally, prosodic deficits of persons with RHD are not attributable to a language disorder as the persons generally have intact language skills.

Prosodic deficits are severe in a few cases, mild in most. Blake et al. (2002) observed prosodic deficits in only 19.5% of individuals with RHD. Such deficits are typically observed clinically, but acoustic analyses have not shown significant impairments (Weed & Fusaroli, 2020). Effects on communication may include:

- **Monotonous speech.** Lacking in prosodic variations, the speech may fail to indicate different kinds of sentences by intonational variations (e.g., persons may fail to show a rising intonation to suggest a yes/no question). A limited fundamental frequency of voice is a part of this monotonous speech.
- **Impaired stress patterns.** Instead of changing pitch to stress words, the persons may change amplitude (loudness).
- **Reduced speech rate.** A slow rate may also contribute to the monotonous quality of speech.
- **Flat affect.** The speech may fail to convey emotional tones and meanings.
- **Impaired comprehension of emotional prosody.** The persons may have difficulty understanding the emotional tone of speech they hear.

Persons with RHD may be aware of their prosodic deficits. Some persons, surmising that their emotional tone is not clear to the listener, may then resort to direct verbal expression of their emotion (e.g., a person might say, “If it is not clear to you, I am very angry”).

Difficulty With Abstract Language

While the LH seems to react more quickly in understanding direct and concrete meaning of words, the RH may react more slowly in understanding alternate, abstract, metaphoric, or multiple meanings of words (Blake et al., 2013; Daque et al., 2021; Gold et al., 2012; Rinaldi et al., 2004). For instance, normally, the LH may quickly grasp the concrete meaning of the phrase *open book*, but the RH may take a bit more time to grasp its metaphoric, abstract, or alternative meanings (e.g., something transparent).

In neurologically healthy individuals, both the hemispheres are activated during metaphor production. In individuals with RHD, however, metaphor interpretation (comprehension) is impaired especially when lesions are located in the temporoparietal areas; lesions elsewhere may not produce difficulty with metaphor comprehension, accounting for individual differences in this skill (Duque et al., 2021).

The persons with RHD tend to exhibit the following kinds of problems with abstract language:

- **Difficulty understanding abstract meanings.** Implied, suggested, or nonliteral meanings may be missed. Individuals may find it difficult to understand the meanings of metaphors, irony, humor, proverbs, idioms, and similes that others express in their speech.
- **Impaired understanding of the overall significance.** Persons with RHD may not grasp the broader and overall meaning of situations, events, stories, or story pictures. Missing the central message of a communication or the theme of a story picture shown, the persons may name individual elements of that picture with no understanding of the inherent theme or story, thus showing lack of integration of individual elements.

- **Difficulty with abstract category names.** The persons may name individual items in a category (e.g., carrots, beans, spinach) but may not name the category (vegetables).
- **Failure to detect logical errors in sentences.** Detection of logical errors also requires abstract reasoning, which may be difficult for persons with RHD.
- **Difficulty interpreting metaphors.** A metaphor equates two unrelated events, objects, or persons. *She is a shining star* and *he is a black sheep in the family* are metaphors. Some individuals with RHD may not suggest the meaning of metaphors, whereas others may suggest too many meanings. While a difficulty with metaphor comprehension is fairly well documented in persons with RHD, evidence on an apparently related skill, *idiom interpretation*, is contradictory and inconclusive. Idioms are culture- and language-specific statements that mean something other than the literal (e.g., *once in a blue moon*).
- **Difficulty understanding sarcasm and irony.** Limited evidence suggests that persons with RHD may have difficulty understanding sarcastic or ironic statements.

It is possible that persons with RHD may perform better in understanding multiple or alternative meanings in real-life contexts than in clinical assessment situations. Most studies have tested abstract language comprehension with contrived tests without social communication contexts. Larger and real contexts of communication may help them deduce such meanings.

Other Deficits

Persons with RHD may also exhibit a few other deficits that vary across individuals, as all other deficits do. Clinicians need to be open to the possibilities of following related deficits in individual cases.

- **Dysarthria.** If the upper motor neurons in the RH are affected, the person may have unilateral upper motor neuron dysarthria. This type

of dysarthria affects not only speech production problems but also speech prosody (Freed, 2020).

- **Impaired memory.** Persons with RHD are likely to exhibit memory problems, but data are limited. LH damage affects memory more than RHD (Blake, 2018). Nonetheless, memory should be minimally screened in persons with RHD.
- **Impaired planning, problem solving, and organizing.** Individuals with RHD may find it difficult to organize daily living activities and schedules, plan their activities ahead, and have difficulty with abstract reasoning and thinking. These deficits have not been adequately researched in the context of RHD.

In summary, attention, left-side stimulus and facial recognition, visuospatial perception, emotional expression and comprehension, narration and discourse, prosodic features of speech, and comprehension of abstract language are more prominently impaired than other speech-language features. Because of significant individual differences, clinician should make a comprehensive client-specific assessment. Clinicians may expect atypical symptoms.

CLASS NOTES

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7

ASSESSMENT AND TREATMENT OF RIGHT HEMISPHERE DISORDER

Chapter Outline

- Initial Screening
- Standardized and Nonstandardized Assessment Tools
- Treatment of Persons With Right Hemisphere Disorder
- Treatment Targets and Strategies
- Social Communication Skills
- Visual Neglect
- Impaired Attention
- Deficit Awareness
- Abstract Language
- Other Communication Disorders
- References

Learning Objectives

After reading the chapter, the reader will:

- Describe screening procedures for right hemisphere disorder (RHD)

- Describe the procedures for making a diagnostic assessment of individuals with RHD
- Describe and evaluate procedures for treating communication deficits in clients with RHD

PRINTED NOTES

Research on assessing right hemisphere disorder (RHD) is limited. There are no agreed-upon or research-based protocols or recommendations for assessing communication and related disorders of persons with RHD (Ramsey & Blake, 2020). Therefore, a practical and client-centered assessment outline is given in this chapter. The outline is based on typical professional practices on clinical appraisal of behaviors of interest. Until evidence-based assessment methods become available, the clinician will most likely design their own procedures and supplement them with available standardized tests.

The standard aspects of assessment including case history, hearing screening, interview of the person and the family members, orofacial examination, language sampling, and administration of standardized and client-specific procedures apply.

The case history should ascertain the circumstances of hospitalization. Often a stroke and, in some cases, traumatic brain injury or a brain tumor may bring the person to the hospital. Information on the premorbid intellectual and communicative skills of such an individual will be helpful in assessing the current deficits. In many cases, this information may have to be obtained in subsequent contacts with the family members.

Individuals with RHD are an unreliable source of their own problems because of the diminished deficit awareness described in the previous chapter. Nonetheless, asking the person to describe the problems being experienced will give a clue to that person's degree of awareness of deficits. The person may describe only a few or no problems being experienced. Interview of family members and other caregivers will provide additional information that may reveal problems the person is unaware of. This interview should include questions about the person's orientation, memory, and attention.

Family members, too, may be limited in giving valid information. Some of them may not appreciate the subtle communicative problems of the person

with RHD, whereas others may be keenly aware of them. Difficulty in facial recognition and lack of awareness of deficits may suggest to family members that the person is confused. Giving family members a brief background information on RHD will help them understand and better report on the person's behavioral problems. Multiple members of the client's family may be interviewed to get a more complete picture.

Initial Screening

Initial screening of major deficits associated with RHD will help plan a detailed assessment. The clinician may first see the client soon after hospitalization, although, at this stage, the client may not be ready for the demanding assessment ordeal. Therefore, the initial screening may be done at the bedside to get an idea of the major deficits that need to be assessed further at an appropriate time. Family members may have been interviewed prior to screening or may be interviewed soon thereafter.

The individual is interviewed during screening. The clinician may ask why the person is in the hospital and ask for a description of the events leading to hospitalization. Questions will be asked about his or her family, work, hobby, daily activities, current personal problems, and so forth. The clinician may note the person's inattention, poor orientation, inadequate comprehension of questions asked, rambling, and irrelevant speech.

An extended interview (conducted with breaks if necessary) will help gain an initial impression of the person's communicative and cognitive deficits that may be assessed as soon as practical. The clinician may find that the person fails to maintain eye contact, abruptly shifts from one topic of conversation to the other, uses gestures inappropriately, speaks in an emotionally flat tone, shows signs of unilateral neglect, and so forth.

As a part of screening, the clinician should sample the person's narrative skills. Story picture descriptions that are included in several tests of aphasia (e.g., the "Cookie Theft Picture" of the Boston Diagnostic Aphasia Examination-3; Goodglass et al., 2000) may be helpful in evoking descriptions that may reveal various kinds of discourse problems. The person's descriptions

may be incomplete, inaccurate, or both. The individual may fail to make correct inferences about the events that led to the current situation and the events that are likely to follow.

Screening the person for neglect is an important task. Neglect may be initially screened with a few simple tests. For instance, brief tests of cancellation may be useful in screening neglect. Most paper-and-pencil cancellation tests contain a page on which short lines, small squares, or a combination of two kinds of stimuli (e.g., flowers and snowflakes) are drawn. The person is asked to cross out or draw a line through all stimuli or a specific kind of stimuli (e.g., only the flowers that are mixed with snowflakes). On most such tests, the person may fail to cancel out the targets on their left side. Hegde and Freed (2022) provide a line and letter cancellation protocol that clinicians may use.

Other informal tests of neglect include copying line drawings, freehand drawing, and line bisection. When persons are asked to copy simple line drawings shown to them, they may fail to copy parts of the picture displayed to their left. They may make a similar mistake when asked to draw freehand. An individual's drawing of a flower or a man may have missing parts to the left. Line bisection involves making a mark in the middle of a printed line that is at least 20 cm long. Persons with RHD tend to make the mark more toward their right, thus missing the midline because of left neglect. Details of such tasks and descriptions of specific standardized tests of neglect are given in the next section. Clinicians may find such drawing and line bisection tests in Hegde and Freed (2022).

A few standardized screening tests are available, but most are not specifically standardized on persons with RHD. They are general cognitive tests that have been used with individuals with all kinds of central nervous system impairments. Therefore, they may miss critical features of RHD, including left neglect. Positive results on such tests are informative, but negative results may not rule out RHD. If the individual's history, overall clinical impression, and medical tests and reports suggest RHD, a detailed assessment is warranted. With these caveats, the clinician may administer one of the following to screen persons with RHD.

The Mini Inventory of Right Brain Injury-2 (Pimental & Knight, 2000) may be used to screen four general domains affected in RHD: (1) visuo-perceptual/visuospatial and attentional processes (visual scanning, integrity of gnosis, integrity of body image), (2) lexical knowledge processing (visuoverbal processing, visuosymbolic processing, integrity of visuomotor praxis, and higher-level language skills), (3) affective processing (emotional expression), and (4) general behavior. The test includes data on typical and right and left brain injuries.

The Burns Brief Inventory of Communication and Cognition (Burns, 1997) is designed as a screening test of RHD, aphasia, and TBI as well. The test helps evaluate scanning and tracking (e.g., single-word tracking), visuospatial skills (e.g., recognition of familiar faces, visuospatial construction, and spatial distribution of attention), expressive and receptive prosody, inferences, and metaphoric use of language. Performance of individuals with RHD and of those with other neuropathological conditions, including left hemisphere damage, may be compared.

Mini Mental State Examination, Second Edition (Folstein et al., 2010), is a screening test for attention, orientation, memory, visual perception, and language. The test has been widely used to screen persons with any kind of brain injury, including those with aphasia, traumatic brain injury, and RHD.

Cognitive-Linguistic Quick Test (Helm-Estabrooks, 2001) is a general screen for attention, orientation, memory, visuospatial skills, language, auditory comprehension, and executive skills.

Montreal Cognitive Assessment (Nasreddine, 2021) is a screening test for orientation, memory, attention, visuospatial perception, and language.

The Oxford Cognitive Screen (Demeyere et al., 2015) targets unilateral neglect, memory, language, calculation, and executive functions.

Standardized and Nonstandardized Assessment Tools

There are only a few standardized and nonstandardized tests to evaluate RHD. Assessment instruments may contain only a few or more items to evaluate communication (language) skills. Available standardized tests may be lacking in sufficient reliability, validity, and comprehensiveness (Blake, 2018).

The Right Hemisphere Language Battery, Second Edition (Bryan, 1995) helps assess comprehension of spoken and printed metaphors, comprehension of inferred meanings, appreciation of humor, discourse analysis (including turn taking, formality of language and behavior, greetings, complaining, demanding, and criticizing), matching spoken words with printed words (lexical semantic recognition), and emphatic stress production (prosody). The test includes a right-left differential scale.

RIC Evaluation of Communication Problems in Right Hemisphere Dysfunction-3 (Halper et al., 2010) is a comprehensive tool to assess the typical skills of interest (e.g., visual scanning and tracking, awareness of illness, attention, orientation, writing, and metaphoric language). Pragmatic language skills and nonverbal communication (e.g., intonation, eye contact, facial expression, topic maintenance, and narrative skills) also may be assessed. A protocol is provided for interviewing the client and recording observations of the client's interactions with family and caregivers.

Montreal Protocol for the Evaluation of Communication (Joanette et al., 2004) is a test for RHD, traumatic brain injury, and dementia. The test helps assess conversational speech, comprehension and expression of emotional prosody, deficit awareness, and interpretation of indirect questions. The test has French, German, Italian, Spanish, and Portuguese versions.

Ross Information Processing Assessment Geriatric, Second Edition (Ross-Swain, 1996) evaluates listening comprehension, memory for immediate and recent events, information recall, temporal and spatial orientation, categorical vocabulary, problem solving, and reasoning skills.

A few instruments target a single impairment associated with RHD. For instance, the **Bells Test** (Gauthier et al., 1989) helps assess neglect. The test contains printed pages with target and nontarget drawings (e.g., bells and other objects scattered across the page). The person who is asked to circle only the bells may neglect to cross out the bells on the left side. Another test,

Aprosodia Battery (Ross et al., 1997), targets prosodic deviations involving affect. The person is required to identify and discriminate between sentences spoken in varied affective tones. **Prosody-Voice Screening Profile** (Shriberg et al., 1990) helps screen such prosodic variables as phrasing, rate of speech, stress, pitch, vocal loudness, and vocal quality in spontaneous speech.

Several rating scales and tests target specific aspects of impairments associated with neurological impairments in general, not necessarily specific to RHD. For instance, The Rating Scale of Attentional Behavior (Ponsford & Kinsella, 1991) allows for observing and rating inattentive behaviors. Test of Everyday Attention (Robertson et al., 1994) helps assess various forms of attention in daily life, including sustained attention and attention switching. Behavioral Inattention Test (Wilson et al., 1987) and Catherine Bergego Scales (Azouvi et al., 2003) may be used to assess various forms of neglect. For a comprehensive list of various kinds of tests that may be used in assessing persons with RHD, see Blake (2018).

Treatment of Persons With Right Hemisphere Disorder

SLPs looking for effective treatment procedures for persons with RHD will be disappointed. “There are simply not enough well-designed studies to effectively inform clinical practice,” wrote Ramsey and Blake (2020, p. 743). They also listed only five or six treatment studies involving persons with RHD. Consequently, the advocated techniques are based largely on clinical experience, untested theories, expert opinion, or positive outcomes in treating aphasia or traumatic brain injury (Blake, 2018; Blake et al., 2013; Payne, 2014). Recommendations in the literature about treatment targets or strategies are strong, but the supportive experimental evidence is weak or nonexistent.

Ramsey and Blake’s (2020) survey of SLPs in the United States paints a disappointing picture of clinical practice with RHD. Most SLPs do not prioritize treatment for communication and prosodic impairments in persons with RHD. Instead, they mostly treat attention and other cognitive skills. However, only SLPs can treat communication and prosodic deficits, whereas

psychologists and occupational therapists treat cognitive and executive functions. This practice may be due partly to an overemphasis in speech-language pathology on *cognitive processes* that presumably underlie communication skills.

There is disagreement about treatment targets themselves. Some clinicians prefer to design treatments for specific skills (e.g., conversational skills or abstract language) while others prefer to address the cognitive processes (e.g., information processing in the brain) presumed to underlie observed behavioral deficits. Targeting underlying processes is at best indirect therapy. There is no convincing controlled experimental evidence to show that treating unobservable underlying processes is effective or that it results in generalized learning of related skills, especially social communication. Until evidence shows that process training is better than direct skill training, the latter is both clinically practical and theoretically parsimonious. Clinicians can better document positive changes or lack thereof during treatment only when they target measurable skills rather than assumed theoretical processes.

Treatment Targets and Strategies

Treatment targets need to be individualized because of two reasons. First, as noted, behavioral deficits differ across individuals. The same skill may be typical, deficit, or excess across individuals. For instance, on discourse tasks, some individuals say too much, and others say too little; still others perform as well as those without RHD. Second, rehabilitation needs of individuals differ, because of symptom variability and personal factors. Therefore, individuals and family members should help select treatment targets that are important to them. While most persons and family members would desire improvement in functional and social communication skills, not all may prioritize metaphor comprehension or deft line bisection. Deficits that affect everyday living the most in a given individual may be prioritized. For example, severe left neglect may adversely affect everyday living in one person, making this an urgent treatment target. For another individual, reducing excessive and incessant rambling may be a treatment priority.

Because SLPs cannot pick procedures of demonstrated effectiveness, they need to take steps to objectively document improvement in their clients. To do so, they may establish reliable baselines of treatment targets, measure the frequency of targeted skills in all treatment sessions, and modify procedures that do not produce positive changes. Clinicians may use behavioral treatment procedures that are otherwise known to be effective. They may model the target behaviors, withdraw modeling, prompt responses, shape complex skills in graded steps, offer positive reinforcement for correct responses, and give corrective feedback for incorrect responses. A few studies on prosodic deficit treatment in individuals with RHD have produced good evidence. These studies have shown that behavioral treatments involving modeling, imitation, and positive reinforcement are effective (Benedetti et al., 2021; Durfree et al., 2021; Hargrove, 2013; Rosenbek et al., 2004).

The range of treatment targets for persons with RHD include (a) social (pragmatic) communication, (b) prosodic variations including emotional tone, (c) visual neglect, (d) impaired attention, (e) impulsive behavior, (f) deficit awareness, and (g) abstract language. Only a few individuals may need treatment for all of these deficits. Therefore, the client-specific target selection is important.

Social Communication Skills

Treatment of social communication skills should be a priority for SLPs. Unfortunately, there are no controlled and replicated studies on intervention for social communication skills in persons with RHD. Clinicians will have to apply procedures that have been better researched in the rehabilitation of persons with traumatic brain injury or left hemisphere stroke (Blake, 2018; Blake et al., 2013; Dahlberg et al., 2007).

Reliable baselines of the individual's intact and impaired social communication skills will be useful (a) in documenting improvement under treatment and (b) to select deficits for modification. Brookshire (2015) recommends a videotaped baseline measure of a person's conversational skills before treatment is started. Conversation between the person and the clinician

or between the person and another person (e.g., a family member) may be video recorded. The latter videos may be played during treatment sessions to draw the client's attention to desirable conversational skills the family member exhibits. These skills may serve as models during treatment.

Several pragmatic (social) communication skills that have been the treatment target for persons with TBI (Dahlberg et al., 2007) may also be appropriate for a given person with RHD. The clinician may select those that are of intervention priority. Among the skills, topic initiation, topic maintenance, conversational turn taking, typical prosodic variations, eye contact, more precise and less rambling speech, appropriate amount of speech, and speech relevancy may be intervention targets. The following treatment descriptions are based on expert opinions and procedures that are expected to be effective. Research, when available, is cited.

Topic maintenance. To teach this skill, the clinician may start with simple topics and move to more complex ones. Viewing the previously recorded videos may help increase the person's awareness of problems in maintaining a topic of discourse. Initially, the clinician may set a brief duration during which the person is asked to maintain discourse on a simple topic. As the person becomes more proficient in maintaining simpler topics of discourse for specified periods of time, both the duration and the complexity of the topics may be increased. At each stage of treatment, the clinician should reinforce the person for staying on the topic of discourse. The clinician should stop the client at the earliest sign of departure from the topic. The clinician may then give such verbal prompts as "You were saying that . . ." to help the person get back on the topic of discourse. Density of reinforcement and verbal prompts should be reduced to encourage topic maintenance in the absence of such explicit behavioral contingency.

Turn taking. Reviewing the video recorded interactions might help identify the person's deficiencies as well as appropriate turn-taking behaviors during conversations. To increase turn taking, the clinician and the client may role-play. Selecting simple discourse topics, the clinician and the client may engage in a structured conversation. Each partner may say only a few words and yield the floor to the other. Such verbal instructions as "it is my turn" or "it is your turn" may help regulate smooth turn taking in conversation.

Prosodic variations. Specific factors that may be clinically modified to improve speech prosody include pitch, loudness, stress, syllable duration, and rate of speech. Modeling, imitation, and positive reinforcement for correct identification and production of prosodic features have been documented to be effective in both individual studies and research reviews (Benedetti et al., 2021; Blake et al., 2013; Durfee et al., 2021; Hargrove, 2013; Rosenbek et al., 2004, 2006; Rodriguez, 2009). Prosodic recognition and improved prosodic production have been successfully taught to individuals with RHD.

Rosenbek and colleagues (2004, 2006) have investigated a method to improve production of affective prosody. Using single-subject experimental designs, they have demonstrated that their method is effective in promoting sentence productions with proper prosodic variations that may be described as *happy*, *sad*, *angry*, or *neutral*. The training effect generalized to an untrained emotion (*fear*). The method has multiple steps, all clearly described. Clinicians may read the original reports for details (also given in Hargrove, 2013). Essentially, the clinician models a sentence with the prosodic feature to suggest a target emotion, shows appropriate facial expressions, and asks the client to imitate it on consecutive trials. In the next stage, the clinician covers the face to fade the influence of facial expression. Next, the clinician produces an affectively neutral sentence and asks the person to produce the same with a specific affect (happy, sad, or angry). Finally, the clinician asks the person to produce sentences with affective prosody that suggests different and untrained emotions. This is a practical procedure that clinicians might use.

Differential *recognition* of affective prosody in individuals with RHD has not been adequately studied. In one of the few published studies, Durfee et al. (2021) trained individuals to recognize *happy*, *sad*, *angry*, and *fearful* emotions in spoken sentences. They also reinforced correct identification of emotions and gave corrective feedback for incorrect response. Training was limited to a single session with improved results, but most participants were still aprosodic.

Production training for prosodic deficits that Rosenbek and colleagues (2004, 2006) have demonstrated may be more parsimonious than *receptive (recognition) training*. It is likely that persons who learn to produce affective prosodic variations in their own speech would begin to recognize them in people who speak to them. Controlled data, however, are needed to test this

possibility. After training productive affective prosody, the clinician may probe receptive skills. Treatment for receptive prosody may be offered if production training has not generalized to receptive prosodic skills.

Narratives and discourse. Research on impairments on storytelling (narratives) and discourse (a brief speech) on an objective topic has produced conflicting results. A few studies have reported problems; others have not (see Blake, 2018, for a review of studies). There are no replicated experimental data on treating narrative and discourse deficits in persons with RHD; treatment offered to others with neurological involvements (such as those with TBI) are recommended (Blake, 2018). Therefore, if a person with RHD needs intervention for narrative and discourse skills, the clinician may use the procedures described in [Chapter 5](#) on aphasia treatment and [Chapter 9](#) on TBI treatment.

Eye contact. In some clients, eye contact during conversation may need only frequent verbal reminders (e.g., “look at me”). Other clients may need a shaping procedure in which the duration of eye contact is progressively increased. Initially, the duration might be brief; in gradual steps, the duration may be increased. Verbal praise for maintaining eye contact may be provided. The reinforcement should initially be continuous and intermittent later on.

Training family members and other caregivers. To sustain the clinically reestablished conversational skills, those who frequently interact with the person should be trained in maintenance strategies. Professional caregivers may be offered suggestions on stopping the client for inappropriate speech and reinforcing appropriate behaviors. Family members may be asked to initially observe treatment sessions and they may be subsequently trained to provide corrective and positive feedback to the person.

Visual Neglect

Occupational therapists are likely to treat visual neglect through daily activities. Persons with moderate to severe left neglect need intervention to accomplish such tasks as dressing, grooming, eating, and walking without the negative consequences of their left neglect. Speech-language pathologists treat

this problem as well because left neglect negatively affects reading and comprehension of read material. Left spatial neglect may also reduce communicative effectiveness in group interactions because the person fails to attend to people talking on the left side. Therefore, left neglect is a communication treatment target.

Most of the available treatment procedures are designed to reduce neglect in reading. As with the treatment procedures designed to improve attentional skills, some of the techniques include such indirect skills as tracking moving objects, responding to lights that flash across a computer monitor, letter or other kinds of stimulus cancellation, and so forth. Once again, in the absence of controlled evidence that indirect (and nonfunctional) target training improves reading and reading comprehension, it may be best to spend treatment time and energy on treating visual neglect in the context of reading tasks themselves.

Forcing a person's attention to the left side of the visual field and then positively reinforcing that behavior are the two essential elements of treating neglect. In reducing reading errors due to left neglect, the clinician may use:

- **Discriminative stimuli on the left margin.** To draw attention to the beginning of each printed line of text, the clinician may draw thick and brightly colored vertical lines on the left margin. Colored dots may be placed on the left margin, at the beginning of each sentence. The person may be reinforced for not missing words on the left side of the page.
- **Verbal prompts.** Frequently given verbal prompts to “look to the left” and “go to the beginning of the line” as the person reads aloud may be effective in themselves or when combined with other discriminative stimuli (e.g., colored vertical lines or dots in the left margin). Verbal prompts may also draw attention to the specific stimuli in the margin. For instance, the person may be asked to “notice the line (or the dot) to your left.”
- **Finger on the left margin.** The person may be asked to keep a finger on the left margin as a reminder to go to the beginning of each line.

This strategy, too, may be combined with verbal prompts to go back to the beginning of the line.

- **Pointing of the beginning of each line.** With a pencil, the clinician may point to the beginning of each line as the person comes to the right end of each line.
- **Positive reinforcement for left attention.** In each procedure, the clinician should positively reinforce either requested, special stimulus-driven (e.g., colored margins) or spontaneously exhibited left attention.
- **Corrective feedback for left neglect.** Persons should be stopped every time they miss words on the left side. Prompts to look to the left side should then be repeated.
- **Fading the special discriminative stimuli.** All discriminative stimuli (colored vertical lines, dots, finger on the left margin, verbal prompts) should be faded to promote maintenance of left attention in natural settings.
- **Left neglect in group interactions.** In group interactions, the person may neglect conversational partners sitting on the left side. To reduce or eliminate this problem, the clinician may arrange group discussion sessions (preferably involving the family members) in which the person is prompted to pay attention to the conversational partners on the left side. The partners also may be taught to draw attention to themselves before speaking. For instance, the partners who are in the person's left visual field may say, "Please look at me, here. I want to tell you something . . . " to draw the person's attention to themselves. Both the clinician and the conversational partners will verbally reinforce the person for paying attention to those on the left side.

Research on the efficacy of procedures described in reducing visual neglect is extremely limited. Clinicians should watch for treatment efficacy studies and use techniques that gain experimental support.

Impaired Attention

Impaired attention is lack of responses to relevant stimuli and responses to multiple irrelevant stimuli. Such response patterns are an impediment to effective communication. Attention may be treated as an exclusive (independent) target or may be integrated with other verbal or nonverbal skills.

When attention is an exclusive treatment target, clinicians tend to use paper-and-pencil tasks (e.g., cancellation of a target letter printed among nontarget letters). Computerized tasks vary. The person may be asked to press a key when dots appear or disappear against a background of dots. Game-like computer programs in which the client is asked to drive a car while maintaining the speed or avoiding crashes are other kinds of examples. On repeated trials, the clinician may give corrective feedback for missing specified targets and positive reinforcement for canceling the targets or reacting in some way to the specified stimuli (such as pressing a key on a computer keyboard).

The results of research on attention training are both contradictory and disappointing (see Peach et al., 2019, for a review of studies, most of them on aphasia). Research on attention in persons with RHD is not particularly informative to the SLPs. Persons who efficiently detect disappearing dots on a computer screen may still be inattentive during social interactions. Even if it is consistently shown that attention training improves attention (e.g., maintaining speed of the car on the computer screen), there is no compelling reason for the SLPs to spend their treatment time on such attention training when the same time could be more efficiently used to train communication skills. In their controlled treatment research study, Peach et al. (2019) demonstrated that language-related attention training is more effective than direct attention training that does not involve language. Therefore, it may be more efficient to avoid teaching nonfunctional attending actions and instead reinforce attending behaviors in social interactions and other realistic environmental contexts.

Clinicians who treat attention in the context of communication frequently draw attention to treatment stimuli, give specific directions to follow, repeat such directions throughout treatment, and reinforce attention during discourse

training. Clinicians may make attention a part of communication treatment. Some examples follow.

- **Attention to topic maintenance in conversation.** When the client begins to wander away from the selected topic of conversation, the clinician may say, “Please stop.” The person then may be prompted to get back on track (e.g., “you were talking about your camping trip”). The person may be reinforced for picking up the interrupted topic and for talking on the same topic for progressively longer durations. The reinforcement may be continuous in the beginning and intermittent as the person makes progress in sustained attention. To promote maintenance in natural settings, interactions between family members and the person may be targeted for treatment. Initially, the family members may be asked to watch the clinician’s treatment procedures. Subsequently, the family members may be trained to give corrective and positive feedback for inattentive and attentive behaviors, respectively.
- **Attention as a part of eye contact.** To encourage eye contact, the clinician might use the same strategy used in strengthening topic maintenance skills. The clinician may reinforce the person for maintaining eye contact and stop inattentive behaviors that may result in loss of eye contact.
- **Attention to give appropriate responses to questions.** To decrease inappropriate responses to questions asked of the person (possibly because of lack of attention), the clinician may first draw the attention of the person before asking a question. For example, before asking a question, clinician may say, “I am about to ask you a question. Please listen carefully.”
- **Attention to treatment stimuli.** To decrease inappropriate responses to treatment stimuli, the clinician might say, “I am about to show you a picture and ask some questions. Please look at the picture carefully before you say something.”

In each of the tasks described, the clinician may reinforce progressively longer durations of sustained attention. By systematically training the family

members and professional caregivers, the clinician may help promote better attentional skills in everyday situations. Furthermore, as the person becomes more proficient in paying attention and staying on target skills, progressively more complex tasks may be introduced. For instance, the client may be reinforced for appropriately shifting attention from one topic to another. Or, the client may be reinforced for paying sustained attention to the same topic when such distractions as extraneous comments, background conversations, and noise are present.

Impaired Inference

An inference is a conclusion drawn from the given verbal and nonverbal contexts. The context or contexts serve as a cue to a generalized response, based on past experiences. Some individuals draw reasonable conclusions based on limited contextual cues, whereas others need more and stronger cues. Based on an available physical stimulus array (such as a story picture) or a verbally narrated scenario, typical individuals may draw conclusions as to what has happened (antecedent inference) and what will happen next (predictive or consequence inference). Thoughts and feelings of characters in story pictures or verbal stories may also be inferred.

Impaired inference has been considered a part of RHD, but it is not a common feature of it. Only some individuals with RHD may have difficulty making inferences, especially when the supplied contextual cues and the reasonable inferences that could be made are both multiple (Blake, 2018; Blake et al., 2013). If the assessment of a given individual shows inference problems, then the treatment may include this target.

Replicated experimental evidence for inference deficit treatment is lacking. Clinicians may design their own treatment with story pictures and verbal stories or descriptions of scenes. The following examples suggest strategies that may be used and modified to suit individual clients:

- **Few cues, simple, single inference.** The picture shows a man packing a duffel bag. Next to the bag stands a pair of hiking boots.

The clinician may show the picture and ask such questions as “Where do you think this man is going?” “*He is going on a hiking trip*” would suggest a correct inference. This is a simple context with a good probability of evoking a single predictive inference. Correct inference may be verbally reinforced. If the inference is incorrect or delayed, the clinician may draw attention to the act of packing and the sight of the hiking boots. If there is no response, the clinician may model the correct inference, reinforce the imitated response, and withdraw modeling on subsequent trials. Training on multiple exemplars with different picture scenarios may be trained for generalization to occur.

- **Multiple cues, multiple inferences.** The picture shows a man packing his duffle bag with a pair of hiking boots on the side. A woman is standing next to a fancy suitcase and folding a nice dress, apparently to put it in the suitcase. Next to the suitcase is a pair of high-heeled shoes. The clinician may ask, “Where are they going?” The man and the woman may be going to different places, man to hike, the woman to a wedding, a party, or a business conference. Cues are multiple, and possible predictive inferences are several. The clinician may prompt, model, and reinforce multiple inferences.
- **Brief stories.** The clinician may tell brief stories and then ask questions to evoke implied information. Simple inferences may be provoked by such brief stories as the following:

While attending a business conference away from home, a man went to a restaurant to eat dinner. While eating, he noticed that several members of a family were seated around the next table. Soon a bunch of waiters and waitresses gathered around the table and began to sing “Happy Birthday.” The man then suddenly realized that it was a special day for someone at his home and that he had forgotten to call that important person in his life.

The clinician can then ask the person questions such as “What was the special day?” or “Whom do you think he was going to call?” Necessary prompts may help make the correct inference (*His wife’s*

birthday, he was going to call his wife). The clinician may use the same treatment procedures as suggested in the other two examples.

Deficit Awareness

Being unaware of illness and indifferent to one's own deficits may be difficult to treat, partly because persons who are not aware of their disabilities may not accept treatment, work hard in the sessions, complete treatment assignments, and may not take suggestions or help from family members. Those who accept treatment without much complaint may do so only halfheartedly and offer confabulated reasons for their failure to cooperate fully (Brookshire, 2015).

From the research literature, it is not clear how much direct work is needed on deficit awareness training. There are no specific treatment studies on it. Possibly, there is no need to spend much time on increasing deficit awareness. When specific communication target behaviors are trained, clients are likely to gain awareness of their impaired behaviors. For instance, when eye contact is the target, the clinician draws attention to lack of eye contact, talks about its importance in social communication, and sets a target duration of eye contact and reinforces it. Significant improvement in eye contact is likely to increase awareness. Efforts at directly increasing a person's awareness of personal deficits by repeatedly telling them about their limitations are a counterproductive negative approach, especially in the absence of positive steps to reduce the deficits. A potential negative effect of successful therapy that makes clients keenly aware of their deficits is discouragement or even depression. Instead of trying to increase the awareness of deficits, one might target reduction in deficits themselves. Awareness should (and most likely will) increase as the deficits decrease.

Behavioral awareness is typically a function of response-contingent feedback. When the SLP models and reinforces specific skill (e.g., affective prosody or conversational turn taking) and gives corrective feedback for incorrect responses, it is nearly impossible to not to become aware of one's deficits. Also, a videotaped baseline session that the client can watch with the clinician to jointly identify the intervention targets may also promote positive

awareness of what is lacking and increase motivation for treatment. If necessary, any time a new target is selected during treatment, the clinician may give a brief overview of the deficit in an objective tone and discuss the need for intervention with the client.

Encouraging self-evaluation of one's own performance under treatment and giving an opportunity to self-correct before giving corrective feedback is another strategy that might enhance positive awareness of deficits. Clients may count or chart their errors. The clinician may remind missed errors. During a brief pause, the clinician may ask for self-evaluation (e.g., "How have you done in the session so far?" "What is your opinion about progress today?"). The clinician may share her own evaluation with the client. If self-evaluation skills are strengthened during treatment, *self-monitoring* outside the clinic may also improve, leading to better generalization and maintenance.

Abstract Language

Intervention for abstract (nonliteral) language skills described in this section targets comprehension and discrimination of logical and absurd statements or situations and comprehension of metaphors. If needed, comprehension of such other nonliteral language skills as idioms and proverbs may be improved with strategies similar to what are suggested.

Impaired recognition of absurdities. Persons who have difficulty recognizing absurdities will misinterpret situations, leading to ineffective communication. Therefore, procedures designed to help persons recognize absurdities may be helpful. To promote this recognition, the clinician may use a few strategies.

Discriminating logical and absurd statements. The clinician may describe situations that permit both absurd and logical statements and ask the person to pick the absurd ones. The clinician may use scenarios like the following:

On my way home from work yesterday, (a) I felt so hungry that I stopped by a movie theater and watched a movie; (b) on my way

home from work yesterday, I felt so hungry that I stopped at a restaurant and ate dinner before reaching home.

Explanation of absurd statements. The clinician may present only absurd statements and ask the person to explain recognized absurdities. A statement such as (a) in the previous example may be presented with a request to explain its absurdity.

Pictorial absurdities. The clinician may present pictures that represent absurd situations (e.g., a bird chasing a dog) along with those that represent logical situations (a dog chasing a man). The clinician then may ask the person to alternately point to the absurd ones and logical ones.

Explanation of absurd pictures. The clinician may present only those pictures that represent absurd situations and ask the person to state the absurdities and explain them.

Impaired comprehension of metaphors and proverbs. A metaphor is an expression that suggests an implicit and nonliteral equation of two objects, events, or persons. Such sayings as *he is an early bird* or *she is a tower of strength* suggest nonliteral meanings. Proverbs are popular sayings that capture some truth or wisdom. They also suggest implied and nonliteral meanings. Such sayings as *a stitch in time saves nine* and *slow and steady wins the race* are proverbs.

Persons who cannot understand metaphors and proverbs miss the implied meanings because they tend to interpret such statements literally; resulting verbal responses may be inappropriate. To reduce this problem, the clinician may select metaphors and proverbs that are appropriate for the person's educational level and ethnocultural background. The clinician may state metaphors and proverbs and ask the client to select printed or verbal statements that give literal (incorrect) or metaphoric (correct) meanings. Initially, more common metaphors and subsequently more complex ones may be targeted. Modeling and prompting correct responses and offering positive reinforcement for correct interpretations may be needed at all stages of treatment.

As noted earlier, much treatment research is needed to establish the efficacy of suggested procedures. Clinicians should watch for treatment research that

supports or refutes procedures described in the literature as well as in textbooks such as this. Besides, the clinicians should collect systematic baseline and treatment data to document improvement and to modify treatment when data warrant it.

Other Communication Disorders

A few other communication problems may need to be modified in some individuals with RHD. Excessive verbal output (rambling); excessively limited speech; tangential, irrelevant, and inappropriate comments; and impulsive behavior may need intervention. These kinds of problems vary across individuals; some may not need intervention, others may, and the need may be minimal in most cases.

As it is typical with RHD, there are no controlled studies on diminishing those problems in persons with RHD. The clinician may use simple, straightforward behavioral techniques in treatment sessions that target other skills, such as social communication, described earlier. The clinician may promptly terminate attention when the person begins to ramble, becomes tangential, or makes inappropriate comments. The clinician also may use the time-out procedure as described in [Chapter 9](#) on assessment and treatment of TBI. Each time the clinician asks a question or makes a request (e.g., “What do you see in this picture?” Or “Tell me about your favorite camping trip”), the clinician may quickly add, “Please try to be as brief and specific as possible.” The clinician may give a hand signal to stop talking when the client begins to ramble, inappropriate, or tangential. Systematic application of such contingencies might be effective.

Hasty responses to treatment stimuli increase the chances of errors when asked to describe something, narrate an experience or story, or answer questions. To reduce such errors, the clinician may give a verbal prompt to slow the reaction time. For instance, after asking a question or making a request, the clinician might say, “Please think for 10 seconds before you answer me. You can begin speaking when I say *now*.” Any duration judged appropriate may be used, and if the duration initially used is longer, it may be reduced

gradually to what appears typical to the clinician. To reduce hasty interruptions, the clinician might use such verbal prompts as “Please wait until I am finished.” In a group situation, the clinician might ask the interrupting person to “Please listen to the other person” and repeat them. Quiet listening may be positively reinforced.

Excessively limited speech is more likely to be revealed in narration and discourse. Such verbal prompts as *say more*, *give me details*, *what about this*, *what about that*, *what happened then*, *what did she do*, *why did he do that*, and so forth might help evoke more elaborate speech. Targeting more specific structures (e.g., *tell me in a longer sentence* and *this time, please use more words*) may also be tried. Any improvement should be promptly reinforced with verbal praise. An aphasia treatment program called the Response Elaboration Training (Kearns, 1985) and its modified version (Wambaugh et al., 2013), known to be effective, may be tried with persons with RHD to increase the amount of their speech. See [Chapter 5](#) for details on this procedure.

In summary, it should be clear that SLPs need controlled experimental treatment research that offers more effective procedures than those described in this chapter. Clinicians should be on constant watch for new treatment research studies on RHD to use methods that controlled treatment evaluations support.

CLASS NOTES

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PART IV

TRAUMATIC BRAIN INJURY



8

TRAUMATIC BRAIN INJURY: CAUSES AND CONSEQUENCES

Chapter Outline

- Incidence and Prevalence of TBI
- Common Causes of TBI
- Types of Brain Injuries
- Primary Effects of TBI
- Secondary Effects of TBI
- Variables Related to Recovery
- Neurobehavioral Effects of TBI
- Overview of Communication Disorders in TBI
- Impaired Memory, Thinking, Reasoning, and Planning Skills
- References

Learning Objectives

After reading the chapter, the reader will:

- Define TBI and describe its prevalence and incidence
- Specify the causes of TBI and the kinds of damage or injury they produce

- Distinguish the causes and consequences of penetrating and nonpenetrating TBI
- Describe factors that affect recovery from TBI
- Distinguish and describe the various neurobehavioral and communicative deficits associated with TBI

PRINTED NOTES

Traumatic brain injury (TBI) is damage the brain sustains due to physical trauma or external force. It is a type of acquired brain injury, but not all acquired forms are TBI. **Acquired brain injury (ABI)** includes nontraumatic brain damage due to strokes, tumors, infection, progressive neurological diseases, metabolic disturbances, toxic agents, and inherited or congenital conditions (Centers for Disease Control and Prevention, 2018–2019; Hegde & Freed, 2022; Najem et al., 2018; Pavlovic et al., 2019). Neurological impairments due to these conditions are not forms of TBI. Also, some neurological diseases (Alzheimer's disease) are progressive (degenerative), whereas the effects of TBI are not. Many individuals with TBI improve over time, especially with rehabilitation that includes communication training or retraining.

Head trauma is injury to the head, caused by damaging external force applied to the head. In head trauma, the facial structures may be injured, but the brain may or may not be. Therefore, head trauma cannot always be equated with TBI. Only when the brain is injured does a long-term concern of multidisciplinary rehabilitation arise. Therefore, this chapter is about persons who have sustained brain injury.

Concussion, more commonly observed among athletes who indulge in certain kinds of sports, is also brain injury but a milder form of it (Jackson & Starling, 2019). Therefore, it is called **mild traumatic brain injury (mTBI)**. Some experts do not make that distinction because of the serious long-term effects of mTBI (e.g., increased risk of dementia in later years). Concussion or any form of mTBI produces transitory immediate effects that include loss of consciousness that does not exceed 30 minutes and posttraumatic amnesia that does not last more than 24 hours.

Incidence and Prevalence of TBI

Incidence figures of TBI vary by wide margins, mostly due to methodological problems of epidemiological studies. Definitions of TBI and its grading for severity vary across investigations. Some investigators include individuals with any kind of head injury (even without brain injury), whereas other investigators include only those who have sustained brain injury. Some investigators may exclude cases of mild injury or those who did not survive their injuries, whereas others may include all such cases. Mild concussion sustained on sports fields may or may not be reported. The Centers for Disease Control and Prevention (CDC) (<http://www.cdc.gov/traumaticbraininjury/index.html>) does not count head injuries treated in primary care, emergency rooms, and urgent care centers; some of them may have had mild forms of TBI. And then there also are individuals who do not receive any medical care for their TBI and therefore do not get counted.

Several sources suggest an annual incidence of about 131 to 250 TBI cases per 100,000 persons in the U.S. general population (Burns & Houser, 2003). Incidence is generally higher for mild TBI than it is for more severe TBI. According to the CDC data (<http://www.cdc.gov/traumaticbraininjury/index.html> and <https://www.cdc.gov/traumaticbraininjury/data/>), there were 223,135 TBI-related hospitalizations in 2019 and 64,362 deaths. Each day in 2020, 176 Americans died of TBI. Mortality rates are higher for the health uninsured than for those with adequate insurance. In 2019, 15% high school students self-reported sport- or recreation-related concussion. Annually, about 1.7 million persons sustain TBI in the United States. Eighty percent of TBI cases receive emergency room treatment. More than 70,000 people with moderate to severe injuries may have some degree of permanent disability; from 2.5 million to 6.5 million Americans live with permanent disability.

Prevalence of TBI varies depending on the age. The highest prevalence rate is found in the age group 15 to 24 years, especially the age group 15 to 19 years (400 to 700 per 100,000, depending on the study). Children in the age range of 0 to 4 years also are highly susceptible to TBI. The prevalence picks up

again for those who are 65 years and older (300 per 100,000). TBI poses a high risk of additional injuries.

TBI is three to five times more common in males than in females. Males are also more likely than females to die from TBI. People living in high-density urban areas and of low socioeconomic and ethnic minority status may have a higher incidence of TBI, which is partly due to neighborhood gun violence (Evans, 2006; Leonard et al., 2015). Ethnic minorities have a higher death rate, mostly due to lack of health insurance and, consequently, absent or lower-quality health care.

Prevalence of concussion on school and professional playgrounds is typically underreported and, therefore, the actual number may be alarmingly high (Institute of Medicine of the National Academies, 2013). Annually, as many as 3.8 million children and adults may sustain concussion on playgrounds (National Institute of Neurological Disorders and Stroke, n.d.; Pierpoint & Collins, 2021). Sports-related concussion has received much media attention in recent years, and the reporting of such events may have increased as a result. Concussion (mTBI) is a common occurrence in battlefields. It comprises up to 85% of all brain injuries sustained by U.S. armed forces.

Common Causes of TBI

Most reports on TBI do not adequately document the cause or the causes; therefore, frequently reported figures are estimates. Although the figures across sources and years vary, there is general agreement that certain factors are consistently associated with TBI (https://www.cdc.gov/TraumaticBraininjury/get_the_facts.html; [Traumatic Brain Injury.com](https://www.traumaticbraininjury.com)).

Falls are a major cause of TBI in the general population, accounting for about 35% of all such injuries. Among persons aged 75 years and older, falls are the leading cause of TBI and reasons for a majority of emergency room treatments.

Automobile accidents account for 17.3% of all cases of TBI, almost a third of which is due to drunk driving. About 70% of injuries sustained in all automobile accidents are TBIs. Wearing a seatbelt may reduce severe or fatal injuries due to automobile accidents by 22% to 54%. Most drunk drivers who have accidents will not have worn seatbelts.

Pedestrians risk brain injury. Up to 15% of pedestrians in the United States may sustain TBIs. As pedestrians, school-age children are especially likely to sustain TBI.

Motorcycle and **bicycle riding** increase the chance of brain injury. The death rate among motorcycle riders due to accidents is about 15% higher than the rate for occupants of passenger cars. Head injury accounts for 50% of all those who die of motorcycle accidents. In U.S. states where helmet laws exist, that death rate is 30% lower than in states without such laws. About 170,000 people who ride bicycles may be treated annually in emergency rooms for their bicycle-related injuries. More than 35,000 sustain some form of head or brain injury and a quarter of them need hospitalization.

Head striking objects or **being struck** is the third leading cause of TBI. The head may strike a stationary object or may be struck by a moving object. These events are involved in about 16% of TBI cases.

Interpersonal violence accounts for about 10% of TBIs sustained in rural areas and up to 40% in urban areas. In large inner cities with high density of population, TBI related to assault and firearms may exceed that of TBI due to transportation accidents (Kraus & McArthur, 2000).

Intimate partner violence is a significant cause of TBI among women. In addition to a blow to the head and face, attempted strangulation also causes brain injury due to oxygen deprivation. Typically, one in three women is affected by intimate partner violence, but a Canadian study reported a 30% increase during the COVID-19 pandemic (Toccalino et al., 2022). Nonetheless, fewer women were served in victim services clinics and organizations, likely due to their confinement and isolation.

Sports-related concussion is a leading cause of TBI among students, as well as professional athletes (Stenson & Stradling, 2017; Theodom et al., 2020). Annually, 1.6 million to 3.8 million individuals sustain sports-related

TBI. On high school and college playgrounds, football, ice hockey, lacrosse, wrestling, and soccer are associated with greater risk of mTBI among males. Soccer, lacrosse, and basketball in both high school and college settings, plus ice hockey in the college setting, are associated with a greater risk of mTBI among female players.

Blast trauma on battlegrounds and dangerous city streets is another significant cause of TBI. Mild and nonpenetrating injuries due to blasts cause up to 85% of all TBIs among combat veterans (Defense and Veterans Brain Injury Center, n.d.). More frequent use of explosive devices in combat situations in Afghanistan, Iraq, and other countries has caused a significant number of TBIs associated with explosions.

Alcohol abuse is another contributor to TBI. At the time they receive treatment for their TBI, 30% to 50% may be intoxicated. The relation between alcohol abuse and motor vehicle accidents was noted earlier. Alcohol abuse and TBI are interactive; abuse increases chances of TBI, and TBI increases the possibility of alcoholism (Weil et al., 2018).

The monthly trend in the incidence of TBI varies, with the highest being recorded during the months from May through October. Outdoor activities that predispose people to injuries probably account for the increase in the frequency of TBI during these months.

Types of Brain Injuries

Brain injuries may be classified on the basis of severity (e.g., mild, moderate, and severe), general etiology (e.g., penetrating vs. nonpenetrating), type of injury (e.g., diffuse or focal), pathophysiology (primary vs. secondary injuries), and so forth. No single classification captures all the dimensions. Each classification will have to include information from the others. A common classification of brain injuries includes *penetrating injuries* (also called *open head injuries*) and *nonpenetrating injuries* (also called *closed head injuries*) (de Lanerolle et al., 2015; Najem et al., 2018; Vakil & Singh, 2017). This classification may include relevant information from other classifications. Nonpenetrating brain injuries cause the most long-lasting effects that require

extensive rehabilitation efforts. Therefore, we will briefly review penetrating brain injuries and consider nonpenetrating injuries in greater detail.

Penetrating Brain Injuries

Penetrating brain injuries involve a crushing or penetrating agent that causes fractured or perforated skull, torn brain coverings (meninges), and brain tissue damage. Either the penetrating object or the fragments of fractured skull due to the crushing force may damage the brain tissue. Penetrating objects such as bullets, nail guns, lawn darts, knives, crossbows, and so forth are described as *missiles* because they travel through air and strike the head.

The basic cause of penetrating brain injuries is the piercing of the skull by an external object. The piercing object may also pass through the head. Depending on the speed with which a penetrating object strikes the head, the injury may be of *high velocity* (impact) or *low velocity* (impact).

Generally, such agents as an arrow, nail gun, knife, and handgun; blows to the head; and automobile accidents produce **low-velocity injuries**. In the United States, bullets shot from handguns cause most of the low-velocity penetrating brain injuries. Penetrating gunshot wounds tend to cause the highest mortality rate. In the United States, gunshot wounds to the head kill 32,000 to 35,000 civilians annually (Vakil & Singh, 2017). Military weapons, rifles, and other automatic assault weapons produce **high-velocity injuries**.

Effects of Penetrating Brain Injuries

The effects of penetrating brain injuries depend on several factors (Currie et al., 2015; de Lanerolle et al., 2015; Vakil & Singh, 2017). The following are the major determining factors:

- **The projectile's entrance velocity.** The higher the speed with which the object penetrates the skull, the greater is the extent of injury.
- **The size of the missile.** The larger the penetrating object, the greater is the injury to the brain.

- **The degree of the projectile's yaw.** Yaw is the tendency of moving objects to change their course. An object that enters the skull and moves in a straight line produces less damage than the one that follows a zigzag path within the brain.
- **Missile fragmentation.** The bullet or any other object that shatters within the brain causes greater damage than the one that stays intact as it moves within the brain.
- **The number of wounds.** Objects that cause multiple wounds cause more damage than those that cause fewer wounds.

Of these variables, the velocity at entrance is the most critical factor that determines the extent of penetrating brain injuries. Among the different kinds of bullets, those that are steel-jacketed (military issue) tend not to fragment in the brain. Hollow-point ammunitions tend to fragment and cause extensive damage (Currie et al., 2015; de Lanerolle et al., 2015; Vakil & Singh, 2017). Depending on the factors just described, the effects of penetrating brain injuries vary but include the following:

- **An upsurge in intracranial pressure.** There is an instantaneous increase in intracranial pressure when the impact occurs. The pressure waves tend to spread to all areas of the brain and the spinal cord. A subsequent and second increase in intracranial pressure takes place within 2 to 5 minutes of impact.
- **Death.** Death may occur immediately after the TBI, soon thereafter, or sometime later. Penetrating brain injury is survivable, although the death rate is high, largely due to the increased intracranial pressure. Seventy percent of those who suffer gunshot wounds die immediately or soon thereafter. Respiratory and cardiac failures due to injury to the brainstem may cause later death. Mortality rate for low-velocity injury is lower (about 25%) compared to the high-velocity injury (about 47%).
- **Fluctuating blood pressure.** Blood pressure drops immediately after the impact, rises above the normal limit after 5 to 10 minutes, and falls again.

- **Reduced cerebral blood flow.** Blood flow to the brain may remain depressed for up to 12 hours because the drop in cerebral metabolic rate and oxygen consumption.
- **Destruction of brain tissue.** Evident on the projectile tract, the amount of destruction will depend on the previously listed variables (e.g., the velocity, yaw, and mass of the missile).
- **Further brain damage and infection.** Such penetrating particles as bone fragments, hair, glass, and metal may cause infection and additional tissue damage.
- **Bleeding, infection, swelling, and hydrocephalus.** Bleeding, secondary infection, edema of the brain cells, and water accumulation in the cranial vault soon follow the TBI to complicate the recovery process.
- **Physical, cognitive, and language deficits.** These are typically long-term effects, requiring extensive rehabilitative efforts, described in subsequent sections.

Nonpenetrating Brain Injuries

Brain damage with *intact meninges*, even if the skull is fractured, is a **nonpenetrating brain injury**. Because of indirect impact, the brain is damaged with or without skull fractures. No foreign substance enters the brain, however (de Lanerolle et al., 2015). Compared to penetrating injuries, nonpenetrating injuries induce more complex and long-lasting symptoms requiring extensive rehabilitation programs.

Two kinds of forces cause nonpenetrating brain injuries. In the first kind, an external object forcefully strikes a stationary head. In the second, the head moves back-and-forth because of a force acting elsewhere on the body. Some specific causes that produce these forces include:

- **Various kinds of accidents.** Industrial, domestic, or sports-related accidents in which force is applied to a stationary head are frequent causes of nonpenetrating brain injury. The collapse of an automobile

on the head of a mechanic lying on their back and working under it is an example of force applied to a stationary head.

- **Falls.** When a person sustains a fall, the head may hit a stationary object such as the floor or furniture. As noted, falls are a common cause of TBI among the very young as well as the older individuals.
- **Blunt blow to the head.** Often an instance of personal and intimate person violence, a blow to the head with a blunt object or instrument is a common cause of nonpenetrating brain injuries. In this case, the head may uncontrollably move, causing *acceleration/deceleration* damage to the brain, described later.
- **Automobile accidents.** Two kinds of forces may operate in automobile accidents. In the first kind, the force is applied directly to the moveable head, which results in rapid back-and-forth movement of the brain within the skull. In the second kind, the force is applied elsewhere in the body (as in whiplash injuries in accidents) and, as a result, the head moves back and forth, causing internal brain injury.
- **Abuse and interpersonal and intimate person violence.** Various kinds of interpersonal and intimate person violence and abuse may cause nonpenetrating brain injuries.

Biomechanics of Nonpenetrating Brain Injuries

Nonpenetrating brain injuries may be of two kinds: acceleration/deceleration injuries and nonacceleration injuries. These injuries are a function of biomechanical forces acting on the skull and the brain inside (Barth et al., 2001). Movement of the head and of the brain inside are related but separate events. When the head begins to move, the brain does not; when the head stops moving, the brain inside the skull may keep moving. The difference in the movements of these two structures causes varied injuries to the brain. The effects of nonpenetrating brain injuries depend on several biomechanical factors (Barth et al., 2001; Currie et al, 2015; de Lanerolle et al., 2015; Vakil & Singh, 2017), including the following:

- **Acceleration.** Increasing and then decreasing movement of the head and the brain inside it are important biomechanical events that cause

injuries collectively known as **acceleration/deceleration injuries**. These injuries occur when (a) a moving object strikes an unrestrained head and thus propels the head or (b) physical forces set the head itself into motion, which then strikes a stationary object. The head may accelerate even when nothing strikes it, as in the case of the chest hitting a steering wheel in an automobile accident, setting the head into rapid, forward motion. Rapid acceleration of the head results in brain injury even if the head does not strike anything. Acceleration/deceleration injuries are up to 20% more severe than nonacceleration injuries. The type and severity of injuries that result from acceleration depend on whether it is linear or angular.

- **Linear acceleration.** A force striking the head midline causes linear acceleration of the head. In linear acceleration, the head moves back in a straight line (linear) and sets off a chain of events as follows:
 - The head suddenly begins to move linearly. The brain, however, remains still for a few milliseconds because of its inertia.
 - At the point of impact trauma, the brain is injured because of the compression of the skull; this injury is called **coup injury** (pronounced *coo*, which means blow or impact in French).
 - When its inertia is overcome, the brain begins to move at the rate the head is moving.
 - When the head decelerates or stops altogether, the brain keeps moving for a few milliseconds and hits the inside portion of the now rapidly slowed or stationary skull, resulting in another injury to the brain; this injury is called **contrecoup** (pronounced *contra-coo*); thus, the coup injury occurs at the point of impact and the contrecoup occurs at the opposite side of the brain.
 - Coup and contrecoup injuries cause focal damage to meninges, cortex, and subcortical structures.
 - When the brain accelerates and then decelerates, its structures at the base of the cranium rub against sharp bony projections, causing damage to soft tissue. The basal portion of the brain in

the frontal and temporal lobes are especially likely to sustain the most damage of this kind.

- **Angular acceleration.** A force striking the head off-center will cause the head to move in an angular (nonlinear) direction and sets off a chain of events as follows:
 - The force that is applied to the head at an angle causes movement and rotation of the head away from the point of impact.
 - For a few milliseconds after the head begins to move, the brain remains still due to inertia; after the head stops moving, the brain keeps moving for a few milliseconds.
 - The brain that remains still when the head began its angular movement will sustain twisting and shearing damage. This type of damage is called **diffuse axonal injury**, described later in the section on the primary effects of TBI.
 - The brain then begins its angular movement in the direction of the skull (away from the point of impact). When the head stops moving, the brain continues its rotational movement, causing additional twisting and shearing damage of the axial structures.
 - Because of rotation and twisting forces, angular acceleration produces more severe damage than the linear acceleration.
- **Deceleration.** Although an external force may cause the head to accelerate, structures that hold the head and the neck will cause it to decelerate (decrease in speed). The rapidly moving head will quickly slow down because of the restraining forces of the vertebrae and neck muscles or because it hits a stationary object (e.g., the windshield in an automobile accident or the ground in the case of a fall). As noted, injuries due to initial increasing speed and subsequent slowing down are collectively known as *acceleration/deceleration* injuries.
- **Impression (impact) trauma.** When a moving object strikes the head or the moving head strikes a stationary object, the initial point of contact results in impression or impact trauma, which deforms the skull at the point of impact.

- **Nonacceleration.** Certain other kinds of forces cause neither acceleration nor deceleration of the head and the brain and yet cause injuries. **Nonacceleration injuries** occur when a moving object hits a restrained head. Forces that do not cause acceleration or deceleration of the head deliver a crushing blow. For instance, when a person is standing against a wall, lying on a firm surface, or sitting with a rigid head support, a force may strike the head. Injuries resulting from workplace accidents, such as an automobile crashing on the head of a mechanic lying under it, are examples of nonacceleration injuries. Because there is little or no acceleration of the head in such cases, there usually is no significant deceleration either.

Nonaccelerating injuries occur less frequently than accelerating injuries. Also, such injuries produce less severe consequences for the brain. In many cases, there may be no or only a few neurological symptoms. Skull fractures are the main danger of nonacceleration injuries.

Primary Effects of TBI

The crushing effect or the biomechanical force that a head receives causes both primary and secondary injuries. The **primary effects** occur at the time of trauma to the head and include injuries due to linear or angular acceleration, deceleration, and nonaccelerating crushing forces. The primary effects of head trauma include the following (Elbourn et al., 2017; Kaur & Sharma, 2018; Khellaf et al., 2019; Ordookhanian et al., 2018; Pavlovic et al., 2019):

- **Lacerations or fracture of the skull.** Lacerations are torn or jagged wounds on the skin surface. More serious than lacerations, fractures of the skull are seen in 80% of fatal cases. Skull fractures may lead to intracranial hematoma and increased risk of infections.
- **Diffuse axonal injury (DAI).** This is a seriously damaging primary effect that occurs at the moment of closed head injury. DAI consists of torn nerve fibers in widespread areas of the brain's white matter. A

direct result of the trauma, DAI is not a secondary consequence of hypoxia, increased intracranial pressure, or edema. Nonlinear (angular) acceleration/deceleration forces produce DAI. Focal lesions in the corpus callosum and parts of the brainstem may also accompany DAI.

- **Primary brainstem injury.** Primary brainstem injury may occur without significant diffuse damage. Other parts of the brain (e.g., the midbrain) may also be involved to some extent. Individuals who sustain primary brainstem injury will lapse into a coma at the moment of trauma.
- **Diffuse vascular injury.** This involves small and widespread ruptures in the brain's blood vessels that cause multiple hemorrhages in the brain, some of which may be obvious upon autopsy, whereas others may need histological examination. Diffuse vascular injury leads to death within hours or even minutes of the trauma.
- **Primary focal lesions (injury).** Depending on the type of force involved, TBI may produce focal (localized) lesions as well. Often described as *contusions*, linear impacts are more likely to produce focal injuries than are nonlinear impacts. As mentioned previously, the skull maybe depressed or fractured at the locus of impact — an example of focal skull damage. Primary focal brain lesions include different kinds of injuries. Previously described coup injury and contrecoup injuries are focal. Abrading injuries are those that result from the brain moving back and forth along the rough surfaces or projections of the skull. Other types of primary focal injury include damage to the hypothalamus, corpus callosum, and other specific structures.

An injured brain is susceptible to additional effects that further complicate the clinical picture. These somewhat delayed effects are called secondary effects of TBI.

Secondary Effects of TBI

Secondary effects (injuries) are consequences of primary injuries and occur sometime after the trauma has taken place. Prompt and competent management or prevention of secondary effects is crucial for speedy recovery and a favorable long-term outcome (Elbourn et al., 2017; Kaur & Sharma, 2018; Khellaf et al., 2019; Ordookhanian et al., 2018; Pavlovic et al., 2019):

The secondary effects of primary brain injury include the following:

- **Intracranial hematoma.** This is an accumulation of blood from hemorrhage within the skull or the brain. Intracranial hematoma is commonly associated with a fractured skull and is a frequent cause of death. Epidural (extradural) hematoma, often found in the temporal regions of persons with fractured skulls, is the accumulation of blood between the dura mater and the skull. Automobile accidents frequently cause this type of bleeding. Subdural hematoma is an accumulation of blood between the dura and the arachnoid caused by lacerated cortical blood vessels, as those induced by falls. Subdural hematoma is common and deadlier than epidural hematoma, with a mortality rate that may exceed 60%. Intracerebral hematoma is an accumulation of blood within the brain itself, commonly occurring in temporal and frontal lobes, and leads to diffuse axonal injury due to linear acceleration. This is a frequent cause of coma and death.
- **Increased intracranial pressure.** As a secondary effect, TBI can cause increased intracranial pressure. Accumulation of blood, water, or cerebrospinal fluid causes increased pressure within the cranium and restricts cerebral blood flow; extreme pressure can cause death.
- **Ischemic brain damage.** This is brain damage from lack of oxygen to the tissue (hypoxia) because of reduced or blocked blood supply, which may be a result of breathing difficulties (leading to poor oxygenation of blood), hypotension (reduced blood pressure), and constricted cerebral blood vessels.
- **Seizures.** Approximately 5% to 7% of persons with TBI experience seizures as a secondary consequence. The rate is higher among soldiers who receive missile injuries in combat (30% to 50%). Repeated seizures cause additional trauma and damage to the brain.

- **Infection.** Various kinds of infections can threaten the life of the person with TBI. The head wound (open, depressed, skull fractures) may be infected. Meningitis (inflammation of the membranes of the brain or the spinal cord) may be especially serious. Prompt antibiotic treatment will help control the infectious sequelae.
- **Increased risk of dementia.** Those who sustain multiple sports-related concussion are vulnerable to dementia later in life.

Variables Related to Recovery

Persons with TBI recover from their symptoms to varying degrees. Because several factors affect recovery, it is difficult to predict its course in individual cases. Some recover most of their skills relatively fast and to a greater degree, whereas others recover less at a slower pace. Advances in emergency medical treatment and trauma management have resulted in a lower death rate due to TBI with an attendant better prognosis (Khellaf et al., 2019). Nonetheless, 50% of those who sustain severe head injuries may die. Among those who survive severe head injuries, a significant number may have permanent memory deficits, limited attention span, more or less subtle language deficits, discourse problems, poor emotional control, and drug and alcohol abuse (Elbourn et al., 2017; Gauthier et al., 2018; Le et al., 2022).

Recovery from more serious injuries, injuries that are diffuse and involve diffuse axonal injury, is slower than from milder and focal injuries. Subarachnoid hemorrhage, brainstem injuries, intracranial hematoma (formation of a blood pool due to a ruptured vessel), and increased intracranial pressure signal a poor prognosis. Persons with secondary injuries and a history of drug and alcohol abuse recover more slowly and to a lesser extent than those with no such injuries or histories. Surprisingly, individuals who sustain head injury in high-speed automobile accidents tend to fare better than pedestrians who are hit by vehicles. Younger age, higher education, better levels of consciousness in the acute stage, and absence of coma are also good prognostic indicators (Elbourn et al., 2017; Kaur & Sharma, 2018; Khellaf et al., 2019;

Ordookhanian et al., 2018; Pavlovic et al., 2019). Race and gender, on the other hand, have no significant effect on recovery rates.

Neurobehavioral Effects of TBI

Behavioral changes are both immediate and long term and depend on the severity of the trauma. The more common behavioral effects include the following (Elbourn et al., 2017; Hegde & Freed, 2022; Kaplan et al., 2018; Kaur & Sharma, 2018; Khellaf et al., 2019; Ordookhanian et al., 2018; Pavlovic et al., 2019):

- **Altered consciousness.** Depending on the severity of the injury, levels of consciousness vary. A person may be simply dazed but still be conscious in the case of mild injury. **Stupor** is a state in which the individual is generally unresponsive, but pain or other strong stimuli may arouse the person for a brief period. **Coma** is an unconscious state of being unresponsive to most or all external stimulation and may last a few days or a few weeks. It often results from a diffuse trauma to the brain and the brainstem; some persons may die without recovering from it.
- **Survival with no recovery of consciousness.** Some individuals who survive TBI may not recover consciousness at all and exist in a vegetative state. In this state, the persons are mostly unconscious, although they may experience a sleep-wake cycle and be alert for short periods of time. Reflexive responses may be intact. Those who do not recover from a vegetative state within 30 days are said to be in a *persistent vegetative state* from which recovery is possible within the first 6 months; adults have a 50% chance of recovery and children have a 60% chance. Chances of recovery diminish beyond 6 months and are very low after the first year. Those who recover late are permanently disabled.
- **Confusion and disorientation.** Persons in the process of regaining consciousness or who are dazed may experience confusion and disorientation with respect to time and place. Additional symptoms

may include headaches, lightheadedness, dizziness, blurred vision, ringing in the ears, fatigue, lethargy, impaired attention and thinking, and mood changes.

- **Memory problems.** Persons who emerge from coma are likely to experience loss of memory for events preceding and following the trauma. Such persons may fail to register events happening around them. Memory skills are categorized and divided into various components; these are described in the next chapter.
- **Speech disorders.** Persons who are in the process of recovering from TBI may have speech problems, described in the next section. These problems may be more or less permanent depending on the extent of brain injury.
- **Dysphagia.** Posttrauma, some 25% to 75% of persons with TBI may have difficulty swallowing (dysphagia). Intracranial bleeding, midline shift, and brainstem injury are likely causes of dysphagia. Intubation and tracheostomy may worsen the swallowing problem.
- **Other neurological symptoms.** Moderate to severe injury may cause seizures, vomiting or nausea, dilation of pupils, weakness or numbness in the extremities, loss of coordination, restlessness, and agitation.
- **Behavioral and psychiatric changes.** A variety of temporary or longer-lasting behavioral changes have been noted in persons who sustain TBI (Kaplan et al., 2018). A small percentage of persons may experience auditory hallucinations, confabulations, and delusions. Posttraumatic stress disorder is a common observation, especially in combat veterans with brain injury. Some persons who have had a history of psychiatric problems or substance abuse tend to be depressed following TBI. Apathy, a less severe reaction than depression, may be found in some persons. Even after functional recovery, some may continue to experience insomnia, anxiety, poor emotional control, social withdrawal, irritability, childishness, and unreasonable behavior.

Overview of Communication Disorders in TBI

The extent of injury, the specific areas of the brain that are injured, and the premorbid speech-language skills of the person are among the variables that affect the types and severity of communication disorders (Douglas, 2017; Elbourn et al., 2017; Gauthier et al., 2018; Hegde & Freed, 2022; Shorland et al., 2020). Individuals greatly differ in their symptom complex. Most persons with minimal brain injury do not experience severe speech and language problems. They might be dazed, give an occasional irrelevant verbal response, and may take time to respond verbally. While generally oriented with respect to place, some may be disoriented to time. Most of these effects may be temporary. Postrecovery from the initial trauma, the most significant problems SLPs notice are social communication impairments and dysarthria (Freed, 2020; Steel & Togher, 2019). Persons may be indifferent to their listeners and may have blunted emotional response (Murphy et al., 2021; Power et al., 2020; Sohlberg et al., 2019; Steel et al., 2017).

Lesions in the left temporal lobe affect conversational speech and speech comprehension, lesions in the left frontal lobe decrease speech fluency, and lesions in the right parietal lobe disrupt comprehension and reasoning skills (Gauthier et al., 2018). Those who have sustained injury to their cerebellum, brainstem, or peripheral nerves tend to have dysarthria (Freed, 2020).

Communication deficits associated with TBI may be summarized as follows:

- **Mutism.** Some persons may not talk during the acute stage following injury or recovery from coma. Duration of mutism varies across persons. Severe and diffuse TBI leads to persistent communication impairments.
- **Confused language.** Soon after the injury, persons who are conscious and talking may be confused in their language usage. Although syntactically correct, the person may speak in an irrelevant, circumlocutory, incoherent, and confabulatory manner.
- **Dysarthria.** A motor speech disorder due to disturbed muscular control of the speech mechanism, dysarthria may demand greater

clinical attention than language problems in persons with TBI because of its severe effects on speech sound articulation and speech intelligibility. Due to pathologies of the central nervous system, peripheral system, or both, dysarthria affects all aspects of speech production, including articulation, prosody, respiration, phonation, and resonance. The spastic dysarthria is more common than other types in persons with TBI. Imprecise consonant production, abnormal stress patterns, hypernasality, impaired speech rate and fluency, reduced breath support for speech, reduced phrase length, prolonged intervals in speech, limited pitch and loudness variations, and arrhythmic speech are among the dominant dysarthric features.

- **Difficulty in social communication and conversation.** Once the initial recovery takes place and confused language recedes, impaired conversational skills become the most pronounced, severe, and persistent problem. Persons with TBI find it difficult to initiate social interaction and conversation, take turns speaking during conversation, select a topic of conversation, and maintain conversation on a topic. They tend to ramble on inappropriate topics and fail to yield to conversational partners.
- **Naming problems.** Confrontation naming may be especially difficult. Brain-injured persons tend to be hasty in naming objects shown, resulting in misnaming. Generally, naming errors are not as prominent or debilitating as in persons with aphasia. Some persons with TBI may, however, have an independent aphasia.
- **Perseveration of verbal responses.** Some persons with TBI may repeat multiple times their own utterances.
- **Reduced word fluency.** This problem may be especially evident on a timed test in which the person is asked to recall as many words as possible in a category (e.g., names of animals). Reduced word fluency may partly be due to naming problems and inattention. Most people with TBI, however, may talk excessively and inappropriately.
- **Impaired discourse and narration.** Storytelling and discourse on a topic may be impoverished with limited information, sentence variety and complexity, and story or topic details. Event sequence

may be confused or disorganized. Persons with TBI may not provide adequate contextual cues for their utterances, and their narrative may lack cohesion, though verbose. False starts may be common. Because these effects are long lasting, individuals may find it hard to maintain their jobs.

- **Problems in being concise and direct.** Persons with TBI tend to speak in vague and inaccurate terms. Frequent circumlocutions may add to this problem.
- **Difficulty in nonverbal communication.** Understanding gestures, facial expressions, and other nonverbal means of communication tends to be impaired. Individuals may have a pronounced difficulty understanding fear, anger, disgust, and sadness others reveal through facial expressions while talking. Missing the subtle cues others give to stop talking, individuals continue their rambling and irrelevant talk.
- **Auditory comprehension problems.** Most affected persons have trouble comprehending spoken language. Comprehension of complex, abstract, and rapidly spoken material may be especially difficult. Persons may fail to understand the emotional tone and indirect meaning of prosodic features of speech they hear.
- **Difficulty with abstract language.** Literal interpretation of abstract language expressions, failure to appreciate irony and humor, and misinterpretation of proverbs and metaphors are typical of individuals with TBI.
- **An associated aphasia.** Some individuals with TBI may also have a coexisting aphasia.
- **Problems in reading and writing.** The persons may have difficulty understanding extended texts they read. Writing problems may be evident especially when the persons are asked to write coherent, precise, and extended paragraphs. Some of these difficulties may reflect the other effects of TBI: lack of sustained attention, difficulty following a sequence, and so forth. Research on reading and writing problems of persons with TBI is limited, however. Obviously, the premorbid reading and writing skills are a significant factor to consider in evaluating those skills in persons with TBI.

Social communication difficulties of persons with TBI affect their family members, friends, and all conversational partners. Consequently, it may be difficult for them to maintain friendship and intimate spousal and partner relationships (Douglas, 2017). When they return to work with residual communication deficits, coworkers may be less inclined to interact with them.

Impaired Memory, Thinking, Reasoning, and Planning Skills

Difficulties in remembering, logical thinking, reasoning, and planning are basic characteristics of brain injury. Memory, thinking, and reasoning skills are considered *cognitive* and planning skills are described as *executive* (execute a certain plan or action). Positron emission tomography and diffusion tensor imaging studies (see [Chapter 2](#)) have correlated cognitive deficits to decreased glucose metabolism at rest in the prefrontal and cingulate areas of the brain. Functional magnetic imaging studies have shown reduced activation in the brain networks involved in attention and executive functions. Extensive brain networks mediate cognitive functions (see Azouvi et al., 2017, and Gilmore et al., 2022, for reviews of brain imaging studies in TBI).

Most individuals with mild TBI recover these functions, but residual deficits may persist in those with severe trauma (Azouvi et al., 2017; Elbourn et al., 2017; Gilmore et al., 2022; Rabinowitz & Levin, 2014; Shorland et al., 2020). Generally, the more abstract the reasoning or planning task, the greater is the difficulty. Also, the more severe the brain injury, the greater is the difficulty with abstraction (Hegde & Freed, 2022; Goncalves et al., 2018; Payne, 2014; Rabinowitz & Levin, 2014). Specific problems including the following:

- **Memory problems.** These are found in up to 75% of persons with TBI, especially when the fornix, hippocampus, mammillary bodies of the diencephalon, and the thalamus are affected. Common memory problems include retrograde amnesia (memory loss for events preceding the brain injury, including personal events and

experiences), anterograde amnesia (lack of memory for events following the injury), impaired prospective memory (forgetting what to do at certain times), impaired immediate memory (forgetting just experienced events), and impaired procedural memory (forgetting how to perform such actions as driving or cooking). Persistent memory problems are a frequent complaint of persons with TBI who have recovered most of their skills. Their family members corroborate those complaints. Persons with a history of TBI also have difficulty imagining possible future events in their lives.

- **Slower and inefficient learning.** Persons with TBI, even with recovery, may need additional time to learn new material. Their learning may be disorganized and hence inefficient. Learned material may soon be forgotten.
- **Impaired understanding of proverbs.** For example, the meaning of *a stitch in time saves nine* may be fuzzy to a person with TBI.
- **Difficulty describing similarities and differences.** Persons with TBI may have difficulty describing how objects or events are similar as well as different. For example, the difference between a *table* and a *chair* or the similarity between a *rose* and a *gardenia* may be difficult for the person to specify.
- **Problems matching printed geometric patterns.** A person with TBI may be unable to complete incomplete patterns and sort objects based on shape, color, or other characteristics.
- **Difficulty describing common actions.** Persons with TBI may not give a reasoned answer when certain situations are described to them. For example, the person may not answer correctly when asked, “What would you do if you found someone’s credit card on the floor of a store?”
- **Failure to detect logical inconsistencies in statements.** For instance, the person may be unsure what is wrong with the statement: “She was not hungry, so she ate two large sandwiches.”
- **Difficulty understanding sequential arrangements.** Persons may be confused about how to arrange sequentially a storytelling set of pictures presented randomly.

- **Failure to detect absurdities in pictures.** The person may not point out what is wrong with a picture that shows a deer chasing a tiger.
- **Inability to find missing elements in pictures.** Persons may not point out the missing body parts in drawings of people or animals.
- **Difficulty planning and implementing activities.** Described as *executive functions* (EFs), persons with TBI may be unable to describe fully and sequentially how to book a hotel room, plan a birthday party, or cook a specific meal. The person may then be confused when actually performing such actions. EFs and language skills are correlated; the poorer the EF scores, the worse the language skills. These long-lasting impairments may severely affect everyday living skills and the quality of life of persons with a history of TBI.

TBI-related pattern of communication is similar to that found in right hemisphere disorder, described in Chapters 6 and 7. Rambling and confused speech bears some similarity to the speech of persons with Wernicke's aphasia.

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9

ASSESSMENT AND MANAGEMENT OF TRAUMATIC BRAIN INJURY

Chapter Outline

- Overview of Assessment
- Initial Screening
- Diagnostic Assessment
- Assessment of Consciousness and Responsiveness
- Assessment of Memory
- Assessment of Reasoning and Planning Skills
- Assessment of Communication Disorders
- Overview of TBI Treatment Research
- Specific Treatment Targets and Strategies
- Teaching Compensatory Strategies
- Skill Maintenance Program
- Group Therapy
- Promoting Community Reentry
- Cognitive Rehabilitation
- References

Learning Objectives

After reading the chapter, the reader will:

- Describe the assessment techniques used to evaluate consciousness, responsiveness, memory, and reasoning and planning skills
- Describe and evaluate techniques for assessing communication disorders in persons with TBI
- Give an overview of clinical management strategies for persons with TBI

PRINTED NOTES

Assessment of communication skills and general behavior in persons with traumatic brain injury (TBI) is a repeated effort because the symptoms, with or without treatment, change over time. Changes are typically positive but may be negative if the person experiences adverse events (e.g., a subsequent stroke or seizures). Those changes should be measured continuously to modify treatment procedures that are relevant to the current status of the person.

TBI assessment is also a team effort, including the emergency care physician, neurologist, radiologist, psychologist, speech-language pathologist, physical therapist, nurse, and other professionals. The team is initially concerned with the individual's general health, levels of consciousness and awareness, and medical or surgical intervention that may be urgently needed. Preliminary assessment of cognitive and behavioral skills is also important at this initial stage. As the individual's physical condition improves, an in-depth assessment of the consequences of injury may be made to develop a relatively short-term intervention plan. A long-term program includes social communication, community reintegration, and return to education or workplace as found relevant (Hegde & Freed, 2022).

The speech-language pathologist (SLP), although mostly concerned with the assessment and treatment of communication skills and dysphagia, needs to have a good understanding of medical assessment and treatment of TBI. SLP's assessment and treatment work depends on the individual's physical condition, medical and pharmaceutical treatment used to stabilize the person, radiological procedures that help document the changes in the brain health, the course of recovery, and the effects of the injury.

Overview of Assessment

All specialists in the team contribute to the initial assessment of TBI. The team needs information on the person, the family, and the circumstances surrounding the brain injury. Assessment begins with a quick evaluation of the individual's status (Hegde & Freed, 2022).

- **Case history.** A detailed case history documents the conditions and events that led to the head trauma. Information is also obtained on the individual's premorbid behavior, verbal skills, education, literacy skills, employment and the tasks performed there, family relationship, interests and hobbies, social skills, general health, substance abuse, alcoholism, and long-term prescription medication.
- **Interviews.** At least one family member is interviewed to build a premorbid profile of the person that may be used to make various clinical decisions. Interviewing the witnesses of the scene of trauma (e.g., automobile accident, workplace, or playing fields) will help ascertain the details of the incident. Law enforcement and emergency medical staff who arrived on the site of injury and those who attended to the patient upon arrival at the hospital may supply valuable information.
- **Person's medical condition.** The individual's current physical condition, organ injuries, medications, alertness, and responsiveness may be evaluated and documented as soon as feasible.
- **Medical tests.** The physician in charge may order radiological tests, blood tests, brain scanning, and other tests to assess the extent of skull and brain injury and its consequences on the individual.
- **Assessment of consciousness and cognition.** This is a critical initial task. Standardized scales and observations of the individual will help the team determine the level of responsiveness to environmental stimuli and the general level of alertness.
- **Effects of current medications.** All professionals should be aware of the medications the person is on and their effects on whatever they wish to assess, including communication. Obviously, the sedated person does not respond well. A person's slurred speech may be due to the antianxiety drug being administered, not necessarily to the sustained TBI.

- **Assessment of memory and reasoning skills.** Psychologists and SLPs evaluate memory, thinking, reasoning, and planning (executive) skills of the person. This will also help establish the initial baseline of such skills.
- **Assessment of communication skills.** Initially, the SLP makes a quick screen of the person's communication status. A detailed evaluation of speech and language skills, social and functional communication, executive (planning) skills, reading, and writing may be done when appropriate.
- **Assessment of dysphagia.** Assessment of swallowing disorders is primarily the responsibility of SLPs. In some cases, dysphagia assessment may take precedence over communication assessment.
- **Self-assessment and caregiver assessment.** It is informative to know how the individuals rate themselves on health, well-being, responsiveness to their surroundings, and communication. Individuals may either under- or overestimate their health and skill levels. Caregivers' perspective is also just as important. Because they are in direct and constant contact with the person, the caregivers may offer reliable information on changes in the health status and functional communication.
- **Integration of information.** The team integrates information from the case history, specific assessment procedures, medical evaluations, communication and cognitive assessment, self- and caregiver assessment, and any other source of information available.

Initial Screening

Soon after hospitalization or a visit to an outpatient clinic, the person with TBI may be screened for the level of consciousness; orientation to time, place, and person; and basic communication. Experienced clinicians may screen with a few questions about when and how the injury took place; the time, day, and date of injury; the person's name and that of family members; home address; the current location, and so forth.

Clinicians may also administer a screening test. The most frequently used include the following:

- **The Brief Test of Head Injury (BTHI)** (Helm-Estabrooks & Hotz, 1991) helps screen orientation, naming, following directions, visuospatial skills, and reading comprehension.
- **The Mini Mental Status Examination (MMSE)** (Folstein et al., 1975) is designed to screen attention, orientation, calculation, name recall, pointing to objects named, repeating names, following commands, and so forth.
- **The Galveston Orientation and Amnesia Test (GOAT)** (Levin et al., 1979) may be used to screen orientation (time, place, and person) and memory for events preceding and succeeding the traumatic event that caused the TBI.

Diagnostic Assessment

A detailed assessment is done as soon as feasible. Assessment targets include the individual's consciousness, memory, alertness to the surrounding environment, orientation to time and space, general responsiveness to stimuli and people, functional and social communication, speech production skills (especially dysarthria), and dysphagia.

SLPs and other professionals administer standardized or semistandardized instruments. Caregivers may rate the person for various speech-language skills, and some rating scales allow for self-rating of communication skills. Tests may be roughly classified as those of memory, reasoning, communication, and executive functions (e.g., remembering how to perform a certain task).

Standardized tests of communication skills vary in reliability, validity, and ecological and functional relevance to the individual being assessed. After a comprehensive review of communication assessment tools for TBI, Sohlberg et al. (2019) concluded that most tests fall short of one or more criteria of reliability, validity, diagnostic precision, ecological relevance, and so forth. They emphasized the need for better standardized tests. Clinicians should

carefully evaluate the test manual to judge its appropriateness to assess a given individual, considering the person's ethnocultural background.

Assessment of Consciousness and Responsiveness

Impaired consciousness and inconsistent responsiveness to verbal and nonverbal stimuli is an initial symptom of TBI that needs to be assessed. The clinician may select one of the standardized assessment tools to formally assess consciousness and responsiveness to stimuli.

- **The Glasgow Coma Scale (GCS)** (Teasdale & Jennett, 1976) helps evaluate the individual's initial condition. It is a subjective rating scale whose results should be interpreted with caution and supplemented with systematic behavioral observations. Assessed responses include eye opening (spontaneously, when commanded, and no response), motor responses (ranging from obeying verbal commands to no response), and verbal responses (ranging from good conversation to no response). A limitation of the test is that untestable behaviors cannot be scored.
- **The Glasgow Outcome Scale (GOS)** (Jennett & Teasdale, 1981) is useful in evaluating outcome for, or course of recovery from, TBI. The scale describes five potential outcomes for persons with TBI (death, vegetative state, severe disability, moderate disability, and good recovery).
- **The Rancho Los Amigos Scale of Cognitive Levels (RLAS)** (Hagen, 2000) is widely used to evaluate the cognitive and behavioral levels of individuals with TBI. The clinician may rate behaviors and the amount of support needed at 10 levels, beginning with no response to purposeful, modified, and independent status. It helps assess the degree of dependence and independence.
- **The Galveston Orientation and Amnesia Test (GOAT)** (Levin et al., 1979) may be used to assess the status of a person just recovering from coma. Test contains questions to assess the individual's

orientation to time, place, and persons (including self) and memory for events preceding and succeeding the injury.

- **Comprehensive Level of Consciousness Scale (CLOCS)** (Stanczak et al., 1984) helps assess behaviors in eight categories: posture, resting eye position, spontaneous eye opening, other ocular movements, pupillary reflexes, motor activities, responsiveness, and communicative effort.
- **Disability Rating Scale (DRS)** (Rappaport et al., 1982) may be used to rate eye opening, communication ability, motor responses, feeding, toileting, grooming, level of functioning (dependency), and employability.

TBI affects several other nonverbal skills typically described as *cognitive functions*. Besides memory, these include reasoning and executive (planning and task completion) activities. In many cases, impairments in cognitive skills may be even more pronounced than the verbal skills (e.g., naming, repetition, grammar).

Assessment of Memory

Memory skills have been categorized and broken down into smaller components in numerous ways. **Retrospective memory**, which is memory for past events, is subdivided into *declarative memory* and *procedural memory*. Declarative memory is remembering what has been learned in the past about things, places, and events in general. Questions about the individual's general knowledge will be useful in evaluating declarative memory. Procedural memory is remembering how to perform actions (e.g., driving or cooking). These skills are often included under *executive functions*, described later.

Prospective memory is remembering to do certain things at particular times (e.g., keeping appointments or checking the mail at a certain time). Questioning the person about such upcoming events as dinner time, doctor's appointment, bedside visits, medication times, and so forth will help assess this type of memory.

Posttraumatic memory loss (also called *anterograde amnesia*) is difficulty remembering events following the TBI. Person-specific questions about known events that took place after the injury will help assess this type of memory. For instance, the clinician can ask about who brought the person to the hospital, activities that followed admission, and so forth. Interview of family members or police and rescue staff who brought the person to the hospital will help corroborate the individual's descriptions.

Pretraumatic memory loss (also known as *retrograde amnesia*) is difficulty remembering events that preceded the trauma. Questions about known events that preceded the injury will give an idea of this deficit. For instance, persons may be asked where they were or what they were doing before the injury. Interview of individuals associated with the person (e.g., family members) and those associated with the events surrounding the trauma (e.g., other passengers in a car that crashed and the police or rescue staff who arrived on the accident scene) may provide corroborative evidence.

The clinician may use one of the standardized tests that predominantly assess memory:

- **The Ross Information Processing Assessment-2** (Ross-Swain, 1996) is a general cognitive test of recent and remote memory, spatial and environmental orientation, recall, abstract reasoning, problem solving, auditory comprehension, and so forth.
- **Comprehensive Assessment of Prospective Memory** (Shum & Fleming, 2008) helps assess prospective memory in the context of daily living skills (e.g., forgetting to lock the front door when leaving, forgetting to buy an item at the grocery store).
- **The Rivermead Behavioural Memory Test–Third Edition** (Wilson et al., 2008) targets verbal and nonverbal episodic, spatial, prospective, and procedural varieties of memory and includes a subtest for facial recognition and another for new learning.

Most other standardized tests of TBI, including tests of communication, consciousness, and orientation, include at least a few questions to assess memory skills. For instance, the Galveston Orientation and Amnesia Test

(GOAT) (Levin et al., 1979) has items to assess memory. The clinician may pool the data from different tests to make clinical judgments about the person's memory skills.

Assessment of Reasoning and Planning Skills

Impaired reasoning and planning skills may be assessed informally as well as with standardized instruments. Asking the person to state the meaning of a few proverbs (e.g., “A stitch in time saves nine.”) and the differences and similarities between objects or words (e.g., “How are pencils and pens similar? How are they different?”) will give a rough idea of abstract language skills. To informally assess the planning (executive) skills, the clinician may ask how the person would cook a meal, fix a leaking faucet, prepare a lecture outline, plant flowers in the garden, and so forth.

Executive functions is a poorly and variably defined construct, although research on it is extensive (Mueller & Dollaghan, 2013). Researchers have swept under the term *executive function* almost every impairment that may be observed in persons with TBI (Hegde & Freed, 2022). Some test items are so subjective that they are of little clinical use and perhaps apply to almost any person (e.g., *the patient does the first thing that comes to mind, a bit over the top, says one thing does something else, loses temper*). Some practical tests target proper sequencing of everyday actions such as mailing a letter, writing a check, or making a grocery list. Other functions that are difficult to measure in a meaningful manner include self-regulation, self-directed behavior, impulse control (behavioral inhibition), and planning for the future. All these tend to be impaired in persons with TBI. In their review of executive functioning assessment, Mueller and Dollaghan (2013) concluded that most standardized tests lack strong evidence for diagnostic accuracy, content validity, and clinical utility. The reviewers also pointed out a lack consensus on the clinical value of executive function assessment for communication treatment.

Impaired executive functions are correlated with lesions in the prefrontal cortex (Snyder et al., 2015), although other regions, including the parietal cortex and subcortical structures, also may be involved.

Standardized tests for assessing memory, reasoning, and executive skills have overlapping test items as the skills are interrelated. For instance, reasoning and memory are both important for describing how a certain action, such as making a peanut butter sandwich, is performed. The clinician may select one of the following tests for assessing reasoning and planning skills in persons with TBI.

- **Functional Assessment of Verbal Reasoning and Executive Strategies** (MacDonald, 2005) uses simulated real-world scenarios to assess verbal reasoning, comprehension, discourse, and executive functions.
- **The Scale of Cognitive Ability for Traumatic Brain Injury** (Adamovich & Henderson, 1992) consists of five subtests to assess perception and discrimination, orientation, organization, recall, and reasoning.
- **The Woodcock-Johnson Test of Cognitive Abilities–Fourth Edition** (Schrang et al., 2014) is a test of general intellectual ability and includes subtests to assess reasoning, cognitive processing speed and efficiency, quantitative reasoning, short- and long-term memory, and several other intellectual skills.
- **Delis-Kaplan Executive Function System** (Delis et al., 2001) targets behavioral initiation and inhibition, impulse control, divided attention, abstract thinking, problem solving, and several other executive functions.
- **Executive Function Performance Test** (Baum et al., 2007) includes such ecologically valid tasks as making some oatmeal, using the telephone, paying bills, taking medication, and so forth, and it evaluates correct sequence and task completion.

Assessment of Communication Disorders

Social communication deficits are dominant in TBI. Therefore, a detailed syntactic and morphological assessment of persons with TBI is less critical

unless the initial observations suggest a need. Adequate samples of conversation, narration, and discourse will serve multiple purposes, including an assessment of social communication, grammatical structures, speech production problems, any coexisting fluency or voice disorders, and abstract language skills. Conversation with friends and family will be especially helpful in documenting communication deficits in natural settings.

Clinicians may also design their own client-specific protocols or consider using a comprehensive set of protocols provided in Hegde and Freed (2022) that may be modified to suit individual clients.

Assessment of Functional and Social Communication

Initially, functional communication assessment takes precedence over social communication. The person's need in the acute and postacute stage is to get the basic needs fulfilled. Pain, hunger, thirst, and a need to visit the bathroom will have to be communicated to the caregivers in any mode possible. Direct observations, interviews, and questionnaires are the main means of assessing functional communication. More complex social communication includes conversation, narration, and discourse.

Several functional communication assessment instruments were described in [Chapter 4](#) on assessment of aphasia. As most of them are applicable to assess functional communication in individuals with TBI, the reader is referred to [Chapter 4](#) for details. [Table 4–3](#) in [Chapter 4](#) lists commonly used tests of functional communication. Particularly useful among those described in [Chapter 4](#) are the Brief Test of Head Injury (Helm-Estabrooks & Hotz, 1991), Functional Assessment of Communication Skills for Adults (ASHA FACS; Frattali et al., 1995), Communication Activities of Daily Living–Third Edition (Holland et al., 2018), and Functional Communication Profile–Revised (Kleinman, 2003). In addition, the clinician also may administer one of the tests that are more specific to individuals with TBI:

- **Social Communication Skills Questionnaire Adapted** (Dahlberg et al., 2006) requires individuals with TBI to rate their own social communication skills. The client may rate *always, often, sometimes, rarely, or never* such items as *I am able to keep the conversation going* or *I am able to change the subject smoothly*.
- **The LaTrobe Communication Questionnaire** (Douglas et al., 2000) has two forms, one for the client to fill out and the other for a familiar person to fill out. The test evaluates conversational tone and flow, engagement, partner reactions, and attention to conversation.
- **Montreal Evaluation of Communication** (Joanette et al., 2015) helps assess topic shifts, inappropriate comments, imprecise language, comprehension of humor, and the tone of voice as observed in a sample of conversation between the client and the clinician.
- **The Adapted Kagan Scales** (Togher et al., 2010) consist of two scales, Adapted Measure of Support in Conversation and Measure of Participation in Conversation. With a conversational speech sample involving the person with TBI and a partner, these scales help assess pragmatic language skills of the client and the type of support needed and offered (or not offered) by conversational partners.
- **The TBIBank of the TalkBank System** is a protocol for discourse analysis on a web-based resource (<http://tbi.talkbank.org>). It contains picture stimuli to evoke narration and discourse. With the help of a faculty, students can gain access to the resource.

Assessment of Other Language Skills

The following more complex speech and language skills may be assessed when the individual is ready:

- **Conversation, narration, and discourse.** To be more naturalistic than the similar tasks included on standardized tests, the person may be engaged in a typical conversation, asked to narrate a story or a

personal experience, and give a brief monologue-like speech on a topic of interest (a discourse on a topic, e.g., a pandemic, presidential election, local fair or festival). The recorded samples will help evaluate relevancy, accuracy, precision of speech, rambling, confused speech, out-of-sequence narratives, or inadequate discourse. Conversational skills (e.g., topic initiation and maintenance, turn taking) also may be assessed. Discourse analysis may be useful in evaluating executive (planning) functions as well.

- **Assessment of verbal expression.** Client-specific procedures or subtests of standardized tests help assess such skills as naming, word fluency, correct sentence construction, and correct production of grammatic morphemes. Although grammatical deficits are not primary in TBI, they may be assessed if necessary. (See [Chapter 5](#) on assessment of aphasia for some of the tests.) A writing sample will help assess impaired writing.
- **Assessment of language comprehension.** Conversation with the person and formal tests may be used to assess language comprehension. The clinician may request the person to silently read passages from a newspaper and ask questions to assess comprehension. Tests that assess comprehension include the Peabody Picture Vocabulary Test (Dunn & Dunn, 1997), the Revised Token Test (McNeil & Prescott, 1978), and the test for Auditory Comprehension of Language, Third Edition (Carrow-Woolfolk, 1999).

Assessment of Dysarthria Associated With TBI

A few questions asked will help screen dysarthria and speech intelligibility in the acute stage. To assess persisting dysarthria, SLPs may use the conversational speech samples, specialized dysarthria test batteries, and traditional tests of articulation (Freed, 2020; Hegde & Freed, 2022; Yorkston et al., 2010). Three useful tests are:

- **Frenchay Dysarthria Assessment** (Enderby & Palmer, 2008) is designed to evaluate respiration, articulation, resonance, phonation, and reflexive aspects of the motor speech mechanism that affect intelligibility. The results help differentiate the forms of dysarthria.
- **Assessment of Intelligibility of Dysarthric Speech** (Yorkston & Beukelman, 1981) is a widely used test of articulation and speech intelligibility, assessed through recorded productions of single words and sentences.
- **The Speech Intelligibility Test for Windows** (Yorkston et al., 2020) is a computerized version of the Assessment of Intelligibility of Dysarthric Speech that automatically analyzes speech rate, intelligibility, and production of speech sounds.

Clinicians may consider comprehensive assessment protocols provided in Hegde and Freed (2022). The protocols may be modified and individualized.

TBI may impair swallowing. A brief bedside screening for **dysphagia**, observation of the person at mealtime, and caregiver interview will help determine the need for assessment. For procedures, the reader is referred to dedicated sources on dysphagia (e.g., Leonard & Kendall, 2019).

Assessment of Quality of Life

In spite of completing a rehabilitation program, individuals with moderate to severe TBI may continue to experience more or less debilitating deficits, some long-lasting. At the time of discharge from an intervention program, it is essential to measure how the persistent deficits affect the individual's quality of life.

Quality of life is assessed by asking individuals a set of questions about how satisfied they are with various aspects of their postinjury life. Generally, the greater is the communication skills and social participation, the higher is the expressed quality of life. The procedures and instruments for measuring quality of life in persons with aphasia described in [Chapter 4](#) are applicable to persons with TBI.

Quality of Life After Brain Injury (von Steinbuchel et al., 2010) is specific to persons with TBI. On this questionnaire, individuals respond to questions about their lives, including their remaining cognitive and physical problems, daily life and autonomy, and social relationships.

Quality of life may be reassessed at the time of follow-up meetings. Persistent problems may warrant professional services, including medical and psychological (or psychiatric) interventions for pain, depression, and other evident problems. Reinstatement of communication therapy may be considered if functional communication skills have not been maintained.

Overview of TBI Treatment Research

Clinical management of TBI is a team effort. In each case, the rehabilitation concerns change as the time passes. The initial concern is to stabilize the individual's physical condition and improve the chances of survival by offering emergency medical treatment.

Evidence on the effectiveness of specific treatments or rehabilitation of individuals with TBI is mixed. Cicerone et al. (2019) reviewed 121 studies, most of them with single-subject experimental designs, on cognitive rehabilitation for persons with acquired brain injuries. The reviewers concluded that the evidence supports intervention for (a) attention and social communication deficits, (b) compensatory strategies for mild memory problems, (c) social (pragmatic) communication deficits including metaphor interpretation, (d) computerized cognitive rehabilitation programs, and (e) self-monitoring and self-regulation therapies for impaired executive functions. However, there is no evidence for programs that teach recalling (a) decontextualized words, (b) names, (c) digits, and (d) paper-and-pencil based reasoning and cognitive tasks. Such interventions do not improve everyday living skills.

Cicerone et al. (2019) also concluded that effective components in individual, group, computerized, or traditional treatments include (a) client-selected treatment targets, (b) repeated treatment trials, (c) intensive within limits, (d) increasing the task difficulty as the performance improves, (e)

systematic feedback, and (f) simulation of real-life events and experiences. These translate to familiar behavioral treatment procedures of repeated discrete trials, skill shaping in gradual steps, positive reinforcement for correct responses, corrective feedback for incorrect responses, and individualized (personally meaningful) treatment targets that the clients themselves help select. Generally, interventions started sooner posttrauma produce a better outcome than those started much later.

On the other hand, several treatment research reviewers have generally concluded that the effects are at best modest, are often of small size, do not reliably generalize to everyday communication, and are typically not maintained over time, and most procedures are *promising* but not proven (Finch et al., 2016; Le et al., 2022; Messinis et al., 2019; Meulenbroek et al., 2019; Wiseman-Hakes et al., 2020). Investigators of individual studies generally fail to either measure long-term effects of their treatments or report lack of maintenance (e.g., see Rietdijk et al., 2022). Ponsford et al. (2014) reported that persons with a history of TBI exhibited limitations 10 years after they completed their rehabilitation programs. Nonetheless, all treatment reviewers support social communication treatment for individuals with TBI. The most fruitful and practical strategy for the clinician is to prioritize social communication training and integrate other goals (such as attention and memory) with it.

While lack of generalization, maintenance, and limited follow-up are real problems, treatment effects produced in specific studies cannot be dismissed. Treatment study reviews, too, are subjected to methodological limitations and personal biases. There is a tendency among treatment research reviewers, especially those who make meta-analyses of studies, to overemphasize the importance of large-scale randomized clinical trials and altogether exclude or undervalue single-subject experimental studies. Cicerone et al. (2019) included single-subject experimental studies but rated them lower than randomized clinical trials, a method riddled with limitations and contradictions (Hegde, 2007). The problem is compounded by the questionable levels of evidence hierarchy generally accepted (Hegde, 2010). The well-known Cochrane Database of Systematic Reviews is especially vulnerable to this criticism; reviewers generally limit their search to group randomized clinical trials. See, for instance, Kumar et al. (2017), who reviewed only the randomized clinical

trials and concluded that cognitive rehabilitation is ineffective in returning individuals with TBI to work.

Reviewers are unified, however, in issuing a call for better designed studies, more effective treatment strategies to produce outcomes that are functional (meaningful) to the individual, precise description of treatment procedures, systematic assessment of generalization, and long-term follow-up for maintenance. Treatment researchers should answer this call.

Specific Treatment Targets and Strategies

Taking a different approach to treatment review, Meulenbroek et al. (2019) have examined the most effective social communication *treatment elements* common to different procedures. Their review was restricted to clinicians' direct treatment in individual sessions. All direct treatment studies Meulenbroek et al. reviewed were behavioral and used the standard behavioral procedures including modeling, shaping, positive reinforcement, corrective feedback, repeated trials, multiple exemplar training, and so forth.

There is no single, comprehensive treatment procedure that the SLP can apply wholesale. Treatment research is about specific targets or procedures. Therefore, the SLPs have to build their own comprehensive treatment program for their clients. Based on the Meulenbroek et al. (2019) review and controlled treatment studies that have documented the effects of several treatment components, the clinicians may build a client-specific treatment program. The SLP may (a) select target communication skills that are functional to the person, (b) select stimulus items that are relevant to the person, (c) use discrete trials in the initial individual sessions, (d) target more naturalistic conversational format in the latter stages, (e) build some but not too much variety into the treatment tasks, (f) teach multiple exemplars of each target skill, (g) integrate cognitive training with communication training, (h) model and prompt the target skill to minimize errors, (i) shape the skills in gradual steps, and (j) reinforce correct responses and offer corrective feedback to incorrect responses.

The SLP may modify the following outline to suit the individual person and implement it:

- **Orientation to place, person, and time.** Ask skill-specific questions, model the target responses, and provide positive reinforcement.
 - Ask questions like the following: “Where are you now?” “Who am I?” “Is this morning or evening?” “What time is it?”
 - Model the answers if the person does not respond or responds incorrectly. For instance, soon after asking, “Where are you now?” model the response, “Say, I am in a hospital.”
 - After a few correctly imitated responses, withdraw modeling and prompt the correct response. For instance, after asking relevant questions, give such prompts as “This is not afternoon, but this is . . .” and “I am your speech . . .”
 - Reinforce correct or approximately correct responses.
 - Post written signs of the day of the week, date and month, and the name of the hospital. Frequently point them out to the person.
- **Attention to the surroundings and communication partners.** Use the following procedure.
 - Frequently draw the individual’s attention to surrounding events and people.
 - Often name the individuals involved in conversation or professional care.
 - Prompt the person to comment on the surrounding events and individuals.
 - Verbally reinforce any attempts at paying attention or saying something relevant.
- **Alternative or augmentative devices.** During the initial stages of rehabilitation, provide for needed alternative or augmentative (AA) assistive devices to promote functional communication.
 - Select a simple and effective device such as a communication board to help with basic communication needs.

- Fade the use of the device as the individual's oral communication improves.
- Continuously assess the need for AA devices and provide for them in later stages of therapy because of evident residual deficits.
- **Memory for daily routines.** As the individual's orientation and alertness improve, begin work on improving the individual's memory for daily routines.
 - Post written signs or the list of daily activities to remind the person of daily routines. Reinforce the person for frequently consulting the signs or the list.
 - Go over the list with the person and ask the person to describe the events or routines he or she is expected to follow.
 - Ask the person to write a list of daily activities from memory. Prompt the name and the correct sequence of the activities.
 - Ask the person to read the list silently and then describe the activities without looking at the written material. Prompt and reinforce the correct responses.
- **Memory for the names of individuals significant for the person.** Improve memory for names of health care staff and family members.
 - Request the family members to supply their pictures; showing the pictures, ask the person to name them; initially model or prompt the correct names.
 - During daily care activities, repeat the names of the health care staff and ask the person to name them. Prompt the names and reinforce the correct responses.
 - Have the person write the names of family members, caregivers, friends, colleagues, teachers, and significant others.
- **Comprehension of spoken language.** Increase the individual's attention to communication partners and topics to promote better comprehension of spoken language.
 - Before beginning a conversational exchange, draw the individual's attention by saying, "Listen carefully, now," "I want

to say something to you,” and so on.

- Repeat such statements throughout conversational exchanges as found necessary.
- Fade such statements as the individual's attention to the conversational topic improves; continue to reinforce the client for paying attention to conversation.
- Introduce new topics of conversation and avoid abrupt shifts. Tell the person that you are going to change the topic or will be talking about something different. Name the new topic or subject; repeat instructions to pay attention.
- Ask the person to restate what you just said, give the main points of a discussion, or summarize the conversation. To minimize errors, prompt and reinforce the correct responses.
- Reinforce behaviors that suggest comprehension or engagement in conversation. For example, reinforce eye contact, smiling, and nodding; however, frequently assess comprehension to ensure you are reinforcing comprehension.
- Teach the client to ask questions when something said is not understood. Give complex instructions or say something ambiguous and then model repair strategies for the person to imitate: “What do you mean?” “I do not understand that,” “Tell me more about that.” Reinforce the question-asking behavior.
- **Irrelevant, inappropriate, or tangential responses.** Withhold attention from such responses to decrease their frequency.
 - Do not respond, change the topic. Ask a simple question for which a more relevant response is likely.
 - Look away from the person for 5 seconds (time-out) to decrease irrelevant, inappropriate, and tangential behaviors.
 - Reinforce all desirable responses.
- **Speech production problems (dysarthria).**
 - Treat dysarthria and related speech production problems if the speech intelligibility is impaired significantly. Target all aspects

of speech production (respiratory, phonatory, articulatory, resonatory, and prosodic features) that may have to be modified. Consider postponing intervention if intelligibility impairment is mild.

- Consult other sources for details (Freed, 2020).
- **Narrative, discourse, and other social language skills.** Teach these skills in graded steps. Integrate such pragmatic skills as topic maintenance and topic initiation into teaching and practicing narrative and discourse skills.
 - To teach narrative skills, tell a brief and simple story and ask the person to retell it. Prompt the responses as well as correct temporal sequence of events. Increase the length and complexity of the stories told in gradual steps. Frequently assess comprehension of stories or episodes told to the person.
 - To teach discourse skills, ask the person to describe such events as planning a vacation, buying a gift for someone, or going on a camping trip. Prompt the words and names the person has difficulty with; reinforce correct narratives.
 - Ask the person to narrate personal experiences. Gradually reduce the frequency of prompting. Continue to reinforce acceptable responses. Withhold attention from inappropriate responses.
 - Ask the person to “Say more” or “Give details” to extend the narration on a personal or objective story. Give such hints as “What about this . . . ?” or “What about that . . . ?” to suggest missing details.
 - Prompt new topics for conversation to teach topic initiation. Fade such prompts as the person begins to initiate new topics for conversation.
 - Teach turn taking in conversation by ignoring interruptions and reinforcing the person for speaking at appropriate times in conversation. Use such verbal devices as “It is your turn” and “It is my turn.”

- Select the client-specific work or school-related vocabulary and expressions and make them a part of such skill training as discourse and narration.
- **Fade all special stimuli.** To promote generalization and maintenance, fade special stimuli that control behaviors.
 - Pair the written signs, prompts, and modeling with natural stimuli for behaviors: typical reminders, questions, and natural conversational exchanges.
 - Fade the written signs, prompts, and modeling while maintaining the behaviors with typical stimuli.
- **Self-monitoring skills.** To promote better generalization and maintenance of target skills, teach self-monitoring skills at all levels of training.
 - Teach persons to catch themselves when their attention wanders.
 - Teach persons to self-reinforce when they give correct, relevant, or extended responses (“I think I did well on that!”; “this time it was okay!”).
 - Teach them to stop talking when confused about something. Initially use a raised finger to signal the person to stop talking. Fade the finger signal as the clients become more proficient at terminating speech when confused.
 - Provide your clients a video of the treatment session and encourage them to view it with a family member at home. Alert the family member to reinforce the person for target behaviors shown on the video.

Teaching Compensatory Strategies

Simpler low- or no-technology alternative and augmentative communication (AAC) devices may be especially useful to individuals in acute stages. AAC approaches emphasize functional and effective communication with any mode

or means possible. During the hospitalized acute stage when verbal communication is difficult, individuals may gesture and point to pictures, verbal phrases, or icons (e.g., for food, drink, bathroom) posted on communication boards or displayed on computer screens.

AAC research on adults with TBI is disappointing. Children, not adults, have been the typical recipients of AAC. Very few treatment studies have been made on individuals with TBI. In a review of 12 studies on AAC for adults with complex communication needs, Andzik and Chung (2021) found that just one study had one participant with TBI. In their review of AAC research with TBI, Brunner et al. (2017) found that cognitive rehabilitation, not communication skills, was the primary target. Advocacy and informational papers predominate (Manzoor & Vimarlund, 2018). Furthermore, most digital devices and applications written for persons with TBI include basic requests (mands), not more complex conversation or social communication (Tincani et al., 2020). Treatment research has not demonstrated systematic, reliable, and clinically significant improvement in communication when persons with TBI have used AAC devices; most studies do not meet the evidence quality criteria (Andzik & Chung, 2021; Brunner et al., 2017; Tincani et al., 2020). There are no clinically verified guidelines on how to promote the use of AAC by individuals with TBI (Brunner et al., 2017).

Many individuals with TBI have expressed frustration with high-technology AAC instruments that are difficult to use and nonfunctional. Individuals may prefer the more mainstream devices (e.g., computers and smartphones to send emails or read news) and social media platforms to interact with other persons (Paterson & Carpenter, 2015). There is the theoretical expectation that social media may help individuals connect with other persons (with or without TBI) and enjoy social participation, but there are no data on the effects of social media use on meaningful communication skills (Brunner et al., 2021). A study that surveyed smartphone use revealed that individuals with TBI used their smartphone mostly as an aid to memory (e.g., keeping appointments) and organization. The study also reported that 90% of those with TBI had not received any training from the clinician on using the smartphone for any specific communication purposes (Wong et al., 2017). Prior smartphone users typically find it difficult to use their device after the brain injury but may not receive training.

Lack of sufficient quality evidence does not mean that AAC is ineffective in meeting the communication needs of individuals with TBI. It only means that more systematic research is needed. Though weak, the available evidence on the effectiveness of AAC training in individuals with TBI is positive, which may be interpreted as justification for using AAC. Therefore, SLPs may make clinical decisions based on an individual's severity of communication impairments and need for an external aid. Smartphones have been reported to be popular with individuals with TBI, although its capabilities may be somewhat limited (Brunner et al., 2017). The clinician can select one or more software programs and load them on the device and train the person on their use (Beukelman et al., 2007; Brunner et al., 2017). The clinician may consider the following guidelines:

- The individual should help select communication targets, devices, and the applications that are functional.
- Selected device should be simple to use, affordable, reliable, adequate to meet the needs, and capable of expansion as the individual's skills improve.
- The required skill to use the device may be currently low, but it should be teachable relatively quickly in motivated individuals.
- The AAC may be completely nontechnological, such as a communication board, a picture book, writing on a piece of paper, systematic (trained) use of gestures, and manual signs such as those of the American Sign Language.
- The treatment team members should adequately train and consistently support the individual. The caregivers must know how to use the approach and the device.
- Repeated assessments must be made to determine if a more complex device, such as a tablet or a laptop computer that is more capable, may be tried.
- Online texting (e.g., emails) may give more control to individuals on what is communicated; they may edit and refine their messages before sending them off; individuals may have been competent computer users before TBI.

- The application loaded on to the device should prompt or notify appointments, messages received, and things to do at certain times. Applications may help learn new skills, manage personal affairs, and so forth. The needs of the individual and the plans for future will help select the applications.

After selecting a suitable device and an application for it, the clinician may teach the individual to use the device effectively. The individual may not learn to use all the functions of the device immediately, but the most basic ones may be learned as soon as possible.

The clinician may also teach the individual all strategies to communicate with the caregivers, including family members:

- **Write down steps involved in performing daily activities.** For example, the client may write down, with help, if necessary, the directions for fixing a breakfast or washing clothes.
- **Request information.** The person may learn to ask about time or date and request clarification of unclear statements.
- **Request others to modify their speech.** The person may learn to request others to speak slowly, repeat statements, and give background information.
- **Rehearse important information.** For instance, the person may first self-talk about how to perform such activities as squeezing juice from oranges.
- **Write down important information.** The individual may write down instructions, appointments, daily schedules, medications, directions to shopping centers, and phone numbers.
- **Request written instructions.** The person may ask for phone numbers, appointments, instructions, and directions in writing.
- **Use electronic devices.** The person may learn to use (a) digital watches that display time and date and signal appointment times and important actions with alarms; (b) databank watches that store messages and appointments; (c) electronic pill boxes that have alarm reminders for taking medications; (d) devices such as smartphones or

tablet computers to record lectures, directions, and even discourse; (e) handheld electronic spell checkers; (f) notebook computers for more complex information management and electronic communication; (g) any other assistive device that may be useful in compensating a specific residual deficit.

- **Keep possessions in specific places.** For instance, the person may have an invariable place for car and house keys, pens, shoes, and jackets.
- **Use strategies of self-cueing.** For example, clients may learn to use alarms as reminders of appointments and notes placed at prominent places in the house as reminders of activities.

The reader may also consider devices and applications described in [Chapter 5](#) for persons with aphasia. Most of them designed for persons with aphasia, summarized in the same chapter, may be useful for persons with TBI because of common communication needs.

Some individuals with TBI may have swallowing disorders that need to be addressed. Dysphagia management is not addressed in this chapter. Therefore, the clinician should consult other sources (Leonard & Kendall, 2019; Ward & Morgan, 2009). The Ward and Morgan text specifically addresses swallowing problems associated with brain injury.

Skill Maintenance Program

A carefully designed and implemented program is essential to promote skill maintenance in natural settings. Research on maintenance of clinically established skills is limited. Skill maintenance requires additional procedures, not just measurement. Based on general treatment research and research on maintenance (Hegde, 2018), the clinician may consider the following steps in promoting maintenance:

- **Teach personally significant skills.** Communication skills that are functional, and those that are socially, educationally, and

occupationally useful, are better maintained.

- **Train the significant others.** Teach persons who regularly interact with the client to prompt, model, and reinforce desirable behaviors to sustain them in natural environments.
- **Shift training to more natural settings.** Move treatment out of the therapy room and into more informal settings, including the client's home, classroom, or workplace.
- **Use natural reinforcers.** Verbal praise, smile, and informative feedback (e.g., a chart showing progress made in treatment) are effective. Work with family members to obtain natural and valuable consequences (e.g., a desired gift that the family can provide for certain skills maintained at home).
- **Teach self-monitoring skills.** To promote greater independence, teach the client to count appropriate and inappropriate behaviors in treatment sessions and to make entries in a notebook of successful or failed attempts at communication in natural settings.
- **Arrange for follow-up and booster treatment.** Measure response maintenance during follow-up sessions and offer booster treatment if necessary.

Group Therapy

Group therapy is advocated on the assumption that it provides social support, helps dispel feelings of social isolation, creates a structure for prompting and reinforcing conversational skills, offers an opportunity for self-evaluation of strengths and limitations, promotes generalization and maintenance, and so forth. By reintroducing the person with TBI to a community of similar individuals, group sessions may help ease the person's anxiety about transition into community reentry. Finally, group treatment is more cost-effective than individual therapy.

Other than its cost-effectiveness, most of the likely advantages of group treatment benefits have remained intuitive and theoretical. There is only limited evidence that group treatment may improve communication skills of

persons with TBI, but there is little or no evidence that the improvement generalizes to everyday communication (Keegan et al., 2019).

To complicate it for the clinician, there are no practical protocols for conducting group treatment sessions. The actual treatment variables in group treatment studies are described in global and vague terms, making it difficult for the practitioners to replicate the variables. Even though a “treatment program” is reported to be effective, which treatment components are effective may not be clear to the clinician. Furthermore, whether treatment was indeed applied as described (or as intended, because the descriptions are poor) has not typically been assessed in study reports (Behn et al., 2019, 2021; Keegan et al., 2019). Often, the outcomes of group treatments are not reported in terms of frequency of operationally defined behaviors during baseline, treatment, and posttreatment. Instead, ratings (e.g., *expected*, *better than expected*) and standardized test scores are reported as evidence of improvement (Behn et al., 2019). These are subjective judgments the clinicians would not know how to evaluate.

Treatment target skills continue to be described in such general terms as *a variety of communication skills*, *a range of cognitive skills*, and so forth (Behn et al., 2019). Clinicians would like to see changes in frequency of behaviors (e.g., the number of conversational turns taken, number of topics initiated), which are often not measured at all. Generally, the clinician is expected to play a supporting, not a directing, role within the group. But in practice, group treatments often regress to clinician-directed procedures in which specific deficits become treatment targets, just as in individual therapy sessions (Kovarsky et al., 2007).

Not all individuals with TBI at every stage of their recovery benefit from group therapy. Therefore, group therapy candidacy must be carefully assessed. Individuals who are still confused or disoriented are not good candidates for group therapy. Group treatment may be most helpful if the probability of appropriate communicative behaviors occurring in the group setting is good so that such behaviors may be naturally and socially reinforced. If such behaviors have to be taught, then individual sessions are most effective.

To document improvement of communication in the group format, SLPs should measure the frequency of behaviors they target for group intervention,

although this is one of the several challenges of group therapy. It is difficult to reinforce and measure behaviors of multiple individuals. Group therapy with no reinforcement for social communication skills is not only of questionable value but also may unravel the skills established in individual sessions.

In setting up and conducting a practical, specific skill-oriented group therapy, the clinician may:

- **Consider *when* and *what* format:**
 - Transition from individual to group sessions when targeted social communication skills are stabilized (better than 75% accuracy, or another defined criterion).
 - Form a smaller, rather than larger group; decide on the number of individuals whose speech you can effectively monitor (measure, prompt, reinforce, or offer corrective feedback).
 - Individualize the target skills (e.g., one person's target may be quiet listening, another person's may be topic initiation).
 - Remind or model skills, if necessary, but keep these to a minimum; do not use a discrete trial-like format to teach specific skills in a group.
 - Moderate but do not direct too much; do intervene when someone begins to ramble or make inappropriate comments.
 - Let the group members describe their target skills so everyone understands each other's targets.
- **Target operationally defined social communication skills:**
 - Target measurable conversational skills.
 - Target conversation about the day's events.
 - Target narratives (storytelling) and discourse to teach such skills as turn taking, topic initiation, topic maintenance, listening to other speakers, and so forth.
- **Use appropriate intervention techniques:**
 - Ask a person to initiate one new topic of conversation in a 30-minute group session; gradually increase the number.

- Ask the speaker to maintain the same topic for a specified time (e.g., 2 minutes); use a timer; increase the duration gradually.
- Encourage group members to ask for clarification (conversational repair) each time when someone's speech is not understood.
- Teach each person to use a different set of words, phrases, or sentences when someone asks for clarification.
- Teach members to listen but not interrupt (zero interruption).
- Train persons to say, "It is your turn" after talking for 1 minute (use a buzzer or a hand signal).
- Train persons to begin talking when a group member says, "It is your turn," and talk only for 1 to 2 minutes (a buzzer or a hand signal to end talking).
- Teach each member to narrate a brief story or a personal experience with correct chronological sequence for no more than 3 minutes (or any such time limit).
- Teach them to tell the group how they might book a hotel room or an airline ticket in 3 minutes.
- Invite each person to make comments on other member's story narration, discourse, or other kinds of speech.
- **Reinforce and give corrective feedback:**
 - Give subtle, natural, social consequences (smile, approving head nod, thumbs-up gestures, such verbal statements as *I agree, that is true, I think so*, and so forth).
 - Give subtle and natural corrective feedback (a hand gesture to stop talking, or such verbal statements as *please let someone else talk now*).
 - Teach the group members to reinforce each other (e.g., *you talked exactly for 2 minutes! Thank you for giving me a chance to talk! Thank you all for not interrupting me! I liked your story!*); count the reinforcing comments each person makes in each session.
 - Record the session and measure the frequency of behaviors.

SLPs may use the format to target other social communication skills. They may operationally define the targets and treatment procedures so as to measure the frequency of targeted communication skills and to document improvement.

Promoting Community Reentry

Rehabilitation of individuals with TBI should end with a program of community reentry. Research suggests that social communication skills are a key element of successful community reentry, especially returning to work (ANCDS TBI Writing Committee et al., 2022). Excessively slow speech, slow reaction time, difficulty understanding instructions and directions, abrupt topic switches, difficulty appreciating the tone of voice, and such other social communication impairments impede successful occupational transition (Douglas et al., 2016). Before a reentry plan is developed, the clinician may reassess the client's social communication skills and address any remaining deficiencies in individual or group therapy sessions.

To facilitate a relatively smooth transition from rehabilitation settings to home, school, and office, the clinician needs to prepare both the person and the people in the natural environment for reentry. The SLP may design an individualized program based on the skills that are needed to reenter the social, educational, or occupational world and succeed in it. The SLP may:

- **Assess the reentry needs.** Consider whether the person plans on returning to work or education; if retired, consider the needs for socialization and family integration; find out about the family members with whom the individual expects to interact the most (e.g., spouse, children, grandchildren, other relatives); ask the person about friends and former colleagues with whom social interactions are desired; learn about the types of such interactions (e.g., meet for lunch and talk; play cards or golf; take a walk together).
- **Let the client help develop the plan.** Select targets and general approaches the client and the family members desire. Suggest important elements the client or the family members miss.

- **Prepare the person for reentry.** In the final stages of treatment, emphasize self-help skills and independent living skills. Discuss the individual's interest in academic, professional, social, and home activities. Incorporate discussions of such activities into discourse training.
- **Target literacy skills.** If such skills as reading, writing, emailing, and other literacy activities are important, provide practice in the final sessions. Have the person write a daily work schedule or *to do list* and ask the person to describe the activities on the list.
- **Develop a list of work-related terms.** If return to work is planned, have the person make a list of occupational terms and activities, and describe them to the members of the group. Reinforce correct and coherent discussions and give corrective feedback on missing details and poor organization of descriptions.
- **Role-play.** Role-play any social role the client does not seem to perform well. Social greetings and asking about how others are doing may be especially useful to role-play.
- **Re-create important scenes.** Try to re-create scenes as much as possible and have the person play the expected role. For instance, the client might call the clinic in and have a phone conversation; call a family member, friend, or colleague as you observe the conversation; if practical, the clinician may have lunch with the client in the rehabilitation facility's cafeteria and use the opportunity to promote social communication skills.
- **Educate others.** Update family members, teachers, and supervisors on what the person can and cannot do, the kinds of compensatory strategies the person uses, and the kinds of support needed.
- **Teach others communication strategies.** Teach family members, teachers, colleagues, and supervisors the strategies the individual needs. For instance, they may have to speak slowly, speak in simple sentences, repeat instructions, give instructions in writing, post signs, remind about appointments and tasks, and prompt to speak or do something.

- **Teach others to minimize the demands.** For instance, the teacher may have to give reduced amount of homework or extra time to take a test; the supervisor may assign new tasks only when the individual signals readiness.
- **Let the family understand oppositional behaviors.** Educate the family members and others to recognize reasons for oppositional behaviors so they can take appropriate actions. For instance, many oppositional behaviors may be due to difficult task demands; if the task is simplified, more appropriate behaviors may follow.

Speech-language pathologists may consider other suggestions outlined in ANCDs TBI Writing Committee et al. (2022) for how they can help with a vocational reentry program involving other rehabilitation professionals, especially occupational therapists.

Cognitive Rehabilitation

Cognitive rehabilitation (CR) of persons with TBI has attracted much research attention. CR is a collection of multiple methods to improve attention, memory, learning, problem solving, executive function, visual scanning, affect (comprehension and expression of emotions), and social communication (Cicerone et al., 2019; Messinis et al., 2019; Schutz & Trainor, 2007). Research reviewers have found that CR is variably effective in treatment sessions, but the evidence for generalization to natural settings and long-term maintenance of treated skills is limited and weak (Cemich et al., 2010; Cicerone et al., 2019; Institute of Medicine, 2011; Koehler et al., 2012; Kumar et al., 2017; Messinis et al., 2019). A Cochrane Database Review of CR and TBI concluded that there is insufficient evidence to conclude that (1) CR is better than no treatment, (2) CR is better than non-CR (conventional) treatment, (3) some types of CR are better than other types, and (4) CR promotes independent living skills, community integration, or return to work (Kumar et al., 2017). Nonetheless, most reviewers recommend CR and call for more rigorous research to better document its effects (Cicerone et al., 2019; Messinis et al., 2019).

Most cognitive training programs psychologists and other professionals implement do not target direct and extensive communication training. Of late, all forms of speech-language therapy SLPs offer to adults with neurological disorders is described as *cognitive communication training* with no particular advantage derived from that terminological change.

Pure cognitive training may target attention, visual processing, and memory drills. Training cognitive skills without the contextual communication skill training is of questionable value. For instance, training the person to push a button every time a number is heard or count numbers backwardly may not result in meaningful gains in personal, social, or communication skills. Similarly, visual processing training that includes copying geometric shapes from memory or scanning an array of symbols to match a target may not have any effect on communication. Memory training drills to remember letters, numbers, pictures, geometric shapes, or isolated words also may not affect functional communication.

Teaching and stabilizing communication skills may also seem to improve memory (Marquardt et al., 1990). For instance, a person who keeps an appointment because of training received to use an alarm watch is said to have improved *memory* for such tasks. Attempts to improve attention may be better handled in the context of communication training. For example, in treating such functional communication skills as maintaining eye contact during conversation, reading printed material, and talking on a topic for an extended duration, reinforcement may be made contingent on paying attention to the task on hand for increasingly longer durations (Giles & Clark-Wilson, 1993). Paying attention to instructions, suggestions, modeling, and various treatment tasks maybe highlighted and frequently reinforced.

Gilmore and colleagues (Gilmore et al., 2019, 2022) have researched a method they call the Intensive Cognitive and Communication Rehabilitation (ICCR) for college students with a history of TBI. ICCR targets clearly defined behaviors (e.g., listening to a lecture, taking notes, studying for quizzes, and answering discussion questions). Individuals were recruited for an immediate treatment and a deferred treatment control group. The treatment group members watched videorecorded classroom-style lectures on four college-level courses, attended group and individual therapy sessions, and

received computer- and application-based training (see the two reports for procedural details). In this intensive training program, the individuals attended 6 hours a day, 4 days a week, for at least one semester (12 weeks). The treatment significantly improved the targeted skills. The measures were test scores on aphasia standardized tests. No target behavior frequencies were measured before and after treatment, a common drawback of many CR studies. If this could be modified and adapted to clinical practice, more clinicians may use this desirable program for training functional communication targets in naturalistic settings. The modification should include behavioral frequencies before, during, and after treatment and a clearer specification of contingent consequences for the target behaviors. A less intensive but equally effective program may also be more practical.

CLASS NOTES

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PART V

THE DEMENTIAS



10

THE DEMENTIAS: PREVALENCE, CAUSES, AND TYPES

Chapter Outline

- Decreasing Incidence and Increasing Prevalence of Dementia
- Prevalence of Dementia in Varied Populations
- What Is Dementia?
- Reversible and Rapidly Progressive Dementias
- Dementia of the Alzheimer's Type
- Vascular Dementia
- Dementia With Lewy Bodies
- Frontotemporal Dementia
- Parkinson's Disease
- Huntington's Disease
- HIV-Associated Neurocognitive Disorder
- References

Learning Objectives

After reading the chapter, the reader will:

- Define dementia and describe the characteristics of various forms of dementia
- Discuss the epidemiological and ethnocultural variables that affect the dementias
- Describe the neuropathology of the dementias and distinguish one type from the other
- Specify the behavioral, cognitive, and communication characteristics of individuals with different types of dementias

PRINTED NOTES

A general decline in intellectual functions caused by neurological diseases, dementia is a major health problem affecting mostly persons 65 years and older. The Centers for Disease Control and Prevention estimated that in the year 2020, 5.8 million Americans had dementia, mostly due to Alzheimer's disease. The number is expected to increase to 14 million by the year 2060 (<https://www.cdc.gov/aging/aginginfo/alzheimers.htm#burden>).

Generally, the higher the age group, the greater the prevalence of dementia. After age 65, the prevalence doubles every 5 years. The prevalence rate for people in their 60s is less than 1%, 14% for those 71 and older, and 30% for those 85 and older (Robinson et al., 2017). Nonetheless, dementia is not a part of normal aging. Older age is a risk factor only because it is vulnerable to neurological diseases that cause dementia.

Decreasing Incidence and Increasing Prevalence of Dementia

Incidence is the percentage of a healthy population in whom annually a disease or a disorder is likely to develop. Wolters et al. (2020) analyzed aggregated data on the incidence of dementia between 1988 and 2015 in individuals 65 years and older living in the United States and some European countries and found a 13% decrease over those 27 years. Langa et al. (2016) found that in the year 2000, 11.5% of the older U.S. population had dementia, but in the year 2012, the percentage had dropped to just about 9%.

The reported decline in the incidence of dementia in the United States and other high-income countries in the past 25 years is attributed to rising levels of education and better treatment for cardiovascular diseases. In the 1970s, 12% of 30- to 34-year-olds had a college degree in several high-income countries,

whereas in 2010, 27% of them had a college education. The increase was from 3% to 11% for the same age group in lower-income countries (Langa, 2015). In 2019, 5% of individuals 70 years or older with a college education in the United States had dementia, compared with 18% of individuals in the same age group who had less than 12 years of education (<https://www.prb.org/resources/fact-sheet-u-s-dementia-trends/>). Higher education and strong and sustained intellectual activities are hypothesized to offer some degree of protection against dementia.

The **prevalence of dementia** — the number of persons who have a disease or a disorder at a given time — has not declined, however. Lower incidence does not translate to lower prevalence. In fact, the prevalence is expected to rise significantly. The World Health Organization (WHO) estimated that in 2020, there were about 55 million people with dementia in the world. Sixty percent of people with dementia live in low- to middle-income countries where longevity is on the rise. WHO estimates that this number will rise to 78 million by 2030 and may reach 139 million in 2050 (<https://www.who.int/news-room/fact-sheets/detail/dementia>).

The same variables that improve the health of persons will increase longevity and, with it, an increased risk for dementia. Prevalence of dementia will keep rising as the economic conditions of people worldwide improve and, as a consequence, more of them live longer. Other factors that might mitigate a declining trend in the incidence of dementia is the rise in several risk factors. For instance, obesity, hypertension, high cholesterol, lack of physical activity, and diabetes — all risk factors for dementia — have been on the increase in most countries, especially in the United States. Better management of these risk factors, along with higher levels of education, may counteract to some degree the potential runaway increase in the prevalence of dementia. The news may be relatively optimistic only for those individuals who take serious steps to receive more education, access prompt treatment for medical risk factors, eat healthier diets, and engage in regular physical activity and exercise.

Prevalence of Dementia in Varied Populations

Data on ethnocultural factors that affect the prevalence of dementia are limited, as well as difficult to interpret. Differential prevalence of dementia in ethnically varied populations may not be due to genetic factors. The health status of individuals and lifelong accessibility to health services may be critical variables. Dementia risk increases with an increased number of multiple cortical infarcts, and such infarcts are associated with the general health conditions of individuals. Ethnic minority people's unaffordability of treatment for diabetes, high blood pressure, and high levels of low-density lipoproteins ("bad" cholesterol) may increase the chance of the major, minor, and repeated strokes that are associated with dementia risk (Armstrong, 2019). Traumatic head injury, more common among African Americans and non-White Hispanics, is an additional risk factor for later dementia. Furthermore, lower levels of education and socioeconomic status, two risk factors, disproportionately affect certain minority groups. The absence of such potential protective factors as estrogen replacement therapy for non-White women, delayed administration of anti-inflammatory drugs, and antioxidant-poor diet may also increase the risk for dementia in minority populations (Payne, 2014).

Research generally suggests that Alzheimer's dementia (AD) is about two times more common among African Americans and Hispanic adults than among non-Hispanic Whites in the United States (Arvanitakis et al., 2019). It is also known, however, that such dementia risk factors as high blood pressure and diabetes are more common among African Americans and Hispanics than among non-Hispanic Whites. The prevalence rates of dementias among Asian Americans are comparable to those among the non-Hispanic Whites (<https://www.prb.org/resources/fact-sheet-u-s-dementia-trends/>).

Cultural and social factors — and factors that affect health care access — may influence the accuracy of counting people who have dementia. In certain cultural groups, early signs of dementia may be thought of as the inevitable result of the aging process. More non-White older persons than comparable White persons receive care at home and, therefore, may not be counted in prevalence studies, leading to an underestimation of dementia in non-White populations (Payne, 2014).

What Is Dementia?

Dementia is a significant decline from an existing level in personal and social behavior commonly associated with neurological diseases in older persons. The disease affects brain functions, and the decline is typically described in cognitive (intellectual) terms. The American Psychiatric Association's (2013) fifth edition of the *Diagnostic and Statistical Manual of Mental Diseases (DSM-5)* renamed dementia as a *neurocognitive disorder*, although the term dementia is still current, and the new name is not especially informative. *DSM-5* considers impaired memory only one of several possible dysfunctions, not a mandatory feature for diagnosis. Decline from a previous level may be evident in one or more skills, including learning and memory, complex attention, executive functions, language, perceptual-motor skills, or social skills (*social cognition*, in the *DSM-5* terminology). Decline may be documented through various assessment tools and concerns the individual, the caregivers, or a professional express. To diagnose dementia, the impairments should significantly affect everyday living skills, including self-management. The decline should not be attributable to delirium or a mental disorder (e.g., schizophrenia, serious depression), transient confusion due to acute cerebral trauma, metabolic disorders, and toxicity. Dementia is distinguished from intellectual or developmental disabilities diagnosed in children by its later pathology and the prior normal intellectual function unless its onset is in persons with intellectual and developmental disabilities.

DSM-5 also classifies dementia into *major neurocognitive disorder* and *minor neurocognitive disorder*. The two categories may share common etiologic factors (such as Alzheimer's disease), and a milder disorder in the early stages may become a major disorder in later stages. A minor or major disorder may be due to a treatable condition (e.g., drug toxicity).

Typical forms of dementia progress slowly, are irreversible, and end in death. There are exceptions, however, as noted in the next section. The early symptoms may be mild as well as difficult to identify. A decade or more may pass before the symptoms become obvious and disabling. The early symptoms of dementia — characterized by the self-reported decline in memory that is documented by assessment results, intact reasoning skills, and unimpaired self-

help and daily living skills — are called **mild cognitive impairment** (MCI) of the *amnestic variety*. When other cognitive skills, including executive functions or language, are impaired, also to a mild degree, it may be called MCI of the *nonamnestic variety*. It is estimated that 10% to 20% of people who are 65 years or older and have MCI may develop dementia (<https://www.nia.nih.gov/health/what-mild-cognitive-impairment>).

A more diffuse cerebral pathology in dementia distinguishes it from aphasia. Also, the typical onset of persistent dementia is gradual, whereas that of aphasia is more acute. In addition, dementia includes deterioration in most aspects of intellectual functioning; aphasia does not. See Appendix B in [Chapter 4](#) on features that distinguish dementia from aphasia.

Reversible and Rapidly Progressive Dementias

In a small number of cases, cognitive impairments that mimic the symptoms of dementia may be **reversible** through medical treatment; the onset tends to be faster than that in typical neurodegenerative diseases. In other cases, even the irreversible dementias that are known to have a slower progression may rapidly progress to death (Anuja et al., 2018; Hermann & Zerr, 2022; Milta & Little, 2018).

Estimates of prevalence of reversible dementia vary greatly, from as low as 5% to as high as 40% (Milta & Little, 2018). Early and effective treatment is critical to reverse the intellectual and behavioral deterioration. Nutritional deficiencies including vitamin B1 and B12 deficiencies, metabolic disorders, and low sodium and low or high calcium may lead to symptoms of dementia. *Uremic encephalopathy*, a form of dementia due to chronic renal failure, has been an established diagnosis. Long-standing lung disease, cardiac disease, and anemia may cause chronic anoxia (oxygen deficiency) that may lead to postanoxic dementia. In some cases, prolonged dialysis may lead to dementia.

Such endocrine disorders as hypothyroidism, hyperthyroidism, and hyperparathyroidism may cause dementia-like symptoms in some cases. Depression also may lead to symptoms similar to those seen in dementia.

These conditions may be treated to reverse the associated dementia-like symptoms.

Reversible dementia-like symptoms may be prescription drug induced. Drugs used to treat mental disorders (e.g., lithium carbonate, used to treat manic-depressive disorder, and tricyclic antidepressants), known as *psychotropic agents*, can cause dementia-like symptoms in some individuals. Anticonvulsants (used to treat seizure disorders) and some antibiotics (e.g., penicillin) may produce similar symptoms. Toxic metal exposures, including lead and mercury, are known to result in symptoms of dementia. Alcoholism is another toxic factor that may induce dementia-like symptoms. Various infections also can cause symptoms of dementia. Some kinds of untreated or not promptly and effectively treated infections (e.g., syphilis) may lead to such symptoms.

Rapidly progressive dementias are defined as those that take less than a few months up to 2 years from symptom onset to dementia (Anuja et al., 2018; Hermann & Zerr, 2022). Metabolic encephalitis may rapidly progress to dementia within weeks of initial symptom onset. Both reversible (treatable) and irreversible types may progress rapidly. In their 8-year retrospective study, Anuja et al. (2018) found three groups of patients with rapid progression. The first group had treatable dementia mostly due to infections (e.g., HIV and neurosyphilis induced) and those due to nutritional, metabolic, and thyroid disorders. The second group had nonreversible infectious dementia associated with the Creutzfeldt-Jakob (prion) disease, which is rare in the general population (Hermann & Zerr, 2022). The third group also had irreversible dementia associated with neurodegenerative diseases (e.g., frontotemporal, Alzheimer's, and vascular or poststroke dementias). In most cases, the symptoms progressed within 2 weeks to a few months. The otherwise rare Creutzfeldt-Jakob prion disease incidence has declined further because of better infection control. **Prion** is a type of protein in the brain that gets misshapen. There is no treatment for the disease. Rapid behavioral decline terminates in death.

Compared to individuals with typical dementia of the neurodegenerative type, rapidly progressive dementias affect younger people and lead to death sooner. A history that suggests accelerated decline in behavior in younger individuals alerts the specialists to possible rapidly progressive dementia.

Inadequately treated epilepsy also may lead to rapid decline in cognitive functions (Miltner & Little, 2018). Although more complete data are not yet available, some individuals infected with COVID-19 that has resulted in delirium and encephalopathies may exhibit symptoms of dementia (Hermann & Zerr, 2022).

Dementias that produce the greatest negative consequences for individuals and their families over a much longer period are all progressive. Among these forms, dementia due to Alzheimer's disease is the most common. Vascular dementia and Lewy body dementia are, respectively, the second and the third most frequently noted irreversible forms. Additionally, frontotemporal dementias, Parkinson's disease, Pick's disease, Huntington's disease, HIV infections, and traumatic brain injury (TBI) in some cases are all associated with irreversible types of dementia. Because the Alzheimer's type is a prototype of dementias, we will consider it in some detail and then summarize the main features of other forms.

Dementia of the Alzheimer's Type

Dementia of the Alzheimer's type (DAT) has been known since late 1907, when the German neuropsychiatrist Alois Alzheimer (1864–1915) described a 51-year-old woman in the Frankfurt asylum for the insane whose symptoms included jealousy toward her husband, rapidly deteriorating memory skills, reading and writing problems, paraphasia, disorientation in her own apartment, and various symptoms of paranoia (Bondi et al., 2017). With no motor dysfunctions, her gait, coordination, and reflexes were normal. She died after 4 and a half years of sickness. An autopsy revealed extensive cortical cell loss and what later came to be called neurofibrillary tangles and neuritic plaques. To Alzheimer's surprise, his mentor, the eminent German psychiatrist Emil Kraepelin, coined the term *Alzheimer's disease* in 1910.

Incidence and Prevalence of DAT

In the United States, about 5.8 million people live with DAT. Of this total number, 17% are in the range of 65 to 74 years, 57% are in the 75- to 84-year range, and 36% are in the 85+-year range (Alzheimer's Association, 2020). Globally, 5 to 7 million older people are *annually* diagnosed with AD. The global incidence is estimated to be 5% to 7% of populations in different countries; the greater the number of older individuals in a society, the higher the prevalence of AD and other forms of dementia (Lopez & Kuller, 2019).

Incidence continues to rise from age 65 until 85. After age 85, especially in the 90s, the incidence is reported to plateau or even decrease to a small extent (Robinson et al., 2017). An autopsy study of a large number of individuals in the age range of 70 to 100 showed that 82.9% had Alzheimer's disease, making DAT the most common form of dementia. Alzheimer disease pathology was observed in an increasing number of cases from age 60 to 90, but there was a small decline after 90. However, the incidence of mixed dementia, in which the Alzheimer's disease was combined with vascular disorders, showed a continued increase throughout the ninth decade of an individual's life. In the oldest-old age group (90 years plus), the most common form of dementia was the mixed type (Jellinger & Attems, 2010).

Sixty-five years and older is the typical age of onset of DAT, but there is an early onset form of it that may begin in persons as young as 30 years. It has a strong genetic component, as noted in the next section. The early onset DAT is more commonly associated with a family history of Down syndrome. Older persons with Down syndrome show brain morphologic changes associated with DAT. Down syndrome is caused by an extra copy of chromosome 21, the same chromosome that also carries a DAT susceptibility gene.

More women than men are diagnosed with DAT, but this is not due to a higher incidence among women but because of their longevity, which puts them in older age groups that are at greater risk.

DAT Risk Factors

The causes of DAT are not known precisely, but research strongly suggests several factors that increase the chances of developing the disease (Breijyeh & Karaman, 2020; Milta & Little, 2018; Robinson et al., 2017; Scheltens et al.,

2021). Armstrong (2019) grouped the risk factors into several categories as summarized:

- **Demographic factors.** These personal factors increase the DAT risk.
 - Age, which is by far the greatest risk factor for DAT
 - Lower level of education, especially 12 years or less
 - Lower level of intellectual (cognitive) activity
 - Simpler and intellectually less challenging occupations
 - Race and lower social class, not to be misinterpreted (see the section on prevalence of dementia in varied populations)
 - Hearing and vision problems
- **Genetic factors.** The prevalence rate among the first-degree relatives of individuals with DAT is about 50%. Genetic mutations lead to *amyloid plaques* that are the essential neuropathological condition of DAT (Scheltens et al., 2021). Genetic testing may reveal the presence of mutated genes but does not help predict the onset of DAT in an individual. Genetic variables may increase or decrease the risk.
 - A genetic variant of apolipoprotein E (APOE) on chromosome 19, known as APOE $\epsilon 4$, increases the DAT risk. The APOE gene makes a protein that helps move cholesterol and fats through the bloodstream. But APOE $\epsilon 4$ inheritance does not necessarily mean the person will have DAT, and many persons with DAT do not have that variant.
 - Another variant, APOE $\epsilon 2$, is a rare gene that may offer some *protection* against DAT.
 - Mutations affecting the gene for amyloid precursor protein may be evident in chromosomes 21 and 14, resulting in an early onset dementia.
 - Abnormalities of genes on chromosomes 1, 3, 10, 12, and 17 have also been suggested.
 - Trisomy 21, which causes Down syndrome, also is a risk factor. Trisomy 21 is a genetic condition in which three free copies of chromosome 21 are involved. Alzheimer's disease in people

with trisomy 21 is likely to manifest in the fourth decade of life.

- The early onset DAT is associated with three single gene mutations: (1) amyloid precursor protein (APP) on chromosome 21, (2) presenilin 1 (PSEN1) on chromosome 14, and (3) presenilin 2 (PSEN2) on chromosome 1.
- The diseases may be evident in one identical twin but not the other, suggesting the influence of environmental factors.
- **Lifestyle variables.** These are the most important *modifiable* DAT risk factors.
 - Abuse of alcohol may increase the risk of DAT (and other forms of dementia).
 - Lack of exercise might increase the risk and regular exercise may decrease it.
 - Lack of cognitive activity (possibly related to lower education, intellectually less challenging occupation, and general lack of interest in cognitive activities) increases the risk and higher cognitive activities may decrease it.
 - Poor diet that includes the consumption of excessive saturated fat and highly processed as well as high-sodium food may increase the risk; the Mediterranean diet with more fruits, vegetables, and fish may decrease it.
- **Medical variables.** Several health and disease factors may increase the DAT risk.
 - Strokes and multiple micro-infarcts are especially potent risk factors.
 - Poorly controlled hypercholesterolemia and type 2 diabetes may affect the brain health.
 - Obesity causes metabolic changes that affect the brain health, resulting in nerve cell death (*necrosis*).
 - Cardiovascular diseases may be an independent risk factor for dementia, but their effects tend to combine with AD pathology to produce mixed dementia.

- Traumatic brain injury and multiple concussions sustained at a younger age increase the dementia risk later in life.
- Epilepsy, especially if inadequately treated.
- Immune system dysfunctions that cause diminished immune response to infections in old age are thought to increase inflammation throughout the body, including the nervous system.
- Infections, including the herpes simplex virus, cytomegalovirus, and Epstein-Barr virus, among others, have been suggested risk factors for AD.
- Malnutrition and poor diet may indirectly increase the risk of DAT by increasing susceptibility to cardiovascular diseases.
- Deficiencies of calcium, vitamin D, and vitamin B12 are strongly correlated with dementia risk.
- Major depressive disorders also are known risk factors.
- **Environmental variables.** Lifestyle factors and several medical conditions (e.g., obesity) are modifiable risk factors (Killin et al., 2016). However, these may be general risk factors for most types of dementia, including DAT.
 - Air pollution, which includes multiple factors; nitrogen oxide; carbon monoxide; ozone; and environmental tobacco smoke are known risks.
 - Metal exposure, especially zinc, copper, mercury, manganese, magnesium, cadmium, and long-term aluminum dust inhalation, may increase the risk for DAT; some of these interact with the gene APOE.
 - Chronic occupational exposure to pesticides, fertilizers, herbicides, and insecticides has been strongly linked to dementia risk.

Although a cause-effect relation has not been established, there is sufficient evidence to suggest that better education, regular exercise, healthy diet, controlled blood cholesterol and blood sugar levels, healthy body weight,

increased cognitive activity, minimized exposure to air pollution and other toxins, occupational safety, stress reduction, and prompt treatment for depression, among others, may reduce the risk of cognitive decline (Rakesh et al., 2017; Sujkowski et al., 2022).

Neuropathology of Alzheimer's Disease

Because there are no conclusive laboratory findings that document the neural abnormalities, only autopsy can be definitive. Routine blood, urine, and cerebrospinal fluid examinations, even if showing certain abnormalities, may not be conclusive. Various brain imaging techniques (see [Chapter 2](#)) do not produce strong evidence for an unequivocal diagnosis of DAT, although such techniques provide supportive evidence (Arvanitakis et al., 2019).

The neuropathology of DAT is better understood than its etiology. Cerebral atrophy, ventricular dilation, neuronal degeneration, neurofibrillary tangles, synaptic loss, amyloid plaques, white matter changes, neural inflammation, and depletion of neurotransmitters are associated with DAT. These changes are most evident in the temporoparietal-occipital junctions and in the inferior temporal lobe. Relatively spared are the frontal lobe, motor and sensory cortices, the occipital lobes, the cerebellum, and the brainstem. To some extent, abnormalities found in the brains of people with DAT may also be found in healthy older persons and otherwise healthy individuals with Down syndrome, as well as those with other types of dementia (Arvanitakis et al., 2019; Atri, 2019; Breijyeh & Karaman, 2020; Hane, Lee, et al., 2017; Hane, Robinson, et al., 2017). It is now well recognized that dementia is the final stage of AD, and the cerebral pathological process begins long before any symptoms are observed (Lloret et al., 2019).

Neurofibrils are filamentous structures in the nerve cell's body, dendrites, and axons. In a form of protein called tau, the neurofibrils are thickened, twisted, and tangled nerve fibers. The tau proteins form unusual loops, triangles, and clumps. The pyramidal neurons of the cortex, the hippocampus, and the amygdala are the most frequent sites of tangled neurofibrils.

Amyloid (neuritic) plaques are minute areas of cortical and subcortical tissue degeneration. These plaques destroy synaptic connections and thus

disturb neuronal transmission of messages. The main chemical constituent of neuritic plaques is called the *amyloid protein*. The primary sites of these plaques are the cerebral cortex, amygdala, and hippocampus (a structure deep within the brain presumably concerned with memory), although they also may be seen in the corpus striatum, amygdala, thalamus, and the cerebral and meningeal blood vessels.

Neuronal loss results in a shrunken cerebral cortex in individuals with Alzheimer's disease. The ventricles of the brain enlarge as the brain tissue is atrophied or shrunk. The shrinkage is most common in the cerebral hemispheres, especially in temporal and parietal lobes. *Synaptic loss*, a particular type of neuronal loss, is associated with memory impairments observed in the early stage of AD (Breijyeh & Karaman, 2020). In high-risk individuals with no symptoms, neuronal loss may begin several years before any cognitive decline becomes evident. The loss may begin in the hippocampus and surrounding areas and then spreads to other structures.

Neurochemical changes accompany the structural abnormalities. Such brain chemicals that facilitate neural transmission as acetylcholine, somatostatin, vasopressin, endorphin, and corticotropin are severely depleted in individuals with AD. Many cognitive deficits may be due to this reduction in neurotransmitters.

Certain **biomarkers of AD** are now available. Cerebrospinal fluid analysis helps determine the abnormal levels of beta-amyloid peptide ($A\beta$, lower than normal), which leads to amyloid plaques and total tau protein and phosphor-tau (higher than normal), which are related to neurofibrillary tangles. This analysis may support both mild cognitive impairment as well as AD. Positron emission tomography (PET; see [Chapter 2](#)) can reveal $A\beta$ load and spatial distribution in the brain, offering further support for a diagnosis of AD. These measures are well correlated with the more definitive postmortem studies. Other types of PET studies help determine the presence of cerebral glucose hypometabolism and synaptic dysfunction, especially in the temporal and parietal regions that suggest AD and the associated cognitive deficit. Magnetic resonance imaging (MRI; see [Chapter 2](#)) can detect cerebral atrophy (reduced thickness), especially in the temporal lobe and the hippocampus, years before the symptoms of AD-associated decline are observed. It is important to note,

however, that A β plaques and tangles are present in 20% to 40% of individuals who are healthy and without AD or any other type of dementia. The presence of amyloid plaques and tangles alone does not justify a diagnosis of dementia (Lloret et al., 2019). Also, differentiating AD in its early stage from other forms of dementia is difficult. In such cases, the diagnosis is considered probable but not certain (Hane, Robinson, et al., 2017).

Symptoms of DAT in Early Stages

In the early stages of AD, behavior, motor functions, and daily living skills may be near normal. The person may manage any subtle problems. Sooner or later, however, the following impairments become obvious, although individual differences are significant (Atri, 2019; Arvanitakis et al., 2019; Hegde & Freed, 2022):

- **Subtle memory deficits.** The person may fail to recall remote events and learn and recall recent events. Being aware of the problems, the individual may take compensatory actions (e.g., writing down things to remember).
- **Impaired visuospatial skills.** Copying three-dimensional drawings, constructing block designs, lacing one's shoes, and simultaneous multitasking may be difficult.
- **Poor reasoning and judgment.** The individual may forget to pay the bills or pay the same bill repeatedly. Poor judgment may alert the family members to a serious problem.
- **Behavioral changes.** The individual may avoid difficult tasks (e.g., cooking or reading books) or ask someone else to perform them (e.g., a tennis player may ask the partner to keep the score). Individuals may stop taking care of themselves and be unconcerned about their social appearance (self-neglect). Social withdrawal and disinhibition may follow (e.g., grabbing food from another person).
- **Disorientation.** In the early stages, disorientation and confusion may be limited to new surroundings.
- **Impaired gait.** Falls may be frequent.

- **Anxiety.** Persons may be anxious in the initial stages.
- **Apathy and depression.** Emotional indifference (apathy) is an early sign. The person may lose interest in favorite activities and fail to initiate actions (e.g., cooking a meal or answering phone calls). Awareness of difficulties may cause depression and frequent mood changes.
- **Subtle language difficulties.** Not apparent to others, but subtle language problems may exist in the early stages of dementia.

Symptoms of DAT in Later Stages

As the disease progresses, the early stage symptoms intensify, and additional symptoms emerge. Symptoms vary in frequency and severity across individuals (Atri, 2019; Arvanitakis et al., 2019; Hane, Robinson, et al., 2017; Hegde & Freed, 2022):

- **Profound memory loss.** Both recent and remote recall may be severely affected. The person may not remember how to use a telephone, a cooking stove, or television remote controller. New learning and retention may be extremely difficult if not impossible.
- **Impaired daily living skills.** Such routine skills as dressing and undressing, eating, walking, bathing, and so forth may be severely impaired.
- **Generalized intellectual deterioration.** Skills in managing daily activities including cooking, shopping, and taking care of personal or family finances may deteriorate. Difficulty integrating new information and problems in making rational decisions will be obvious.
- **Restlessness, agitation, and hyperactivity.** Purposeless pacing and ritualistic or meaningless handling of objects may emerge.
- **Acalculia.** Difficulty with numbers and calculations becomes more acute.
- **Profound disorientation.** Individuals may get lost in their own home or other familiar surroundings and may wander off. Eventually,

the individual may lose all sense of orientation to time, place, and person.

- **Agnosia.** Persons who are unable to name objects or people, but are able to recognize them, may subsequently fail to recognize them. Initially, they may mistake a fork for a spoon but, in more advanced stages, mistake a pencil for a fork and try to eat with it. Family members may be distressed to find the individual unable to recognize them and even accuse them of being strangers, intruders, or impostures.
- **Delusions of persecution.** Individuals may believe that family members and other caregivers are conspiring to harm them.
- **Lack of insight (anosognosia).** Not realizing one's limitations, the person may insist on driving or managing personal finances.
- **Aberrant behaviors.** Persons may act on their delusions; make baseless accusations (e.g., the caregiver is stealing money); steal, hide, or hoard things; and rummage through rooms of other people. Some may hallucinate (see nonexistent people or objects) and engage in purposeless, repetitive actions (e.g., fidgeting with buttons or utensils for hours). Uninhibited behaviors may include inappropriate humor, incongruous laughter, and urinating and masturbating in public places. Apparently trivial incidents may trigger emotional outbursts (catastrophic reactions). Some individuals may be physically and verbally violent and aggressive. Being sleepless, they may wander through the house in the night, walk out, wander, and get lost. They may sleep most of the daytime.
- **Loss of initiative.** Lack of motivation and affect, as well as clinical depression, may be more serious. Individuals may talk about distressing experiences with no emotion.
- **Periodic incontinence.** Control over urinary function may be lost in the more advanced stages.
- **Problems of language and communication.** (Please see the next section for details.)
- **Physical deterioration.** Motor and physical problems appear in the most advanced stage. Generalized muscle rigidity (spasticity),

unstable gait, and frequent falls may force individuals to spend more time sitting or lying down. A bewildered facial expression may be evident. Mostly bedridden, individuals may assume the fetal position, have seizures, exhibit grasping and sucking reflexes, lose weight, and become urine and feces incontinent.

In most cases with Alzheimer's disease, death is due to aspiration pneumonia, heart failure, or infection.

Communication Problems of Persons With DAT

Relatively intact language of the early stages begins to deteriorate progressively. Initially, speech sound production, voice, syntactic and prosodic aspects, and gestures that accompany speech are unaffected, as is automatic speech. But there may be other problems as noted (Atri, 2019; Arvanitakis et al., 2019; Bayles et al., 2020; Hegde & Freed, 2022; Payne, 2014).

Language problems of the **early and middle stages** include the following:

- **Anomia.** Mild naming problems, especially with low-frequency words, are among the very early signs of dementia. Generating a list of words that start with a specific letter may be especially difficult.
- **Verbal paraphasia.** Alzheimer himself noted circumlocution (describing an object without naming it) and verbal paraphasia in his first individual who said *milk pourer* when shown a jug. Anomia leads to such paraphasic expressions.
- **Language comprehension deficits.** Comprehension of concrete language may be intact in the early stages, but that of humor and abstract language (implied and proverbial meanings) may be impaired. Comprehension of all types of language deteriorates over time.
- **Impaired picture description.** Anomia may affect picture description, resulting in paraphasic and circumlocutionary descriptions.

- **Difficulty in topic maintenance.** The individuals may shift topics abruptly and lose track of what they were saying.
- **Reading aloud and writing.** These skills, intact in the early stage, deteriorate in the later stages.

The speech of individuals with DAT is more fluent, resembling the speech of those with transcortical sensory aphasia. Persons with DAT do not show the agrammatic and effortful speech of Broca's aphasia.

Language problems of the **later stages** include the following:

- **Literal (phonetic) paraphasia.** Sound substitutions (e.g., *Lamerican* for *American*) and additions (e.g., *Amelrican* for *American*) may be common.
- **More frequent circumlocution.** The individual may beat around the bush instead of producing direct expressions and may fail to use the correct pronouns.
- **Jargon and empty speech.** Speech may be repetitious, paraphasic, jargon filled, and meaningless.
- **Hyperfluency.** In some cases, speech, although lacking meaning, may be extremely fluent and flowing.
- **Incoherent speech.** In the latter stages, speech may be completely incoherent with disorganized thought processes.
- **Impaired conversational skills.** Most pragmatic language skills may be impaired, resulting in severe problems in maintaining a conversation.
- **Impaired comprehension.** This deficit will be more pronounced as the disease advances and more evident for complex spoken material (e.g., sequential instructions given for task performance).
- **Impaired reading comprehension.** The problems in reading may parallel progressive deterioration in comprehending spoken speech.
- **Impaired oral reading.** This skill may be better preserved until the late stage.

- **Impaired writing skills.** Poor letter formation and other writing problems may be present, as in the case of aphasia.
- **Echolalia, palilalia, and logoclonia.** Repeating what is heard is echolalia, repeating one's own utterances is palilalia, and repeating the word-final syllable is logoclonia.
- **Difficulty initiating conversation.** Reluctance to initiate social interaction.
- **Inattention to social conventions.** Failure to greet people and bid farewell.
- **Lack of meaningful speech.** In the final stages of the disease, speech is uninterpretable, largely because of much paraphasic and confused language.
- **Articulatory problems.** Unless a dysarthria coexists with dementia, speech sound production is not affected.
- **Mutism.** Repetition of nonlanguage sounds or complete mutism may characterize the terminal stage.

A few individuals with DAT may have visual problems that may account for confrontational naming difficulties. Ideomotor apraxia (difficulty performing an act on command that is performed spontaneously) and ideational apraxia (difficulty in demonstrating a sequence of movements, such as filling a glass with water) are evident in the later stage.

Vascular Dementia

Vascular dementia (VaD) is deterioration in cognitive and daily living skills due to ischemic or hemorrhagic cerebrovascular disease, cardiovascular disease, or circulatory diseases that limit the blood supply to brain tissue, causing infarcts (Wolters & Ikram, 2019). Being the second most common form of dementia (after the Alzheimer type), 15% to 20% of individuals with dementia are diagnosed with VaD (Bir et al., 2021; Iadecola et al., 2019; Wolters & Ikram, 2019). One year after a single transient ischemic stroke, 5% may develop VaD, but after a severe stroke, 34% may. Vascular diseases that

cause widespread bilateral damage may be subcortical, cortical, or both (mixed). Similar to AD, the incidence of VaD increases with age.

VaD diagnosis is based on vascular disease in the absence of other etiological factors. However, in some individuals, vascular and AD may coexist and interact with each other to produce *mixed dementia* (Vinters et al., 2018). VaD was also described as *multi-infarct dementia* because multiple small strokes and their cumulative effects are one of its causes. VaD is the currently preferred term.

Associated with a history of hypertension, diabetes, hypercholesterolemia, obesity, and similar risk factors, VaD is more common in men than in women. Individuals with VaD are typically younger and die sooner than those with DAT. Generally, the risk and the protective factors are the same for VaD and DAT (summarized earlier). Effective treatment for the risk factors may reduce the incidence of dementia of vascular pathology and the associated mortality rate.

African Americans are more prone to VaD than Whites because untreated high blood pressure, cerebrovascular diseases, and stroke are more common among them. South Asians living in Asia and abroad also have a high prevalence of high blood pressure and diabetes and thus have a high incidence of VaD (Singh et al., 2018).

Unlike most other forms of dementia, the onset of VaD is relatively sudden, although dementia due to the degeneration of subcortical white matter has a more gradual or stepwise progression (Mahendra & Engineer, 2009). The course of dementia fluctuates and the individual experiences increased confusion during the night.

According to an expert panel (Iadecola et al., 2019), there are four major subtypes of VaD: (1) poststroke dementia that is manifest 6 months after a stroke with no reversal of symptoms, (2) subcortical ischemic vascular dementia, (3) multi-infarct (cortical) dementia, and (4) mixed VaD, often a combination of VaD and AD. To diagnose a VaD, MRI evidence of vascular lesions is required.

Small-vessel and large-vessel diseases cause VaD. Small-vessel diseases (e.g., arteriosclerosis, thickening of the walls of arteries) cause microscopic lesions

due to microbleeds (small hemorrhage) within the brain. Large-vessel infarcts are more obvious than the small-vessel infarcts because they result in familiar symptoms of stroke (Iadecola et al., 2019). Persons who have large-vessel strokes with affected perisylvian cortical regions tend to have aphasia. If the regions supplied by the posterior and anterior cerebral arteries, along with the thalamus and basal ganglia, are damaged, language and memory skills are likely to be affected, resulting in eventual dementia. Subcortical pathologies due to multiple ischemic strokes tend to produce stepwise cognitive decline. Dysarthria and motor problems are associated with such pathologies.

In general, VaD is characterized by memory loss, inattention, emotional lability, aphasia, apraxia, dysarthria, anomia, delayed recall of word list, reading problems with or without writing problems, impaired thinking and judgment, impaired planning (executive) skills, and significant behavior changes (Iadecola et al., 2019). Persons with VaD may perform better than those with AD on verbal memory tasks and category fluency tasks but perform worse on tests of attention. The two groups may be comparable on digit span tasks, block design tests, vocabulary, and language tests. Their discourse may be fluent but empty of meaningful content (Mahendra & Engineer, 2009). Across individuals, symptoms may form different patterns and combinations depending on the site and the number of infarcts.

Dementia With Lewy Bodies

The third most common, Lewy body dementia (LBD) is found in 10% to 25% of all dementia cases. Some sources describe it as the second most common, after AD (Lewy Body Dementia Association, n.d.). The incidence is estimated to be more than 1 million a year in the United States. More men than women are affected. From diagnosis to death is 5 to 8 years but can be longer in some individuals. The course of LBD is more rapid than the one associated with Alzheimer's disease. The LBD age of onset is between 50 and 80 years. Parkinson's disease, REM (rapid eye movement) sleep disorders, and three gene variants (APOE, SNCA, and GBA) increase the risk of Lewy body formations.

Lewy bodies are excessive accumulations of alpha-synuclein protein in the brain, forming clumps or pathological spots within the damaged nerve cells and disrupting the normal brain functions. Damaged cells are typically found in the substantia nigra (a structure at the top of the brainstem). This neuronal pathology is named after the German-born American neurologist Frederick Lewy (1885–1950), who described them when he was working with Alois Alzheimer.

Lewy body dementia and Parkinson's disease (leading to dementia) share the same neuropathological changes. Persons in the two groups develop similar symptoms. The distinction is that in Lewy body dementia, cognitive disorders appear first, about a year earlier than movement disorders. In Parkinson's dementia, movement disorders (e.g., tremors) appear first, about a year earlier than cognitive impairments (Morra & Donovick, 2014).

Major symptoms of LBD include impaired thinking, confusion, memory loss in later stages, visual and auditory hallucinations, paranoid thoughts, rapid eye movement sleep disturbances, mild features of Parkinson's disease (e.g., muscular rigidity and slowness of movement), and transient changes in consciousness leading to repeated falls. Attention and alertness may widely fluctuate in the course of a day. Unlike other dementias, visual and spatial skills, especially in the early stages, may be more severely involved than memory skills. Some individuals may be depressed, and others may have gait problems. Agitation, restlessness, pacing, and hand wringing may also be present (Hemminghyth et al., 2020). A critical feature of these individuals is a hypersensitivity to antipsychotic drugs, which may be prescribed because of the individuals' hallucinations. Negative reaction to such drugs may be so severe as to cause death in some cases.

Language disorders associated with LBD include confabulation, reduced speech fluency, naming difficulties, problems comprehending longer and more complex sentences productions, poor narrative skills, and impaired repetition. Generally, the language performance of individuals with LBD is comparable to that of individuals with Alzheimer's dementia, although the former group may perform better on verbal memory tests but less well on attentional tasks (Payne, 2014). Anxiety is another feature possibly leading them to repeatedly ask the same questions.

Frontotemporal Dementia

In 1892, Arnold Pick (1851–1924), a Prague psychiatrist, described a man with marked language problems and dementia. Pick described two kinds of neuronal abnormalities in this and other patients of his: (1) dense intracellular formation in the neuronal cytoplasm (later named *Pick bodies*) and (2) ballooned, enlarged, or inflated neurons (later named *Pick cells*). The syndrome he described came to be known as Pick's disease (PiD), which results in a form of dementia (Olney et al., 2017).

Currently, PiD is considered a variant of **frontotemporal dementia (FTD)**, also known as **frontotemporal lobar degeneration**. PiD is now more likely to be called the *behavioral variant* of FTD. Degeneration in frontal and anterior temporal lobes characterizes FTD. The expanded diagnostic category of FTD is more common than the classic form of PiD; it may be found in 12% of cases of dementia diagnosed in people younger than 65. The typical age of onset is between 45 and 65, although it has been observed in persons as young as 21 years (Olney et al., 2017). FTD is thus an early onset dementia.

The classification of FTD seems to be in a flux. Newer varieties periodically emerge (Olney et al., 2017). Most experts currently classify FTD in the following manner, although the nomenclature may differ across sources: (1) the behavioral variant of FTD, (2) primary progressive aphasia (PPAs) with three subvariants, and (3) FTD in amyotrophic lateral sclerosis with motor speech disorders. A separate frontotemporal lobar degeneration syndrome lateralized predominantly to the *right hemisphere* has also been described (Gainotti, 2019).

Behavioral Variant of FTD

This variant is so called because the behavioral changes appear first and other symptoms of dementia, including communication deficits, appear later. Behavior changes may be apparent even before brain imaging studies show neural degeneration. The initial behavioral changes are so much like those of

schizophrenia and other psychiatric disorders that the individual may be taken first to the psychiatrist.

Loss of cells in the left and right frontal and temporal lobes is the main neuropathology in FTD (Bahia et al., 2013; Olney et al., 2017). The atrophy (shrinking) may be predominant in the temporal lobe in about a quarter of individuals and in the frontal lobe in another quarter. The rest have atrophy in both the lobes, although it may be left or right dominant. The symptom complex may differ, depending on the differential involvement of the right and left sides and temporal and frontal lobes. The Pick bodies may be found in about 40% of individuals with FTD. In some individuals, atrophy in the frontotemporal regions may show no distinctive histology.

Main symptoms of the behavioral variant of FTD include the following (Bahia et al., 2013; Gossink et al., 2017; Olney et al., 2017):

- **Behavioral changes.** Deterioration in behavior and social conduct, especially in those with a predominant right hemisphere involvement, is the hallmark of this variant. Affected people become entirely different persons with previously nonexistent profanity, irresponsibility, irreverence, and disregard for social conventions. Individuals may go on uncharacteristic shopping sprees and, while there, shoplift. Other kinds of criminal behaviors may emerge for the first time. They may begin to gamble, smoke, and drink excessively. They may be overly friendly with strangers.
- **Impaired emotional behavior.** Depression and withdrawal may alternate with euphoria and excessive jocularity. Apathy, inertia, lack of sympathy or empathy, and exaggerated self-esteem are among the other emotional disorders.
- **Compulsive behaviors.** Persons may imitate the gestures of others compulsively even when told not to imitate. They may repeatedly fold napkins, clap endlessly, rock themselves, brush teeth many times, perform bathroom routines all the time, count and hoard food items, collect cigarette butts, check locks and doors endlessly, and insist on eating the same food at precisely the same time every day.

- **Excessive eating and weight gain.** With a craving for carbohydrates, the individuals may eat bizarre combinations of foods. They may try to orally explore nonedible substances (called *hyperorality*) and try to eat them.
- **Delusions.** About one fifth of persons may exhibit delusions, hallucinations, and suspiciousness.

Intellectual (cognitive) deterioration appears later in the behavioral variant of FTD. Judgment, insight, constructional skills, planning, personal money management, self-correction of mistakes, and the comprehension of abstract language may be affected though not to the extent in persons with AD. Deterioration in memory skills may be subtle in the early stage and significant later. This is in contrast to DAT, in which the intellectual deterioration is observed before significant changes in behavior. When the disease advances, profound deterioration in physical status, behavior, and intellectual skills may be expected. Most will be bedridden and may have seizures and other neurological impairments. Aspiration pneumonia due to dysphagia is a frequent cause of death. Many may die suddenly (van Engelen et al., 2020).

Communication disorders associated with the behavioral variant of FTD, often considered less severe than those found in persons with AD or primary progressive aphasia, may still be significant even in the early stages. Language skills worsen as the disease advances. In their systematic review of research, Geraudie et al. (2021) noted mixed evidence on most language impairments, but a naming deficit was reported more consistently. Verb or action naming, word comprehension, knowledge of word meaning (difficulty defining words), and sentence comprehension may be impaired. Speech sound production skills may be better preserved in most cases. Oral reading may be slow, with frequent, longer, and variable pauses. Writing orthographically ambiguous words may be impaired in some individuals. Proverb and idiom comprehension may be poor. Evidence for prosodic deficits is limited and weak. The discourse skills may appear superficially unaffected, but closer examination may reveal problems in event reporting and connectedness, theme maintenance, and so forth. Prosodic features may be affected, especially those for emotional intonation. Grammar is mostly preserved. As expected, left-

dominant lesions may affect language skills more than the right-dominant lesions. Geraudie et al. (2021) emphasized the heterogeneity of symptoms across individuals with the behavioral variant of FTD. The end stage of the behavioral variant of FTD may be muteness (van Engelen et al., 2020).

Diagnosis of the behavioral variant of FTD is complicated by one particular phenomenon: the **phenocopy syndrome**, which is the simulation of symptoms of another disease and, in this case, the symptoms of the behavioral variant with no neurological evidence to support them. There is no progression to dementia. Family members and others insist on the behavioral aberrations in the person. The causation of phenocopy is unclear (Valente et al., 2019).

Primary Progressive Aphasia

Primary progressive aphasia (PPA) is due to a neurodegenerative disease that terminates in dementia, whereas the more typical forms of aphasia described in [Chapter 3](#) are not progressive and do not lead to dementia. PPA is called as such because the initial symptoms are predominantly aphasic (language impairments), not cognitive or behavioral deterioration. Some studies have suggested that subtle cognitive impairments may be present even in initial stages, but the more obvious behavioral symptoms typical of dementia appear later, in most cases 2 years later (de la Sablonniere et al., 2021; Kirschner, 2014; Montembeault et al., 2018; Olney et al., 2017; Stalpaert et al., 2020). In behavior, cognition, language, and daily living skills, persons with PPA tend to decline faster than those with the Alzheimer's disease. In just 2 years postonset, nearly a third of the persons with PPA were mute (see de la Sablonniere et al., 2021, for a review of studies).

PPA has three variants: a nonfluent variant (nfvPPA), a semantic variant (svPPA), and a logopenic variant (lvPPA). Some 31% of individuals with PPA cannot be subclassified into one of these three. Both nfvPPA and svPPA may be associated with Pick's disease (Karageorgiou & Miller, 2014).

Nonfluent variant of PPA. Constituting 27% of PPA, this variety is associated with lesions in the left perisylvian (language) area of the brain including Broca's area (Olney et al., 2017; Wicklund et al., 2014). Agrammatic

language and effortful, slow, nonfluent, and halting speech were the most prominent symptoms. These are diagnostic of the nonfluent variety of PPA because they are not typical of the other two varieties (semantic and logopenic). Apraxia of speech, another significant feature, may consist of inconsistent sound insertions, deletions, and substitutions. A slower speech rate, impaired repetition, and prosodic impairments may also be observed. Confrontation naming deficit, good comprehension of single words, and impaired comprehension of complex sentences are common to all three variants (Stalpaert et al., 2020). Two years postonset of PPA, symptoms of dementia, including behavioral changes seen in bvFTD and memory and cognitive problems, emerge. Mutism is the terminal stage of all forms of PPA, but the nonfluent variety reaches that stage sooner (de la Sablonniere et al., 2021; Montembeault et al., 2018; Olney et al., 2017).

Semantic variant of PPA. Constituting 7% of PPA, this variant is associated with atrophy of the (mostly) left temporal lobe. Initial aphasic symptoms include progressive loss of word meaning (*semantic problem*, hence the name). Noun naming (including object naming) and category list generation may be more severely impaired than verb naming. Comprehension of the meaning of low-frequency words may be seriously impaired in the earlier stages. Fluency, repetition, articulation, grammar, and reading skills are relatively unimpaired. Irregular word writing may be impaired. Speech may be excessive and disinhibited. Orientation to time and space may be good until the disease advances to dementia. There may be visual agnosia if the atrophy extends to the right temporal lobe. As aphasia progresses to dementia, sentences get shorter, and mutism emerges. Compared to the other two variants of PPA and Alzheimer's, behavioral deterioration appears sooner in the semantic variant. Some studies have reported behavioral symptoms at the initial assessment (de la Sablonniere et al., 2021; Montembeault et al., 2018; Olney et al., 2017; Stalpaert et al., 2020; Wicklund et al., 2014).

Logopenic variant of PPA. Constituting 35% of PPA, the variant is associated with both the Alzheimer neuropathology and temporal lobe degeneration, including especially atrophy in the left posterior superior temporal gyri, the middle temporal gyri, and the inferior parietal lobe (Wicklund et al., 2014). A history of learning disability may be evident in some persons. Initial aphasic symptoms include moderate confrontation

naming problems, slow speech, pauses due to word-finding problems, severe difficulty repeating phrases and sentences, and impaired sentence comprehension. Phonological errors in spontaneous speech is a diagnostic sign. Agrammatic effortful speech and prosodic deficits may be absent. Repetition and comprehension of single words may be unimpaired (Montembeault et al., 2018; Olney et al., 2017; Stalpaert et al., 2020). Speech becomes progressively more paraphasic and empty of content. Behavioral deterioration advances as dementia sets in. Cognitive decline is faster in the logopenic than that in the other two varieties, and memory deficits lead the decline (de la Sablonniere et al., 2021).

FTD in amyotrophic lateral sclerosis with motor speech disorders (ALS-FTD). Amyotrophic lateral sclerosis (ALS) is a progressive motor neuron disease that, among other consequences, produces dysarthria, a motor speech disorder due to sclerosis (thickening and hardening of the motor neurons) (Hegde & Freed, 2022). Upper or lower motor neurons may be affected initially, but eventually both are affected. Voluntary movements, including speech muscle movements, may be severely impaired because of amyotrophic (muscle) atrophy. In the past, ALS was thought to cause only speech production, not dementia, but more recent research suggests otherwise. ALS and FTD share certain genetic, clinical, and pathological factors. With the involvement of the frontal and anterior temporal lobes, changes in language, executive functions, and behavior have been reported in up to 50% of persons with ALS, and in 10% to 15% of these cases, the changes are severe enough to make a diagnosis of FTD (Masrori & Van Damme, 2020). Conversely, 15% of persons with FTD eventually develop a motor neuron disease. Dementia symptoms in persons with ALS may be similar to those found in the behavioral variant of FTD or one of the three variants of PPA (Masrori & Van Damme, 2020). Both language production and comprehension tend to be affected. Limited language output, impaired grammar, perseveration, echolalia, and stereotypic expressions have all been noted. Dementia progresses to mutism and death.

Parkinson's Disease

James Parkinson, the English physician who in 1817 described a brain disease he called *shaking palsy*, did not know that it could lead to dementia. The French physician Jean-Martin Charcot later named the disease *Parkinson's disease* (PD). Dementia may develop in 25% to 35% of individuals with Parkinson's disease. Mild cognitive deficits may be evident in the rest. Because Lewy bodies (described earlier) are a neuropathological feature of PD, fully developed dementia of PD and dementia with Lewy bodies (DLB) may be clinically indistinguishable (Hanagasi et al., 2017). Generally, Parkinson's disease dementia (PDD) is diagnosed when the symptoms of dementia emerge in the context of a well-established clinical picture of PD. DLB is diagnosed when symptoms of dementia emerge well before the symptoms of PD (motor impairments, described later). About a quarter of persons with DLB do not develop Parkinsonian motor symptoms. Alzheimer disease abnormalities also may be found in persons with PDD (Jellinger, 2017).

PD is more common in males than in females, and 1 in 100 individuals aged 50 years and over may have it. About one million people in the United States have PD, and about 60,000 new cases are reported each year (Olanow et al., 2001). The average age at onset is 60 years, although a juvenile form exists. Most individuals survive for about 8 years postonset and longer if the disease is diagnosed in younger people.

Neuropathological factors associated with Parkinson's disease typically include degeneration of nuclei, especially in the brainstem and substantia nigra. *Sulci* in the frontal region of the brain may be widened. *Neurofibrillary tangles* and *neuritic plaques* may characterize the cortical regions. *Lewy bodies* may be present, especially in the basal ganglia, brainstem, spinal cord, and sympathetic ganglia (see Dementia With Lewy Body in an earlier section of this chapter). *Depletion of dopamine levels*, due to neuronal loss, may be evident in the basal ganglia. Dopamine depletion causes a lack of inhibitory potential, leading to excessive cholinergic excitation and Parkinsonian motor symptoms. *Bradykinesia* is the main motor symptoms of PD, characterized by immobility or slow voluntary movements. Movements, especially walking, may suddenly freeze. *Tremor* at 4 Hz to 6 Hz or 8 Hz starts with the hand or foot and may spread to all four limbs. *Rigidity* (increased tone) and resistance to movement are experienced in the muscles of limb and trunk. *Disturbed gait and posture* may cause difficulty standing up from a sitting position. Small and shuffling

steps, a forward-leaning and absence of hand swinging while walking, and festinating gait (taking short, accelerating, involuntary steps while walking) are other movement disorders. Individuals are prone to fall frequently. Finally, individuals are susceptible to *swallowing disorders*, which may be more severe during periods when the levodopa treatment is not effective.

Dementia and Communication Problems Associated With PD

Dementia of mild to moderate severity, more likely in individuals 60 years and older, may be characterized by the following features (Hanagasi et al., 2017; Jellinger, 2017:

- **Psychiatric symptoms.** Apathy, depression, confusion, visual hallucination, paranoid delusions, impaired attention, anxiety, insomnia, and delirium, which may be aggravated by antiparkinsonian medications. Nightmares and social withdrawal also may be present. Possibly related to levodopa, some individuals may suddenly fall asleep (sleep attacks).
- **Impaired visuospatial perception.** Angle matching, figure-ground discrimination, spatial position, and constancy of shape and size may be affected.
- **Impaired executive functions.** Persons may be unable to plan actions or perform daily activities (e.g., cooking or lawn mowing).
- **Difficulty with abstract thinking and language.** These difficulties are similar to the typical symptoms found in all dementias.
- **Communication problems.** Naming difficulties, impaired wordlist generation, hesitations and pauses in speech, poor discourse comprehension, and monopitch and monoloudness may be common in more advanced stages. Hoarseness, breathiness, roughness, and hypernasality may be observed in some individuals. Nearly half of all individuals with PD show motor speech disorders. Parkinson's disease is the most common cause of hypokinetic dysarthria with variable

speech rate, reduced stress, short phrases, short rushes of speech, and imprecise production of consonants (Freed, 2020).

- **Micrographia.** This is writing in extremely small letters.

In dementia associated with Parkinson's disease, aphasia, agnosia, and severe amnesia are rare. In some individuals, cortical changes associated with Alzheimer's disease may be found, producing a combination of symptoms.

Huntington's Disease

In 1872, George Huntington (1850–1916), an American physician, described the case of a Long Island family with an idiopathic degenerative disease of the brain characterized by chorea, psychiatric problems, and cognitive decline. The disease now bears his name.

Huntington's disease (HD) is a genetic neurodegenerative disease. Given autosomal dominant inheritance, half the offspring of an affected person may have the disease. A genetic test is available and provides accurate diagnosis of the disease. The genetic mutation causing HD occurs on the short arm of chromosome 4. A destructive product of the gene mutation is called *huntingtin*, a malformed protein that kills the brain cells that control movement and memory. Loss of neurons occurs primarily in the basal ganglia, more strikingly in the caudate nucleus and the putamen, resulting in reduced levels of such inhibitory neurotransmitters as gamma-amino butyric acid (GABA) and acetylcholine. Not only the basal ganglia but other parts of the brain, including the frontal lobe, may be affected. Gray matter and cortical thinning and reduced cerebral metabolism may affect the functioning of the whole brain (Ghosh & Tabrizi, 2018; Martinez-Horta et al., 2020; Pagan et al., 2017).

The incidence of HD is about 40 to 70 per million persons with equal male-female distribution. The typical age of onset is between 35 and 50 years. In about 5% of the cases, the disease begins before age 20 (Ghosh & Tabrizi, 2018; Pagan et al., 2017).

For many years postonset, most persons with HD may not exhibit disabling signs of dementia. However, MCI may be present at the onset of motor symptoms. Julayanont et al. (2020) reported MCI without functional disabilities in 84% of persons with HD and dementia in 5%. In about 5 years postonset, a majority developed dementia. As the disease progresses, symptoms of frank dementia appear in all who have PD. Thinning of the parietal-temporal and occipital cortices is especially prominent in persons who have HD as well as dementia (Martinez-Horta et al., 2020).

Dementia and Related Problems Associated With HD

Huntington's disease produces a spectrum of motor, behavior, cognitive, and communication impairments. HD and PD produce similar communication deficits (Freed, 2020). Generally, individuals with HD tend to exhibit the following kinds of deficits:

- **Behavioral changes.** Excessive complaining, nagging, eccentricity, irritability, emotional outbursts, and a false sense of superiority are among the earliest signs of the disease and precede chorea in a majority of cases.
- **Psychiatric disturbances.** Mood swings (depression and euphoria), paranoia, persecutory delusions, and hallucinations appear in more advanced stages. Suicide accounts for approximately 8% of deaths of individuals with HD, and attempted suicide is common. Psychiatric disturbances precede chorea in about one third of individuals.
- **Chorea.** The major neurologic (motor) impairment in HD, chorea is the irregular, spasmodic, involuntary movement of the limbs, neck, head, and facial muscles. Chorea begins as restlessness and fidgetiness and develops into debilitating movements.
- **Tic-like movement disorders.** Transient facial grimaces and head nodding, among the early symptoms, become increasingly uncontrollable.

- **Gait disturbances.** Alternating hand extension-pronation and flexion-supination while walking may combine with lurching, halting, and faltering movements, giving the appearance of dancing while walking.
- **Slow movements.** Slow and greatly reduced voluntary movements characterize the advanced stage of the disease.
- **Rigidity.** May be seen in 12% to 14% of individuals.
- **Memory and attention problems.** Impaired memory for both recent and remote events, attentional deficits, and slowness in all intellectual activities follow the onset of chorea.
- **Communication deficits.** Appearing in the later stage of the disease, persons may have difficulty with word-list generation, naming, and recalling low-frequency words. Speech may consist of shorter or simpler utterances and fewer grammatically correct sentences. Comprehension of subtle and implied aspects of discourse may be difficult.
- **Dysarthria.** A prominent characteristic of individuals with HD, the choreiform movements of the lips and the tongue affect speech articulation, resulting in dysarthria of the hyperkinetic type. Abnormal movements of the respiratory muscles and larynx cause excess variations in speech loudness, monoloudness, short phrases, monopitch, and harsh voice quality.
- **Mutism.** The final stage of the disease before death is characterized by mutism, an extreme degree of intellectual deterioration, incontinence, sleep disturbances, dysphagia, confusion, and violent behavior.

Individuals with HD are prone to diabetes, heart disease, lung disease, and infections of various sorts. The typical postonset survival duration is 10 to 20 years.

HIV-Associated Neurocognitive Disorder

The long-term neurocognitive consequences of human immunodeficiency virus (HIV) infection are a type of dementia. However, the introduction of the combination antiretroviral therapy in 1996 has greatly improved the health of persons with HIV infection. The infection is better treated, but the virus has not been eliminated. Also not eliminated are some of the neurocognitive consequences of the HIV infection. The former *AIDS dementia complex* or *human immunodeficiency virus encephalopathy* is currently described as **HIV-associated neurocognitive disorder** (HAND). Other terms used, depending on the context and severity of impairments, include *HIV-associated mild neurocognitive disorder* and *HIV-associated dementia*. Difficulties with concentration, memory, and some executive functions are the main symptoms of HAND. If untreated, HAND may worsen and more typical symptoms of dementia may appear (Eggers et al., 2017).

Severe motor disorders that were formerly associated with untreated HIV infection that included gait disturbances, tremor, seizures, ataxia, rigidity, and so forth have diminished with the treatment. Severe forms of dementia also are rare. Nonetheless, studies report HAND in 25% to 35% of infected and treated individuals (Sacktor et al., 2016). In most individuals, HAND is stable, is treatable, and does not progress to frank dementia (Eggers et al., 2017).

CLASS NOTES

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Chapter Outline

- Assessment of Dementia
- Clinical Management of Dementia
- Intervention for Communication Deficits
- The Caregiver Program
- References

Learning Objectives

After reading the chapter, the reader will:

- Give a brief description of procedures to assess mental status, cognitive functions, and intellectual levels of persons with dementia
- Describe the procedures and tools of communication assessment
- Specify the various management strategies implemented by speech-language pathologists to improve or sustain communication deficits

- Describe the caregiver program and environmental modifications

PRINTED NOTES

Assessment and clinical management of persons with dementia is a team effort, requiring contributions from medical and nonmedical specialists. Clients, their family members, and other caregivers are all involved in planning and implementing assessment and management programs.

Assessment needs to be continuous as the status of the individual with dementia changes. The goals of the management program also change as the individual's health status progressively deteriorates. This chapter is about the contribution of speech-language pathologists (SLPs) to dementia assessment and management.

Assessment of Dementia

The symptoms of moderate to severe dementia are easily recognized, although a detailed documentation of current skills and deterioration from a previous level may pose challenges. Diagnosis of mild dementia requires measurement of subtle changes, which may pose greater challenges. Verbal description of common objects, immediate and delayed recall of events, and word fluency are especially helpful in diagnosing mild and early forms of dementia, while pointing to stimuli and automatic speech tasks (e.g., reciting the days of the week) are least sensitive.

To diagnose dementia, the multidisciplinary assessment seeks to measure intact and impaired skills; establish a baseline of skills against which changes, both natural and possibly due to intervention, can be evaluated; design an intervention plan for the individual; and develop a caregiver program. SLPs' specific goals include thoroughly assessing the person's communication skills, establishing their baselines, and designing a communication intervention program for the person with dementia, the caretakers, and the family

members. Assessment begins with a carefully obtained case history to understand the changes that have taken place in the health and behavior of the person with dementia.

Case History

To better understand the individual's premorbid skills and behavior patterns and initial signs of dementia, the SLP may interview the person and multiple family members. Information on a variety of issues including the following factors may be obtained:

- The reasons for seeking help (the current problems the person, the family members, or both report)
- Mode of onset of the symptoms (sudden, gradual, constant, or fluctuating)
- General health of the person, presence of chronic diseases, and medical and surgical treatments received
- Family constellation and current living conditions
- Information on caregivers, financial situation of the person or the family
- Family history of dementia
- Educational and occupational history of the person, including information on hobbies, leisure activities, religious faiths, and typical daily activities
- The time when the family members noticed changes in behavior
- Medical conditions that could explain the changes, for example:
 - Strokes (single or multiple), brain tumors, head injury, or other neurological problems
 - Infections (e.g., HIV, neurosyphilis, prion disease), high fever, convulsions, drug toxicity, metabolic and thyroid deficiencies, or other medical conditions that might suggest a treatable and reversible form of dementia

- Alcoholism and other drug abuse along with prolonged use of anticonvulsants and certain antibiotics (e.g., penicillin)
- Prolonged dialysis for kidney problems
- Vitamin deficiencies (especially B1 and B12 deficiencies) and endocrine disorders
- Long-standing lung and cardiac diseases, anemia, and anoxia
- Neurological symptoms (e.g., tics, tremors, rigidity, facial gestures, gait disturbances) and the timing of their onset
- History of Down syndrome in the family
- Long-standing signs of such psychiatric problems as delusions, hallucinations, paranoia, anxiety, depression, and mania that may have existed prior to the onset of dementia
- Changes in behavior that the family members may have noticed include altered eating and sleeping patterns; emergence of irritability, eccentricity, emotional outbursts; all forms of atypical behaviors; deterioration in memory and other intellectual tasks; difficulty managing personal finances; and deterioration in self-care and in such daily activities as driving, cooking, and shopping
- Changes in speech, language, communication, reading, and writing

A detailed case history will lead to a preliminary diagnosis of dementia and possibly suggest a differential diagnosis. A firm diagnosis requires the completion of other aspects of assessment, however.

The SLP's main assessment target is communication and its impairments. A related task is to understand the behavioral status, typically described as mental status and cognition. Other professionals, especially the psychologists, may be more concerned with the mental status and cognitive functions. The SLP will have access to the results of physical and neurological examinations and laboratory tests, including brain imaging procedures. A *geriatric review* will have been completed to ascertain the individual's driving skills, continence, balance and mobility, and vision and hearing levels. The clinician will review the medical charts to understand the drugs the person is taking and their potential effects on performance of skills to be assessed. All professionals also

need to know the individual's advanced health care directives so as to honor their end-of-life wishes.

Screening Tests for Dementia

A quick screening test gives a clinical impression of current status and suggests the extent and depth of needed assessment. One of the following may be used:

- **The Mini-Mental State Examination** (Folstein et al., 1975). A widely used screening test on which a score below 30 is of clinical significance. Simple questions help assess orientation to time, naming three objects, recall of names tested earlier, following commands, attention, calculation, writing a sentence, and so forth.
- **7-Minute Screen** (Solomon et al., 1998). Screens recall, category fluency, orientation, and clock drawing. Standardized on persons with AD.
- **The Montreal Cognitive Assessment, Version 3 (MoCA)** (Nasreddine, 2011). Screens visuospatial skills, visuo-construction skills, naming, memory, attention, sentence repetition, verbal fluency, abstraction, delayed recall, and orientation.
- **The Information Questionnaire on Cognitive Decline in the Elderly** (Jorm & Jacomb, 1989). A questionnaire that a caregiver, a family member, or a friend fills out on the behavior of the person to understand how others evaluate the problems.

Other screening tests include the Alzheimer's Quick Test Assessment of Temporal-Parietal Function (Wiig et al., 2002) and the Burns Brief Inventory of Communication and Cognition (Burns, 1997).

Dementia Diagnostic Assessment Batteries

Several assessment batteries are available to diagnose dementia and distinguish its different forms. Clinicians should always select the reliable and valid tests and their latest editions.

- **Addenbrooke's Cognitive Function, Revised (ACE-R)** (Mioshi et al., 2006). Administrable in 16 minutes, this test helps assess attention and orientation, memory, fluency, language, and visuospatial skills. The test may help differentiate Alzheimer's disease from frontotemporal dementia and to assess cognitive deficits related to substance abuse. It is available free of cost in multiple languages.
- **Alzheimer's Disease Assessment Scale (ADAS)** (Rosen et al., 1984). In addition to assessing orientation, memory, language, and praxis, this test also assesses depression, anxiety, and psychosis.
- **Arizona Battery for Communication Disorders of Dementia (ADCD-2)** (Bayles & Tomoeda, 2019). This norm-referenced and standardized test may be administered to assess mental status, episodic memory (story retelling, word learning), language expression (description, naming, concept definition), comprehension (language and reading), and visuospatial construction (drawing, figure copying). The test can help differentiate mild, moderate, and severe cognitive and language impairments. It is available in English, French, and Spanish.
- **Dementia Rating Scale-2 (DRS-2)** (Jurica et al., 2001). This test helps assess attention, initiation and perseveration, construction, conceptualization, and memory. The test administration is discontinued if the person fails the initial items. It has alternative forms to assess changes over time in persons with dementia.
- **The Global Deterioration Scale (GDS)** (Reisberg et al., 1988). This is a 7-point rating scale to measure deterioration in behavior and intellect. It samples the individual's complaints and reactions (e.g., subjective feelings of memory loss) on levels of functioning ranging from no cognitive decline to very severe decline. Clinically significant scores on the GDS range from 3 (mild cognitive impairment) through 4 (mild dementia), 5 (moderate dementia), 6 (moderately severe dementia), and 7 (severe dementia). This scale has been widely

used in treatment efficacy research studies, although it is relatively insensitive to pharmacological effects unless they are very large.

- **Repeatable Battery for the Assessment of Neuropsychological Status Update (RBANS)** (Randolph, 2012). This test evaluates five domains: immediate memory, visuospatial/constructional, language, attention, and delayed memory. Such practical memory skills as remembering appointments, recall of faces and pictures, and story recall are targets of assessment. An available alternate form may be used to repeatedly test the same person to track progress in intervention. A Spanish version is available.
- **Functional Assessment of Daily Living, Second Edition (FADL)** (Holland et al., 1999). This test samples everyday living and communication skills, including nonverbal communication, reading and writing, humor, metaphoric language, and recognition of absurdities.
- **Functional Communication Profile, Revised** (Keilman, 2003). This test targets everyday communication skills with families and caregivers, along with verbal, nonverbal, sign, and augmented modes of communication.
- **Activities of Daily Living Questionnaire (ADLQ)** (Johnson et al., 2004). With the help of a knowledgeable informant, the questionnaire helps assess six functions: self-care, household care, employment and recreation, shopping and money, travel, and communication. It also helps assess decline in daily living skills over time.
- **Specific Memory Tests.** Although scales of dementia assess memory skills to some extent, specific tests may be used to assess memory skills in detail. In assessing dementia, measures of recently learned information are especially important. Specific tests of memory include the Wechsler Memory Scale—4th Edition (Drozdick et al., 2011), the Memory Assessment Scales (Williams, 1991), and the Revised Visual Retention Test, 5th Edition (Benton, 1992).

SLPs may design their own client-specific procedures to assess specific communication skills. For instance, clinicians may show pictures that depict a story or an event to evaluate such language skills as memory for words, temporal sequence, logical connections, antecedents and consequences, grammaticality of sentence structures, topic maintenance, and so forth. Many SLPs use the “cookie theft” picture of the Boston Diagnostic Aphasia Examination (Goodglass et al., 2001), but similar pictures may be used. To test memory for names, subtests of an aphasia test or the Boston Naming Test (Kaplan et al., 2001) may be appropriate. Individuals may be asked to give as many names as possible either within a category (e.g., food items or clothing items) or those that start with a given letter (speech sound). When their results are cautiously interpreted, tests of aphasia help identify communication and memory deficits.

Finally, clinicians may use assessment protocols of the kind provided by Hegde and Freed (2022). These protocols may be modified to suit an individual client, printed out, and used in assessment.

Ethnocultural Considerations in Assessment

Many dementia tests are not standardized on representative samples that include women, minority groups, and people of varied educational and socioeconomic backgrounds. Therefore, the results of such instruments should be interpreted with caution. Detailed behavioral observations and interviews of family members and the client help understand the cultural and communication background and perform an in-depth assessment of specific skills (e.g., memory loss or language deterioration).

Communication is a cultural phenomenon; therefore, a purely *culture-free test* is a misnomer. Translating a test to other languages may not solve the problem; in fact, it may introduce additional limitations because many terms used in dementia or aphasia tests may not be translatable. It has been shown that a simple clock-drawing task used in several tests to assess neurological impairments may not be predictive of such impairments in some ethnoculturally diverse clients (Storey et al., 2004). Nonetheless, there are a

few tests that are relatively culture free and may be preferred. The Mini-Cog (Borson et al., 2000) and the Rowland Universal Dementia Assessment Scale (RUDAS; Storey et al., 2004) may be two such tests. The RUDAS targets skills assessed by the popular Mini-Mental Status Examination described previously, but the items are selected to be valid across cultures, gender, and educational levels (Naqvi et al., 2015).

The SLP faces similar challenges in assessing ethnoculturally diverse persons with dementia and those with aphasia (Payne, 2014). Therefore, the clinician may consider the guidelines offered in [Chapter 5](#) on assessment of aphasia. In addition, several other sources offer suggestions on making a valid assessment of clients from varied backgrounds (Hegde, 2018; Hegde & Freed, 2022).

Clinical Management of Dementia

Although there is no cure for progressive dementia, there have been sustained efforts at developing pharmacological, behavioral, and communication treatment strategies. Different professionals believe that treatment makes a difference. Mild cognitive impairments may be especially receptive to treatment. In their overview of 38 systematic reviews and 142 primary studies of nonpharmacological interventions for persons with dementia, Abraha et al. (2017) concluded that well-designed and implemented behavioral interventions including communication strategies, pharmacological treatment when combined with behavioral interventions, family and professional caregiver training, physical exercises, music therapy and other art-based approaches, dining room and other environmental modifications, person-centered care, and home-based behavior management techniques were beneficial. The effects were mostly modest, however, with room for methodological improvements.

Dementia treatment research is extensive in quantity, limited in quality. It is not uncommon for systematic reviewers of research to find on aggregate databases thousands of published articles and select only a handful of studies that meet the reviewer's quality criteria. Most reviewers reject up to 99% of

studies they find initially because of numerous methodological deficiencies. (Part of the problem lies in the database search engines that return many irrelevant studies.) For example, Hall et al.'s (2021) initial database search on physical intervention for persons with dementia returned an impressive 7240 studies, but they selected just 4 studies (0.0005%) for full review. Kim and Park (2017) initially found an astonishing 18,157 studies on person-centered care and reviewed only 19 (0.001%). Furthermore, individual reviewers of studies draw differing conclusions on the effectiveness of the same procedures. Treatment research reviews are made to find strengths and limitations of individual studies, but now *reviews of reviews* are made to find problems in published reviews or draw still broader or different conclusions (e.g., Abraha et al., 2017; Meyer & O'Keefe, 2020). The strength of evidence the reviewers find for procedures varies across studies. Investigators may implement the same treatment differently or describe the procedures vaguely. Therefore, data from many studies cannot be combined to perform meta-analysis. Clinicians cannot depend on meta-analyses to understand treatment procedures. They need to consult the individual treatment studies for procedures. Unfortunately, many treatment researchers, especially those with a cognitive bent, tend to write extensively about treatment theory and targets but little about procedures. Furthermore, generalization and maintenance of skills that improve during treatment have been inadequately documented. All experts call for better-designed studies that produce stronger evidence (e.g., see Abraha et al., 2017; Aguirre et al., 2013; Carrion et al., 2018; Hopper et al., 2013; Kim & Park, 2017; Machiels et al., 2017; Meyer & O'Keefe, 2020; Murray & Paek, 2016; Swan et al., 2018; Tisher & Salardini, 2019, among many other reviewers). Call should also go out for replicable treatment procedural descriptions instead of underlying theories that are usually speculative and rarely related to the procedures.

Measuring the Effects of Interventions for Dementia

Collecting and evaluating evidence of dementia interventions are complicated. There are multiple personal and methodological variables that confound the

effects of interventions for dementia. The age, the premorbid language and cognitive skills, education, health, and the course of dementia are among the personal variables that may interact with or confound the treatment effects. Natural age-related decline in the skills and decline due to the progressive course of the disease interact with treatment effects and are hard to separate. Most of these intrapersonal variables may make the intervention appear like it was less effective than it actually was. A few of those variables, especially higher level of skills and better general health, may enhance the treatment effects. This is true of both medical and nonmedical interventions.

What to measure and how to demonstrate treatment effects are methodological problems. In the order of most desirable to minimally acceptable effects, one might measure the following: (1) a reversal of the deteriorating process and recovery of normal skills (cure), (2) partial recovery of most skills, (3) partial recovery of some skills, (4) reduction in the severity of symptoms while the progression of the disease is slowed, and (5) reduction in the severity of symptoms while the rate of deterioration is unabated. At least one of these outcomes must be realized to document treatment effects.

There are no standard methods of measuring changes in the skills that are treated. Pre-post changes in standardized test performance are often reported in treatment studies, but changes in test scores are not informative of the changed skill levels in everyday living; frequency of behaviors before, during, and after treatment is more informative. Across studies, social participation, narrative and discourse analysis, improved quality of life, life satisfaction, and improved mood and emotion, among others, are not measured objectively or consistently. Lack of measurement of generalization and maintenance are additional problems of treatment research.

Classification of Interventions

Being a complex set of deficits, dementia requires the expertise of different professions. Interventions are a collection of highly varied procedures that different professions implement with varied levels of expertise and theoretical orientations. Generally, most professions practice within their own scope of

practice, but evidently there are no such boundaries in behavioral interventions for dementia. Psychologists, social workers, nurses, occupational therapists, physical therapists, recreational therapists, aides and assistants, graduate students from different disciplines, and SLPs administer behavioral treatments for dementia. Some treatment research reports do not specify who administered the treatment. In fact, other professions have done more studies on dementia intervention than SLPs.

There is no uniform classification of dementia interventions, and the validity of typical classifications is questionable (Tisher & Salardini, 2019). In general, interventions may be first grouped into two main categories: medical (pharmacological) and nonmedical. The nonmedical approaches may be subcategorized according to the treatment targets, procedures, or (variably) the professions that implement them. Interventions are described as cognitive, behavioral, physical, and communication. *Cognitive intervention* is currently a popular category among researchers, including SLPs, even though much of that intervention is not about communication. But all procedures included under cognition as well as under communication are behavioral. Only the drug interventions are not behavioral. The distinction between behavioral and cognitive interventions is not clear. Naming objects, faces, or things that belong to a category (e.g., food items) is not *cognitive*. Nor are the spaced retrieval or fading treatment stimuli.

First, a brief overview of pharmacotherapy (medical treatment) is provided. Subsequently, SLPs' communication intervention strategies, including augmentative and alternative communication, are described. Finally, the caregiver training programs, including environmental modifications, are described.

Pharmacotherapy

There is intense interest in developing and testing newer and more effective pharmacological treatments for dementia. With increased funding made available for treatment research, many investigators in both the academia and pharmaceutical industry are at work to develop new medical treatments for

dementia. Unfortunately, most newer drugs have failed at clinical trials (Cummings et al., 2019). There is no cure yet for dementia.

The U.S. Food and Drug Administration has approved two classes of drugs for dementia, mostly for Alzheimer's disease: one to delay symptom onset and the other to treat specific symptoms. The drug aducanumab (Aduhelm™) is prescribed to delay symptom onset. It is an anti-amyloid intravenous infusion therapy administered once every 4 weeks to reduce the chances of plaque formations in the brain, the main brain disorder in Alzheimer's disease. Improvements in memory, orientation, language, and daily living skills have been noted with its administration.

Drugs that target specific symptoms include donepezil (Aricept®), rivastigmine (Excellon®), and galantamine (Razadyne®). These three are *cholinesterase inhibitors* that prevent the breakdown of acetylcholine, a neurotransmitter. The drugs may support memory and new learning by increasing the level of acetylcholine in the brain and, thus, neurotransmission. Memantine (Namenda®) regulates glutamate, an excitatory chemical in the brain. This may help memory, attention, and verbal skills. Finally, there is a combination of a cholinesterase inhibitor (donepezil) and a glutamate regulator (memantine), called Namzaric®. Several newer drugs are in various phases of clinical trials (Cummings et al., 2019). Standard drugs may be prescribed to control anxiety, depression, insomnia, seizures and tremors, and other symptoms. Persons with dementia may derive greater benefits with a combination of drugs and behavioral interventions.

Intervention for Communication Deficits

Stabilizing or improving **communication** skills of persons with dementia is the main concern of the SLPs. Other professionals and family caregivers are involved in the process. Understanding their skill deficits, persons in their early stage of dementia may be open to direct treatment. Because of the inevitable decline in skills, it is prudent to include augmentative and alternative communication (AAC) with typical speech-language interventions. In the more advanced stages, the concern is to maintain communication to the extent

possible, supplement it with AAC, and help caregivers adapt to the needs of the persons with dementia and support their efforts at communication with any means possible.

Historically, Alzheimer's disease and, in recent decades, primary progressive aphasia have attracted behavioral treatment researchers' attention. Nonetheless, no single treatment study includes a comprehensive program to address most if not all of the deficits associated with dementia. Investigators concentrate on single skills, such as naming or memory — the two skills that have been researched the most. Clinicians are on their own to put together a comprehensive treatment program for their clients.

Interventions for Naming Deficits

Naming deficits, often described as word-retrieval deficits, are common in most forms of dementia and some forms of aphasia, especially Broca's. Therefore, the **naming treatment** for dementia is essentially the same as that for aphasia, described in [Chapter 5](#). The basic procedure is to present pictures or objects and ask the client to name them. Selected naming targets are personally meaningful to the client (e.g., names of everyday objects or activities). Printed words may accompany pictorial stimuli, in which case, the person may be asked to read it aloud or write it down. Picture stimulus presentation may accompany played audio recordings. As described in [Chapter 5](#), the basic treatment procedures are modeling, cueing, and fading modeling to promote more spontaneous naming. A cue for a word may be phonemic (e.g., "*It starts with an A*" for *apple*) or semantic ("*You find it in the kitchen*" for *pan*). All cues should be person specific to make them more meaningful (e.g., "*It is your favorite breakfast food*" for *oats*). The reader may review [Chapter 5](#) for details.

A variety of naming intervention researched in dementia is called the **face-name association**. A vast number of studies have taught individuals with dementia the names of persons (famous persons, unknown persons, family members, other caregivers, and so forth) with the spaced-retrieval method. A printed photograph, an image on the computer or tablet screen, or a silent

video clip is shown; a question is asked (e.g., “What is her name?”); the name is modeled; and the person imitates the model. If it is spaced retrieval, the intertrial interval is increased. Studies have generally demonstrated that the individuals with dementia could name the pictures weeks and months later (for a sample study, see Viccaro et al., 2019; for a review of studies, see Hopper et al., 2013).

Another variation of naming deficit intervention is called **category naming** or **verbal fluency**. In this variation, the person may be asked to recall names within specific semantic categories (e.g., naming as many *animals*, *vegetables*, or *clothing items* as possible in 1 minute). Cues may be phonemic or semantic as previously described. A picture that shows an item from a category (e.g., an orange for *fruits*) may be shown, with the name written under it. On repeated trials, positive reinforcement for correct responses and corrective feedback for incorrect responses are offered (for detailed procedures, see Beeson et al. 2011; for a review of studies, see Murray & Paek, 2016). Generalization of learned names to conversational speech and maintenance over time have been limited.

Naming cues may be given in a hierarchical manner to evoke correct names. For instance, Henry et al. (2018) provided the next cue in their hierarchy only when the person failed to respond to a previous cue. Showing a picture, the clinician first asks the person to tell something about it; if no response, instructs to write the word; if no response, gives a graphemic cue and then says the word; and so forth. See also [Chapter 5](#) for cueing hierarchies.

The semantic and the phonological cueing hierarchies have been found to be effective in teaching naming skills to persons with PPA, regardless of the specific PPA variety (for systematic review of treatment studies, see Morello et al., 2017; Pagnoni et al. 2021; Suarez-Gonzalez et al., 2021; Volkmer et al., 2020). Generalization has been variable and maintenance has been limited. Interventions have rarely moved beyond single word naming.

Conversational Skills

While most studies have taught only single words to persons with dementia, Arkin (2001), Arkin and Mahendra (2001), and Mahendra and Arkin (2003, 2004) have trained more complex language productions, including story recall and proverb interpretation, combined with physical exercises. These are described in a later section on combined interventions. Gentry and Fisher (2007) taught conversational repair skills to three individuals with Alzheimer's disease. They evaluated the effects of a direct procedure in which the investigator interjected and corrected the person who made unclear statements and an indirect procedure in which the investigator paraphrased (e.g., "So what you are saying is . . . " or "If I understand you correctly . . . "). The indirect repair was more effective than the direct repair.

Script Therapy

Script therapy, which has been used in aphasia treatment, does target verbal expressions that are more complex than word retrieval (naming). A **script** may be a few sentences describing the client's hobby, travel, or some specific experience (see Henry et al., 2018, for details). For instance, independently or with assistance, an individual may write a paragraph about the time spent at the Grand Canyon or the city of Chicago.

Script training moves through a hierarchy of reading the script to answering questions about it. The final task is to produce the whole script from memory. The scripts may be video recorded for the person to repeatedly practice them at home. The procedure has been effective in producing connected and grammatically correct sentences. The skill was maintained up to a year on follow-up (Henry et al., 2018). More research is needed to show that script learning leads to better functional and conversational speech. SLPs also may consider other procedures described in [Chapter 5](#) to expand word and phrase productions into sentences and simple conversational speech.

Spaced Retrieval

Spaced retrieval is a procedure in which the person is taught to recall a piece of information with progressively longer intervals; it is a behavioral shaping method in which the duration between recalls of specific information is gradually increased. Because of the shaping procedure, few or no errors are made; such **errorless learning** may promote better retention. Spaced retrieval has been shown to be effective in improving the use of memory aids by study participants with different kinds of dementia (Bourgeois et al., 2003, for a review).

The spaced-retrieval training may begin with a sequence like the following: “When you want to know what you should do today, you look at your list of daily activities. What do you do when you want to know about what you should do today?” If the person replies, “I look at my list of daily activities,” the clinician reinforces the correct response and proceeds to talk about extraneous topics for a predetermined duration (1 minute, 2 minutes, etc.). At the end of an interval, the clinician asks the question again. If the response is wrong, the clinician models the correct response; the person imitates and gets reinforced. The interval between the prompts (questions) is increased gradually until the person gives correct responses with a 24-hour interval between prompts. When the intervals reach several minutes, the clinician may train other skills during that interval (e.g., having the person sort objects instead of simply talking about extraneous topics).

Reminiscence Therapy

Reminiscence is narrating personal life events to others. Pictures, objects, verbal prompts, and other means may stimulate a person to talk about personal events from the past, memory for which is usually better preserved. The reminiscence therapy procedures, however, are vague. For instance, individuals with dementia may be asked to *talk about dramatizing events in life; review historical terms; narrate life memories; describe past, present, and future; express thoughts and feelings*; and so forth. Judging from reviews of studies, it appears that none were conducted by SLPs. Depending on the reviewers, reminiscence therapy may produce none, negligible, or modest effects that may not be maintained (Redulla, 2020; Saragih et al., 2022; Woods et al., 2018).

Combined Interventions

Combining communication treatments with other forms of treatment, including drug treatment and physical exercise, may be one of the most effective strategies (see Hopper et al., 2013, for a review of studies). For instance, Chapman et al. (2004) combined behavioral communication intervention with the cholinesterase inhibitor donepezil. Individuals with Alzheimer's disease who were taking this drug received training in producing relevant verbal content through discussion and description of life events and getting involved in hobbies and activities at home. The authors reported that their study participants' discourse skills and emotion improved, and their behavioral problems decreased. They did not describe their treatment procedures in operational terms to help replication, however.

Arkin (2001), Arkin and Mahendra (2001), and Mahendra and Arkin (2003, 2004) have combined language stimulation with fitness and strength training sessions as well as recreational activities that involved social interactions. Language stimulation activities included picture and object description, story recall with quiz, category naming, generative naming, problem solving, proverb interpretation, word association, famous name recall, and so forth. Reporting on a 4-year program, Mahendra and Arkin (2003) showed that persons with dementia improved their discourse skills.

Studies done by occupational and physical therapists have suggested that physical exercises may help slow down both physical and cognitive decline (Sujkowski et al., 2022). Evidence suggests that cognitive training alone does not check decline in letter and category word fluency, whereas aerobic exercise increases it (Nocera et al., 2020). Therefore, it might be a useful strategy to combine exercises with speech-language intervention to enhance the effects of the latter, as the studies of Arkin and Mahendra have shown.

Augmentative and Alternative Communication

Because of the progressive nature of dementia, severe impairment in communication (and other behaviors) is inevitable. Therefore, persons with

dementia will experience a need for augmentative and alternative communication (AAC) sooner or later. Some clinicians may wait until the oral and typical communication skills deteriorate to a level that AAC seems like the only option. However, research suggests that AAC should be introduced sooner rather than later (Fried-Oken, 2008). If an AAC device is introduced at the beginning of verbal intervention, the person with dementia will have learned to use the device before the decline that makes learning new skills more difficult. When mastered, AAC may help the individuals maintain a certain level of social interaction and participation. At an advanced stage, AAC may be essential to communicate basic needs to the staff and family members.

May et al. (2019) reviewed research on current AAC strategies used with persons with dementia of the Alzheimer's type and found that most investigators had targeted nonelectronic memory and communication aids to support social interactions. Studies are extremely limited on individuals with other types of dementia and on the use of electronic devices. A few useful strategies that have received experimental support will be described briefly. For the procedural details, the clinicians are encouraged to consult the cited articles.

External memory aids include methods that can trigger appropriate behaviors at the right times. Generally, making the person rehearse the information in the form of memory drills is not effective in prompting behaviors. Of the several kinds of memory aids, the most researched are the memory wallets or memory books. **Memory wallets** are a set of index cards that contain pictures and sentences about familiar persons, places, and events the individual could look at during conversation and talk. *Memory books* contain similar personal and factual information and can be in the paper or digital format. Bourgeois (1992) taught individuals with dementia whose reading skills were still intact to use their memory wallets. The study participants produced more appropriate, elaborate, on-topic, and factually more accurate verbal expressions with the cards than without them. The cards used during conversation reduced errors as well as ambiguous, unintelligible, and perseverative utterances. The study participants had maintained their conversational skills up to 30 months postintervention. Professional or volunteer caregivers may be trained in the use of memory wallets of persons with dementia. The effects of this method have been replicated in multiple

studies since the original report (see May et al., 2019, for review and evaluation of studies).

Many other forms of external memory aids may prompt timely and correct responses from individuals with dementia. Calendars, maps, cue cards, diaries, written to-do lists or daily reminders, weekly planners, watch alarms, timers, vibrating signals, prominently displayed written signs, and orientation boards have been helpful to many individuals.

A low-technology communication board or a high-technology digital display should include words, sentences, icons, pictures, scenes, themes, topics, messages, and so forth that are relevant to the users. In their investigation, Fried-Oken et al. (2015) had 109 individuals with Alzheimer's disease or primary progressive aphasia suggest topics or choose them from a supplied list. A family member or a familiar informant assisted the persons in topic selection. The conversational themes the participants suggested included such personally relevant topics as hobbies, family, work, travel, and places lived. Each of these themes included multiple topics (e.g., specific hobbies or places lived). The list gives general guidelines on topic selection, but SLPs need to find out what interests their individual clients the most. Conversational topics should be ethnoculturally relevant to the individuals; letting them select the themes and topics is the most appropriate strategy.

Only a few studies have investigated the effectiveness of electronic forms of AAC. They include digital memory books, specialized computer application on laptops and tablets with or without audio-recorded sentences, applications for reminiscence therapy, PowerPoint videos, smartphones, and so forth. Digitized voice output devices and applications are fewer still and mostly in the early phase of research.

Wilson et al. (2022) made a systematic review of studies on **smartphone** and **tablet** use by older individuals with and without dementia. The reviewers found that most investigators had trained individuals with dementia to use a smartphone or a tablet to get organized for daily activities, remind them about activities and appointments through audible alerts, send text messages to family members and professional caregivers, make the *to-do* and shopping lists, use the *map function* to plan shopping trips and find their way back home, make emergency calls, keep timetables of public transportation, find word

meaning in online dictionaries, and so forth. Although high-quality studies were limited, the reviewers found that persons with dementia benefited from the use of smartphones and tablets.

Wilson et al. (2022) noted that smartphone or tablet users may feel more secure, more autonomous, and connected with their caregivers. They may feel less fear and anxiety when out of their homes. Both the users and family members reported improved quality of life. Several individuals had maintained their skills at 6-month follow-ups, and some had learned new skills not trained. Many users liked the discrete function and portability of smartphones. On the negative side, the cost of the devices, lack of access to high-speed Internet connections, lack of training, and low level of education and technical knowledge were the barriers.

When severe dementia and motor impairments preclude the use of technological devices, individuals may be taught to use such unaided means as pointing, eye gaze, looking, gestures, and facial expressions to communicate basic needs (May et al., 2019). If feasible, approaches may be combined (e.g., eye gaze and a communication board).

To design an AAC program for an individual, the SLP first needs to consult the individual and the caregivers to select a method (e.g., memory wallets or a smartphone) and the target skills (e.g., appointment reminders, activity schedules). Beyond teaching the method-specific skills (e.g., using a tablet), the SLP may teach the individual to:

- Make a list of what to do each day.
- Write down instructions and directions when memory begins to fade.
- Keep phone numbers and possessions in a particular, invariable place.
- Keep a checklist of things to do before leaving the house.
- Carry a card with the names, addresses, and phone numbers of caregivers.
- Wear a bracelet that contains identifying information and names, addresses, and phone numbers of caregivers.

- Keep a large calendar in the living room and bedroom and cross off the current date every night to help maintain orientation to day and time.
- Keep a map of frequently visited places (e.g., homes of relatives or friends, shops, and restaurants).
- Request that messages, especially more complex directions and instructions, be given orally as well as in writing.
- Consult written messages and instructions frequently enough to follow through.
- Request more time to speak or respond to questions. This might be especially useful to individuals who have word-finding problems or unusually long reaction times.
- Express or gesture failure to understand a message when such is the case. This will prompt the speaker to alter the message to make it more easily understandable.
- Request the speaker to repeat a spoken message when it is not understood.
- Request conversational partners to make periodic reminders about the topic.
- Give descriptions of objects and persons or nonverbally indicate the use of objects when attempts at naming fail.
- Use gestures, signs, facial expressions, and any other form of nonverbal communication when words fail.
- Combine different methods of communication (e.g., memory books and smartphones).
- Use face-to-face computer interactions (e.g., Skype or Facetime).

The Caregiver Program

The caregiver program, as important as the direct intervention provided to the individual, has three components: (1) caregiver education about dementia, (2) teaching the caregiver behavior management tactics, and (3) teaching effective

communication strategies. Because of the progressive nature of the disease, caregivers need to continuously modify their behavior to meet the changing needs of the individual with dementia.

The caregiver program is implemented to help the persons with dementia as well as their caregivers. At home or at a professional setting, dementia care is a stressful occupation. For the family members, it is a day-and-night responsibility. Therefore, a good caregiver intervention program may reduce the stress and give the caregivers the tools and methods to effectively care for the individual so that their own stress level is manageable.

There is no comprehensive and systematically researched caregiver program that applies to all staff members in all settings or all families. Problems that need intervention are likely to vary across staff within and across professional settings, and family members and their home situations. Intervention should be specific to persons involved (caregivers and care receivers). Therefore, SLPs need to design their own programs based on their observations of caregiver-receiver interactions in a specific setting.

Caregiver Education

Most individuals with mild to moderate dementia are cared for in their homes. Family members take care of a person with dementia from 2 to 15 years. Individuals are likely to be institutionalized only when the family cannot cope with the ever-escalating demands of taking care of the person. After spending some time in a nursing home, most people enter an extended care facility, perhaps eventually a hospice, during the final stage of the disease.

Both the individual and the caregivers need behavioral intervention. An individual's family caregivers are prone to depression and chronic stress, especially when they cannot attend to their personal affairs. The degree of stress the caregivers feel is correlated with nursing home placement of their loved one. It is known that the stress the caregivers experience physically harms them: Their immune response may decrease and inflammation may increase, resulting in vulnerability to infections and diseases (Roth et al., 2019). Emotional and social support offered to caregivers may reduce their stress and

improve their physical health. Furthermore, effective support offered to family caregivers may delay nursing home placement of people with dementia.

The caregiver program involves multiple steps as outlined:

- **Educate the caregivers about dementia.** Family members may need and appreciate more extensive information than the professional caregivers. Give the caregivers information on the specific type of neurological disease and the dementia in the individual they are caring for. Point out both the strengths and limitations of the individual. Describe the prognosis and the long-term rehabilitative needs.
- **Review the problem behaviors and their potential causes.** Offer information on how lack of communication skills may lead the person with dementia to behave aggressively or inappropriately. For example, unable to request a drink, the person may grab it from another person.
- **Give an overview of communication deficits associated with dementia.** This is an SLP's unique responsibility.
- **Review available resources.** Give the caregivers a list of social and professional organizations and agencies in their community that might provide volunteer help, respite care, financial assistance, self-help groups, transportation to medical or rehabilitation facilities, and so forth.
- **Counsel the caregivers.** Provide counseling to the caregivers to cope with the demands of caring for a person with a degenerative neurological disease; refer the family member or members to a psychologist or other professionals if found appropriate.
- **Arrange for respite care.** Respite care gives the family members a break from the unrelenting demands of care and gives them an opportunity to relax or attend to other aspects of their lives that have been ignored or postponed.
- **Educate the caregivers about potential abuse.** Nearly half of all those who have dementia may be abused. Older women are especially vulnerable to abuse or mistreatment by caregivers,

especially those who are stressed out or burned out (Payne, 2014; Wigglesworth et al., 2010). Family members should be aware of signs of physical abuse, emotional abuse, sexual abuse, neglect, and financial exploitation at the institutions in which the individual lives.

After baserating the communication patterns between the caregivers and the person with dementia, the SLP may design and implement an intervention program to teach the skills needed for improved interactions.

Caregiver Management of Behavior Problems

Persons with dementia may exhibit serious disruptive, aggressive, and noncompliant behaviors. Prudent management of such behaviors helps secure the individual's safety, health, and comfort. The caregivers also may experience reduced occupational stress and enjoy a better working environment.

Behaviors that are most distressing to the caregivers include emotional outbursts, agitation, depression, hallucinations, egression (leaving the house, wandering), and noncompliance (e.g., refusal to eat or take medications). Psychiatrists may treat such problems as hallucinations and depressions. Nonetheless, caregivers need positive strategies to manage problem behaviors without being punitive. The caregivers may be taught to do the following:

- **Minimize the triggers (antecedents) of disruptive behaviors.** Agitation, emotional outbursts, noncompliant behaviors, and aggressive actions may be triggered by too little or too much sensory stimulation. Too much noise, too many tightly scheduled activities, too hard task demands, general fatigue, uncontrolled physical symptoms, and lack of sleep may trigger problem behaviors. To the contrary, lack of typical sensory stimulation the person is used to (e.g., soothing music, a quiet place to relax, a game or activity the person enjoys) also may trigger disruptive behaviors. The caregivers may be taught to refrain from making impossible demands on the person. They may set achievable targets, such as *quiet sitting for 10 minutes* instead of 1 hour. If too many activities seem to agitate a

person, plan for a few activities the person enjoys the most. If fatigue seems to play a role, give frequent rest pauses between daily activities.

- **Understand that communication deficits may cause problem behaviors.** Persons who cannot express their needs and wants may behave inappropriately. Therefore, the caregivers may be taught to anticipate needs and wants and try to fulfill them. Instead of expecting well-formed verbal expression, they may accept and act on any type of communication (e.g., gestures, single words, nonverbal behaviors that suggest a need to go to the bathroom).
- **Minimize or eliminate the triggers of anxiety and agitation.** Reducing task demands, simplifying daily routines, and following a predictable schedule of activities, among others, may reduce a person's anxiety and agitation.
- **Reinforce alternative behaviors.** If the individual appears to get depressed because of lack of stimulation, create opportunities for activities and habitually enjoyed sensory stimulations. Reinforce and increase the frequency of such stimulating activities as listening to music, playing games, and watching TV.
- **Maximize positive reinforcement.** Praise and offer tangible reinforcers (e.g., a favorite drink, an opportunity to listen to music, play a game) for such desirable behaviors as sitting quietly, engaging in social interactions, watching television, and so forth. Refrain from ineffective criticism and unhelpful explanations as to what is good and bad.
- **Ignore certain problem behaviors.** Behaviors that are not injurious to self or other persons may be ignored. Withholding attention and other positive reinforcers may decrease repeated questioning and stereotypic utterances.
- **Limit choices to avoid confusion.** A choice of only two beverage items at a time (e.g., "Do you want orange juice or milk?"), two clothing items (e.g., "Do you want to wear this blue shirt or this green shirt?"), and so forth will help reduce confusion.
- **Avoid confrontation.** Guide the person, and offer the needed help. It is not useful to ask the person, "Why are you doing this?" Instead,

encourage the person to do what is desirable; model the behavior if necessary.

Effective Caregiver Communication

There is some evidence that caregivers in institutional settings do not communicate much with the individuals they care for. Typical communication consists of controlling instructions *to do or not to do* something. There is little verbal interaction. Professional care staff's tendency to patronize or elderspeak has been noted. **Elderspeak** is characterized by high pitch, exaggerated intonation, simplified grammar, and slow rate (Williams et al., 2003). Although simpler sentences may be helpful to persons in advanced stage of dementia, other features of elderspeak stereotype the persons. Lack of meaningful interactions with the staff may add to the individual's stress and decline. In their review of studies, Machiels et al. (2017) found effective strategies to improve communication between the caregivers and those they cared for. For instance, Sprangers et al. (2015) offered brief training to nursing aides to give short instructions, produce positive speech, and make biographical statements. The staff did increase these target skills and reported reduced stress. But the authors did not define or describe these communication styles in replicable terms. Most studies mention such skills as *shorter sentences* and *active listening* but do not give operational descriptions of communication skills taught to the long-term care staff (for reviews, see Machiels et al., 2017; Morris et al., 2018). Descriptions of training procedures, too, are not specific enough to facilitate study replication or clinical application. Also, many studies report that caregivers learned new communication skills but do not offer evidence that the caregivers or the persons with dementia experienced significant benefits. Therefore, there is still a need to document the caregiver program's effects on both the caregivers and the persons with dementia. Clinical intuition and experience, however, continue to support a well-designed caregiver program. Researchers need to develop better methods to measure the effects of caregiver training on the quality of life of both the individuals with dementia and their caregivers.

For a clear description of communication skills and training methods, SLPs may consult Bourgeois et al. (2004), who did one of the early studies on training nursing aides in a facility for persons with dementia. Their baseline data showed that the staff may give ineffective multistep instructions, make negative statements, ask unhelpful questions, and so forth. They also found that announcing care when entering the resident's room, addressing the person by name, introducing self by name, announcing every activity, talking about the resident's life, waiting for a few seconds before providing physical help, giving short and clear instructions, and providing plenty of positive reinforcement for the resident's desirable behaviors were effective communication strategies.

In the absence of well-researched and comprehensive strategies, SLPs may design setting- and person-specific caregiver communication training programs based on their observations. The SLP may collect systematic data during interventions so that the promising procedures may be continued or modified and ineffective procedures may be abandoned. The following outline suggests what the SLP can train the caregivers to do.

- **Approach the person slowly.** The person with dementia may be apprehensive of sudden events.
- **Establish eye contact.** Place yourself in the individual's visual field before speaking.
- **Have the person use the typical assistive devices.** Make sure the person is wearing the eye glasses, using hearing aids, or holding the tablet or a smartphone routinely used for communication.
- **Secure the individual's attention.** Before speaking, call out the person's name. Bend down or kneel to speak to a seated person.
- **Address the person properly.** Ask whether to use the person's first or last name. Avoid terms of endearment (e.g., "honey" or "sweetheart") or the institutional *we* ("How are we doing today?"). Use your first name if the person responds better to his or her first name.
- **Ask personally relevant questions.** For instance, ask, "What do you do today at 10? Look at your list and then tell me." Such questions help the person remember activities as well as use the memory aid

(the list). Also, ask questions about events that mean something to the individual. For instance, ask, “Who is coming to see you today?”

- **Give a prompt.** The first letter of a word, the first word of a sentence, a picture, or a physical object may be used as a prompt for the expected response.
- **Model when prompts fail.** After asking a question, wait for a few seconds before modeling the response.
- **Listen and watch.** Do not interrupt the person; even with impaired language expression, the person may communicate with a few key words, gestures, or facial expressions.
- **Accept circumlocution.** Let the person talk in indirect and roundabout ways when word retrieval hinders specific expressions.
- **Supplement verbal expressions.** When you talk, use gestures, smiles, posture, and other cues. Give a nonverbal cue before the verbal command. For instance, gesture to *stand up* before telling the person, “It is time to go to the dining room for breakfast.”
- **Demonstrate the requested actions.** For instance, while asking the person to wipe the dining table, demonstrate the action and then hand the piece of cloth to the person.
- **Avoid elderspeak.** Speak clearly and with normal prosodic features, but do not exaggerate the articulatory movements or intonation patterns.
- **Speak in simple and short sentences.** But do not speak too slowly unless you know the person responds better to slower speech.
- **Repeat or modify sentences.** When complex sentences are necessary, repeat them; when not understood, say them differently.
- **Speak softly rather than loudly.** However, consider the individual’s hearing acuity.
- **Specify the referents.** Minimize the use of pronouns (e.g., *he*, *she*, and *it*); instead, use the names of the individuals or the objects being referred to.
- **Ask yes/no questions.** When factual responses are needed, ask direct questions (e.g., “Would you like cereal for breakfast?” instead of

“What would you like to eat for breakfast?”).

- **Ask open-ended questions.** To encourage free expression of narratives, feelings, and opinions, avoid yes/no questions. For instance, ask, “Tell me how you feel this morning,” instead of, “Do you feel good this morning?”
- **Follow the *one thing at a time* rule.** At any one time, ask only one question or one thing to do; follow one-step direction. Model the verbal or nonverbal response as needed.
- **Give manual guidance.** Offer physical support for actions requested (e.g., physically help the person stand up while requesting to do so).
- **Be redundant.** Restate important information and repeat instructions with different words.
- **Talk about familiar topics.** Talk about personal events, happenings in the surroundings, and objects present.
- **Give background information.** Do not be abrupt; when making a new request or introducing a new topic of conversation, give sufficient background as to why and what.
- **Help maintain the conversational skills.** Prompt the person to stay on the same topic, request more details of narratives, listen when others are talking, say something when it is appropriate, and so forth. The caretaker may ask such questions as, “How does the story begin?” “What happened next?” “Who did it?” “What did she say?” “How did the story end?”
- **Ignore repeatedly asked stereotypic questions.** Directly respond one time, and then ignore. Direct the individual’s attention to a message board, a memory aid, a printed list, and such other sources of information; also, try to distract the person with an activity or a new topic of conversation.
- **Touch the person.** Touch to comfort, draw attention, reassure, and strengthen verbal contact. Before doing any of these, make sure the person welcomes it.
- **Respond to nonverbal behaviors.** Respond to the person’s nonverbal behaviors, gestures, postures, and facial expressions,

especially when the communication skills deteriorate.

- **Reinforce abundantly.** Positively reinforce appropriate verbal and nonverbal behaviors (e.g., offer verbal praise for following social conventions or being compliant).
- **Expect unfamiliar expressions in a bilingual person.** Watch for expressions in the primary language that may emerge in a bilingual person. Get help in interpreting such expressions.
- **Greet and say goodbye.** Follow the typical social conventions.

Environmental Modifications

Environmental modifications are changes in the living conditions at homes or institutions to help maintain communication and interaction, reduce undesirable or dangerous behaviors, sustain the general health and safety of the person, and reduce the burden of care on the caregivers (Bayles et al., 2020; van Hoof et al., 2010). This aspect of dementia rehabilitation, however, is not the exclusive province of SLPs. Other professionals (e.g., social workers and occupational therapists) may play a greater role than the SLPs in redesign the living arrangements of persons with dementia.

To modify the client's living arrangement, the caregivers and professionals need to work as a team to perform the following:

- **Make an initial analysis.** Analyze the individual's living environment and identify the barriers and dangerous situations that need to be modified. Check the living room, kitchen, bathrooms, bedrooms, front and back yards, and any other attached spaces for possible threats to an individual's safety and comfort. Use this analysis to design an environmental modification program.
- **Do not overly curtail freedom.** Safety of the person may require certain restrictions, but allowing freedom within the safety limits should be a priority. Excessive restrictions on movement, however, may induce agitation and anxiety. There may be no need to restrict access to items and spaces that do not pose a threat. For instance,

there may be no need to hide a TV remote or access to a room with no hazards. Family members may design a secured courtyard or a garden into which the doors open for the person to walk about.

- **Control egression.** Exiting the home and getting lost may be a dangerous problem in the later stages of the disease. Use locks that need a key to open the door and keep the key in a safe place; use electronic security devices that require a code to unlock the doors.
- **Limit or eliminate access to hazardous materials.** Keep knives, scissors, hammers, screwdrivers, saws, lawn mowers, and such other tools and instruments out of reach. Have the family members lock up or remove guns, poisons, pesticides, paint, and medications.
- **Improve kitchen safety.** Install devices that automatically switch off stoves or induction cooktops.
- **Establish a simple and consistent routine.** Keep the schedule of activity constant and consistent across days, although providing some variety may enhance the quality of the individual's life. Also, consider individual differences; some do well with little structure.
- **Simulate the home setting.** Consider structuring professional settings more like home settings.
- **Reduce strong and distracting stimuli.** These may lead to agitation, disruptive actions, aggressive behaviors, and discomfort. Minimize noise that tends to distress the person. Carpet the living areas and drape windows to minimize sound and noise that bounce off of hard surfaces. When talking to the person, turn off televisions and radios.
- **Use beneficial sounds.** To cue and calm the person, keep the sounds the person is used to. Kitchen sounds that announce dinner time, bird sounds, and music the person enjoys may soothe the person.
- **Use cues to improve awareness and orientation.** Signs, lines drawn or pasted on the floor, night lights, pictures, words, and such other cues may orient individuals to their bedrooms, bathrooms, dining rooms, and other spaces. In a professional setting, a personal item may be placed at the entrance of a person's bedroom or bathroom.
- **Establish an eating routine.** Serve food at a predictable time and in a constant place. Use such stimuli as music that signals eating time.

- **Manage lighting conditions.** Minimize direct and glaring sunlight. Arrange for even lighting in the living areas to avoid dark corners that may frighten some individuals.
- **Use contrast to enhance visual discrimination.** For instance, use tableware that contrasts with the dining table and paste color tapes to signal a change in the floor level.
- **Install safer furniture.** To minimize injury, use sturdy furniture with rounded edges instead of sharp corners.
- **Use rooms for single activities.** Avoid using a single room for multiple purposes.
- **Do not impoverish the environment.** Let individuals retain access to nonhazardous items that they would like to pick up, explore, or use (e.g., a soft ball, books or magazines, games).
- **Let the person keep personal items.** At home or in professional settings, individuals should keep their family pictures, games and other belongings, and religious or culturally significant items in their bedrooms.
- **Let the individual maintain routines and hobbies.** In a professional setting, instead of forcing an unfamiliar task (e.g., basket weaving or painting) on residents, they may be allowed to maintain their usual activities or hobbies (card playing or knitting).
- **Let the person continue to use familiar technological devices.** Persons who have been trained to use smartphones, tablets, or other devices for communication or entertainment should continue to use them.

Bayles et al. (2020) and van Hoof et al. (2010) offer additional suggestions on environmental modifications. It should be clear that such modifications should be based on an analysis of an individual's and caregivers' behaviors and the environmental setting.

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Glossary



Acceleration/deceleration injuries. Brain injuries caused by an accelerating (and then decelerating) movement of the head and the brain inside the skull.

Acetylcholine (Ach). A neurotransmitter that helps improved cognitive function and psychiatric and behavioral symptoms; offered as medical treatment for dementia.

Acute stage. The initial and often medically the most serious stage of an injury (including brain injury) or a sudden attack (such as a stroke).

Aducanumab. An anti-amyloid intravenous infusion therapy to reduce the chances of plaque formations in the brain, the main brain disorder in Alzheimer's disease.

Afferent nerves. Also known as sensory nerves; those that carry sensory impulses from the peripheral sense organs toward the brain.

Agnosia. Difficulty grasping the meaning of certain stimuli in the absence of sensory impairment.

Agrammatic speech. Characteristic of nonfluent aphasias; speech with missing grammatical elements.

Agraphia. Writing problems associated with recent brain injury.

Alexia. Reading problems due to recent neurological impairment; loss of previously acquired reading skills; contrasted with dyslexia.

Alexia with agraphia. Reading and writing problems due to recent neurological impairment.

Alexia without agraphia. Reading problems due to recent brain injury with intact writing skills.

Alternative communication. Nonverbal means of expression for persons who are extremely limited in functional verbal communication; this means may replace verbal communication.

Alzheimer's disease. A neurodegenerative disease due to the formation of amyloid (neuritic) plaques neurofibrillary tangles in the brain; leads to dementia.

Amines. Neurotransmitters that include acetylcholine (Ach), dopamine, epinephrine, histamine, norepinephrine, and serotonin.

Amino acids. Neurotransmitters that include gamma-aminobutyric acid (GABA), glutamate (Glu), and glycine (Gly).

Amnesia. Total or near total loss of memory.

Amyloid plaques. Minute areas of cortical and subcortical tissue degeneration.

Aneurysm. A balloon-like swelling of a weak and thin portion of an artery that eventually ruptures.

Angular acceleration. Movement of the head at an angle (nonlinear), causing brain injury; movement caused by an angular force.

Angular gyrus. A gyrus that lies posterior to the supramarginal gyrus; damage to the angular gyrus is associated with naming, reading, and writing difficulties and, in some cases, transcortical sensory aphasia.

Anomia. Difficulty naming objects or persons; difficulty recalling nouns during conversation; a general aphasic symptom.

Anomic aphasia. A type of aphasia with a persistent and severe naming problem with relatively intact language skills.

Anosognosia. Diminished or absent awareness of one's own illness, symptoms, or limitations.

Anoxia. Oxygen deprivation or deficiency.

Anterior cerebral artery. Supplies blood to mostly the middle portion of the frontal and parietal lobes, including the basal ganglia and corpus callosum.

Anterograde amnesia. Difficulty remembering events following traumatic brain injury; the same as posttraumatic amnesia.

Anticoagulation agents. A group of drugs that help prevent blood clotting in patients whose stroke is due to cardiac embolus.

Antiplatelet agents. Drugs that prevent the formation of the blood platelets that play a role in coagulation (clotting) of the blood.

Antipsychotic (neuroleptic) drugs. Drugs that help control such psychiatric symptoms as delusions, hallucinations, and depression.

Aorta. The main artery of the heart; carries the blood from the left ventricle to all parts of the body except the lungs.

Aphemia. Broca's term to describe language disorders associated with brain lesions; now it is known as Broca's aphasia.

Aphonia. Lack of voice; a voice disorder.

Apraxia. A movement disorder; difficulty executing sequenced and volitional movements in the absence of sensory or neuromuscular problems.

Apraxia of speech. A neurogenic speech disorder caused by difficulty in motor planning of speech and characterized by a difficulty in initiating articulation, effortful articulation, groping articulatory movements, and articulatory inconsistency in the absence of speech muscle weakness or paralysis.

Apraxic agraphia. Writing problems associated with apraxia.

Arachnoid. A thin, semitransparent, nonvascular, delicate, and weblike membrane with the dura mater above and the pia mater below; part of the meninges that covers the brain.

Arousal. General readiness to respond to external stimuli.

Arterial lines. Tubes inserted into arteries to continuously monitor blood pressure of patients.

Arterioles. Smaller branches of larger arteries.

Association fibers. Short or long fibers that connect areas within a hemisphere.

Astrocytes. A variety of glial cells of the central nervous system; small cells with varied branches that occupy the space between neurons.

Astrocytomas. Malignant tumors that arise from astrocytes.

Atherosclerosis. A slowly developing arterial disease process in which the arteries are hardened and narrowed from an accumulation of lipids and other particles.

Auditory agnosia. Failure to understand the meaning of auditory stimuli, with normal peripheral hearing.

Auditory verbal agnosia (pure word deafness). Failure to understand the meaning of spoken words that are normally heard.

Augmentative communication. Modes of nonverbal communication that enhance, expand, or augment the limited verbal communication skills of a person.

Automated speech. Such routine and overlearned speech production as reciting the days of the week, the alphabet, or the annual seasons.

Autonomic nervous system. System that regulates the internal environment of the body; mobilizes under stress.

Axon. The nerve fiber that is longer than dendrites.

Bacterial meningitis. Infected meninges and the cerebrospinal fluid.

Basal ganglia. Structures found deep within the brain; include caudate nucleus, putamen, and globus pallidus; modulate movement.

Basilar artery. The conjoined vertebral arteries.

Binswanger's disease. Also called subcortical arteriosclerotic encephalopathy; associated with leukoareosis, which is atrophy of the subcortical white matter that produces a variety of subcortical vascular dementia.

Blood-brain barrier. A mechanism that prevents the cerebral penetration of harmful chemical substances and infectious microorganisms.

Bradykinesia. A movement disorder; characterized by difficulty initiating movement and generally slow movement.

Brainstem. A structure that includes medulla, pons, and midbrain.

Broca's aphasia. A type of aphasia characterized by nonfluent, effortful, and agrammatic but generally meaningful language production with relatively better auditory comprehension; may be associated with dysarthria.

Broca's area. The left, lower, and posterior portion of the frontal lobe on the inferior frontal gyrus at the juncture of the lateral and central fissures; concerned with speech production.

Buccofacial apraxia. Difficulty in performing buccofacial movements when requested.

Carotid endarterectomy. A surgical procedure in which a thrombus formed in the common or internal carotid artery is removed to improve the blood flow to the brain.

Carotid phonoangiography. A method of assessing the health of the carotid arteries by the characteristics of the sound generated by the blood gushing through the arteries.

Caudal. The portion that refers to the lower or tail section of the spinal cord; it may also refer to the back part of the brain.

Central nervous system. Includes the brain and the spinal cord, both encased in bone.

Cerebellum. A major portion of the hindbrain; regulates motor movements.

Cerebral angiography. A radiographic procedure in which radiopaque contrast material is injected into selected arteries, typically the femoral artery in the groin.

Cerebral toxemia. Cerebral tissue damage resulting from poisoning of the brain.

Cerebral vasospasm. Constriction of the muscular layer surrounding blood vessels.

Cerebral ventricles. A system of interconnected cavities deep within the brain; contain the cerebrospinal fluid.

Cerebrospinal fluid (CSF). Protective fluid within the ventricles of the brain.

Cerebrovascular accidents (CVAs). Strokes; frequent and immediate cause of aphasia.

Cerebrum. The cerebral cortex; the final integrative and executive structure of the nervous system.

Cholinesterase inhibitors. Drug treatment for dementia to prevent the breakdown of acetylcholine, a neurotransmitter.

Chorea. The major neurologic symptom of Huntington's disease; irregular, spasmodic, involuntary movement of the limbs, neck, head, and facial muscles.

Circle of Willis. The anastomosed form of the two carotid and the two vertebral arteries; a common (redundant) blood supply to various cerebral arterial branches.

Circumlocution. Talking in a roundabout manner; beating around the bush instead of expressing directly.

Client-specific measures. Measures of skills or target behaviors that are specific to an individual client.

Cluttering. A disorder of fluency characterized by excessively fast rate and increased number of dysfluencies.

Coagulation. Clotting of the blood.

Cognition. Complex of intellectual functions, including knowledge, memory, and the presumed modes of information processing in the brain.

Coma. A state in which the patient is unconscious and unresponsive to most or all external stimulation.

Commissural fibers. Band of fibers that connect the corresponding areas of the two hemispheres.

Commissurotomy. Surgical severance of the corpus callosum.

Conduction aphasia. A type of aphasia characterized by paraphasic fluency, good comprehension, and impaired repetition.

Confrontation naming. Naming objects or persons when asked to; difficult for patients with aphasia.

Constructional impairment. Difficulty performing such tasks as copying a block design or assembling a car model.

Contrecoup. Injury to the brain at the site that is opposite to the initial impact; caused by a moving brain that hits the inside portion of the skull.

Conversation. Verbal exchange between two or more persons who share roughly equal time talking and listening; see *discourse* and *narration*.

Coronal plane. A vertical cut, resulting in two halves of the brain or other anatomic structure.

Corpus callosum. A broad band of fibers at the base of the hemispheres; connects the two hemispheres.

Coup injury. Brain injury at the point of impact trauma, caused by the compression of the skull at that point.

Cranial nerves. Twelve pairs of nerves that are attached to the base of the brain and emerge from the brainstem; they innervate larynx, tongue, pharynx, and muscles of face, neck, and head.

Craniocerebral trauma. The same as traumatic brain injury.

Creutzfeldt-Jakob disease. A neurological disease associated with diffuse and varied loss of neurons in many cortical areas, the basal ganglia, the thalamus, the brainstem, and the spinal cord; may lead to dementia.

Crossed aphasia. A rare form of aphasia in right-handed individuals who sustain a single right hemisphere lesion.

Cues. A variety of prompting stimuli used to teach naming responses; see *phonetic cues* and *semantic cues*.

DAT-SPECT. Combination of DAT (dopamine-transporter) with SPECT imaging methods to study the brain structures and functions; see *single-photon emission computed tomography*.

Delirium. Impaired consciousness associated with cognitive deficits.

Delusions of persecution. False but stubborn belief that one's own family members and others are conspiring to harm one's self.

Dementia. A significant decline from the typical levels in personal and social behavior commonly associated with neurological diseases in older persons.

Dementia pugilistica (punch drunk). A variation of posttraumatic dementia; seen in professional boxers who sustain repeated head injuries.

Dendrites. Short, unmyelinated nerve fibers that extend from the cell body.

Diencephalon. A structure in between the brainstem and the cerebral hemispheres.

Diffuse axonal injury. Twisting and tearing damage to cerebral axons due to angular movement of the head.

Diffuse vascular injury. Small and widespread ruptures in the brain's blood vessels.

Diffusion MRI fiber tractography. Brain imaging technique to visualize the subcortical white matter fiber pathways that connect different parts of the brain.

Diffusion-weighted MRI. A magnetic resonance imaging procedure that constructs pictures of structures based on its detection of microscopic motion of water protons in the brain tissue.

Disconnection syndromes. Problems in movement, reading, and naming due to damage to the corpus callosum that disconnects the two cerebral hemispheres.

Discourse. Brief talk on an objective topic with minimum audience participation; see *conversation* and *narration*.

Discriminative stimuli. Special stimuli that help evoke target skills (e.g., a colored border on the left margin of a printed page to force attention to the left side of the page).

Disorientation. Confusion about time, space, other persons, and self; a symptom of brain injury.

Disprosody. Impaired pitch and loudness variations and other prosodic features of speech.

Distal. Structures that are farther from a reference structure.

Dominant bilingual. A person who speaks two languages but speaks one with greater proficiency than the other.

Donepezil. A type of cholinesterase inhibitor drug to treat persons with Alzheimer's dementia.

Dopamine. An inhibitory neurotransmitter produced especially by the neuronal cells in substantia nigra of basal ganglia; reduced dopamine is a suspected cause of several symptoms found in Parkinson's disease.

Doppler ultrasonography. Also called ultrasound; helps assess arterial health by measuring the velocity of blood flow.

Dorsal. Refers to the back portion, specifically to the portion that lies in between the superior and posterior portions of the brain (or other structures).

Dorsal stream. A set of subcortical white matter connecting fibers in the brain including the arcuate and the superior longitudinal fasciculi mediating speech production. See also *dual-stream model* and *ventral stream*.

Dual-stream model. A theory that there are two interacting neural pathways in the brain that connect different regions in the brain and mediate language functions; see also *dorsal stream* and *ventral stream*.

Dura mater. A tough and thick membrane with one side adhering to the skull and the other side to the arachnoid; part of the meninges that protect the brain.

Dysarthria. A neurogenic speech disorder caused by muscular weakness that affects respiratory, phonatory, articulatory, resonance, and prosodic features of speech.

Dyslexia. Children's difficulty in learning to read, even though the instruction was adequate.

Dysphagia. Disorders of swallowing.

Echolalia. Repeating what is heard; a parrot-like imitative response.

Edema. Swelling of tissue.

Efferent nerves. Also known as motor nerves; those that transmit impulses away from the central nervous system.

Elderspeak. A stereotypic and demeaning way of speaking to older persons with slow rate and exaggerated articulatory movements and prosodic features.

Electromyography. A method of studying muscle functions by recording the electrical potential the nerves generate when they contract.

Embolism. An arterial disease in which a moving or traveling fragment of arterial debris blocks a small artery through which it cannot pass.

Embolus. A traveling mass that may have been formed farther away from the place where it occludes a vessel.

Empty speech. Generally fluent and grammatically correct speech that does not make sense because of extensive paraphasias.

Environmental modifications. Changing the living conditions of a persons with dementia to reduce distress and increase safety and comfort.

Epidural (extradural) hematoma. Accumulation of blood between the dura mater and the skull.

Epileptic focus. The damaged or pathological brain area that triggers epileptic attacks.

Errorless learning. Behavior shaping in small and gradual steps with frequent modeling so that the learner makes no errors; used in *spaced retrieval method*.

External carotid artery. A branch of the common carotid artery; supplies blood to muscles of the face and neck, nasal and oral cavities, sides of the head, skull, and dura matter.

External memory aid. External aids (such as a set of index cards or a book) that contain information about what to do when.

Fissure of Rolando (central sulcus). A major fissure that runs laterally (from one side of the brain to the other), downward, and forward.

Fissures. Furrows in the brain.

Frontal alexia. Reading skills in patients who have suffered damage to areas in the frontal cortex.

Frontotemporal dementia. A form of dementia associated with degeneration in the right and the left frontal lobe, temporal lobe, or both the lobes with behavioral, semantic, logopenic, and primary progressive aphasia variants; also known as frontotemporal lobar degeneration.

Functional magnetic resonance imaging (fMRI). A procedure in which a contrast material is intravenously administered to the patient before MRI to detect changes in cerebral blood flow as the patient experiences different states or performs different activities.

GABA (gamma-amino butyric acid). An inhibitory neurotransmitter that is reduced in some neurodegenerative diseases, including Huntington's disease.

Galantamine. A type of cholinesterase inhibitor drug to treat persons with Alzheimer's dementia.

Geographic disorientation. Confusion about one's own geographic location.

Glial cells. Nonneuronal cells, also known as glia for glue or neuroglia; provide a structural framework for the neural cells.

Glioblastoma multiforme. The most malignant of tumors in the glia; associated with a high death rate.

Gliosis. An excess accumulation of astrocytes (a neuroglia cell) in the atrophied regions of the brain in patients who have variants of frontotemporal dementia.

Global aphasia. The most severe form of aphasia, which has a generalized (global) effect on communication skills.

Glutamate. An excitatory chemical in the brain.

Gyri. Elevated masses on the surface of the brain.

Hemineglect. Tendency to neglect one side of his or her body; common in patients with right hemisphere syndrome.

Hemorrhagic strokes. Strokes that result from ruptured cerebral blood vessels causing cerebral bleeding.

High-frequency deep brain stimulation (DBS). Implantation of electrodes in target brain cells; the electrodes connected to an externally controllable pulse generator placed over the chest wall just below the skin; a treatment procedure for patients with Parkinson's disease.

High-velocity injuries. Brain injuries caused by military weapons, rifles, and other automatic assault weapons.

HIV-associated neurocognitive disorder. A form of progressive dementia caused by the human immunodeficiency virus (HIV) infection; also known as HIV encephalopathy; formerly known as AIDS dementia complex.

Holistic approach. The view that the brain functions as an integrated unit and that a lesion in one area affects functions of most if not all areas; contrasts with the localizationist view.

Horizontal plane. Sectioning brain (or other anatomic structures) into two halves.

Human immunodeficiency virus encephalopathy. The same as AIDS dementia complex.

Huntington's disease. A neurodegenerative disease associated with loss of neurons primarily in the basal ganglia (especially in the caudate nucleus and the putamen).

Hydrocephalus. Accumulation of the cerebrospinal fluid that causes a dilation of the cerebral ventricles; in children, may be associated with mental retardation.

Hyperfluency. Extremely fluent and flowing speech that may be more or less meaningless.

Hyperkinetic agraphia. A disorder of writing associated with tremors, tics, chorea, and dystonia.

Hypokinetic agraphia (micrographia). A motor disorder of writing with unusually small letters or letters that get progressively smaller in a piece of writing.

Hypotension. Reduced blood pressure.

Idiom. Language- and culture-specific sayings that mean something other than what the words suggest (e.g., *once in a blue moon*).

Impression (impact) trauma. Trauma to a structure (such as the head) at the point of initial contact with a striking object.

Incidence. The number of new cases of a disease or disorder counted on an annual basis per 100,000 persons or any other specified number.

Incontinence. Lack of control on urinary function.

Insight. Awareness of one's own limitations, disabilities, or symptoms.

Internal carotid artery. A branch of the common carotid artery; the major blood supplier to the brain.

Intracerebral hematoma. Accumulation of blood within the brain.

Intracranial hematoma. Accumulation of blood from hemorrhage within the skull or the brain.

Intracranial neoplasms (tumors). Pathological growths within the cranial structures.

Intraoperative language mapping. Locating the brain regions involved in speech and language by electrical stimulation to spare them during neurosurgery.

Intravenous thrombolytic therapy. A form of drug treatment for stroke patients; known as recombinant tissue plasminogen activator (t-PA).

Ischemia. Interrupted blood supply and the resulting deprivation of oxygen.

Ischemic penumbra. A tissue region surrounding the major locus of infarction.

Ischemic strokes. Cerebrovascular accidents caused by occlusive vascular disorders that block or interrupt arterial blood flow to a region of the brain; result in oxygen deprivation that causes infarction.

Jargon. Invented, meaningless nonwords; a symptom of aphasia.

Lacuna. A hollow space, cavity, or gap in an anatomic structure.

Lacunar states. Hollow spaces in the brain that are associated with a variety of vascular dementias.

Lateral cerebral fissure (sulcus). A deep fissure that starts at the lower (inferior) frontal lobe at the base of the brain and moves laterally and upward; also known as the sylvian fissure.

Lateral plane. Structures that lie away from the median plane.

Left neglect. A condition in which the patient is unaware of objects and persons on the left side; often associated with right hemisphere injury.

Lesion-symptom mapping. Correlating brain lesion sites with observed symptoms through high-resolution brain imaging techniques.

Levodopa. A neurotransmitter that is the most commonly prescribed drug treatment for patients with Parkinson's disease; helps movement disorders.

Lewy bodies. Intraneural inclusion granules typically found in the basal ganglia, brainstem, spinal cord, and sympathetic ganglia of patients with Parkinson's disease and those diagnosed with Lewy body dementia.

Lewy body dementia. A form of dementia caused by Lewy bodies; characterized by visual and auditory hallucinations, paranoid thoughts, and mild features of Parkinson's disease.

Linear acceleration. The straight-line movement of the head when a force strikes the head midline.

Localizationist. One who holds the view that particular behavioral functions are strictly controlled by specific structures within the brain.

Logoclonia. Repeating the final syllable of words.

Longitudinal cerebral fissure. A fissure that separates the left and the right hemispheres of the cerebrum.

Low-velocity injuries. Brain injury produced by such agents as an arrow, nail gun, knife, and other such projectiles.

Lumbar puncture. A method of diagnosing infections or hemorrhages in the central nervous system by an analysis of a sample of cerebrospinal fluid.

Magnetic resonance angiography (MRA). A procedure in which the rate and velocity of blood supply to the selected cerebral structures are measured.

Magnetic resonance imaging (MRI). A method of generating pictures of brain structures with the help of a powerful magnetic field that alters the electrical activity of the brain.

Magnetoencephalography (MEG). A high-resolution brain imaging procedure to determine the lateralization of language (usually the left hemisphere).

Median plane. A longitudinal (vertical) section that divides the brain (or other anatomic structures) into a right half and a left half.

Medulla. Also known as medulla oblongata and myelencephalon; the upward extension of the spinal cord.

Memantine. A drug for dementia that regulates glutamate, an excitatory chemical in the brain.

Memory wallets. A set of index cards with written information that stimulates conversation in persons with memory impairments; also known as memory books.

Meningiomas. Tumors that grow within the cerebral meninges.

Metaphor. Expressions that equate two unrelated events, objects, or persons (e.g., *she is a shining star*).

Metastatic intracranial tumors. Tumors that have grown elsewhere in the body but have migrated into the brain, where they begin to grow.

Microglia. A type of CNS glial cells that are elongated, dark staining, with long and branched cytoplasmic processes (thin projections).

Midbrain. Also called mesencephalon; lies above the pons.

Middle cerebral artery. The biggest branch of the internal carotid artery; supplies the entire lateral surface of the cortex, including the major regions of the frontal lobe.

Mild cognitive impairment (amnestic variety). Decline in memory not attributable to normal aging, with intact reasoning, daily living, and self-help skills; an early sign of dementia.

Mild cognitive impairment (nonamnestic variety). Mild decline in language and executive functions not attributable to normal aging; also an early sign of dementia.

Mixed transcortical aphasia. A type that combines symptoms of transcortical motor aphasia and transcortical sensory aphasia (a fluent form of aphasia) with language impairment but retained repetition skills.

Modeling. A clinician's production of a correct response that may lead to imitation by the client.

Motor agraphia. A group of writing disorders that are due to neuromotor problems.

Motor nerves. Neurons that cause muscle contractions (movements) or glandular secretions.

MRI spectroscopy. A scanning method to create images of brain structures by detecting biochemical composition of tissue.

Multi-infarct dementia (MID). A form of dementia caused by multiple strokes, often in bilateral regions of the brain; a form of vascular dementia.

Multiple bilateral cortical infarcts. Repeated strokes in both hemispheres that produce a variety of mixed vascular type of dementia with both cortical and subcortical pathology.

Mutism. Total or near-total lack of verbal expression; sign of severe neurological impairment and some psychiatric disorders.

Myelin. A white, protective, and fatty material that wraps around the nerve fibers.

Narration. Storytelling; telling personal, read, or heard stories; see *conversation* and *discourse*.

Necrosis. Damaged cells.

Nerves. Bundles of axons, dendrites, or both that specialize in certain functions.

Neuritic (senile) plaques. Minute areas of cortical and subcortical tissue degeneration; found in patients with Alzheimer's disease.

Neuroanatomy. The study of structures of the nervous system.

Neurodiagnostic methods. Mostly medical methods of diagnosing neural pathology and its various effects.

Neurofibrillary tangles. Twisted and tangled neurofibrils; found in the brain of patients with Alzheimer's disease.

Neurofibrils. Filamentous structures in the nerve cell's body, dendrites, and axons.

Neurogenic stuttering. A fluency disorder found in some patients with cerebral diseases or trauma; may or may not be associated with aphasia.

Neurohistology. The study of the basic structures of neural cells, tissue, and organs in relation to their function.

Neurology. Medical study of neurological diseases and disorders, their diagnosis, and their treatment.

Neurons. Nerve cells.

Neurophysiology. A branch of neurology; study of the function of the nervous system.

Neurotransmitters. Chemical compounds within the axon terminal buttons that help transmit information across the synaptic space.

Nonacceleration injuries. Brain injury that occurs when a moving object hits a restrained (stationary) head.

Nonpenetrating injury. Closed head brain injuries in which the skull may be fractured, but the meninges remain intact.

Occupational therapists. Specialists concerned with daily living activities including self-care (cooking, bathing, dressing, driving), safety, and other such functional skills.

Oligodendrocytes. A form of glial cells; found in both the gray and white matters of the central nervous system.

Oligodendrogliomas. Tumors found in the oligodendrocytes; typically found in the adult frontal lobes.

Oncologists. Cancer specialists.

Palilalia. Repeating one's own utterances.

Pallidotomy. Ablation of the internal segments of the globus pallidus; a form of surgical treatment for patients with Parkinson's disease.

Paraphasia. Errors of speech characterized by substitution of wrong sounds or words for target sounds or words.

Parkinson's disease. A neurological disease associated with nerve cell deterioration in basal ganglia and the brainstem; the dopamine-producing cells are especially affected.

Penetrating brain injuries. An open wound in the head due to some crushing or penetrating agent that results in fractured or perforated skull, torn brain coverings (meninges), and various degrees of brain tissue damage.

Peptides. Neurotransmitters that include cholecystokinin (CCK), dynorphin, neuropeptide Y, and somatostatin.

Peripheral nervous system. Includes all of the nervous system except for the brain and the spinal cord.

Phonetic cue. The clinician's production of the first letter or sound of the target word as a cue to the patient (e.g., "the word starts with a /b/" to prompt the production of ball).

Phrenology. A pseudoscience that correlated mental and intellectual skills with the shape and size of a person's skull.

Physical therapists. Specialists mostly concerned with improving the physical status of the patient by implementing various physical exercises and other programs to enhance physical strength, endurance, range of motion, and general mobility and balance.

Pia mater. A thin, delicate, and transparent membrane that adheres to the brain surface; the part of the meninges that cover the brain.

Pick bodies. Dense intracellular formations in the neuronal cytoplasm, especially in nonpyramidal cells in the cerebral layers 2, 3, and 6; neuropathological feature found in frontotemporal dementia, including Pick's disease.

Pick cells. Ballooned, inflated, or enlarged neurons, especially in the lower and middle cortical layers; neuropathological feature found in Pick's disease (the behavioral variant of frontotemporal dementia).

Pick's disease. A neurodegenerative disease associated with Pick cells and Pick bodies in the frontal and temporal lobes; a variety of frontotemporal dementia.

Platelets. A small, disk-shaped structure in the blood that is mainly responsible for coagulation.

Pons. A part of metencephalon (which includes cerebellum, not a part of the brainstem); a bridge to the hemispheres of the cerebellum.

Positron emission tomography (PET). A nuclear medical procedure to study activation of brain structures associated with differences in cerebral metabolic rates.

Postanoxic dementia. Dementia due to chronic anoxia (oxygen deficiency).

Posterior. The back portion of the brain (or other structure).

Posterior cerebral arteries. Arteries that supply blood to the lower and lateral portions of the temporal lobes and the middle and lateral portions of the

occipital lobes.

Posttraumatic memory loss. Difficulty remembering events following traumatic brain injury; the same as anterograde amnesia.

Pretraumatic memory loss. Difficulty remembering events that preceded traumatic brain injury; also known as retrograde amnesia.

Prevalence. A head count of the number of people in a defined population that have a disorder or a disease at any given time.

Primary auditory cortex. A structure at the border of the superior temporal gyrus and the lateral fissure; concerned with hearing.

Primary focal lesions (injury). Localized brain lesions.

Primary intracranial tumors. Tumors that originate in the brain.

Primary motor cortex. Mostly the precentral gyrus in the frontal lobe that controls voluntary movements of skeletal muscles on the opposite (contralateral) side of the body.

Primary progressive aphasia. A group of neurodegenerative diseases that begin with the language symptoms of aphasia and advances to dementia with nonfluent, semantic, and logopenic variants.

Prion. A type of cerebral cellular protein that gets misshapen and causes the transmissible and fatal Creutzfeldt-Jakob disease and dementia.

Perfusion-weighted magnetic resonance imaging. A scanning method that produces images based on blood flow variations.

Progressive supranuclear palsy (PSP). A degenerative neurological disorder whose symptoms are similar to those found in Parkinson's disease; associated with atrophy of cells in the basal ganglia.

Projection fibers. Band of fibers that transmit sensory information to the brain and motor information to the muscles and glands.

Prosopagnosia. Failure to recognize familiar faces; part of the right hemisphere syndrome.

Prospective memory. Remembering to do certain things at particular times.

Proximal. Structures that are relatively close to a reference structure.

Pseudodementias. Dementias associated with such psychiatric disorders as depression and schizophrenia.

Pulse oximeter. A small, clamp-like instrument attached to a patient's toe, finger, or earlobe to measure the blood oxygen levels.

Pure agraphia. An isolated writing disorder, with all other language functions, including auditory comprehension and being normal or nearly so.

Random (unrelated) paraphasia. Substitution of words that are semantically or phonetically unrelated to target (intended) words; seen in patients with aphasia.

Rapidly progressive dementia. An atypical form of dementia that takes less than a few months or up to 2 years from symptom onset to dementia.

Recreation therapists. Specialists who manage the daily activities of patients in rehabilitation settings.

Reduplicative paramnesia. Belief in the existence of multiple and identical persons, places, and body parts; rarely found in patients with right hemisphere syndrome.

Regional cerebral blood flow (rCBF). A technique to assess the amount of blood flow in different areas of the brain.

Reliability. Consistency of repeated measures.

Reminiscence therapy. Narrating past events in life; a form of communication therapy for persons with dementia.

Respite care. Care service offered to give temporary relief to family members taking care of a person with dementia and other chronic diseases.

Retrograde amnesia. Difficulty remembering events that preceded traumatic brain injury; the same as pretraumatic memory loss.

Retrospective memory. Remembering past events.

Reversible dementia. Symptoms that mimic those of neurodegenerative diseases but are treatable and reversible as in hypothyroidism, depression, or drug induced.

Reversible ischemic neurological deficit (RIND). Also known as minor strokes; symptoms may last more than 24 hours, but the patient may

recover completely or nearly so.

Rivastigmine. A type of cholinesterase inhibitor drug to treat persons with Alzheimer's dementia.

Rostral or anterior. Specific to the brain, the portion that is in the frontal part of the head.

Sagittal plane. Sectioning the brain with a longitudinal cut into unequal left and right portions.

Schizophrenia. A severe form of psychiatric disorder in which the person is not oriented to reality; hallucinations and delusions characterize the disorder.

Schwann cells. The myelin sheath of the peripheral nerves.

Script therapy. A form of therapy for aphasia and dementia; scripts are connected sentences that describe a personal experience.

Secondary intracranial tumors. The same as metastatic tumors.

Selective attention. Paying attention to specific stimuli while ignoring others.

Semantic cue. Clinician's production of a word that serves as a prompt for the target word in teaching naming (e.g., the clinician may say "it is the opposite of husband" to prompt the word *wife*).

Sensory aphasia. The same as Wernicke's aphasia.

Sensory nerves. The nerves that carry sensory impulses from the peripheral sense organs toward the brain. Because they carry information toward the center, sensory nerves also are known as **afferent nerves**.

Shunt. A tube inserted into a structure in the body to perform such functions as draining excessive fluid and keeping a structure open (such as an artery).

Single-photon emission computed tomography (SPECT). A nuclear imaging procedure that produces three-dimensional images of the brain.

Sinuses. Channels through which the blood or other fluids flow.

Spaced retrieval. A procedure in which patients with dementia are taught to recall a piece of information over progressively longer intervals with few or no errors; a behavior-shaping method.

Spinal cord. A cylindrical bundle of nerve fibers within the vertebral column; a caudal (lower) continuation of the medulla oblongata.

Spinal nerves. Thirty-one pairs of nerves that arise from the spinal cord.

Stem cells. Precursor cells that can differentiate into virtually any kind of body cell.

Stereotactic surgery. A surgical method aimed at subcortical structures; used to map and treat subcortical abnormalities.

Stroke. A syndrome with acute onset and resulting in focal brain damage; caused by disturbed cerebral blood circulation; the same as cerebrovascular accidents.

Stupor. A state in which the patient is generally unresponsive, but pain or another strong stimulus may arouse the patient for a brief period.

Stuttering. A disorder of fluency characterized by excessive amounts or durations of dysfluencies.

Subacute stage. A period following the acute stage when the patient is stabilized and is ready for various rehabilitation efforts; a period that facilitates functional recovery.

Subdural hematoma. Accumulation of blood between the dura and the arachnoid.

Superior. The top portion of the brain (or other structure).

Synapse. The point at which two neurons come in contact with each other; a neural junction.

Tactile agnosia. A disorder in which the patients cannot recognize objects they touch and feel when blindfolded (they do not see the objects) or hear the sounds the objects make.

Telerehabilitation. Clinical services the specialists offer online and the clients accessing them from home.

Thalamotomy. Surgical removal of the ventral intermediate nucleus of the thalamus; reduces the frequency and magnitude of tremors of patients with Parkinson's disease.

Thalamus. The largest of the diencephalon structures; integrates sensory experiences and relays them to cortical areas.

Thrombosis. A vascular disease involving the formation of a thrombus.

Thrombus. A blood clot that restricts blood supply to structures beyond.

Tissue plasminogen activator (tPA). A thrombolytic drug that helps dissolve blood clots in persons with ischemic stroke.

Topic maintenance. Talking on the same topic for an extended and socially appropriate duration; a pragmatic language skill.

Topographic disorientation. Confusion about space; sometimes found in patients with brain injury.

Transcortical direct current stimulation (tDCS). A noninvasive procedure of stimulating the brain with weak electrical currents through a pair of surface electrodes placed on the scalp.

Transcortical motor aphasia. A variety of nonfluent aphasias characterized by agrammatic, effortful, telegraphic speech with preserved repetition skills.

Transcortical sensory aphasia. A type of aphasia characterized by fluent, well-articulated, paraphasic, somewhat echolalic, empty speech in the context of poor auditory comprehension.

Transcranial magnetic stimulation (TMS). Placing an electromagnetic coil in the scalp to induce a magnetic field in the brain of stroke patients to stimulate language activity.

Transient ischemic attack. “Ministrokes” whose symptoms last only a brief period (usually 30 minutes or less) without permanent effects; may be the warning signs of more serious strokes that produce more lasting effects.

Traumatic brain injury. Injury to the brain sustained by physical trauma or external force (not a result of disease or stroke).

Trisomy 21. A genetic condition in which three free copies of chromosome 21 cause a type of mental retardation called Down syndrome.

Tumors. Space-occupying lesions that cause swelling in, and increased pressure on, the surrounding tissue.

Turn taking. Appropriately playing the role of a listener and talker in conversation.

Uremic encephalopathy. A form of dementia caused by chronic renal (kidney) failure.

Validity. Measures that reflect the skill intended to be measured.

Vascular dementia. A form of dementia caused by a variety of diseases that affect the cerebral vascular system.

Veins. A system that drains deoxygenated blood from organs and carries it back to the heart and to the lungs to have the blood reoxygenated.

Ventilator. Also called a *respirator*, an instrument that maintains breathing for a patient who is unable to breathe without external assistance.

Ventral. Lower portion of the brain.

Ventral stream. A large number of connecting fibers in the brain that mediate speech-language comprehension. See also *dual-stream model* and *dorsal stream*.

Venues. Minute branches of veins.

Vertebral arteries. Branches of the two subclavian arteries that emerge from the aortic arch.

Vigilance. Heightened attention to detect changes in stimuli.

Virtual reality therapy. Computer programs that simulate real-life situations in which persons with aphasia and other disorders may become personal avatars and role-play.

Visual agnosia. A rare disorder in which the meaning of objects seen normally is not understood.

Voxel-based morphometry. A type of structural MRI to localize language functions in extremely small areas (voxels) of the brain.

Watershed areas of the brain. A region of the brain that receives blood from the small end- branches (terminal branches) of all three primary arteries that supply blood to the brain — the anterior, middle, and the posterior cerebral arteries; the blood supply could be somewhat inefficient because the region is farther away from the origins of the primary cerebral arteries.

Wernicke's aphasia. A type of aphasia characterized by fluent, paraphasic, empty, and grammatically correct verbal expressions associated with poor auditory comprehension.

Wernicke's area. The posterior two thirds of the superior temporal gyrus in the left (or dominant) hemisphere; concerned with comprehension and formulation of speech.

Yaw. The tendency of moving objects to change their course; objects that penetrate the skull and change their course as they move inside the brain cause severe damage.

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AN INNOVATIVE TEXTBOOK AND COURSE NOTEBOOK IN ONE RESOURCE!

A Coursebook on Aphasia and Other Neurogenic Language Disorders, Fifth Edition is a textbook for courses in aphasia and other neurogenic communication disorders. It is organized in a unique and interactive “coursebook” format that divides pages into columns with written information next to columns with space for note taking. This allows instructors to make lecture notes and students to write class notes on the right half of each page of the text.

The *Coursebook* offers a comprehensive description and critical review of basic and applied research on aphasia, right hemisphere disorder (RHD), traumatic brain injury (TBI), and dementia—the four major language and communication disorders associated with neurological pathologies. The relationship between the brain and language, major features of aphasia and other disorders, their assessment, and treatment have been described in streamlined and clinician-friendly language. Critical review of theories, assessment, and treatment research helps speech-language pathologists distinguish valid from the questionable in the professional and scientific literature. All assessment and treatment chapters give an outline of comprehensive and practical procedures, integrating current practices that clinicians might readily use.

New to the Fifth Edition:

- Part I has been restructured under the heading, “Brain and Language” to describe the neuroanatomical bases of language and language impairments associated with neuropathological variables. The chapter on neurodiagnostics has been updated to include a variety of surgical, radiological, and imaging procedures that help students understand the relationship between the brain structure and function and their involvement in language production and comprehension.
- Part II has been reorganized into three chapters on aphasia. Chapter 3 offers a comprehensive review of aphasia prevalence, definition, and classification. Newer perspectives on intraoperative cortical brain mapping and alternative classification of aphasia, based on recent research on the dual stream hypotheses related to brain and language, have been reviewed with critical evaluation to help clinicians. Chapters 4 and 5 on assessment and treatment of aphasia offer a more comprehensive review of established and newer procedures. The chapter on treatment is

expanded to include telerehab, drug treatment, brain stimulation, and technologically-based interventions. All major language intervention techniques are reviewed with outlined recommendations for clinicians.

- Part III on RHD consists of two revised chapters reflecting current terminology, research, and clinical practice issues. Sections on neglect, deficit awareness, social communication, and abstract language intervention have been updated.
- Part IV offers the most recent research on TBI rehabilitation. Research on teaching compensatory strategies, group therapy, and community reentry has been updated with clinical recommendations.
- Part V's two chapters on dementia include new information on changing incidence and prevalence patterns of dementia, infectious and rapidly progressive dementias, frontotemporal dementias, primary progressive aphasia, and HIV-associated neurocognitive disorders.

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